

(12) **United States Patent**  
**Katine et al.**

(10) **Patent No.:** **US 12,402,324 B2**  
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **CROSS-POINT MAGNETORESISTIVE  
RANDOM MEMORY ARRAY AND METHOD  
OF MAKING THEREOF USING  
SELF-ALIGNED PATTERNING**

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(US); **Lei Wan**, San Jose, CA (US)

(73) Assignee: **Sandisk Technologies, Inc.**, Milpitas,  
CA (US)

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patent is extended or adjusted under 35  
U.S.C. 154(b) by 723 days.

(21) Appl. No.: **17/654,773**

(22) Filed: **Mar. 14, 2022**

(65) **Prior Publication Data**  
US 2022/0223650 A1 Jul. 14, 2022

#### Related U.S. Application Data

(60) Continuation-in-part of application No. 17/477,958,  
filed on Sep. 17, 2021, now Pat. No. 11,631,716,  
(Continued)

(51) **Int. Cl.**  
**H01L 27/22** (2006.01)  
**G11C 11/16** (2006.01)  
(Continued)

(52) **U.S. Cl.**  
CPC ..... **H10B 61/10** (2023.02); **H10N 50/01**  
(2023.02); **H10N 50/80** (2023.02)

(58) **Field of Classification Search**  
CPC .. H01L 27/224; H01L 27/228; G11C 11/1655  
See application file for complete search history.

(56) **References Cited**

#### U.S. PATENT DOCUMENTS

7,474,723 B2 1/2009 Oyama  
7,502,211 B2 3/2009 Gill  
(Continued)

#### FOREIGN PATENT DOCUMENTS

JP 5585212 B2 9/2014  
KR 20130129046 A 11/2013  
(Continued)

#### OTHER PUBLICATIONS

Notification of Transmittal of the International Search Report and  
Written Opinion of the International Search Authority for Interna-  
tional Patent Application No. PCT/US2019/068899, mailed Apr. 26,  
2020, 12 pages.

(Continued)

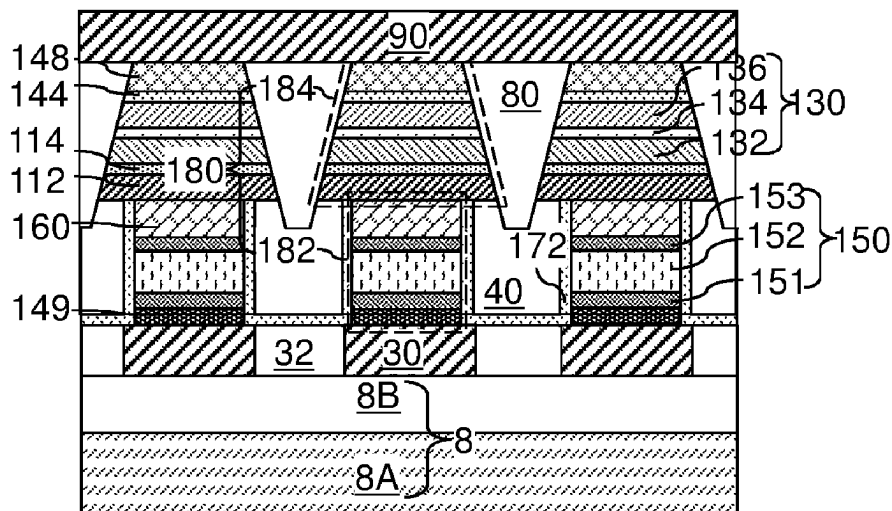
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GROUP PLLC

(57) **ABSTRACT**

A memory device includes a cross-point array of magne-  
toresistive memory cells. Each magnetoresistive memory  
cell includes a vertical stack of a selector-containing pillar  
structure and a magnetic tunnel junction pillar structure. The  
lateral spacing between neighboring pairs of magnetoresis-  
tive memory cells may be smaller along a first horizontal  
direction than along a second horizontal direction, and a  
dielectric spacer or a tapered etch process may be used to  
provide a pattern of an etch mask for patterning first elec-  
trically conductive lines underneath the magnetoresistive  
memory cells. Alternatively, a resist layer may be employed  
to pattern first electrically conductive lines underneath the  
cross-point array. Alternatively, a protective dielectric liner  
may be provided to protect selector-containing pillar struc-  
tures during formation of the magnetic tunnel junction pillar  
structures.

**19 Claims, 116 Drawing Sheets**



**Related U.S. Application Data**

which is a division of application No. 16/666,967, filed on Oct. 29, 2019, now Pat. No. 11,152,425, application No. 17/654,773, filed on Mar. 14, 2022 is a continuation-in-part of application No. 17/590,561, filed on Feb. 1, 2022, now Pat. No. 11,765,911, which is a continuation of application No. 16/401,172, filed on May 2, 2019, now Pat. No. 11,271,035, application No. 17/654,773, filed on Mar. 14, 2022 is a continuation-in-part of application No. 17/354,541, filed on Jun. 22, 2021, now Pat. No. 11,882,706, which is a division of application No. 16/460,820, filed on Jul. 2, 2019, now Pat. No. 11,056,534.

(60) Provisional application No. 62/867,590, filed on Jun. 27, 2019.

**(51) Int. Cl.**

**H01L 43/02** (2006.01)  
**H01L 43/12** (2006.01)  
**H10B 61/00** (2023.01)  
**H10N 50/01** (2023.01)  
**H10N 50/80** (2023.01)

**(56) References Cited****U.S. PATENT DOCUMENTS**

7,733,685 B2 6/2010 Scheuerlein et al.  
 9,252,362 B2 2/2016 Pio  
 9,542,987 B2 1/2017 Naik et al.  
 9,768,229 B2 9/2017 Braganca et al.  
 9,830,966 B2 11/2017 Mihajlovic et al.  
 10,229,723 B1 3/2019 Choi et al.  
 10,249,683 B1 4/2019 Lille et al.  
 10,381,551 B1 8/2019 Lille  
 10,553,783 B2 2/2020 Lille  
 10,726,892 B2 7/2020 Le et al.  
 10,788,547 B2 9/2020 Kalitsov et al.  
 10,797,227 B2 10/2020 Le et al.  
 10,811,596 B2 10/2020 Le et al.  
 10,862,022 B2 12/2020 Le et al.  
 11,056,534 B2 7/2021 Wan et al.  
 11,152,425 B2 10/2021 Wan et al.  
 11,271,035 B2 3/2022 Wan et al.  
 11,271,036 B2 3/2022 Lille et al.  
 2005/0105325 A1 5/2005 Haneda et al.  
 2005/0220150 A1 10/2005 Oyama  
 2009/0290406 A1 11/2009 Xia et al.  
 2009/0290409 A1 11/2009 Xia et al.  
 2010/0054027 A1 3/2010 Xia  
 2010/0110776 A1 5/2010 Yoon et al.  
 2011/0267874 A1 11/2011 Ryu et al.  
 2013/0201757 A1 8/2013 Li et al.  
 2013/0215675 A1 8/2013 Ryu et al.  
 2015/0357376 A1 12/2015 Seo et al.  
 2016/0225423 A1 8/2016 Naik et al.  
 2017/0047510 A1 2/2017 Chen et al.

2017/0117027 A1 4/2017 Braganca et al.  
 2017/0117323 A1 4/2017 Braganca et al.  
 2017/0125078 A1 5/2017 Mihajlovic et al.  
 2017/0244030 A1\* 8/2017 Park ..... H10N 70/231  
 2018/0366642 A1 12/2018 Yang et al.  
 2019/0013353 A1 1/2019 Lee et al.  
 2019/0027201 A1 1/2019 Petti et al.  
 2019/0080738 A1 3/2019 Choi et al.  
 2019/0172871 A1 6/2019 Yang et al.  
 2019/0207088 A1 7/2019 Schabes et al.  
 2019/0287592 A1 9/2019 Matsunami  
 2020/0006431 A1\* 1/2020 Mayuzumi ..... H01L 23/528  
 2020/0006633 A1 1/2020 Lille  
 2020/0220068 A1 7/2020 Hashemi et al.  
 2020/0273910 A1\* 8/2020 Wu ..... H10N 70/066  
 2020/0342926 A1 10/2020 Katine et al.  
 2020/0350364 A1 11/2020 Wan et al.  
 2020/0411589 A1 12/2020 Wan et al.  
 2021/0126052 A1 4/2021 Wan et al.  
 2021/0217812 A1\* 7/2021 Hsiao ..... H01L 23/5226  
 2021/0313392 A1 10/2021 Wan et al.  
 2021/0408114 A1 12/2021 Lille et al.

**FOREIGN PATENT DOCUMENTS**

KR 20200042837 A \* 4/2020 ..... H01L 45/14  
 WO WO 2019/005162 A1 1/2019  
 WO WO 2019/005172 A1 1/2019  
 WO WO 2019/139774 A1 7/2019  
 WO WO 2020/222883 A1 11/2020

**OTHER PUBLICATIONS**

Notification of Transmittal of the International Search Report and Written Opinion of the International Search Authority for International Patent Application No. PCT/US2020/023707, mailed Jul. 28, 2020, 10 pages.  
 Kim, Y. et al. "Multilevel Spin-Orbit Torque MRAMs," IEEE Transactions on Electron Devices, vol. 62, No. 2, pp. 561-568, (2015).  
 U.S. Appl. No. 17/654,760, filed Mar. 14, 2022, SanDisk Technologies LLC.  
 U.S. Appl. No. 17/654,762, filed Mar. 14, 2022, SanDisk Technologies LLC.  
 U.S. Appl. No. 17/654,768, filed Mar. 14, 2022, SanDisk Technologies LLC.  
 U.S. Appl. No. 17/654,777, filed Mar. 14, 2022, SanDisk Technologies LLC.  
 U.S. Appl. No. 17/654,781, filed Mar. 14, 2022, SanDisk Technologies LLC.  
 USPTO Office Communication, Non-Final Office Action for U.S. Appl. No. 17/654,760, mailed Nov. 9, 2023, 22 pages.  
 USPTO Office Communication, Non-Final Office Action for U.S. Appl. No. 17/654,762, mailed Nov. 9, 2023, 22 pages.  
 USPTO Office Communication, Non-Final Office Action for U.S. Appl. No. 17/654,768, mailed Nov. 9, 2023, 22 pages.  
 USPTO Office Communication, Non-Final Office Action for U.S. Appl. No. 17/354,541, mailed Jun. 2, 2023, 19 pages.

\* cited by examiner

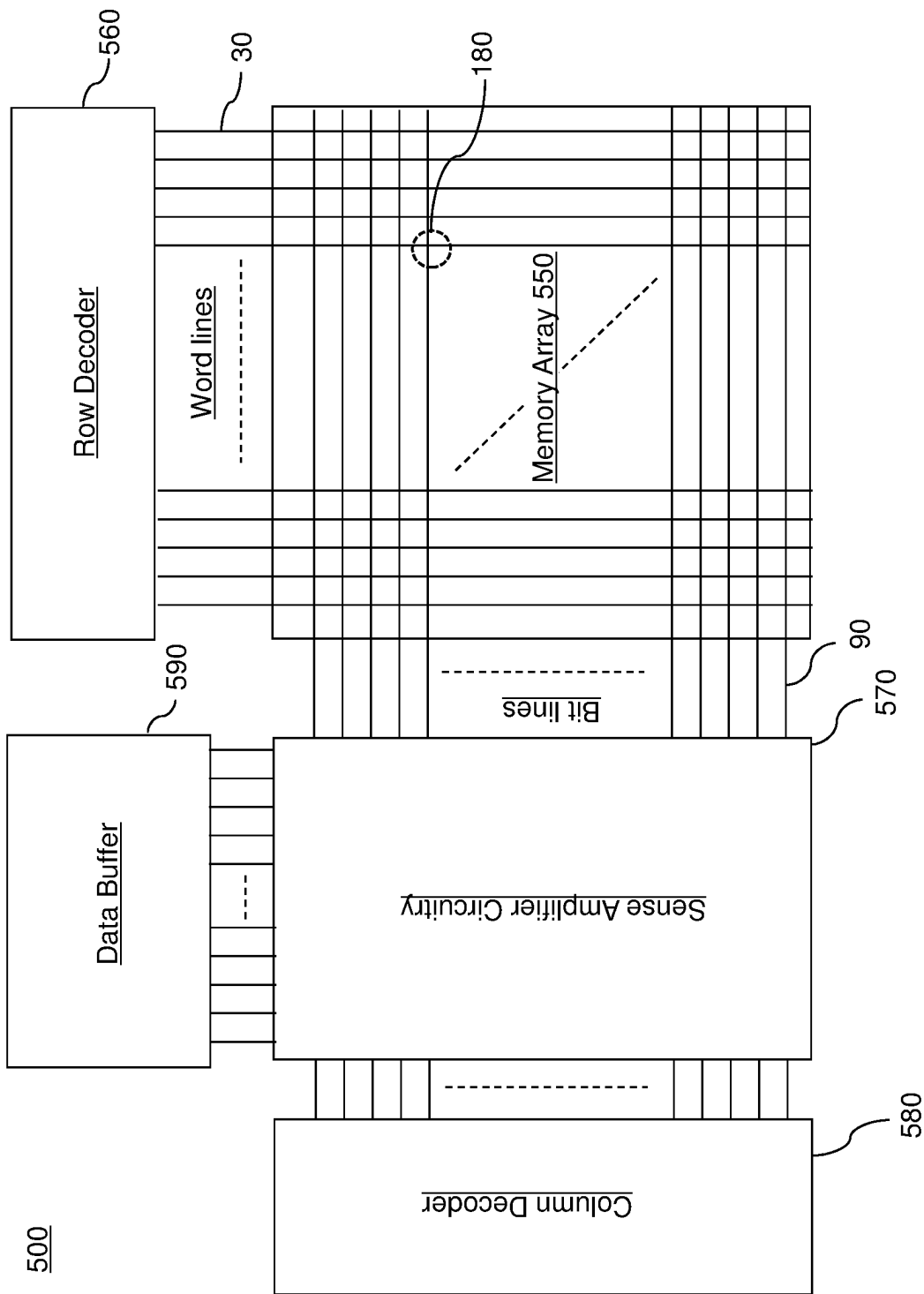
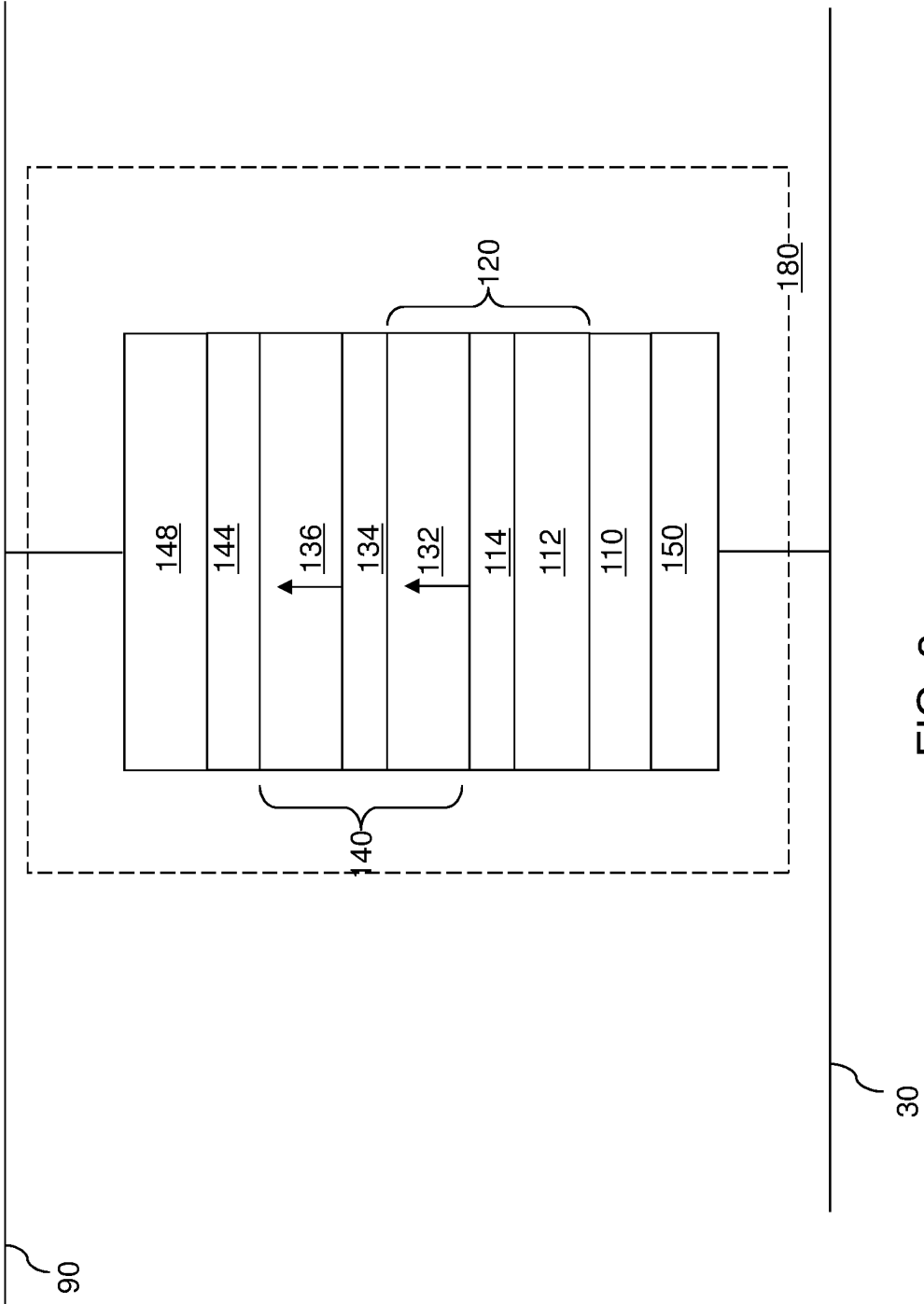
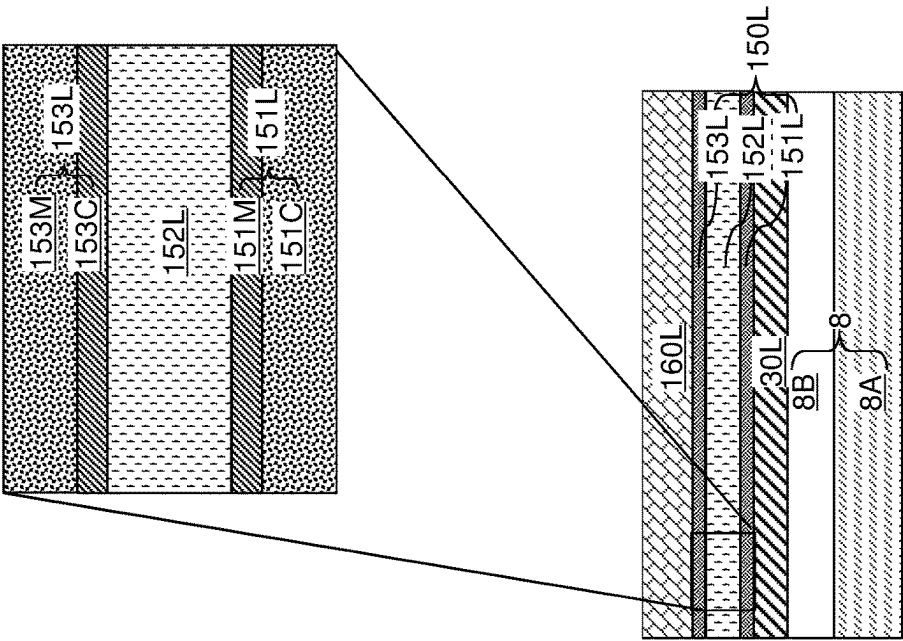
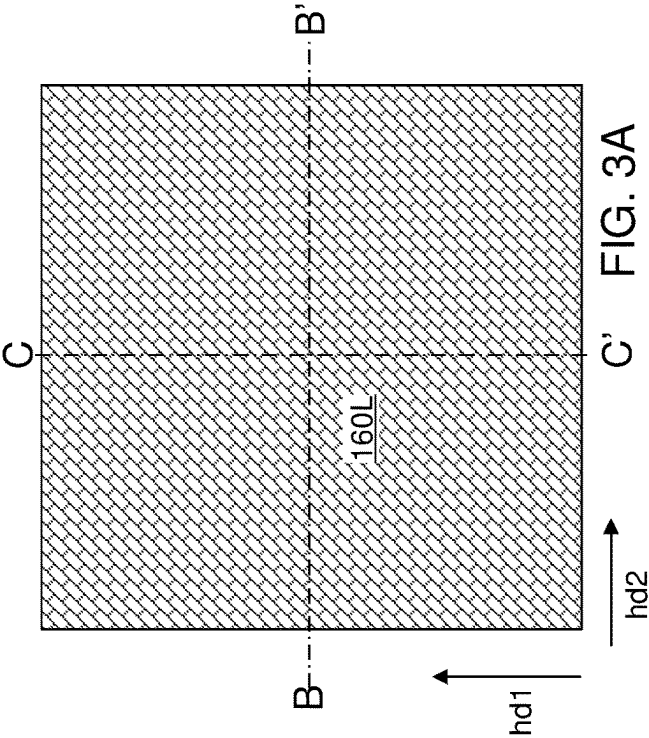


FIG. 1







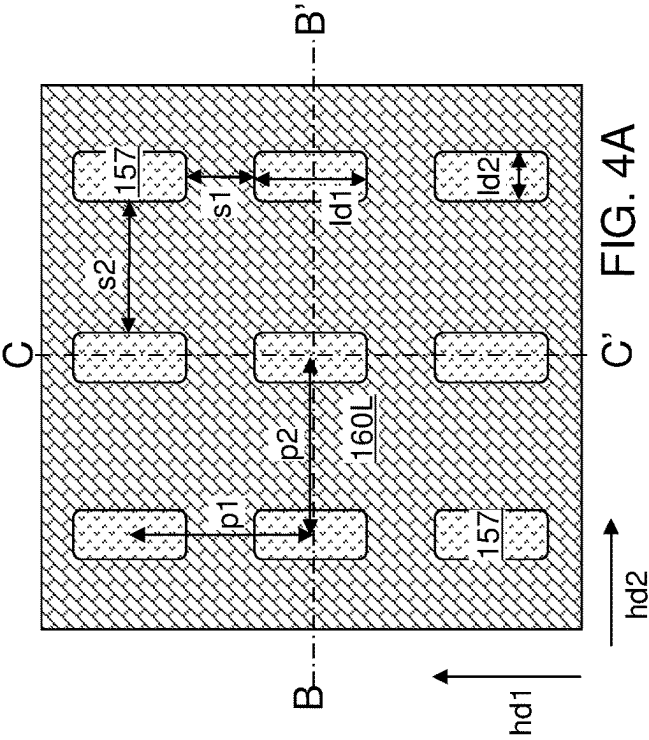


FIG. 4A

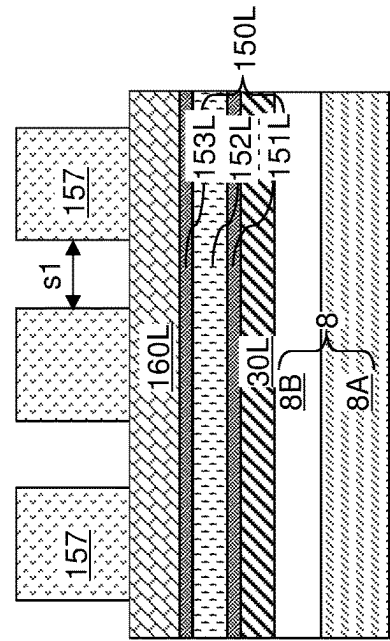


FIG. 4C

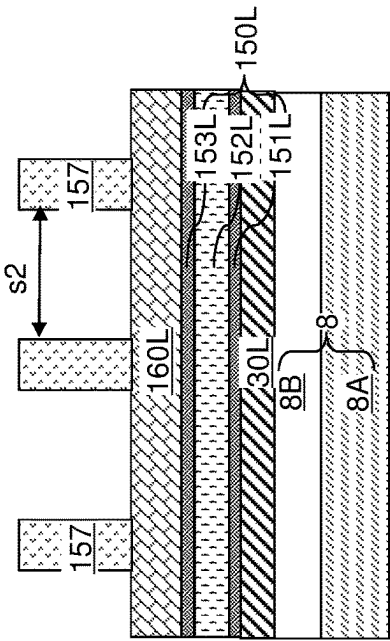
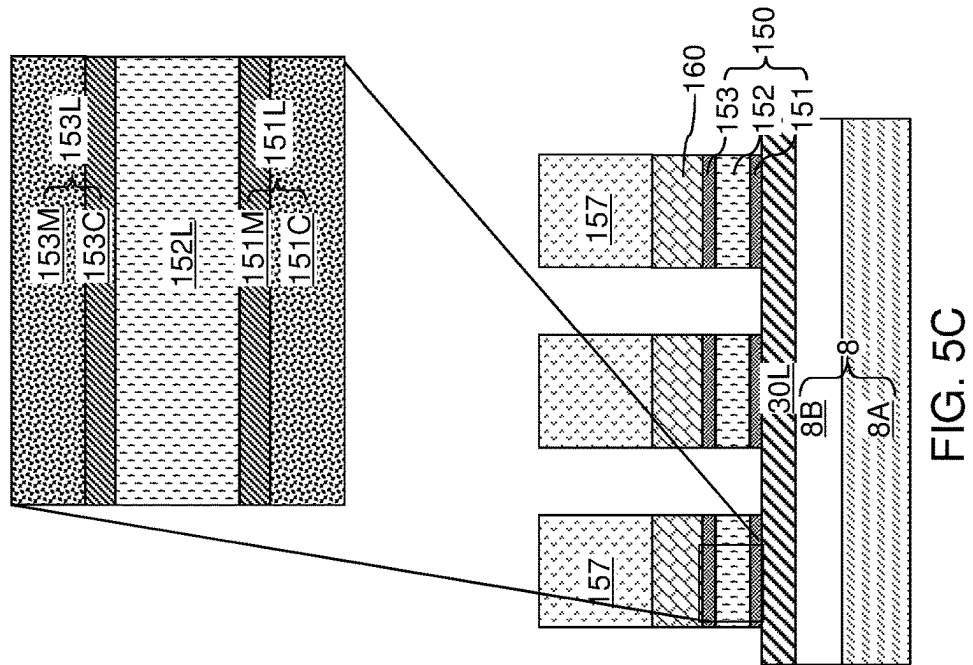
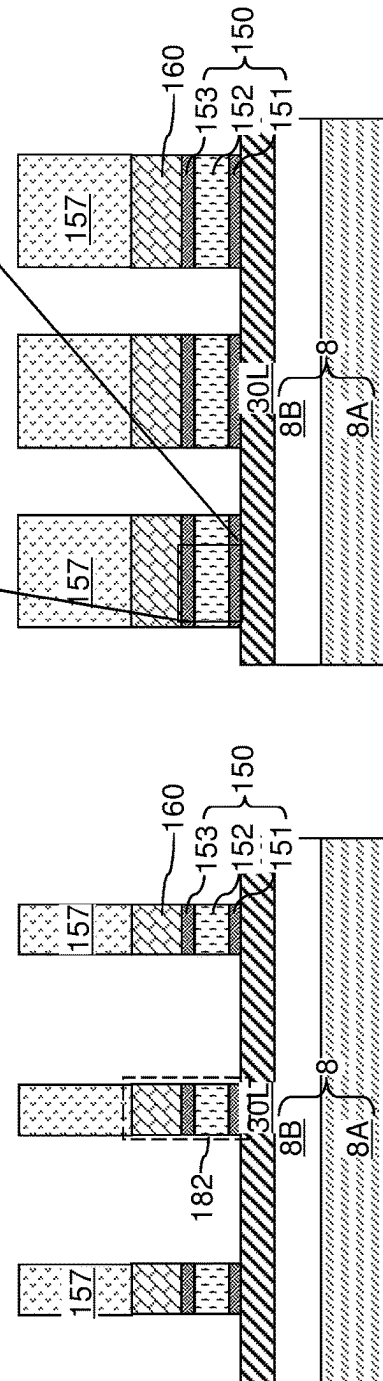
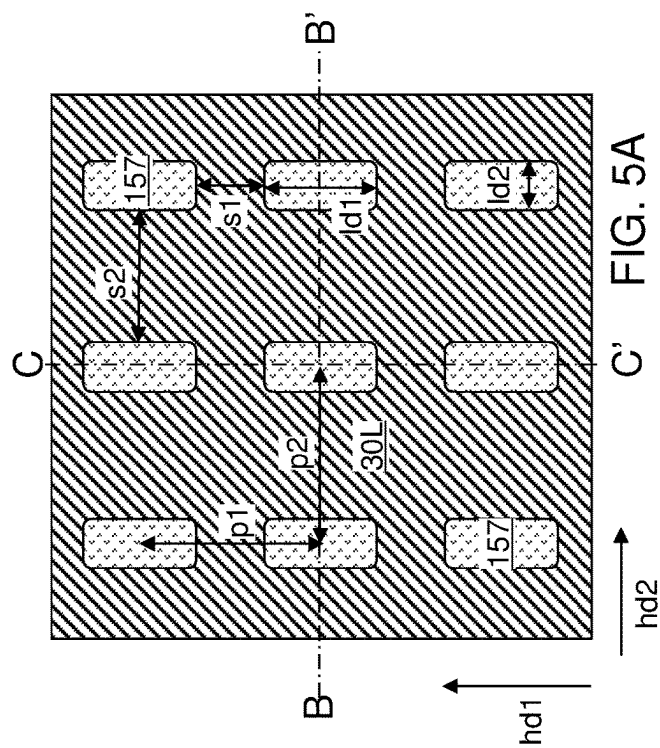


FIG. 4B



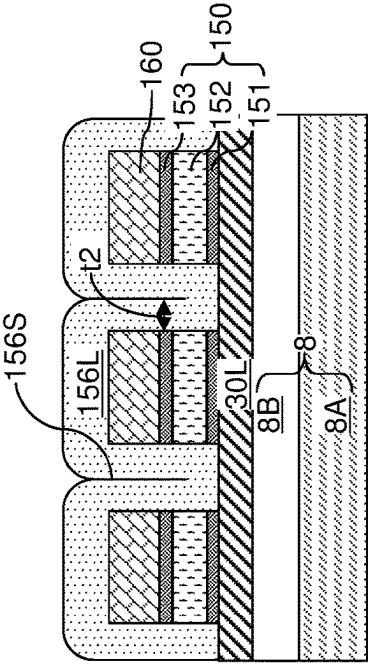
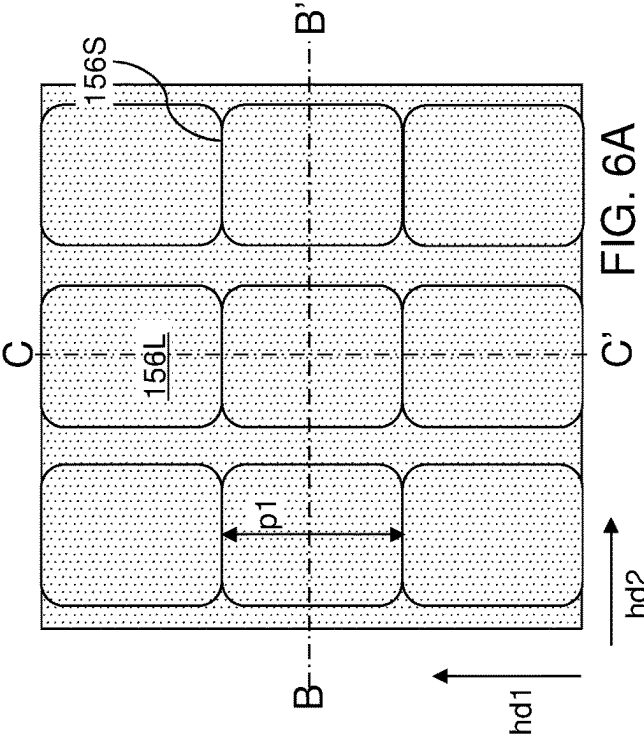


FIG. 6C

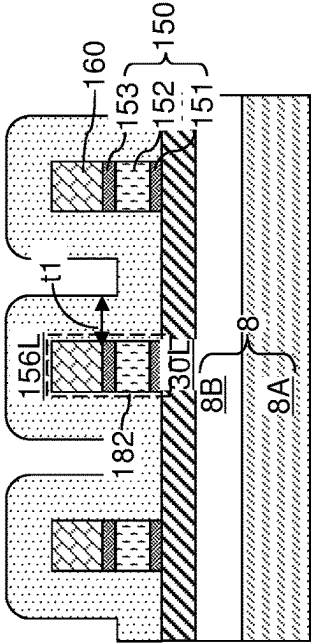


FIG. 6B

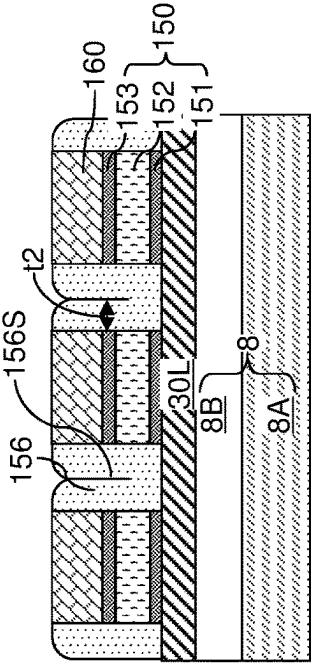
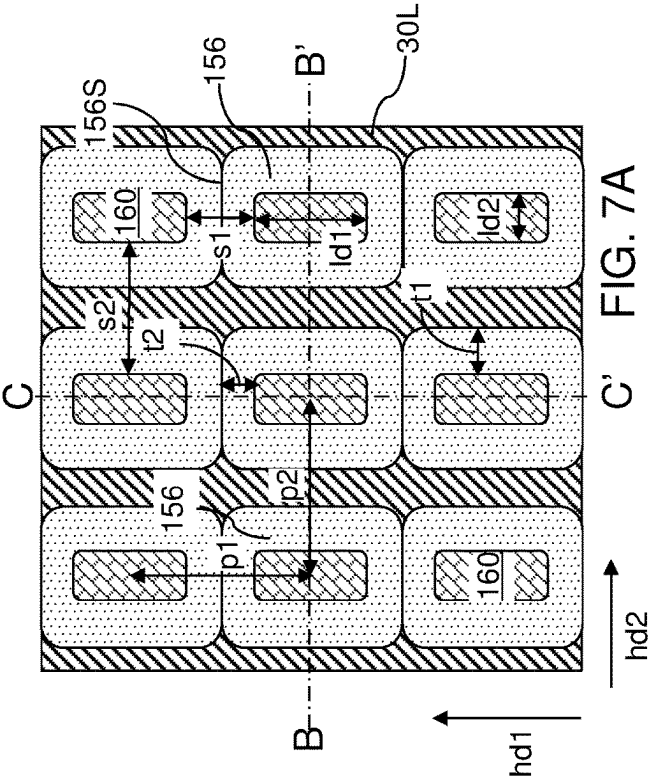


FIG. 7C

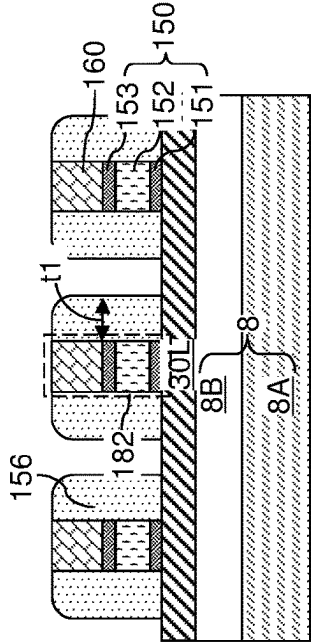
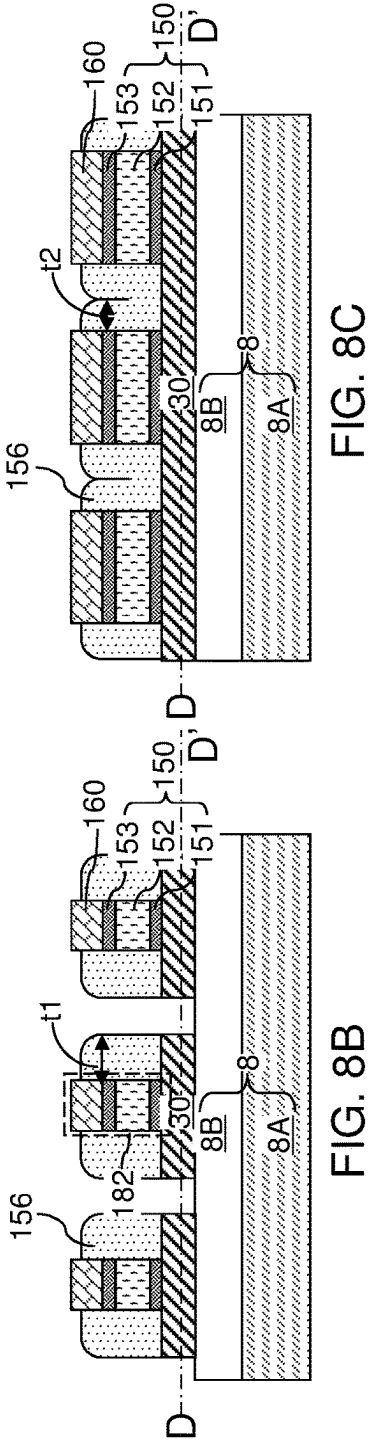
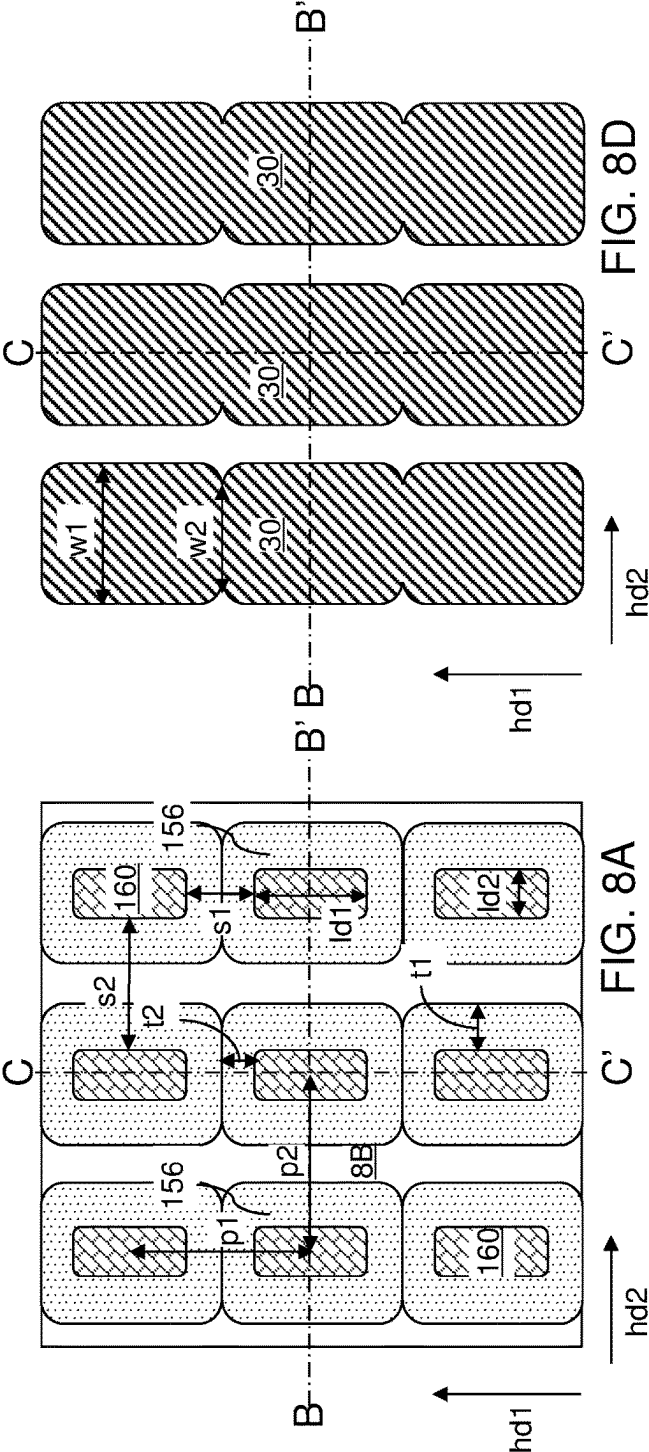


FIG. 7B



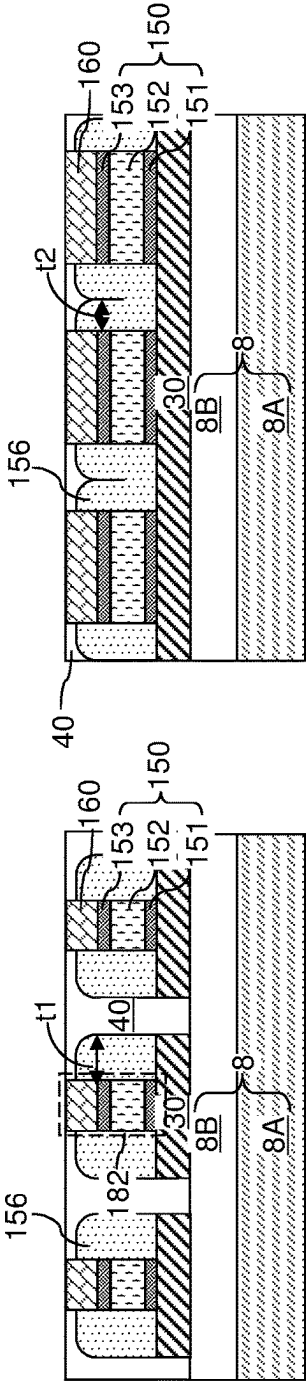
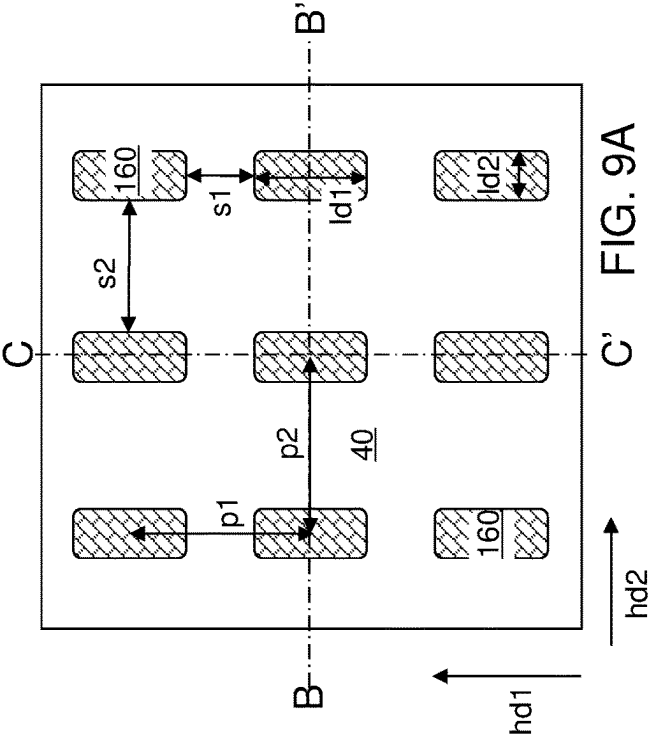
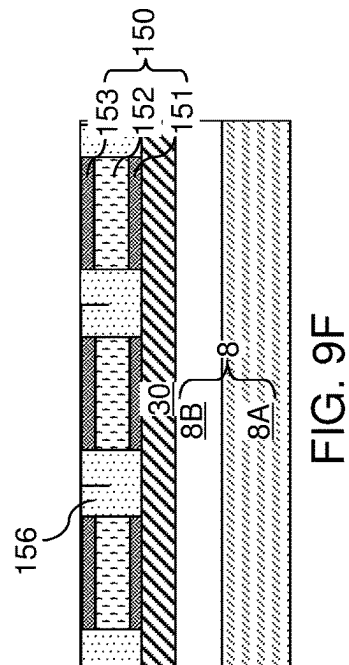
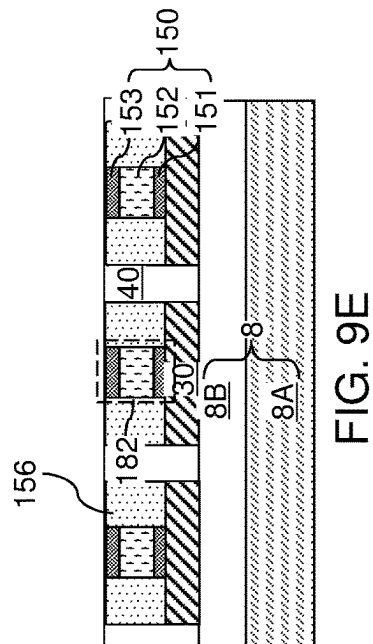
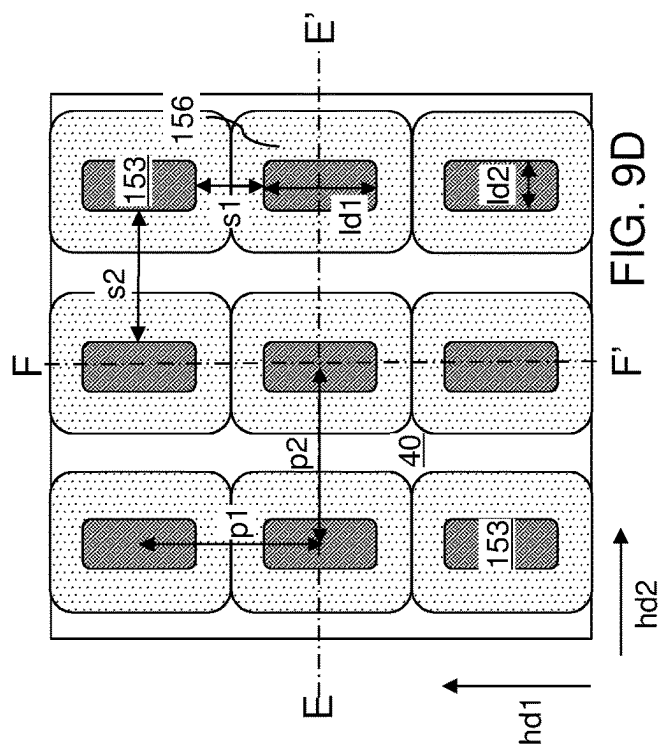
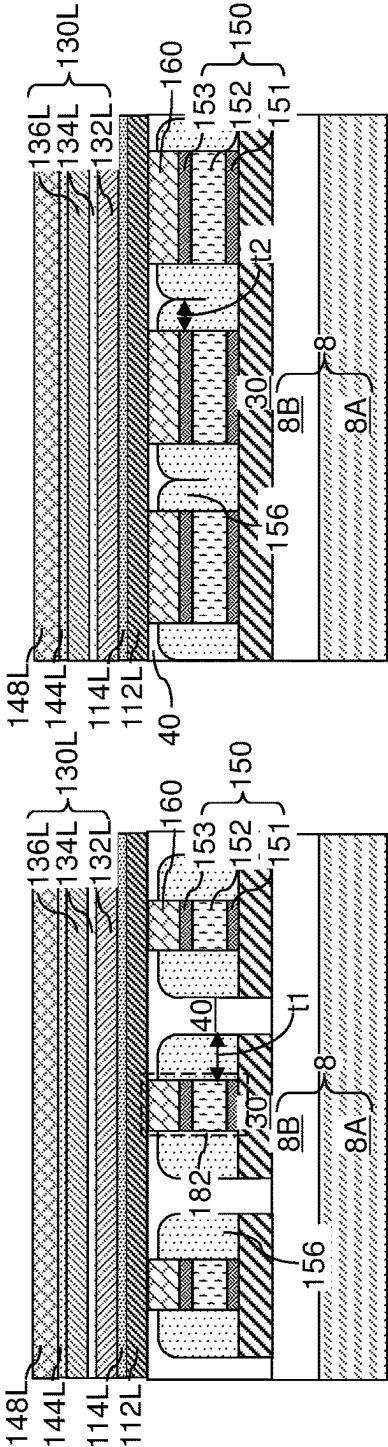
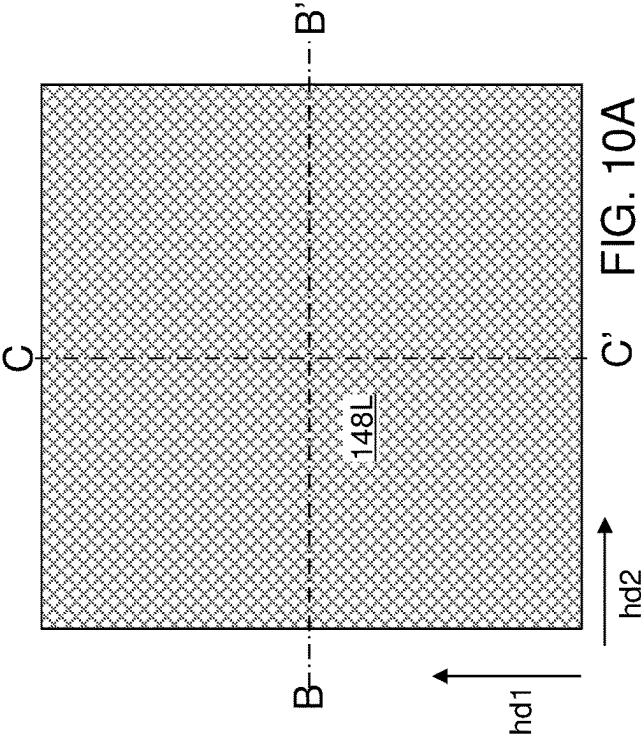


FIG. 9C

FIG. 9B







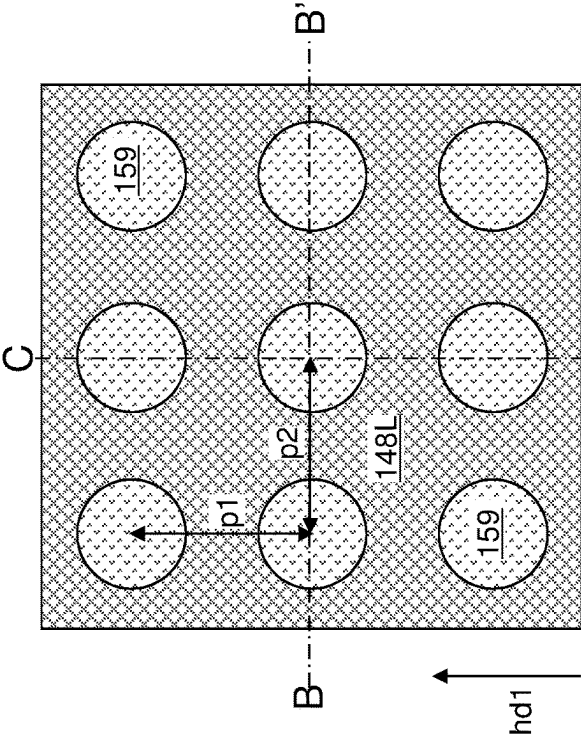


FIG. 11A

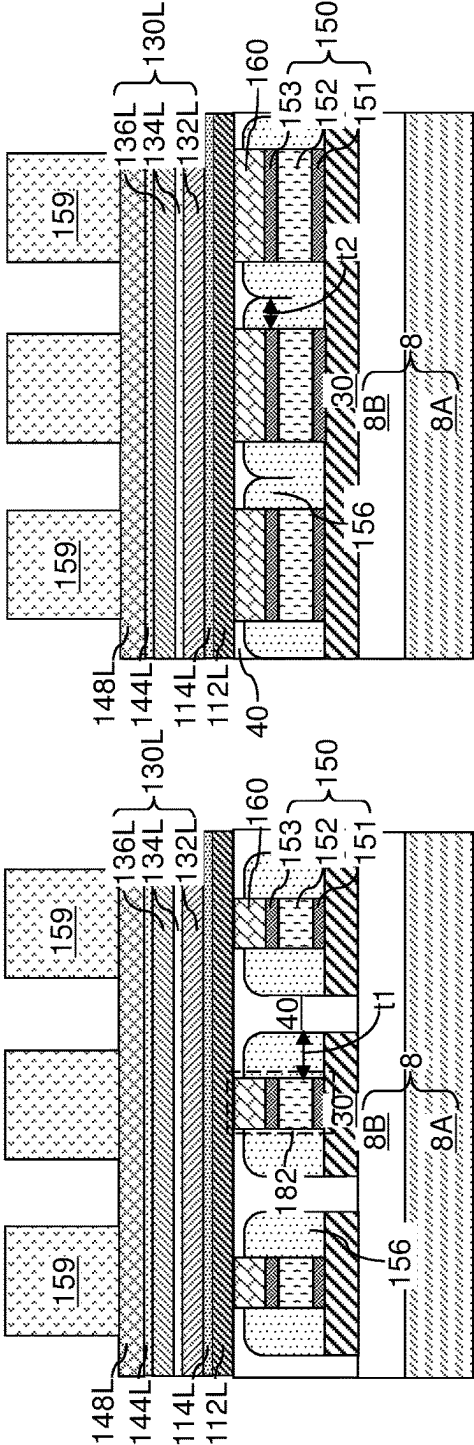


FIG. 11C

FIG. 11B

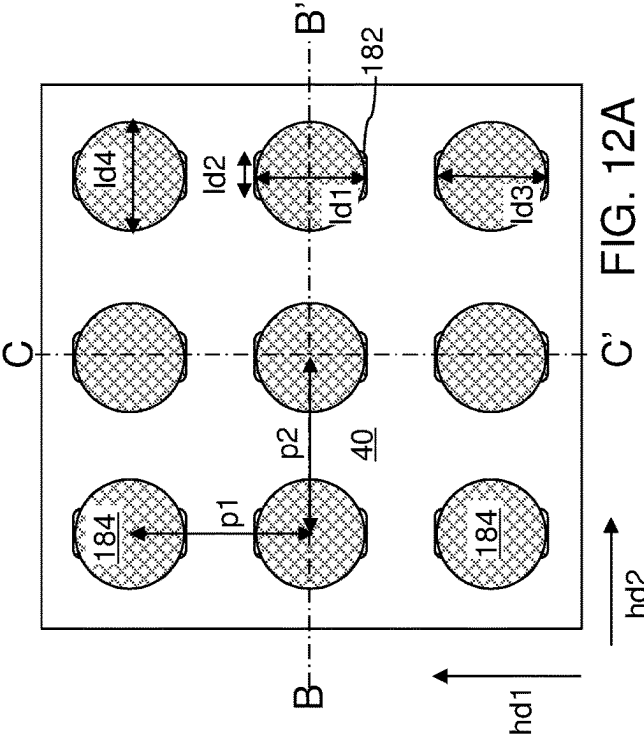


FIG. 12A

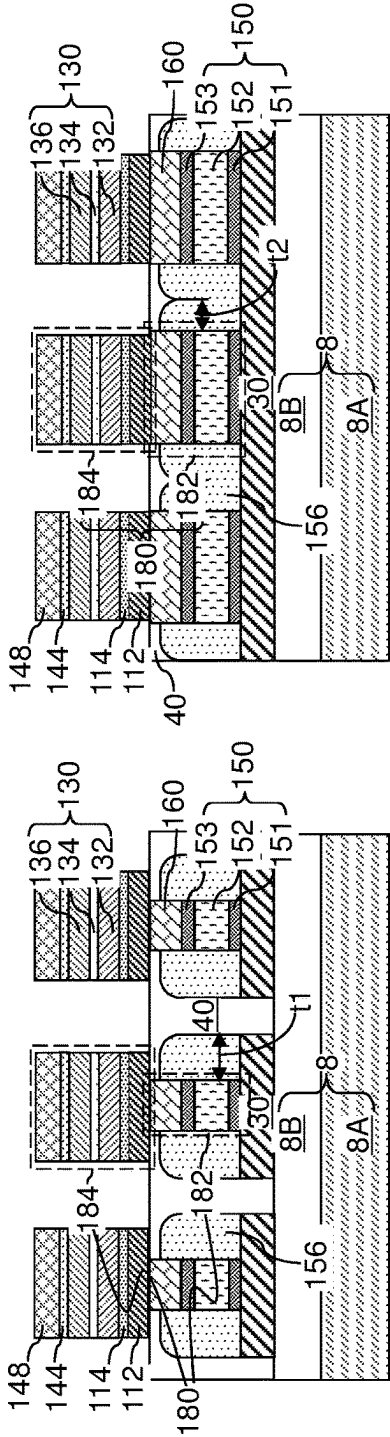
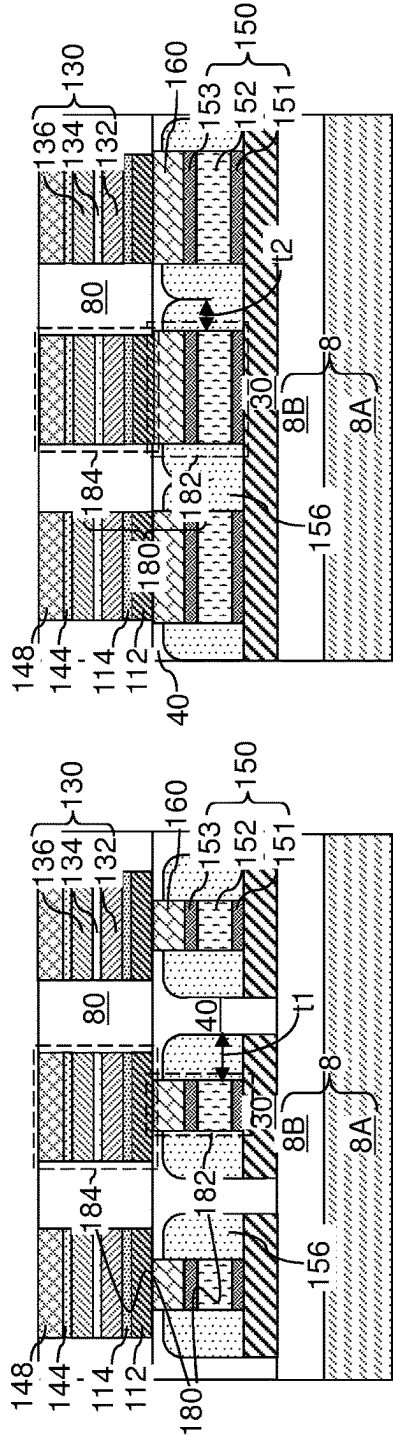
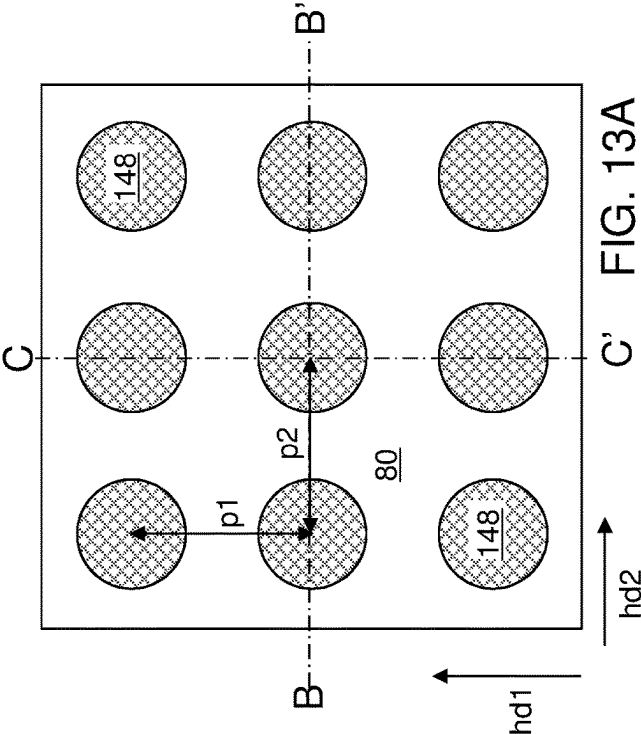
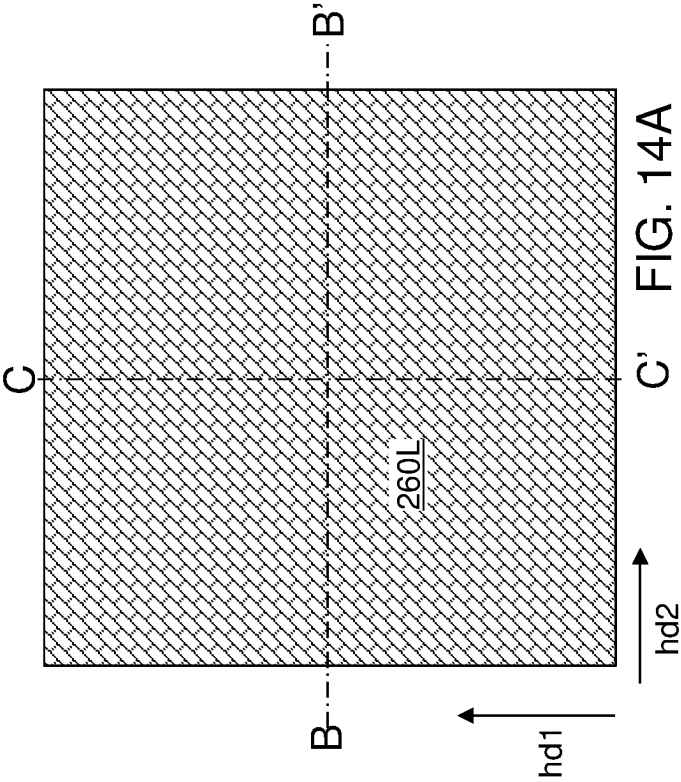


FIG. 12C

FIG. 12B





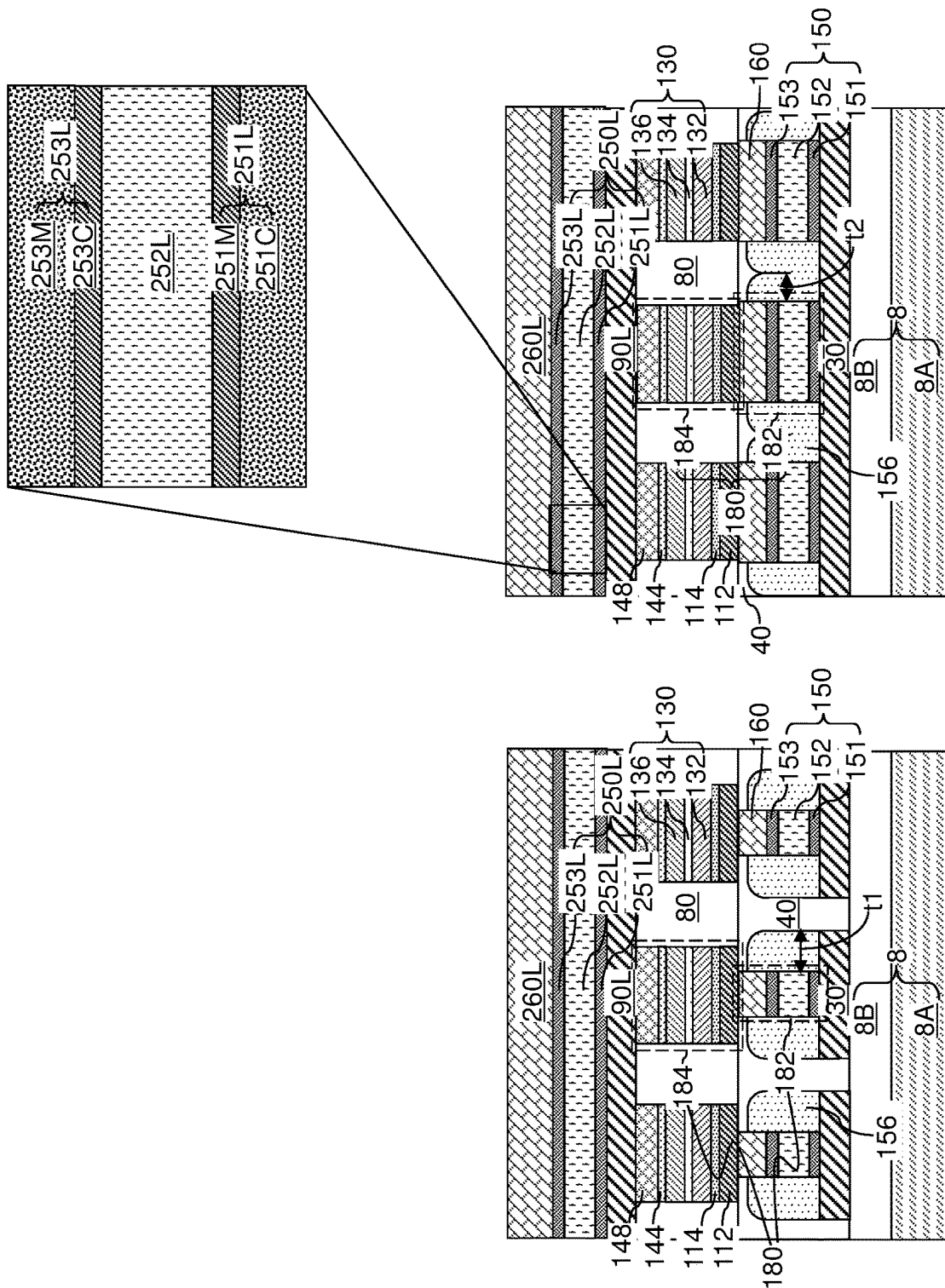
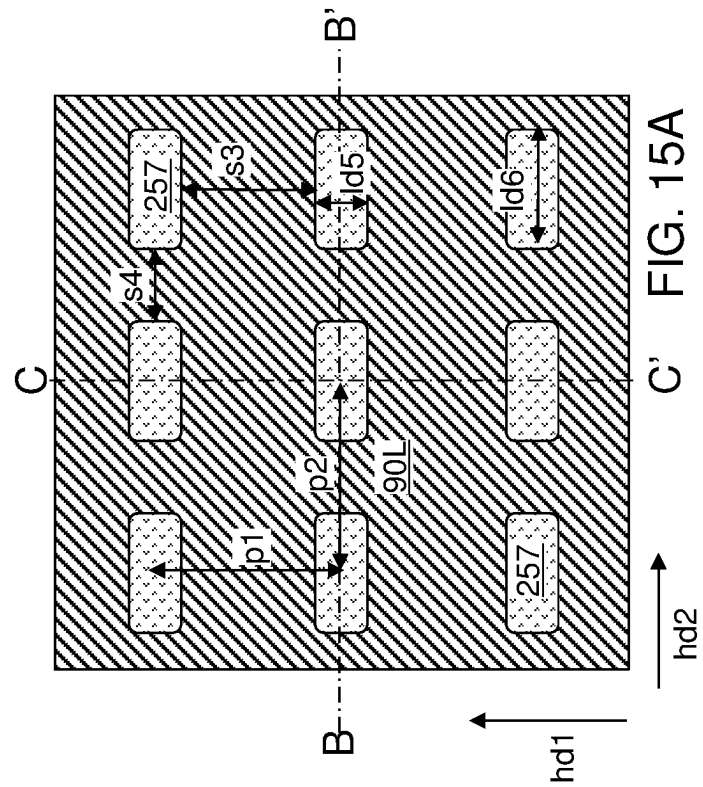


FIG. 14C

FIG. 14B



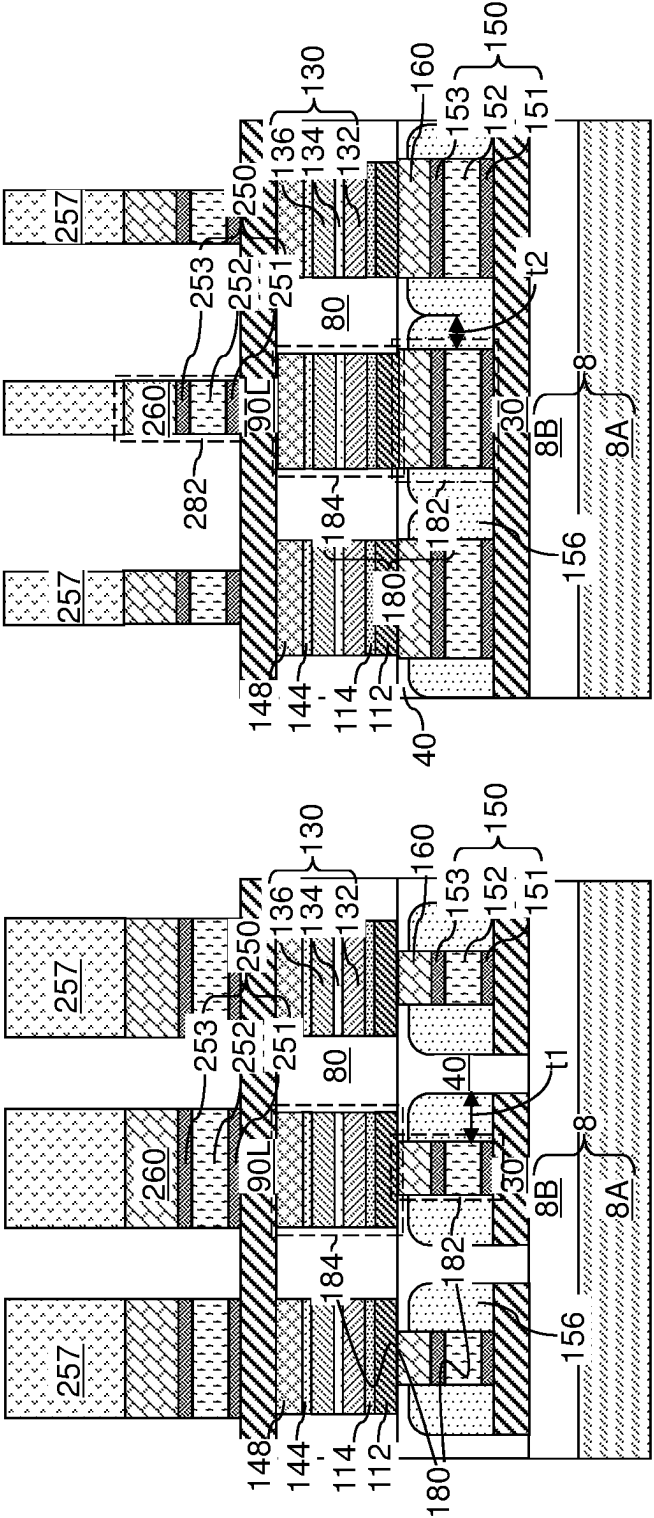
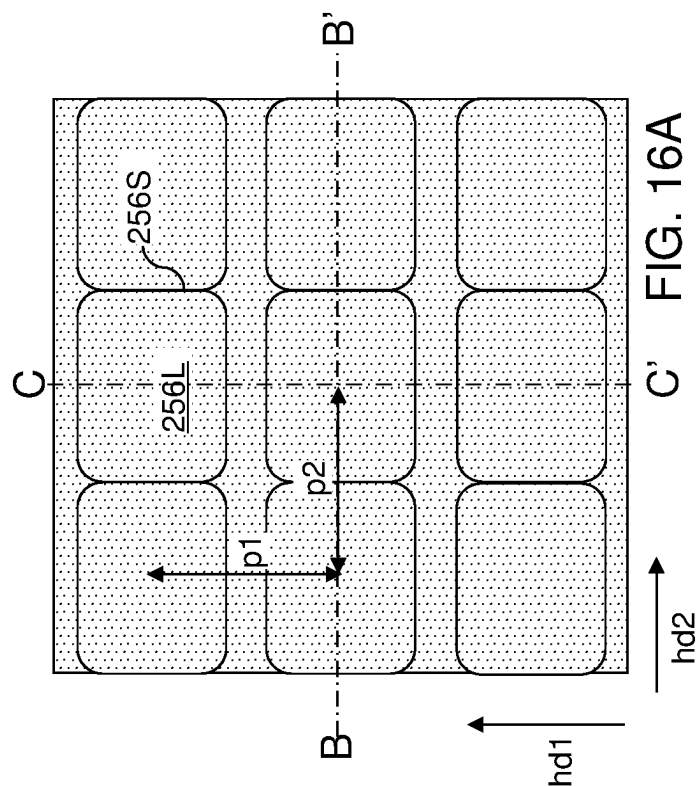
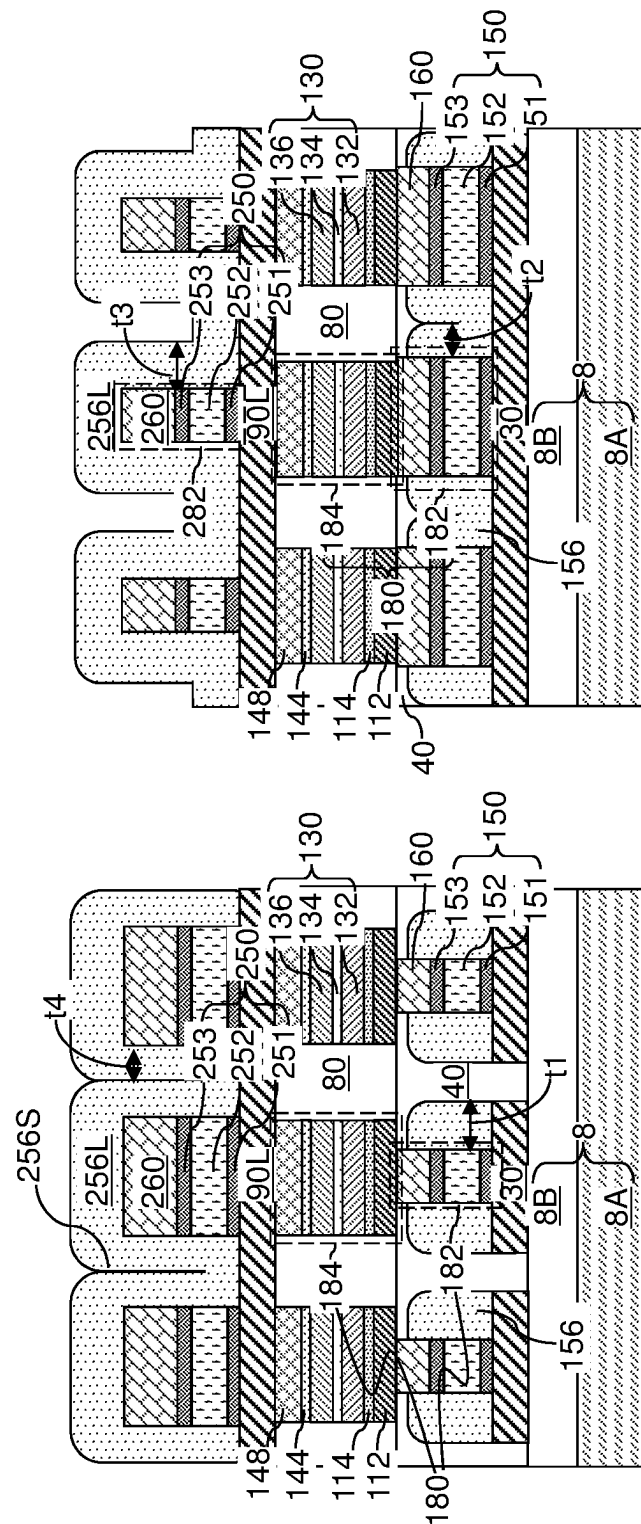


FIG. 15C

FIG. 15B







**FIG. 16B**

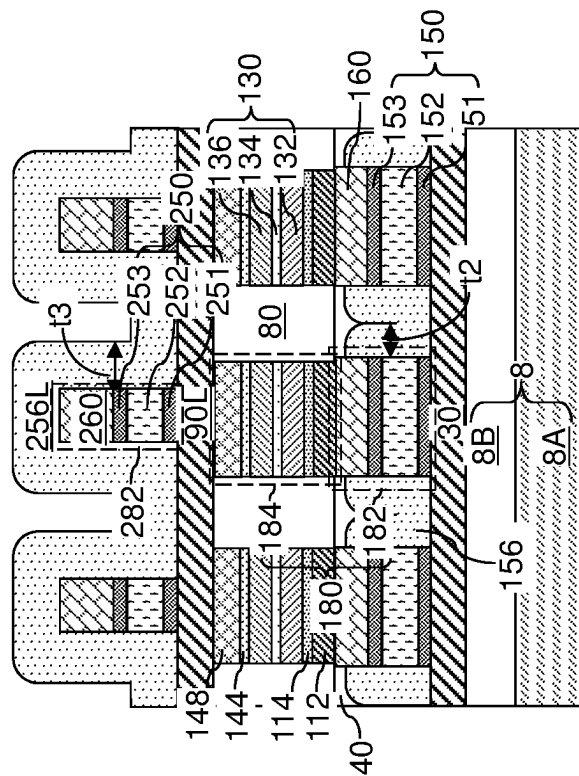
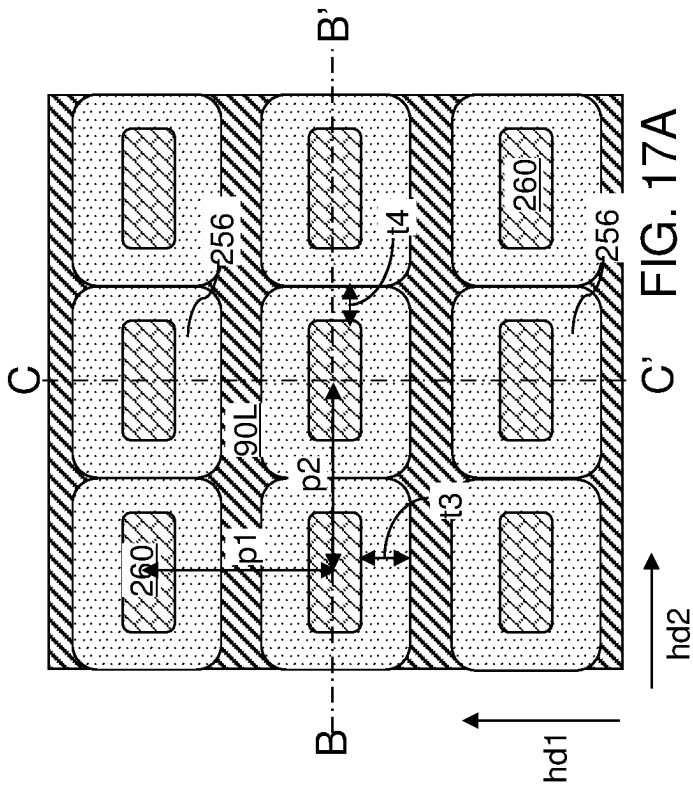


FIG. 16C



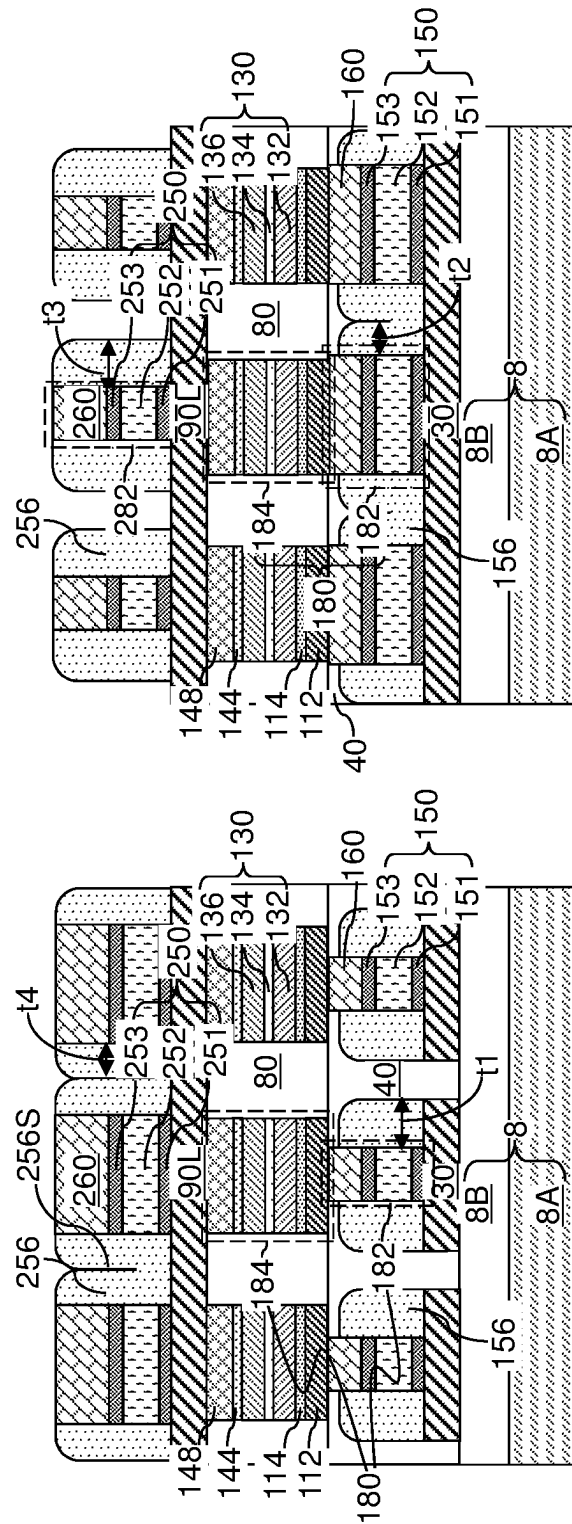


FIG. 17B

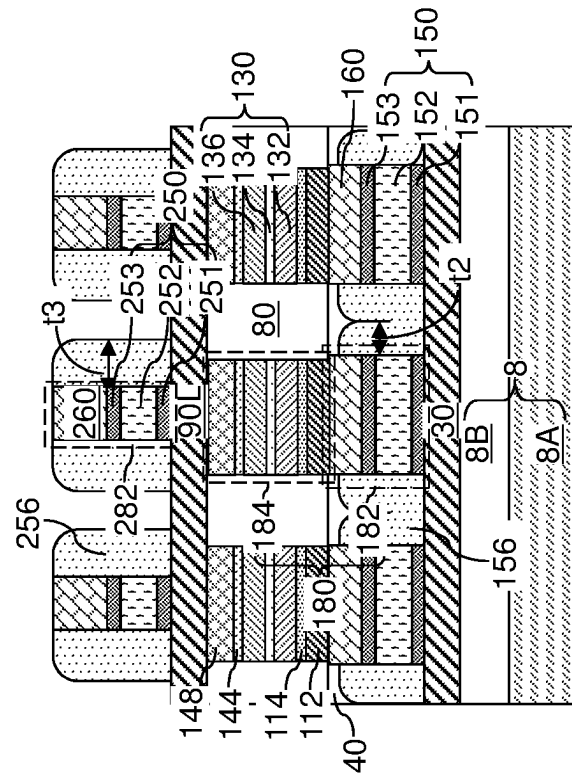
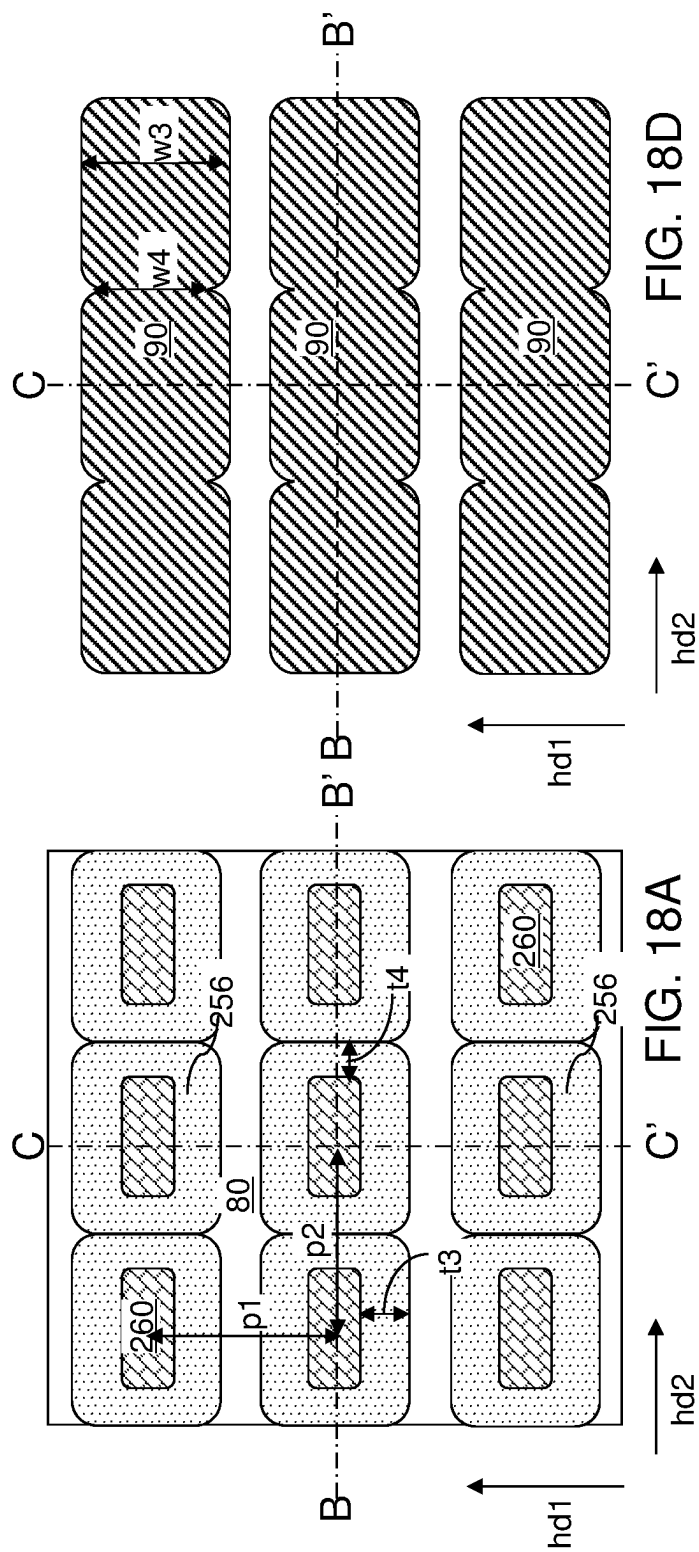


FIG. 17C



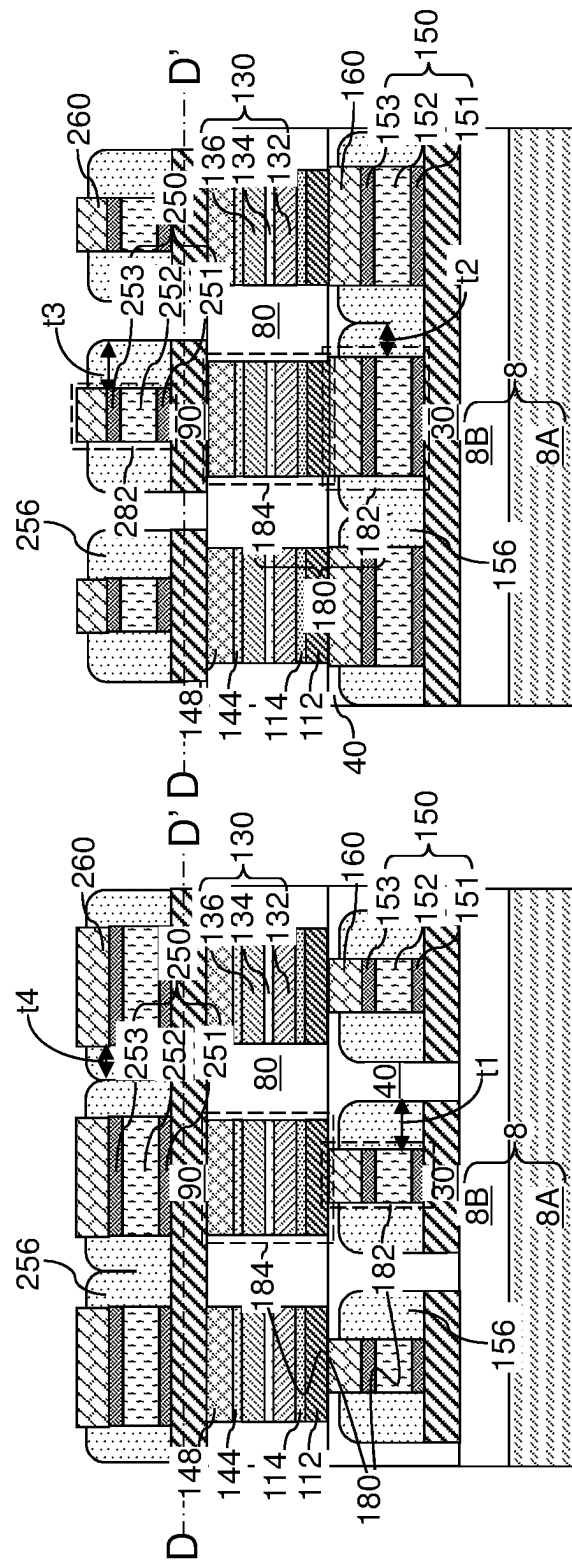


FIG. 18C

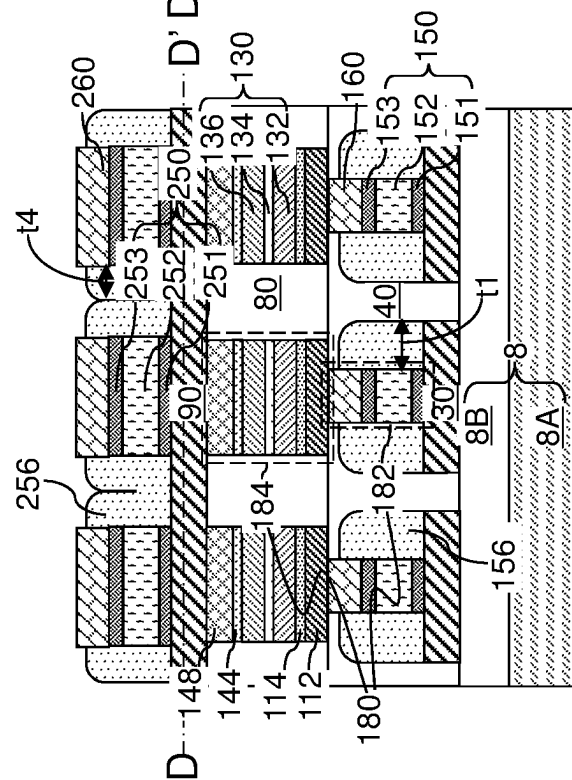
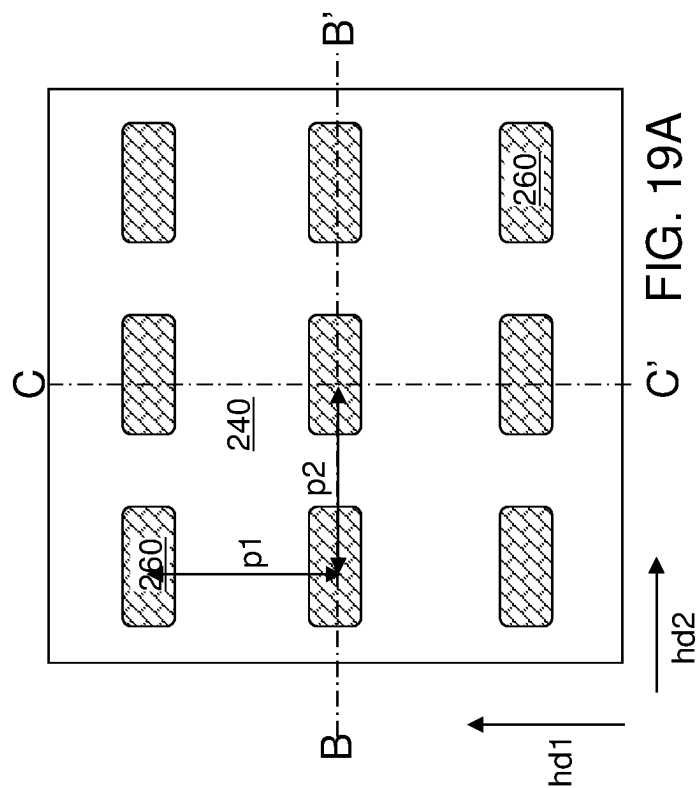


FIG. 18B



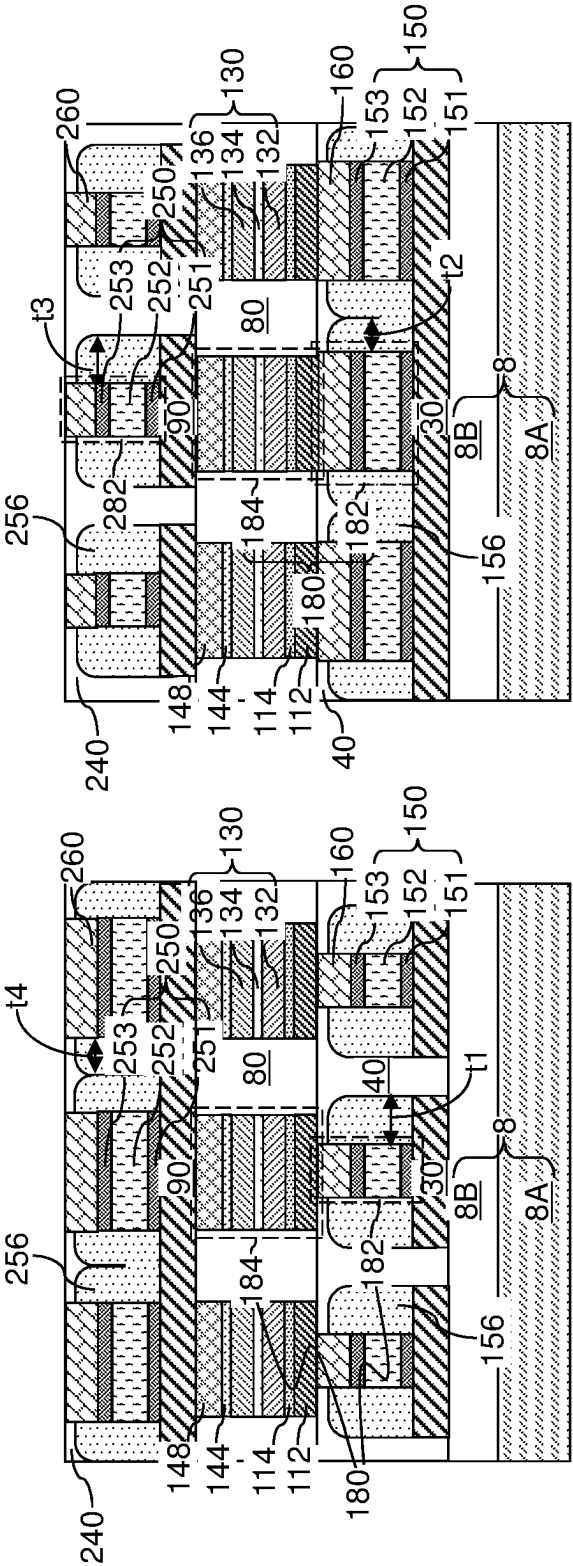
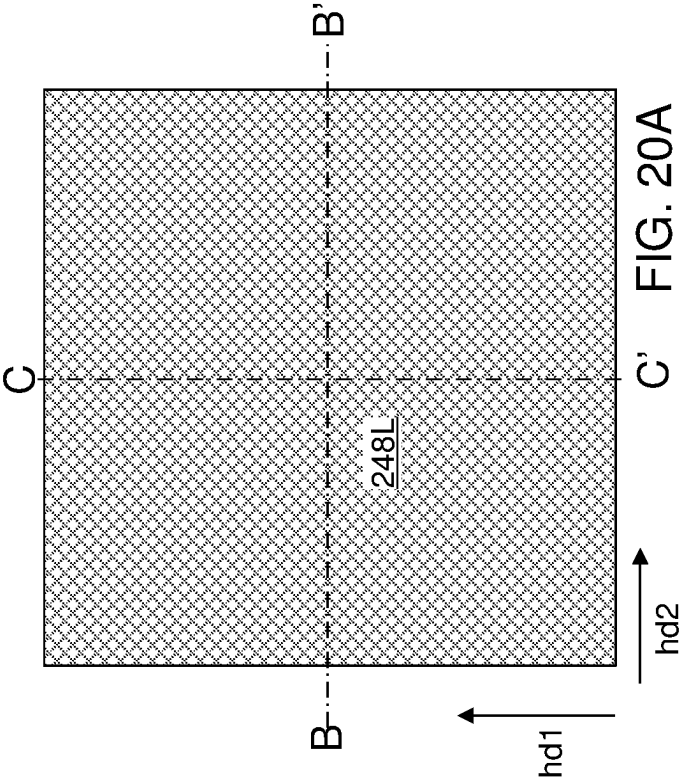


FIG. 19C

FIG. 19B





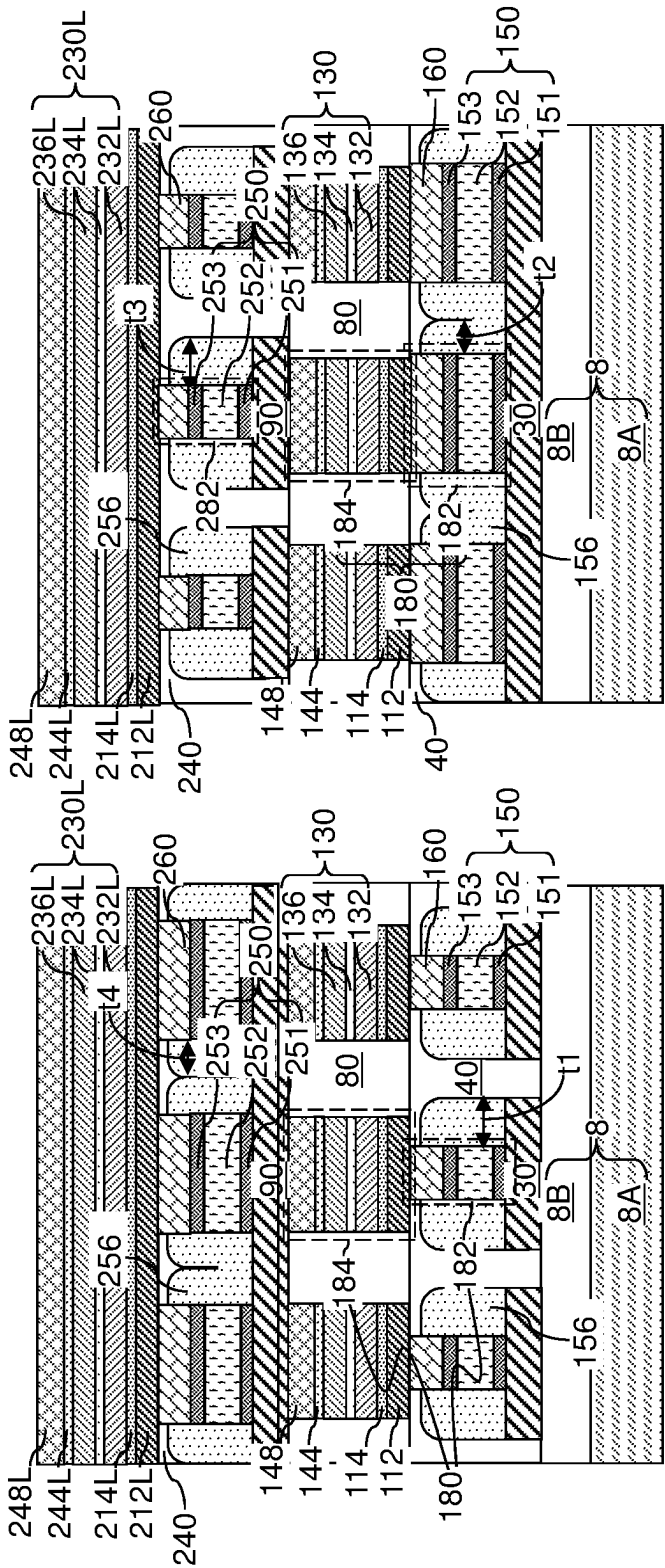
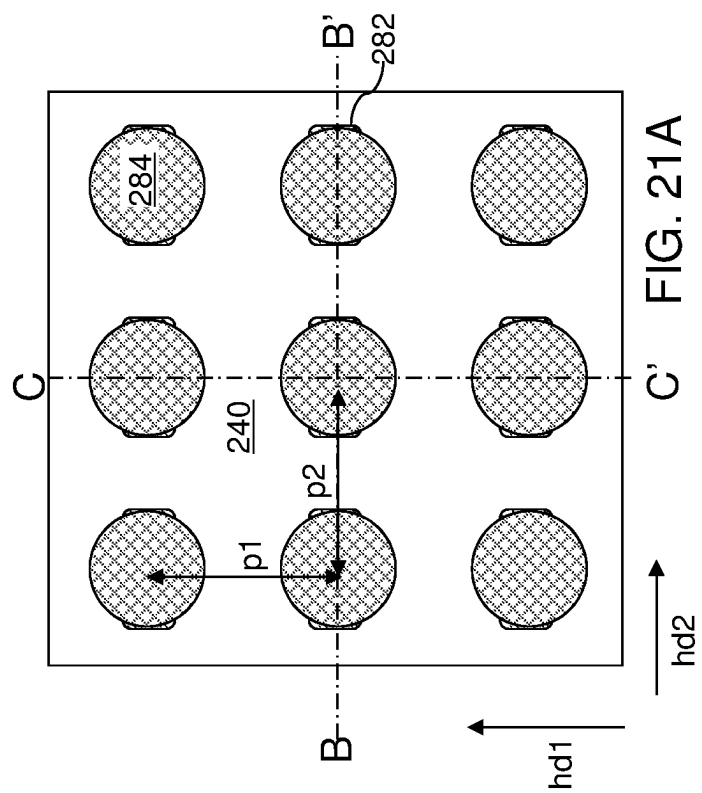


FIG. 20C

FIG. 20B



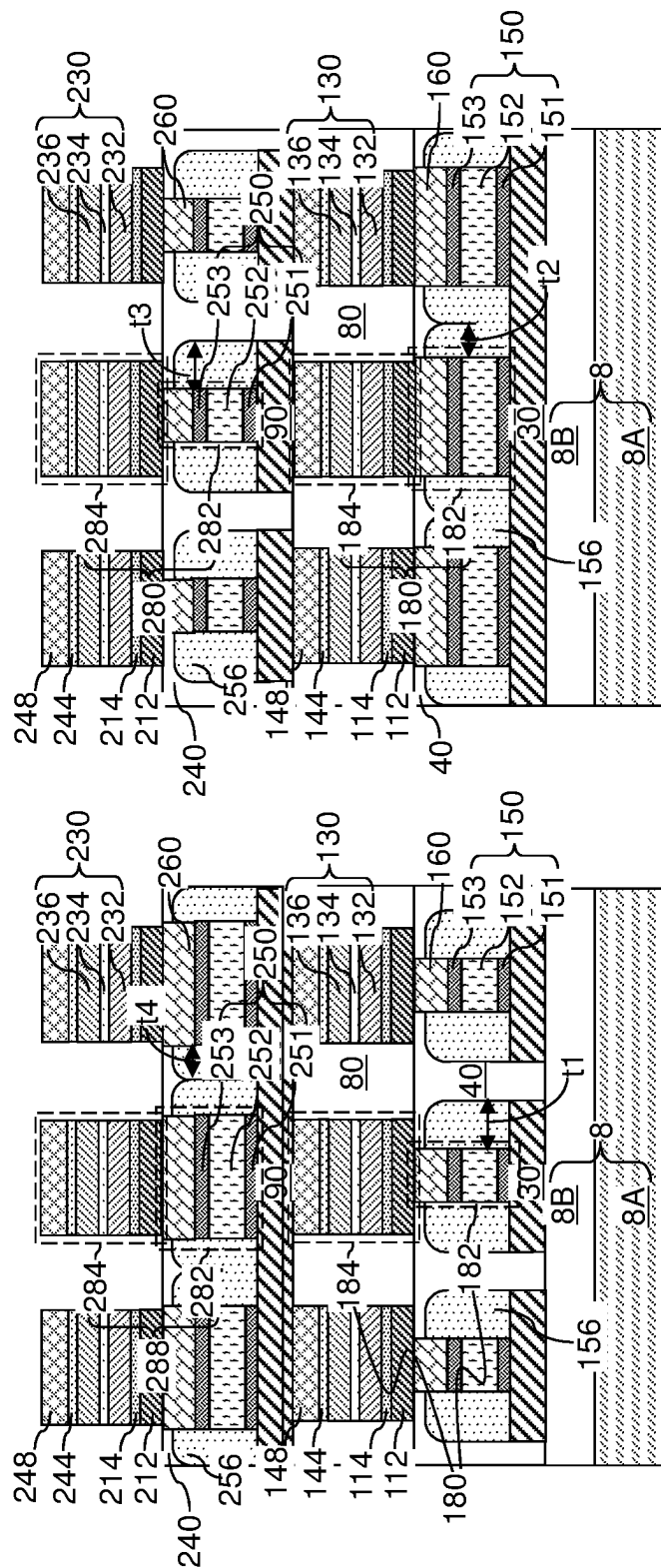


FIG. 21C

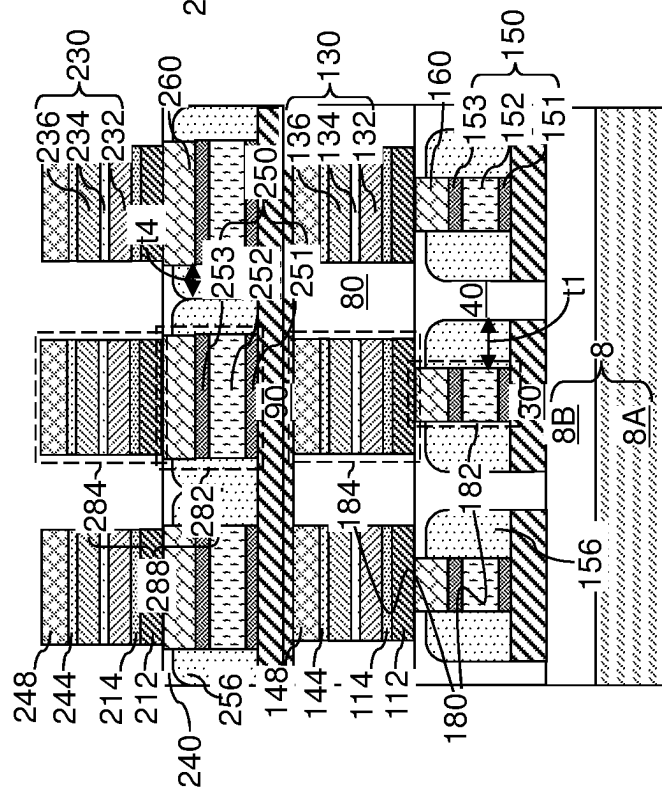
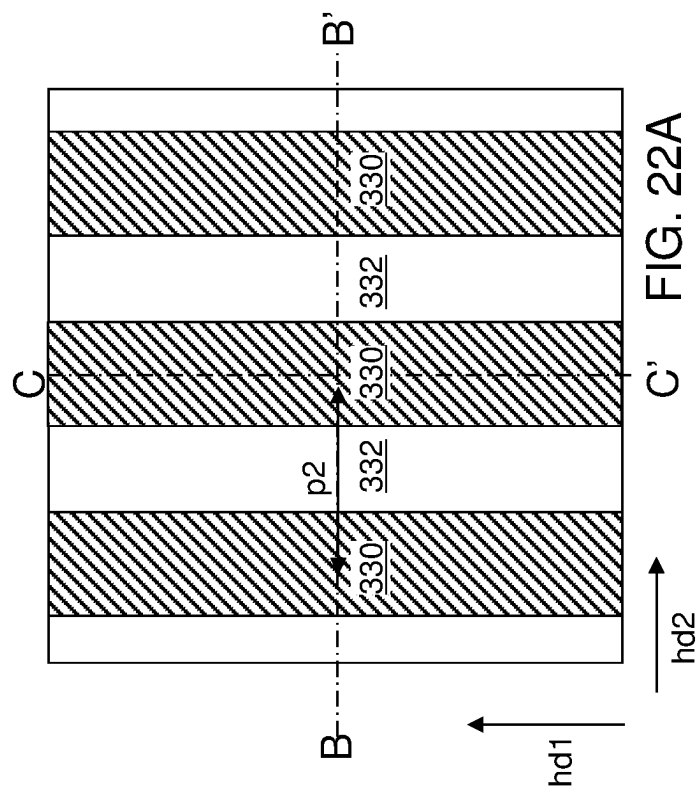


FIG. 21B



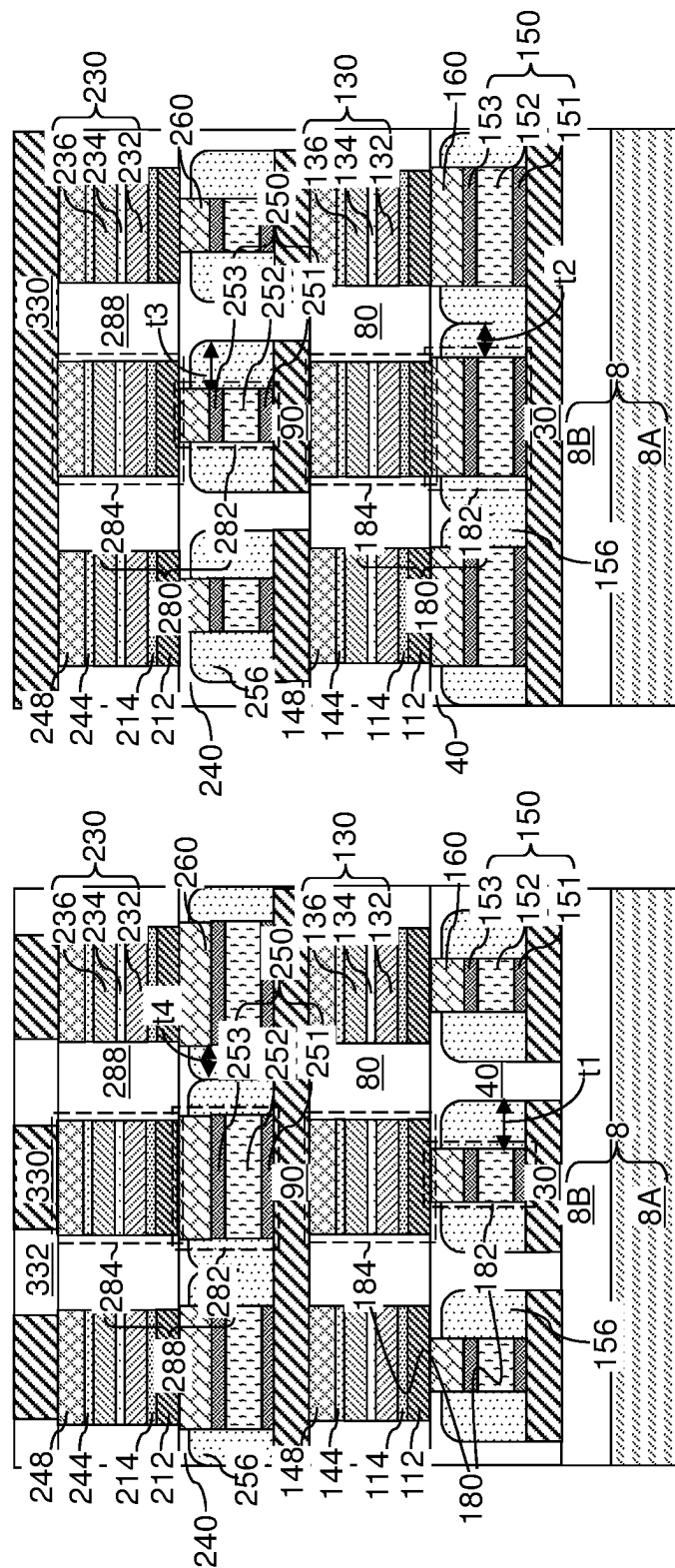
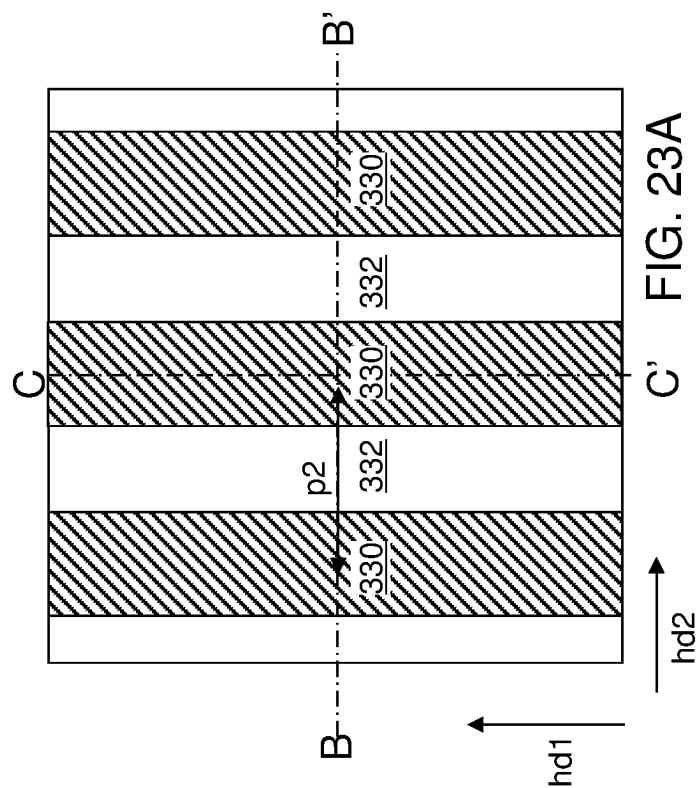


FIG. 22C

FIG. 22B



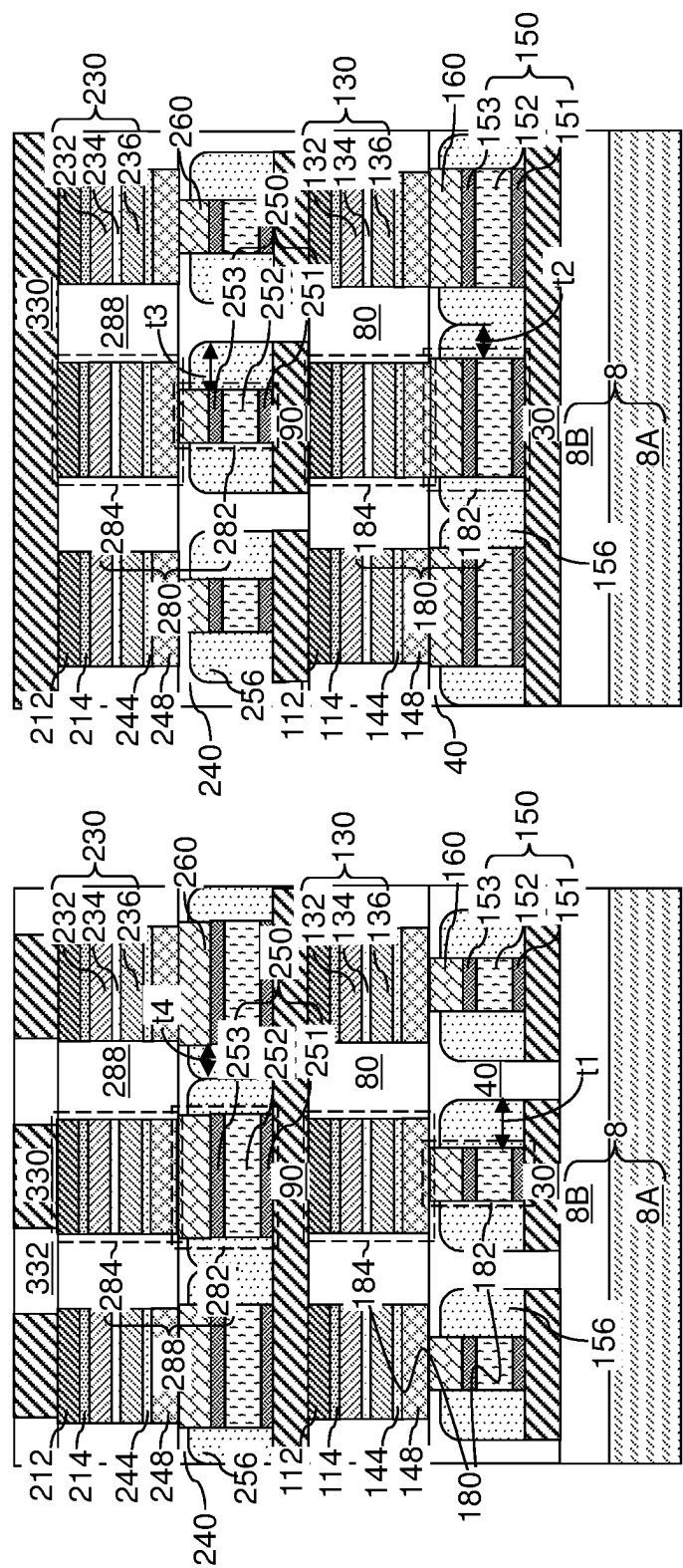


FIG. 23C

FIG. 23B



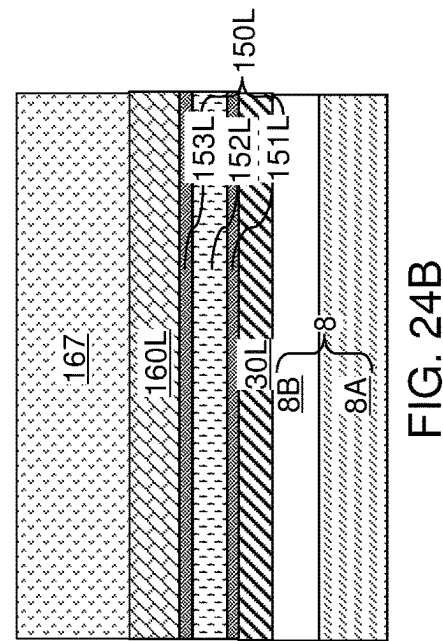
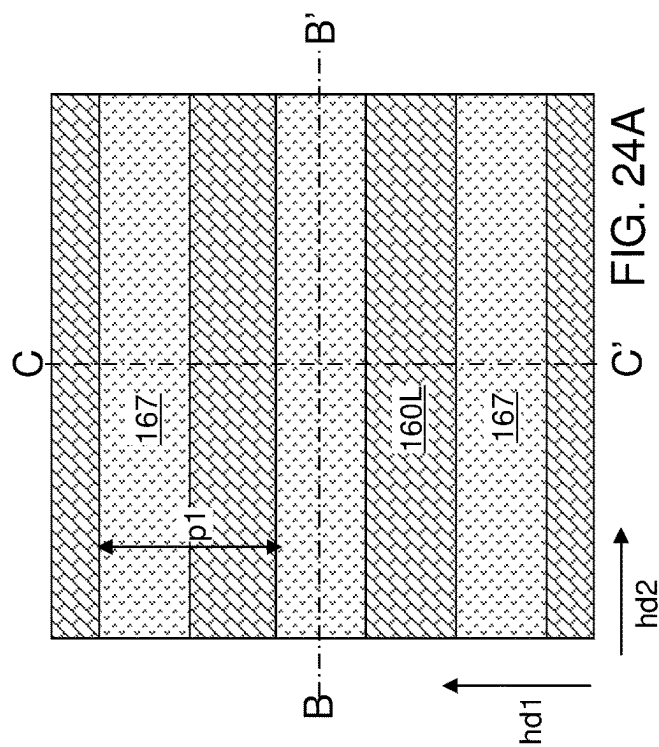


FIG. 24B

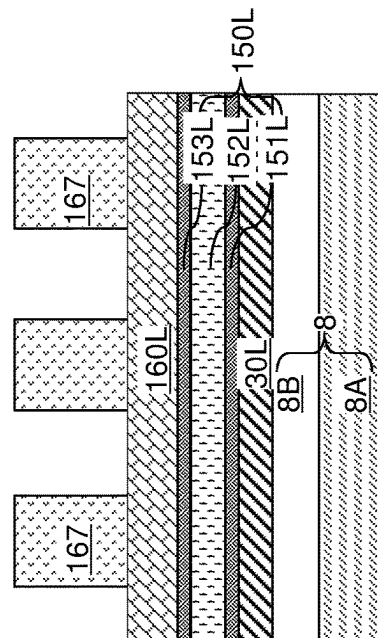


FIG. 24C

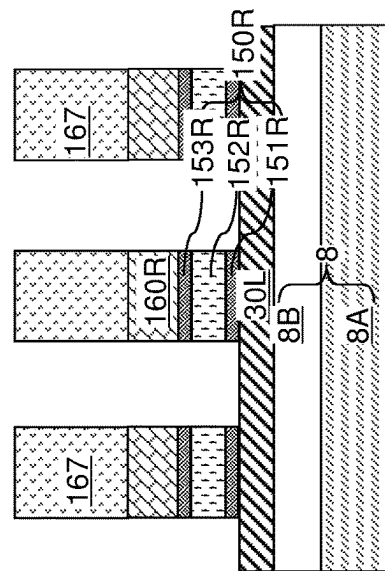
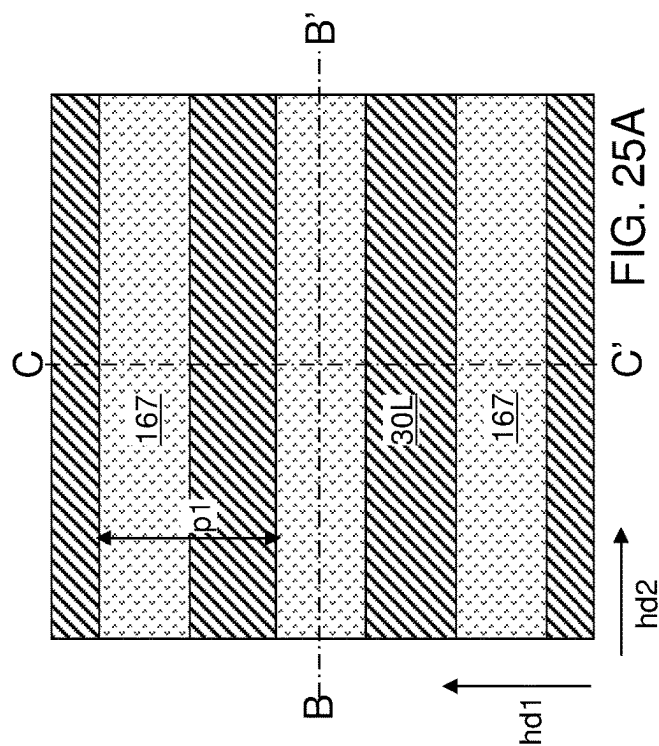
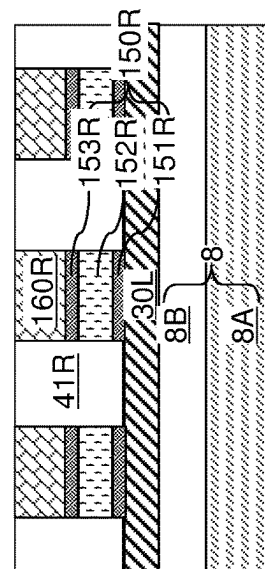
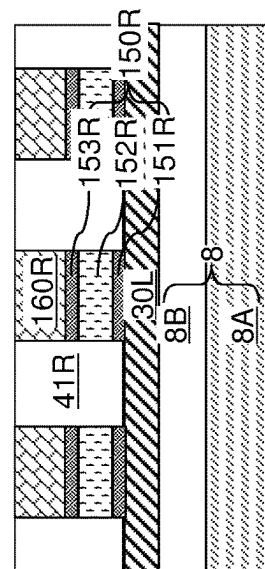
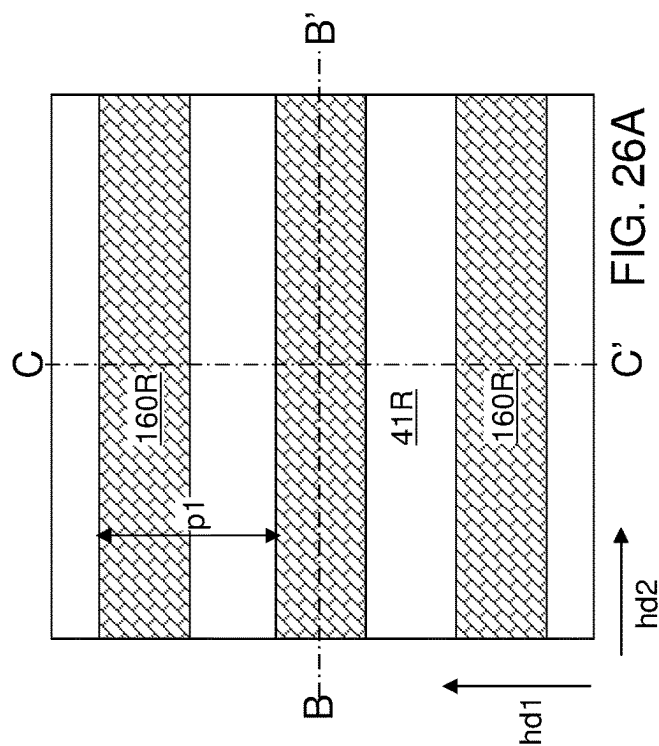


FIG. 25C



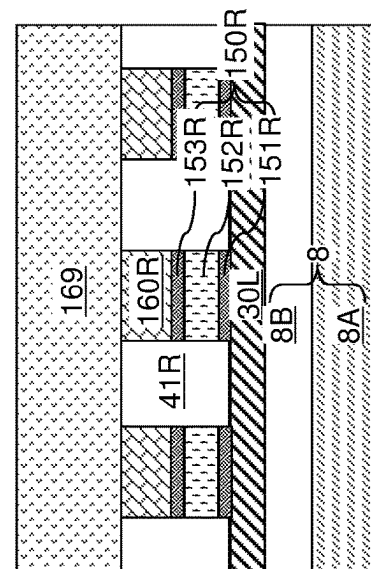
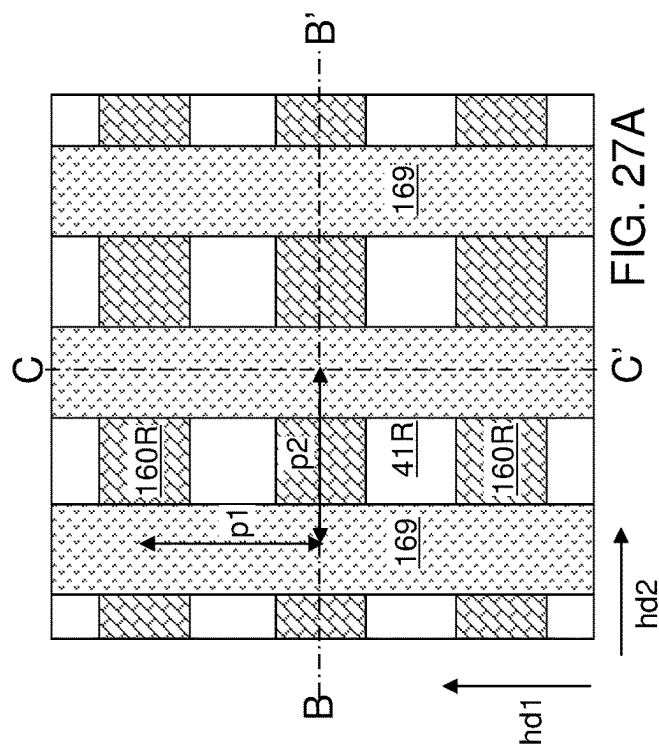


FIG. 27C

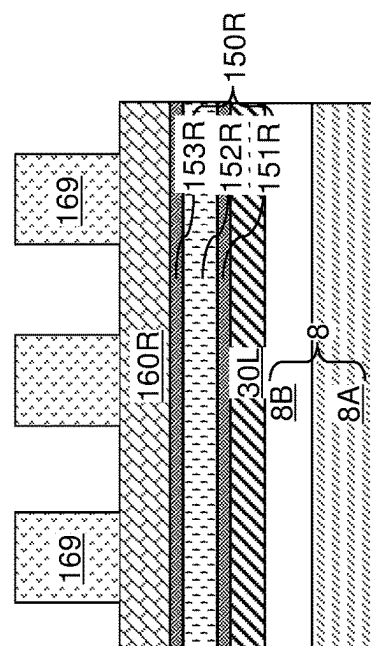


FIG. 27E

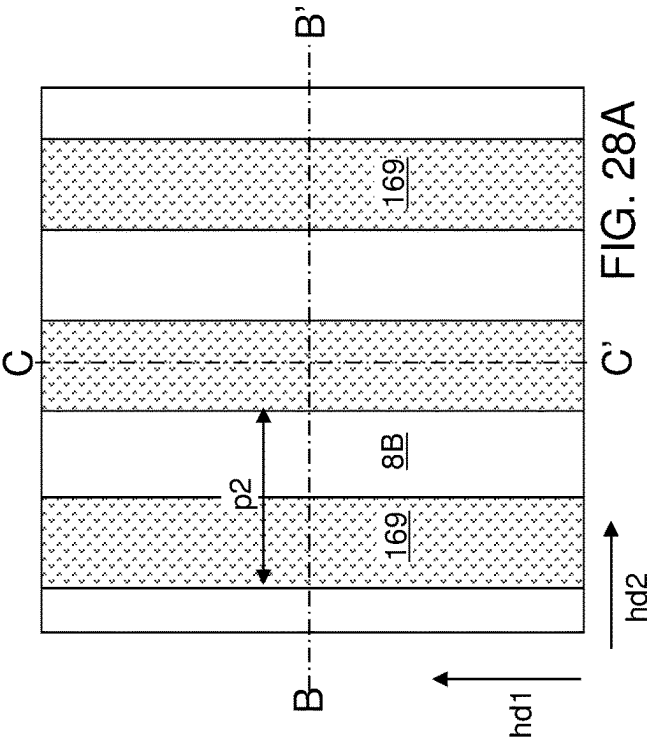


FIG. 28A

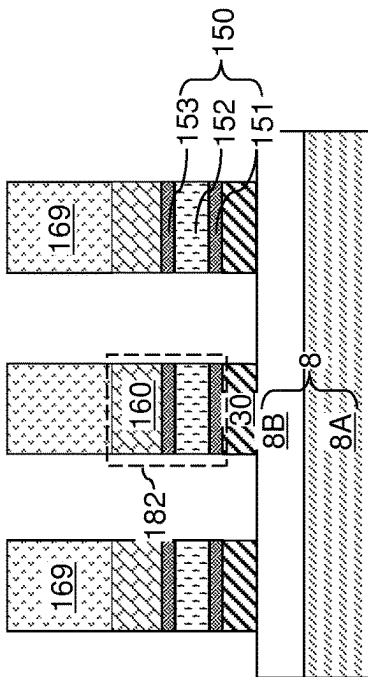


FIG. 28B

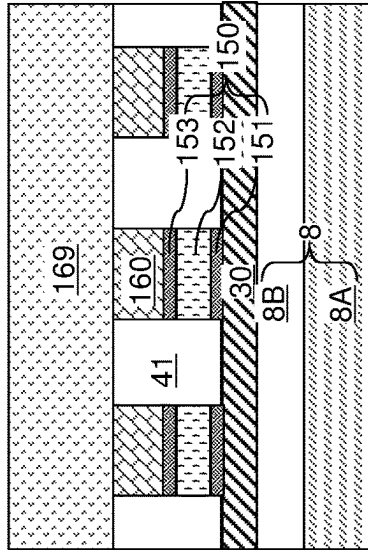
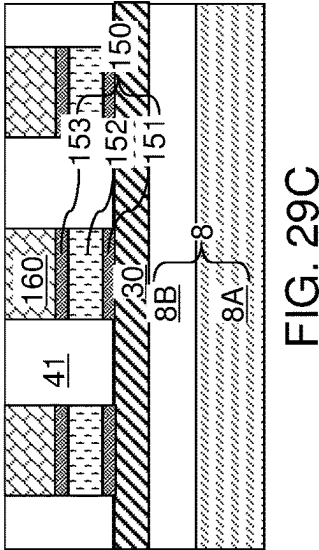
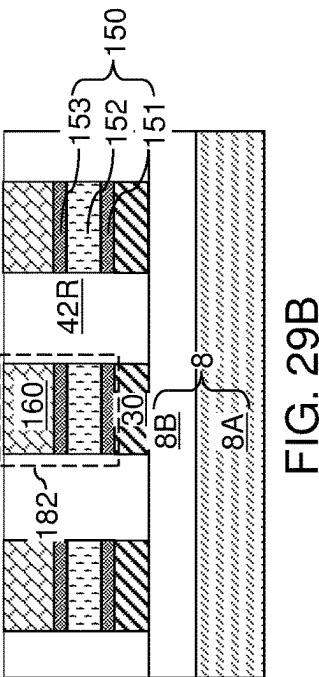
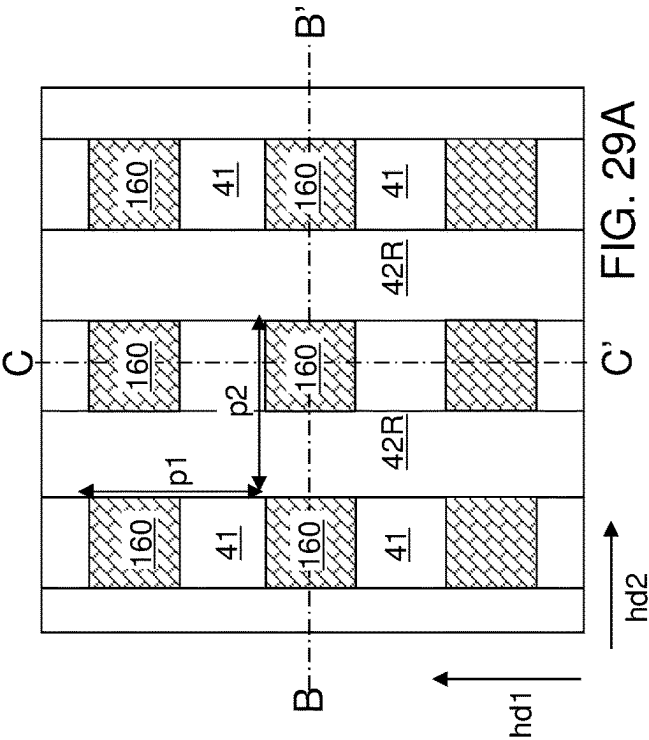


FIG. 28C



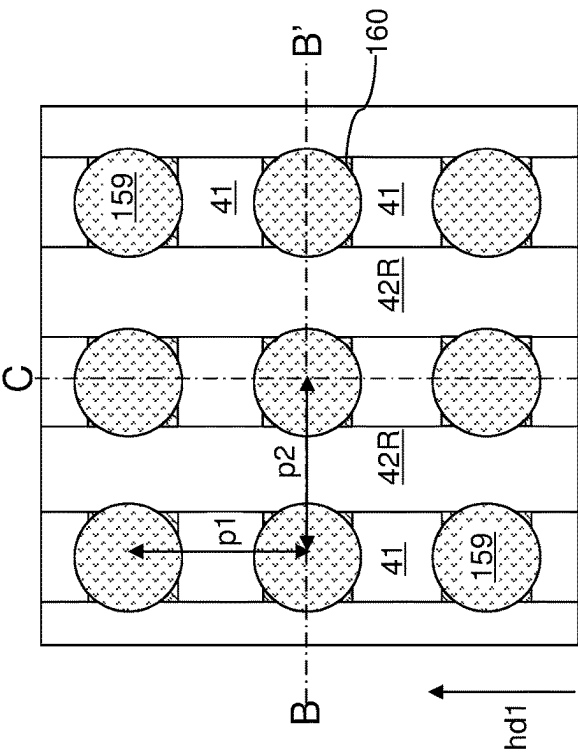


FIG. 30A

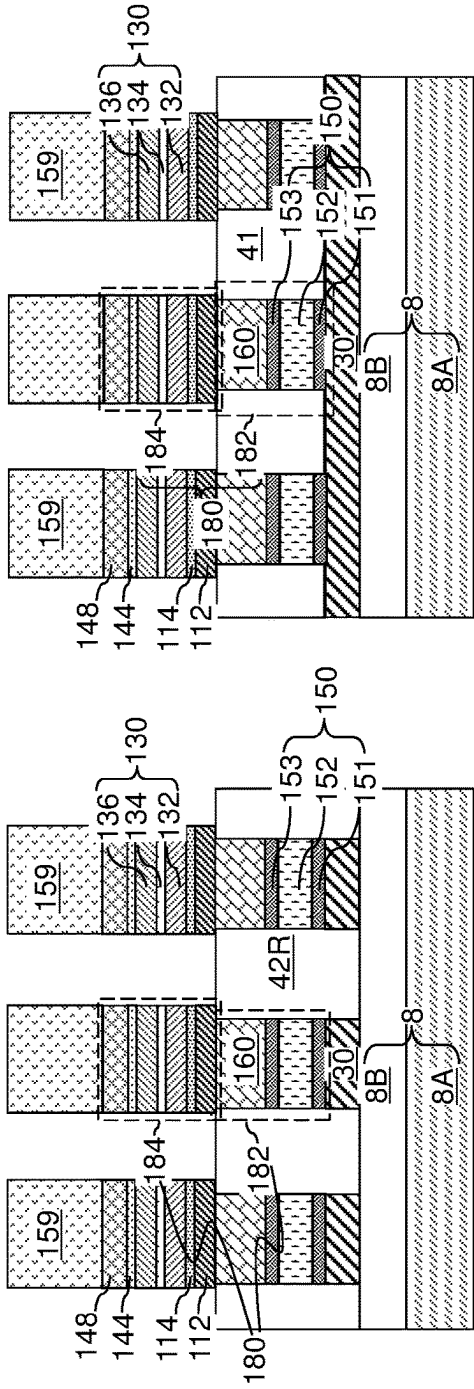


FIG. 30B

FIG. 30C

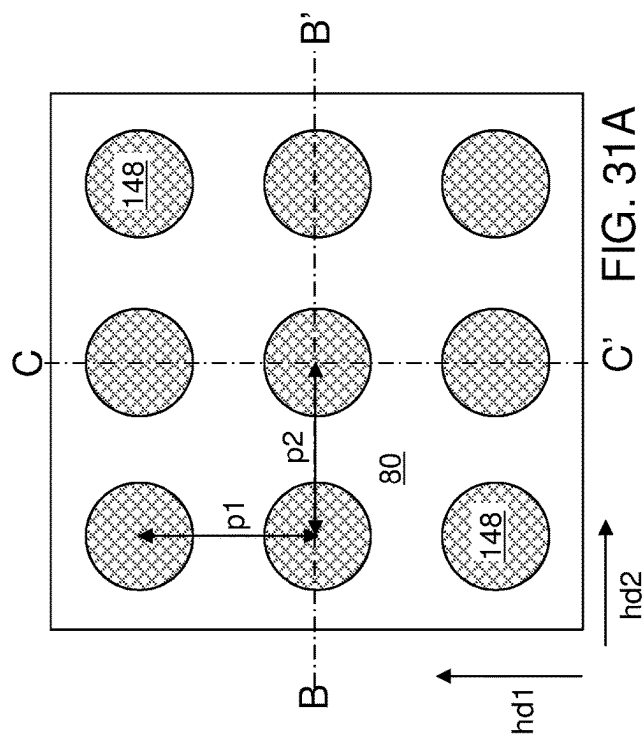


FIG. 31A

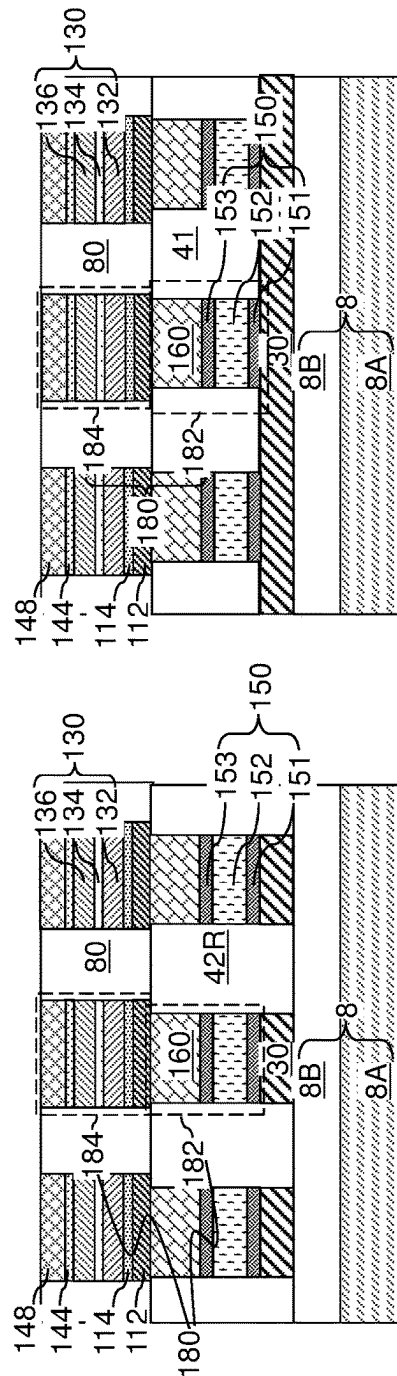


FIG. 31C

FIG. 31B



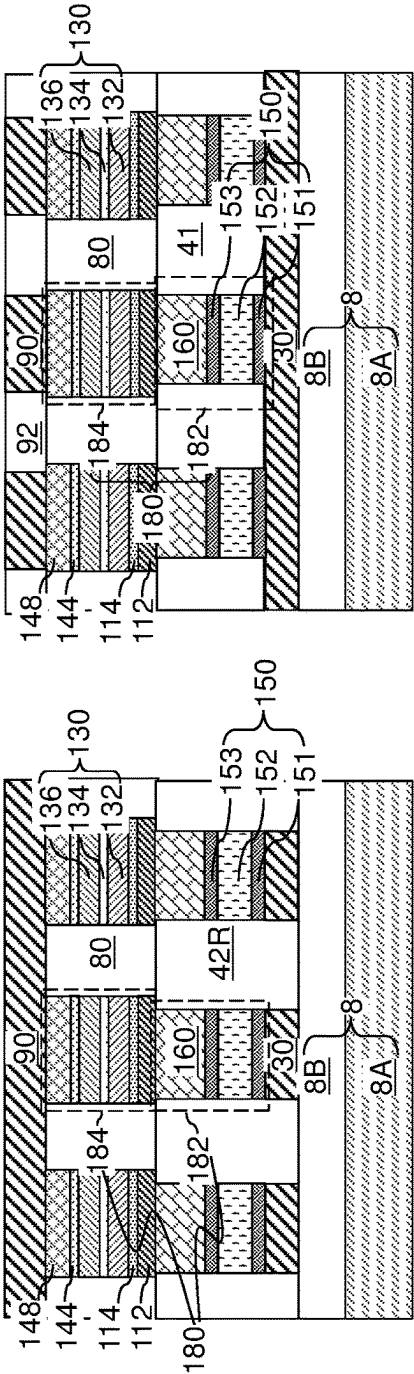
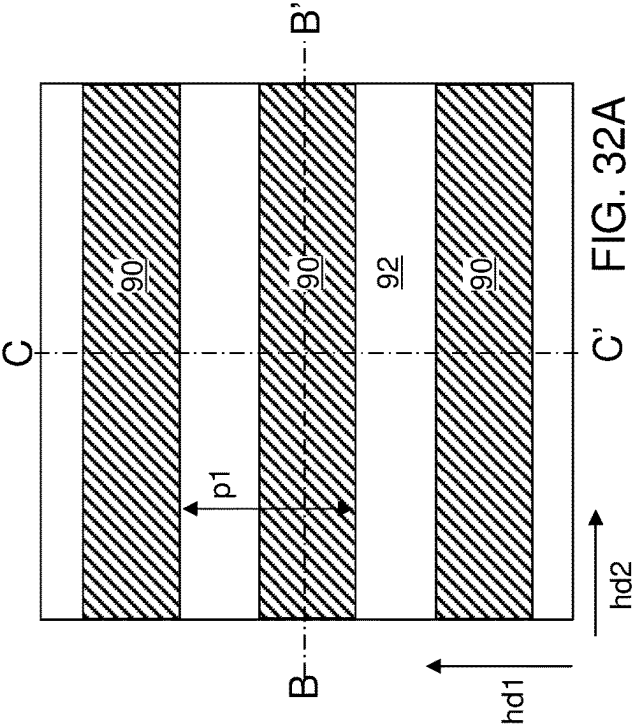
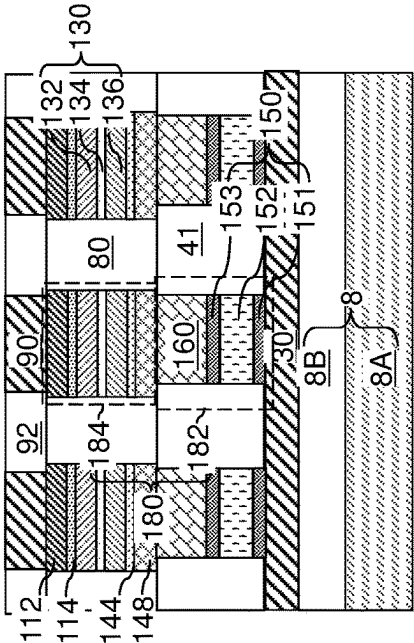
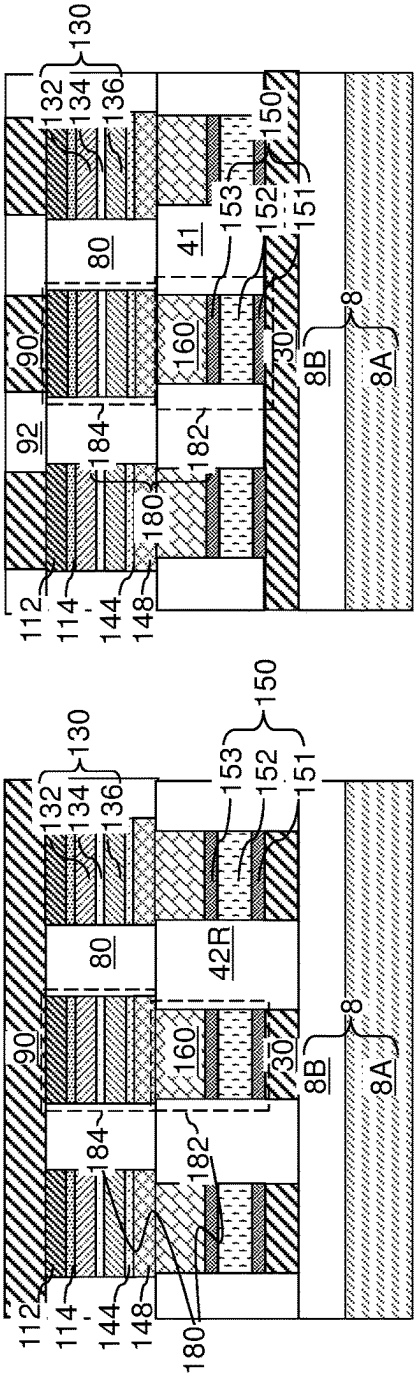
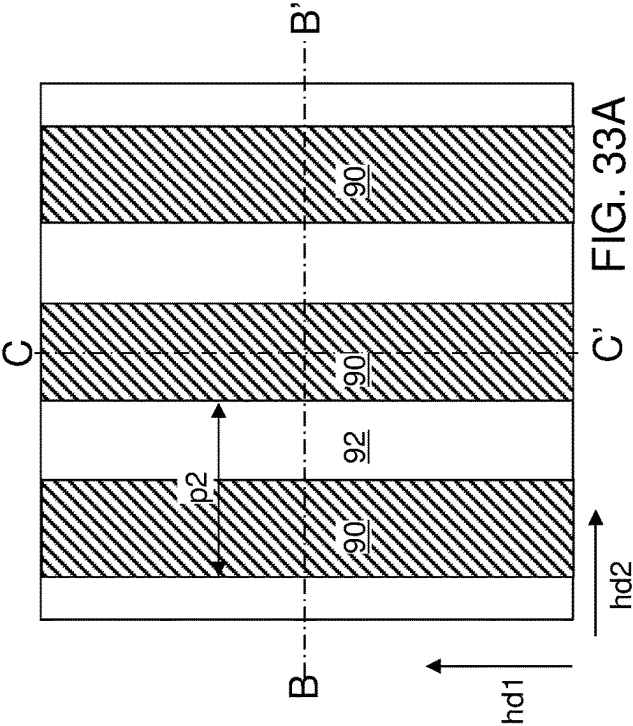


FIG. 32C

FIG. 32B



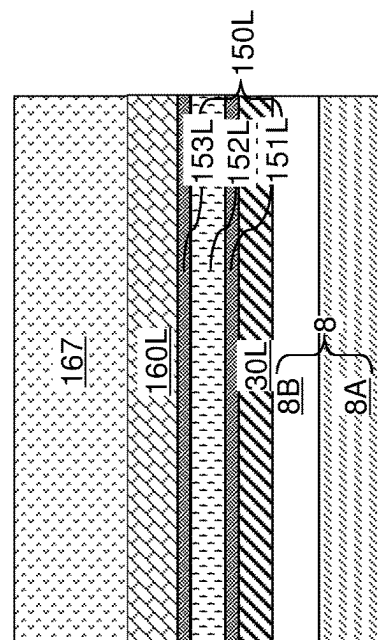
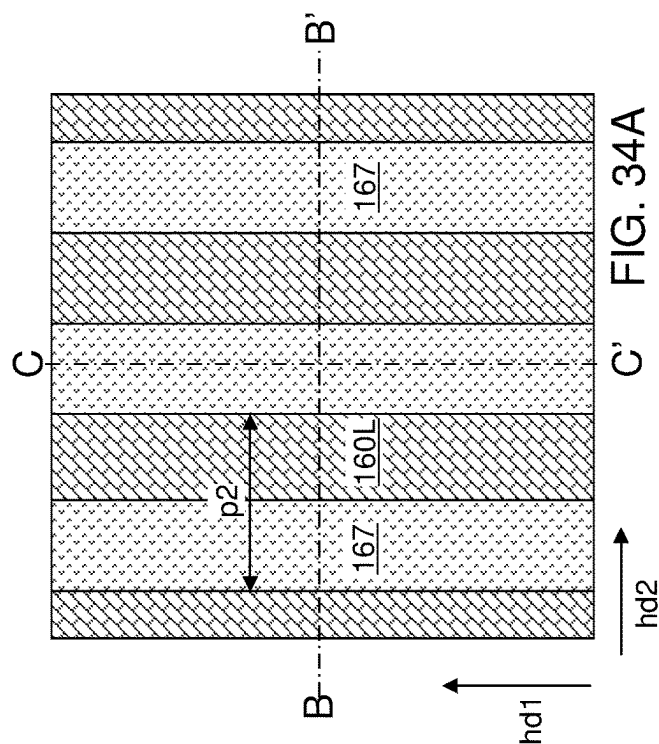


FIG. 34C

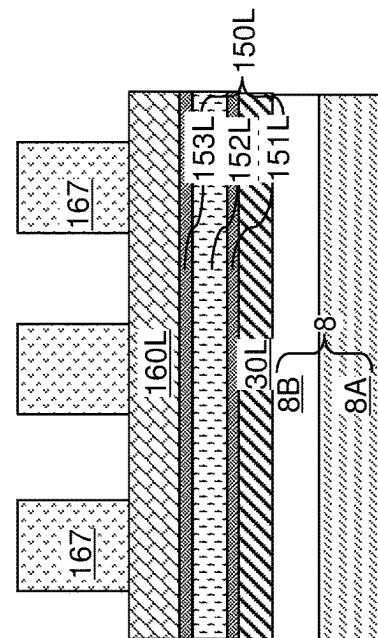


FIG. 34B

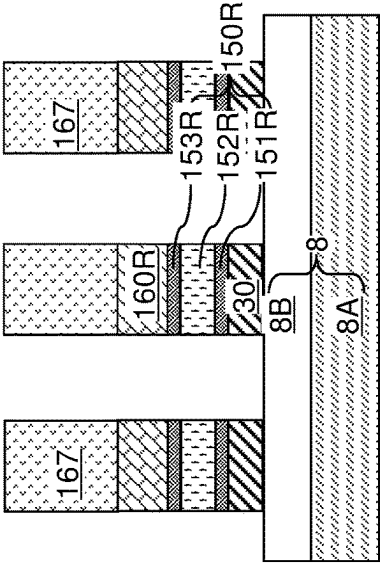
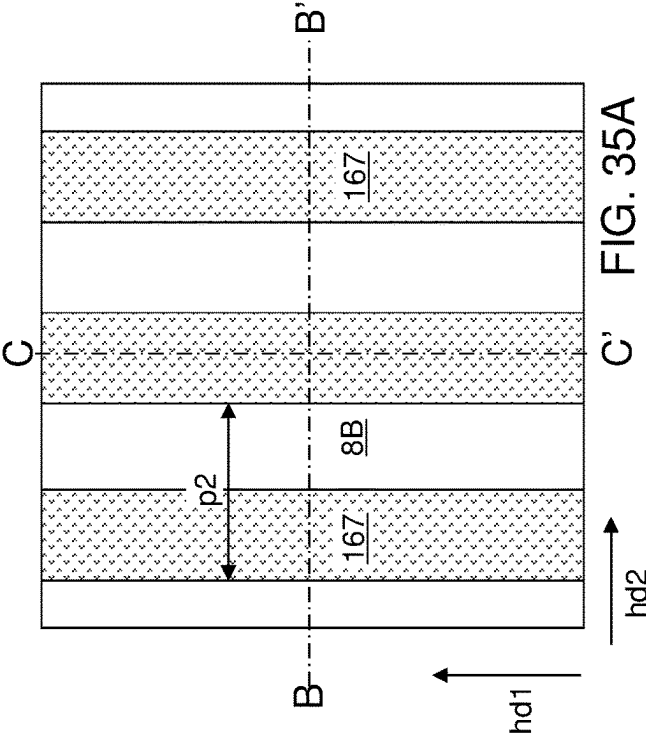


FIG. 35B

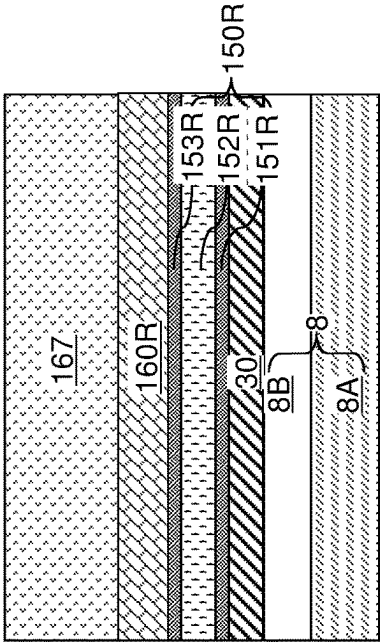


FIG. 35C

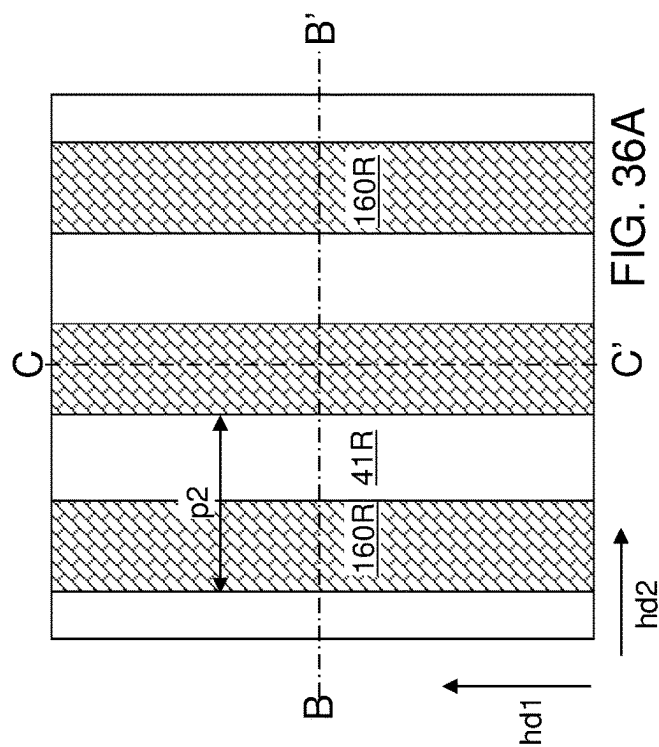


FIG. 36A

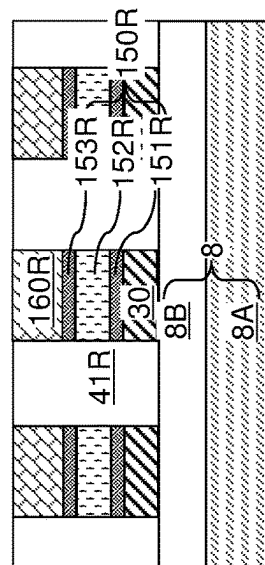


FIG. 36B

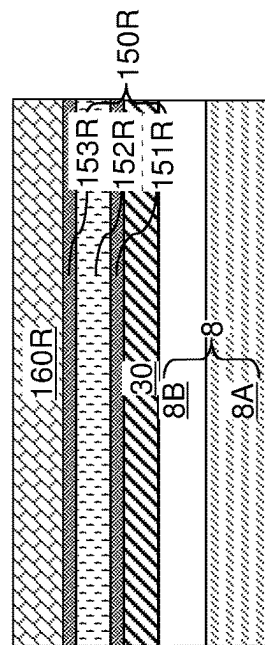


FIG. 36C

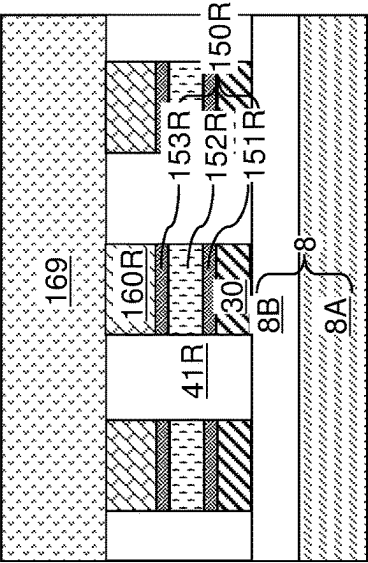
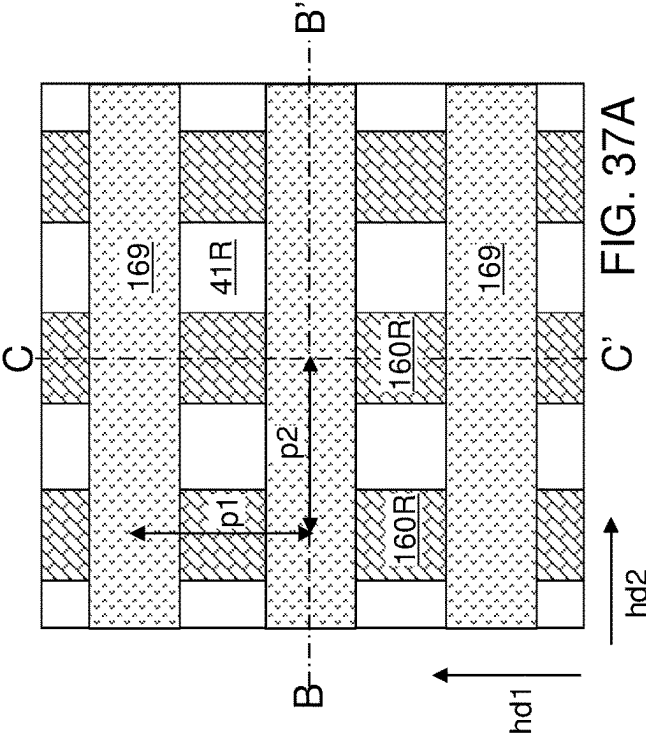


FIG. 37B

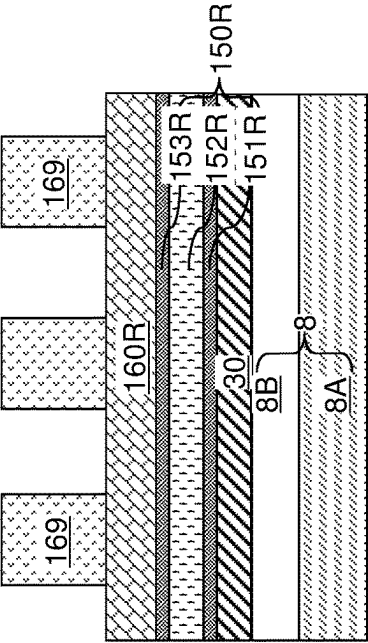
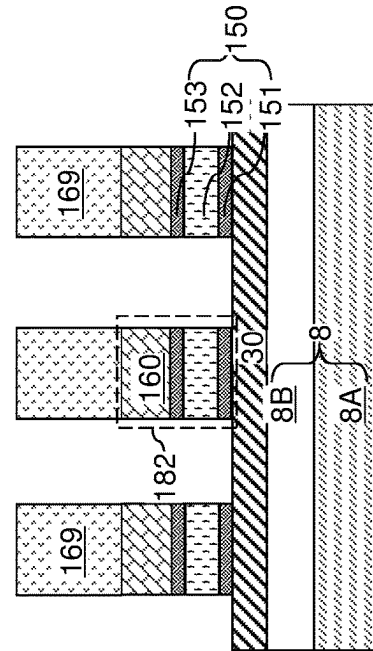
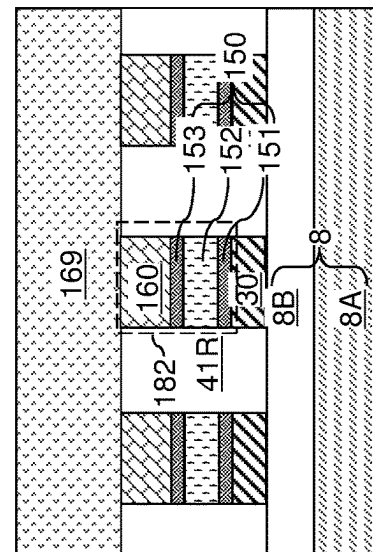
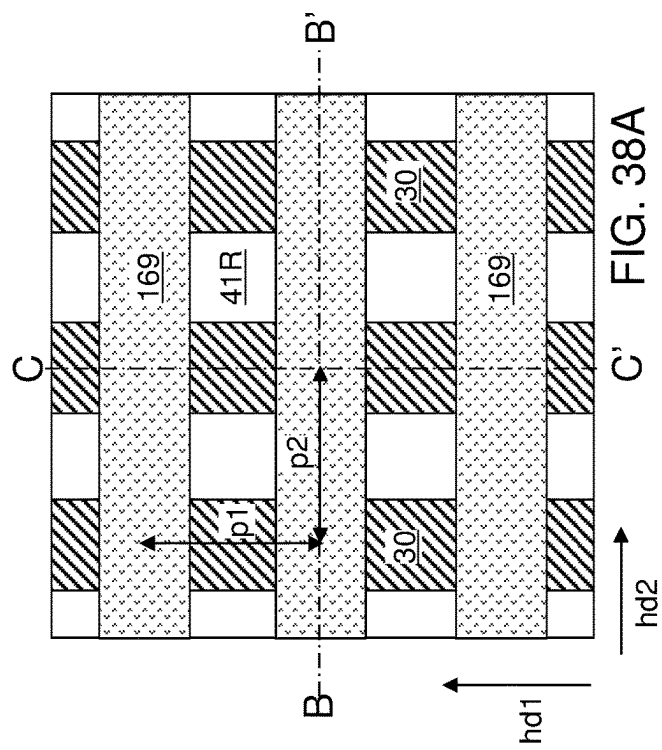


FIG. 37C



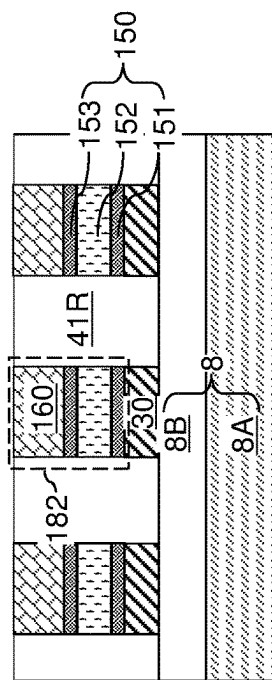
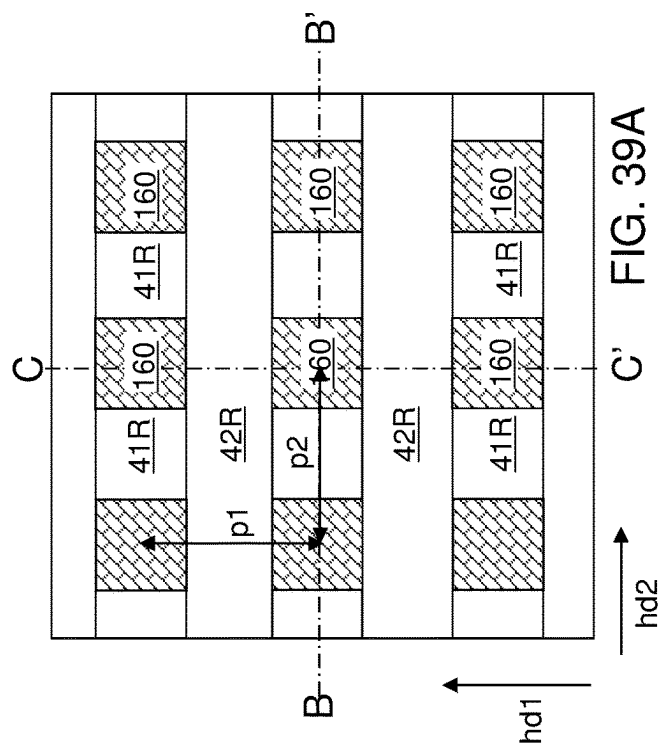


FIG. 39B

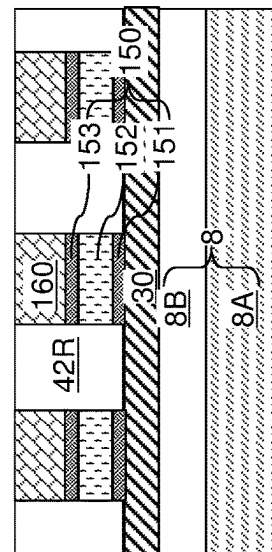


FIG. 39C



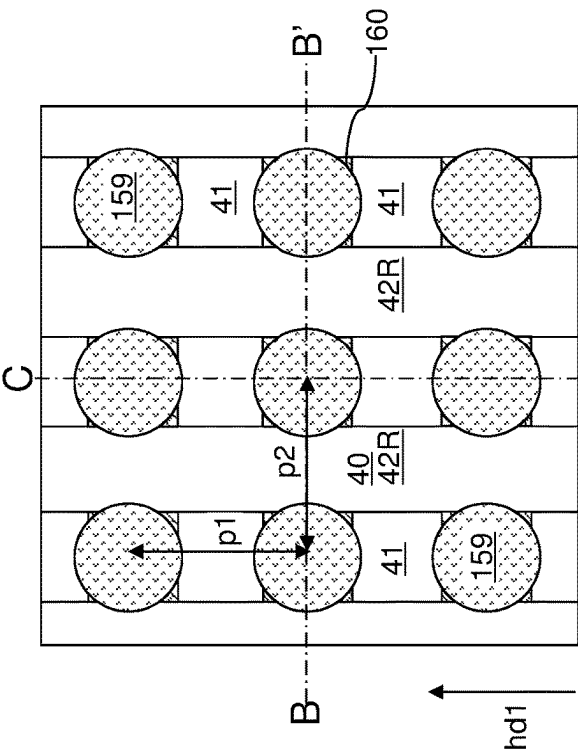


FIG. 40A

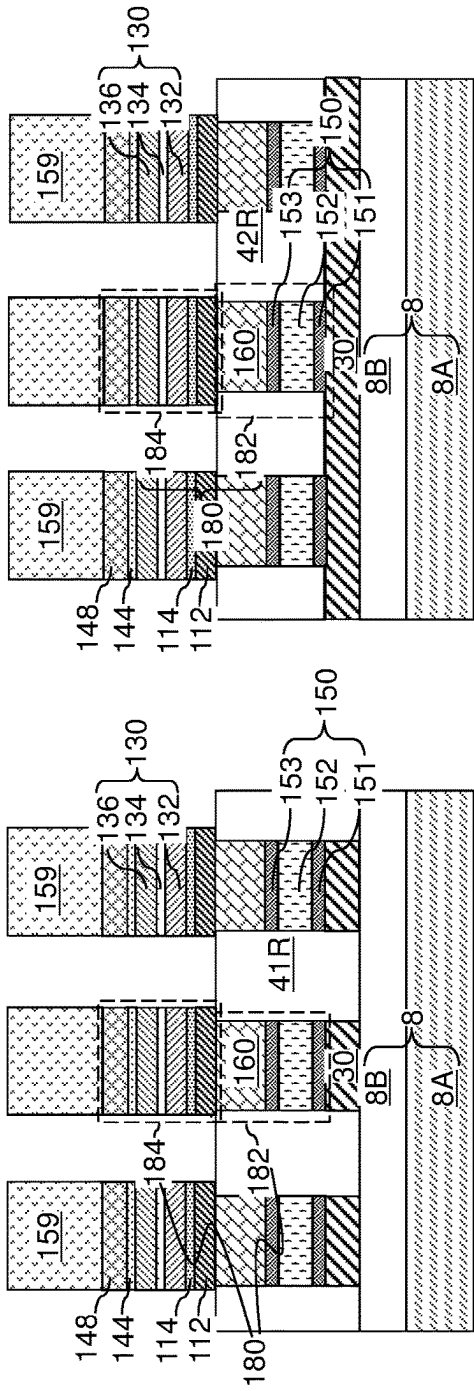


FIG. 40B

FIG. 40C

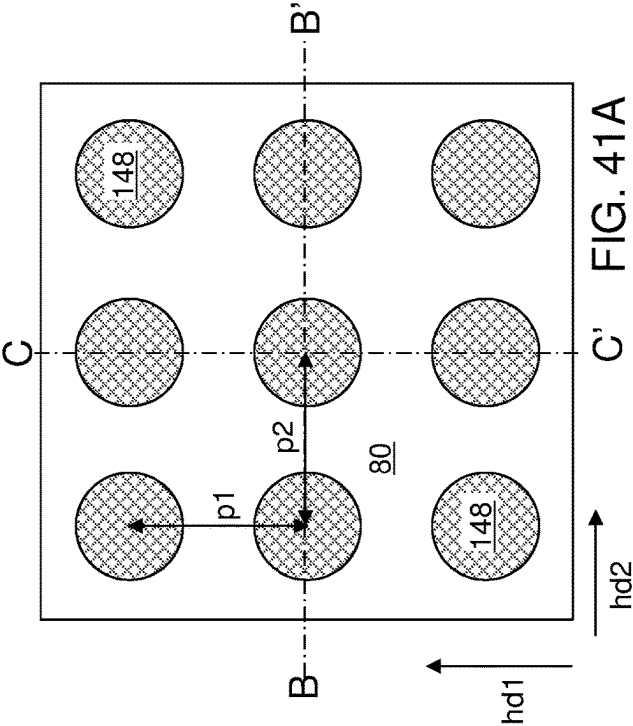


FIG. 41A

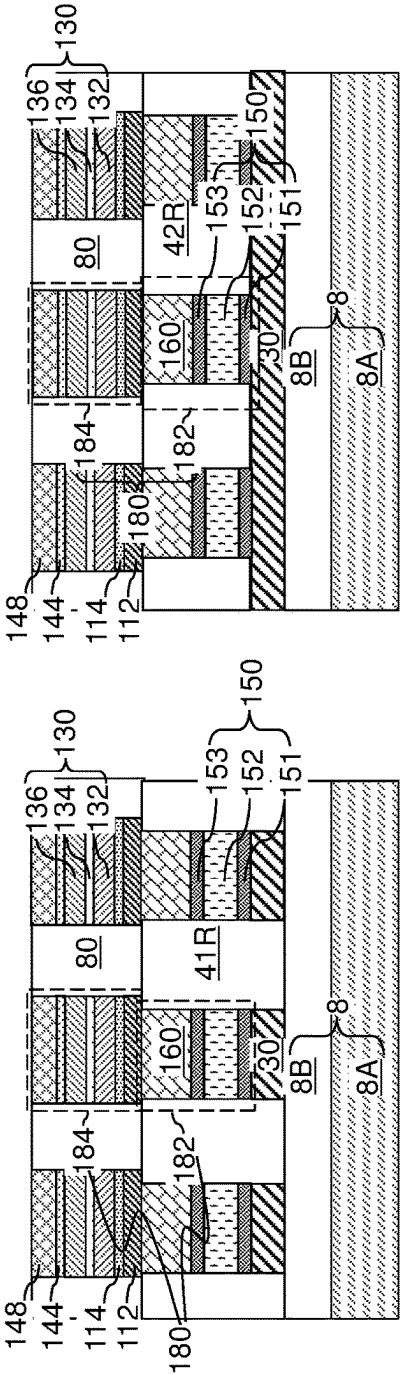


FIG. 41B

FIG. 41C

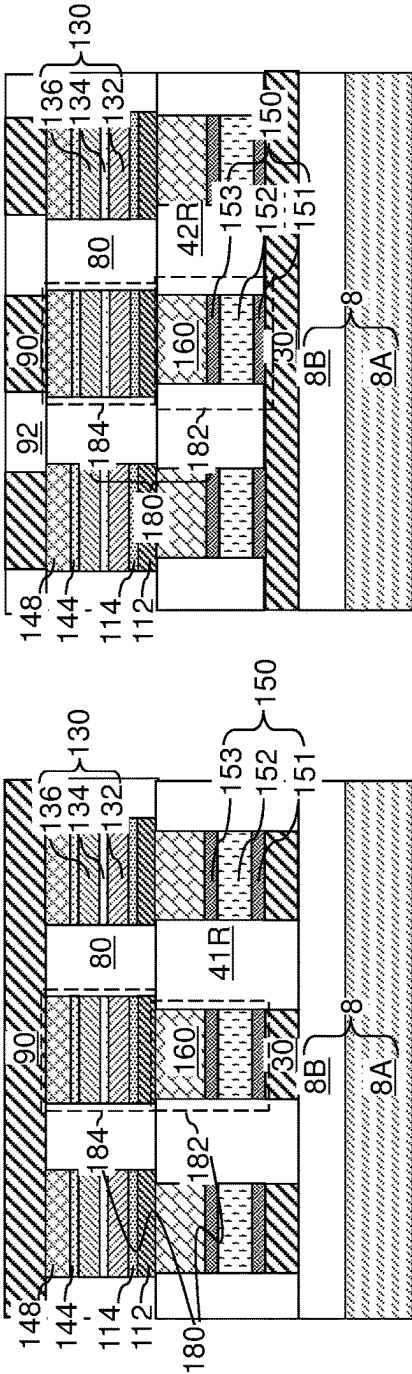
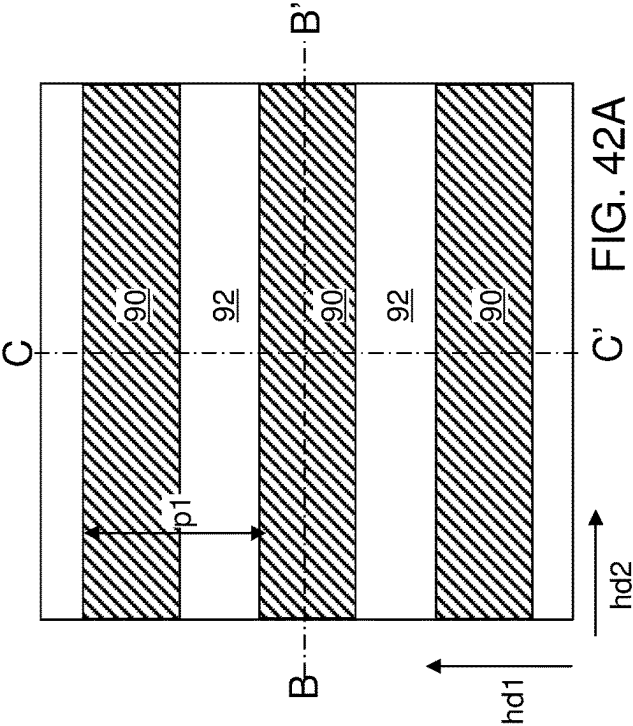
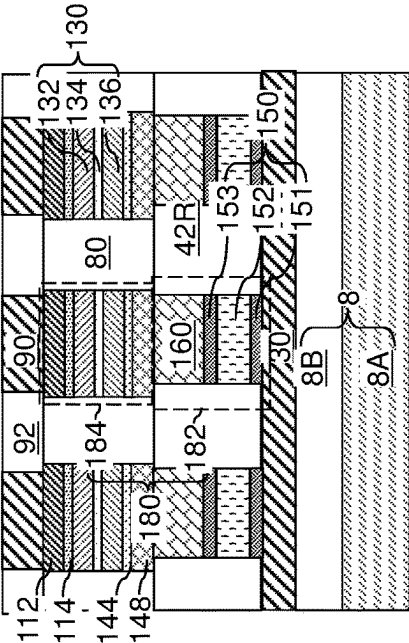
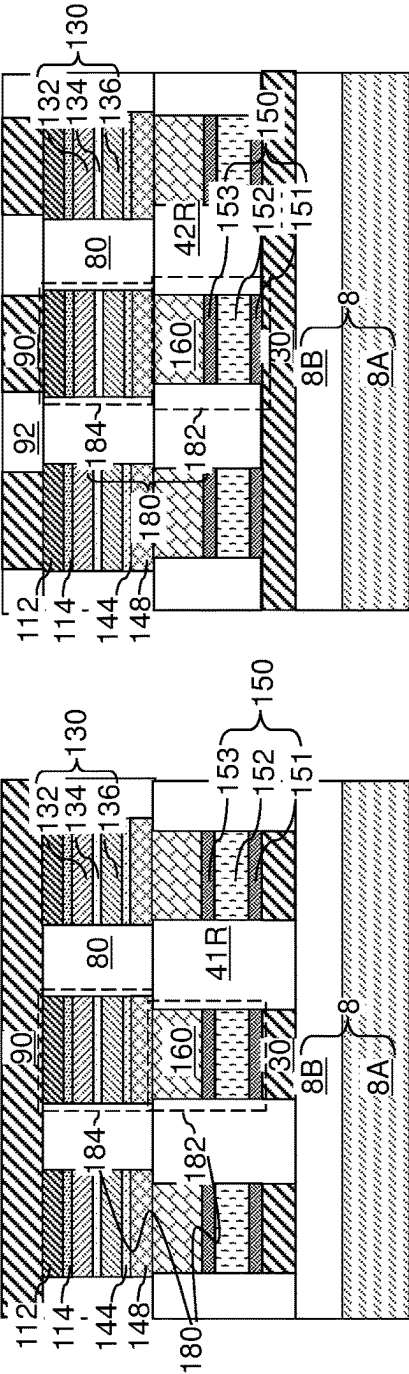
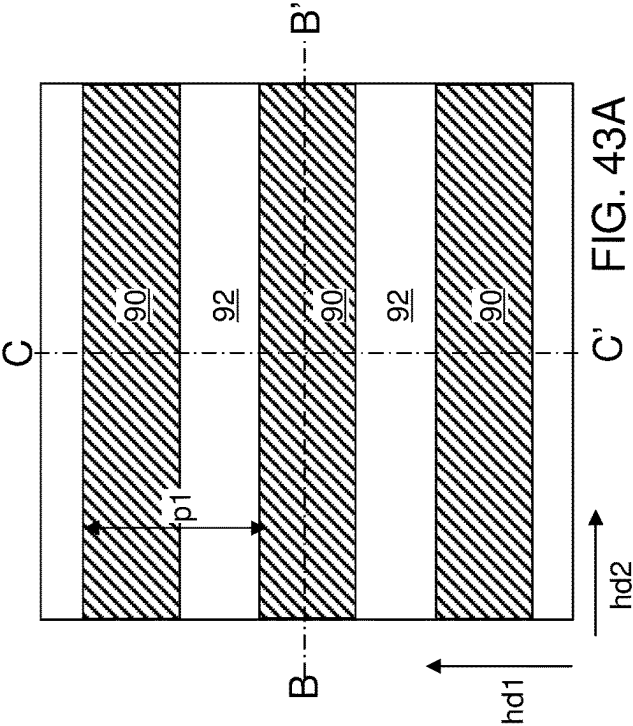
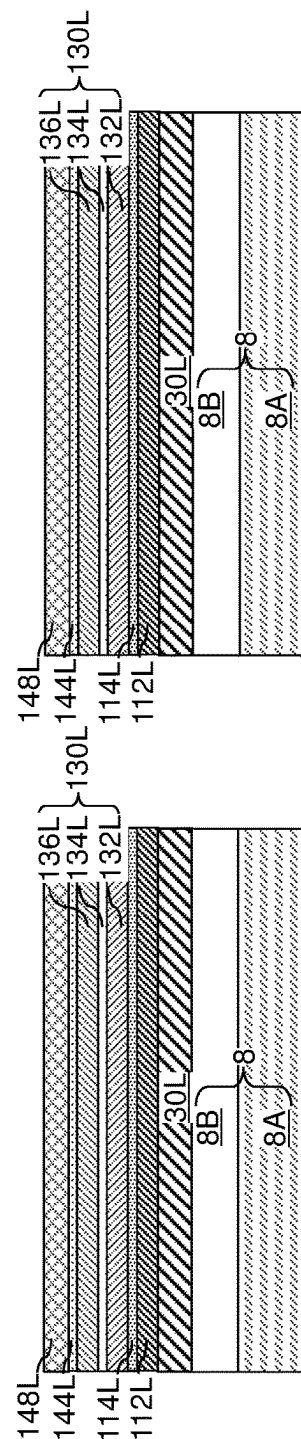
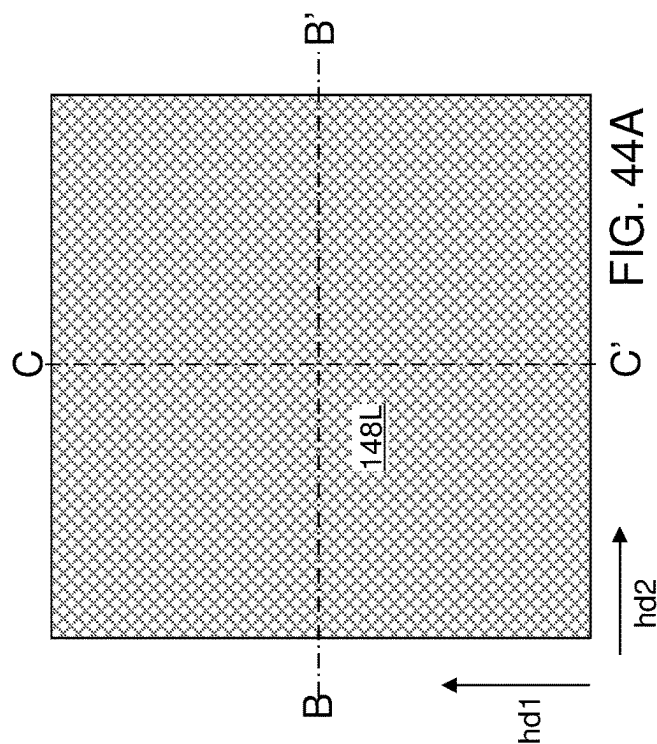


FIG. 42C

FIG. 42B





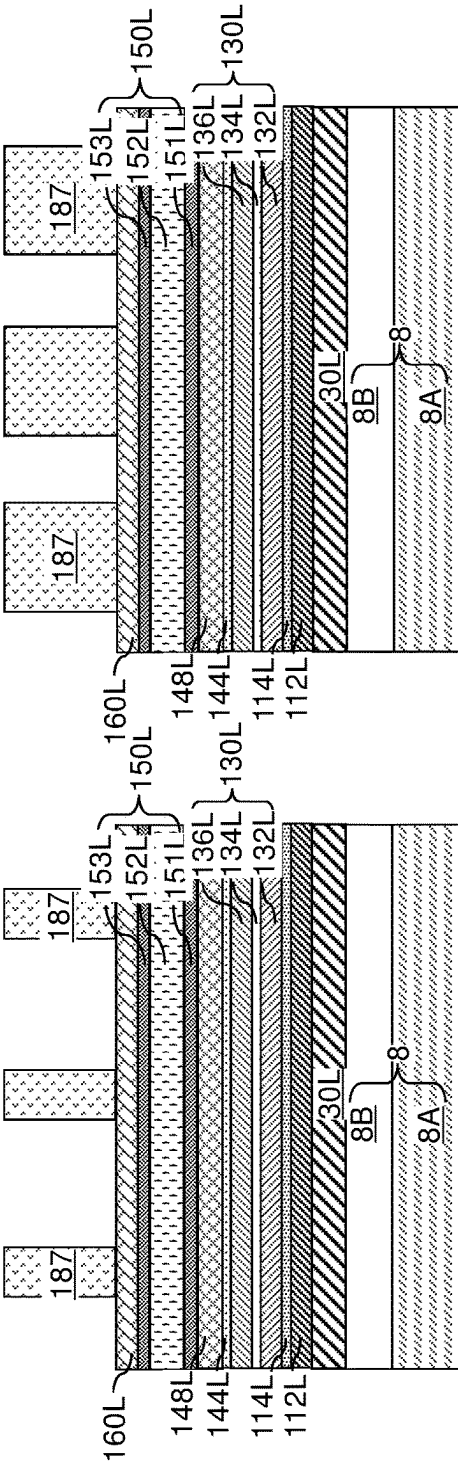
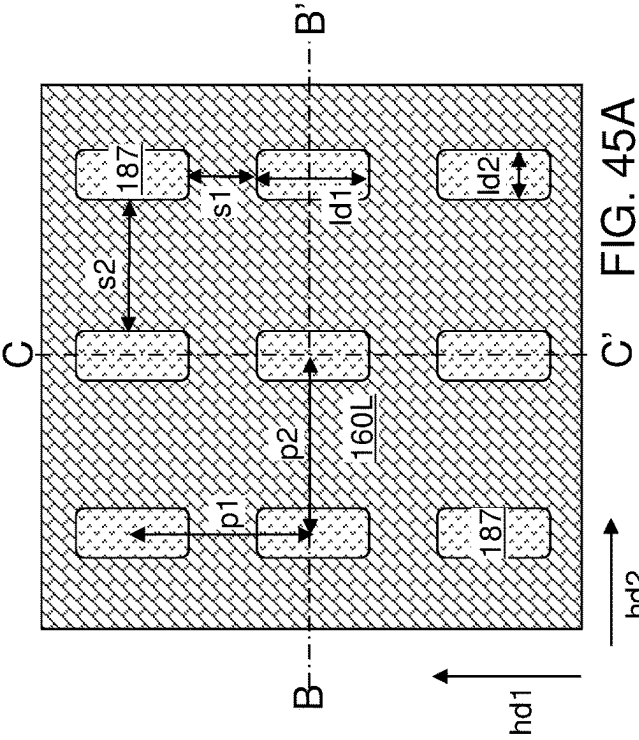
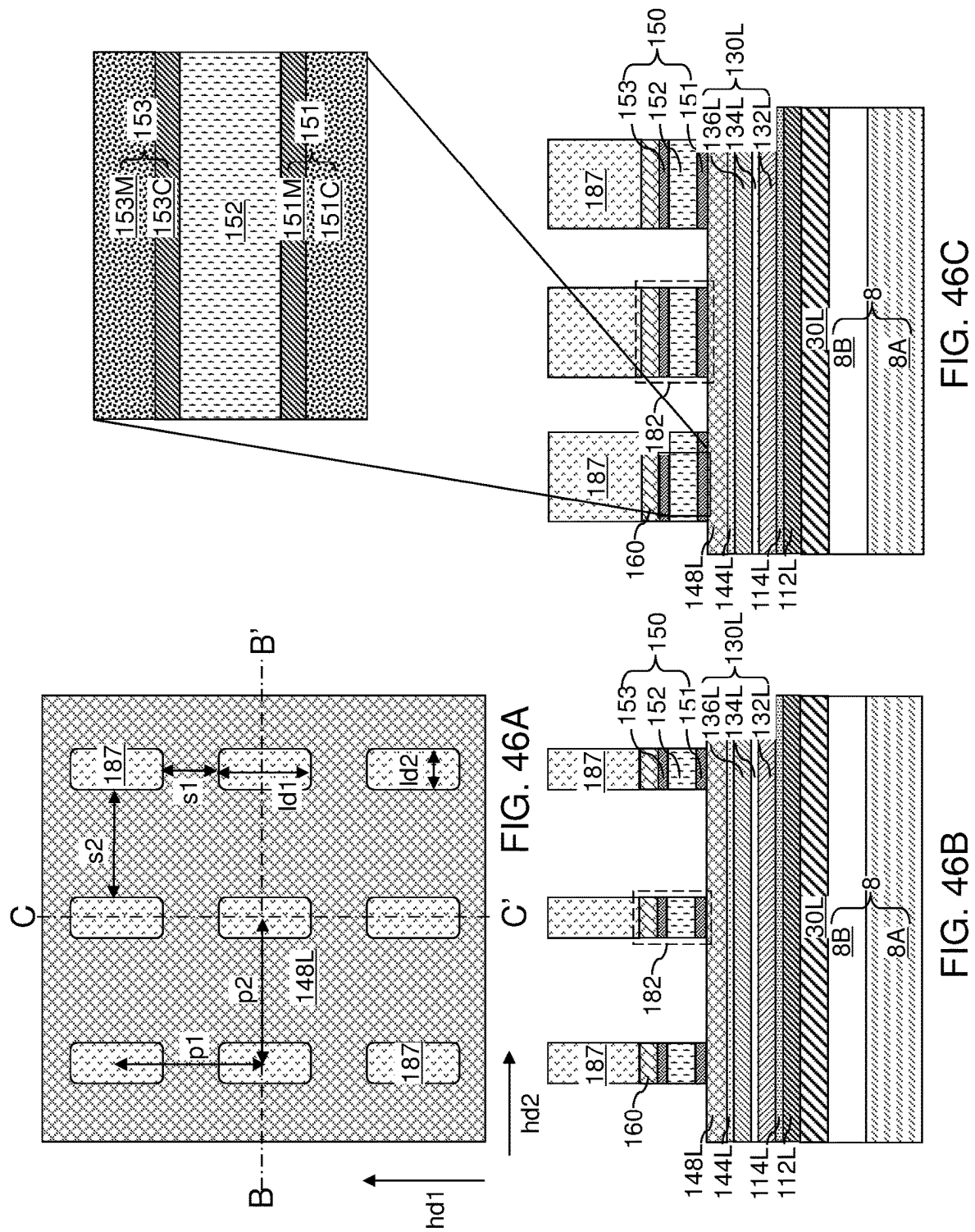
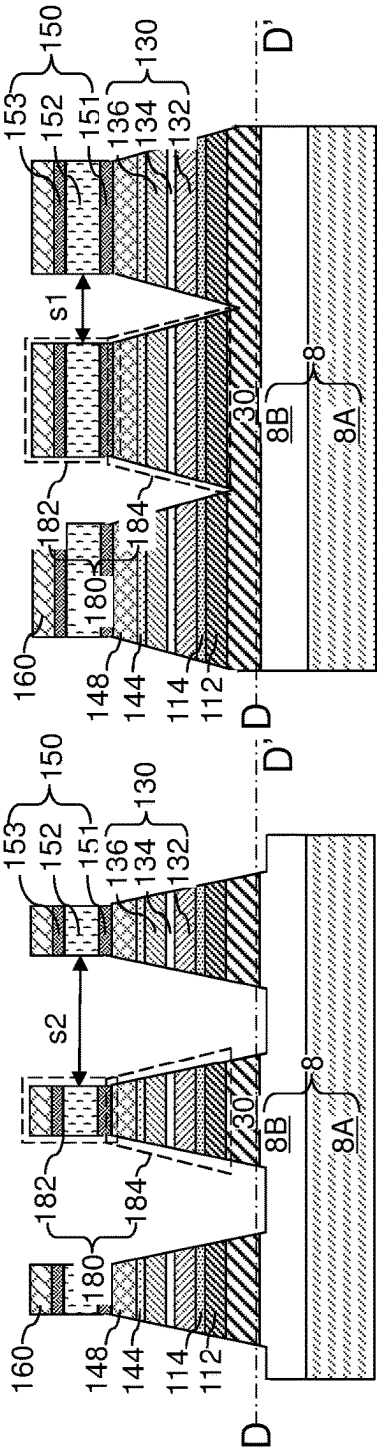
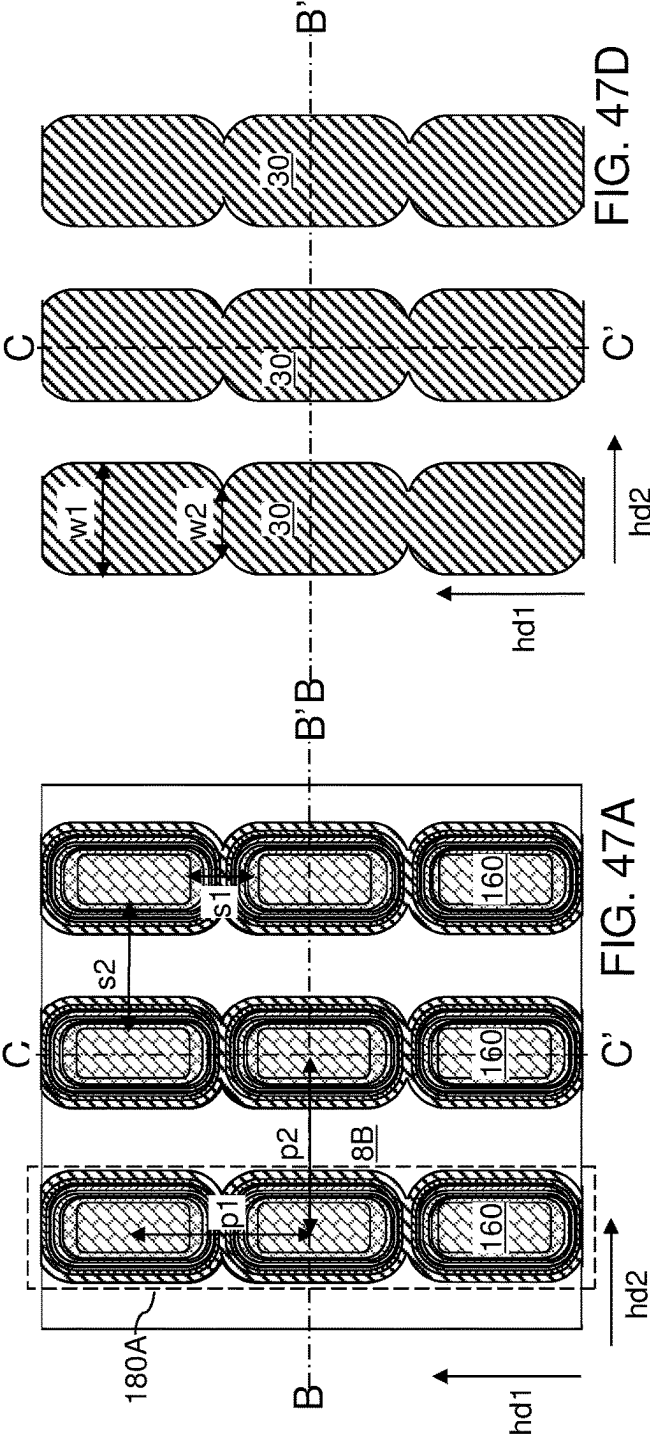


FIG. 45C

FIG. 45B







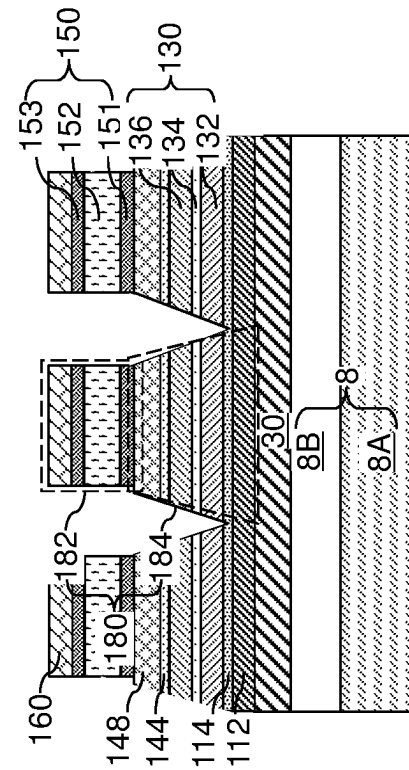


FIG. 47F

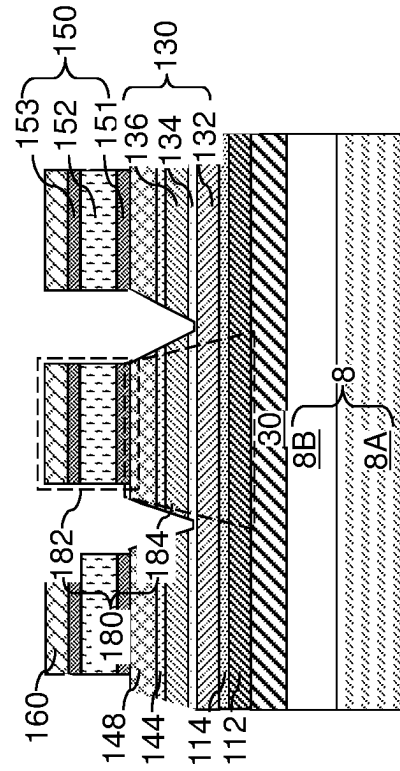


FIG. 47H

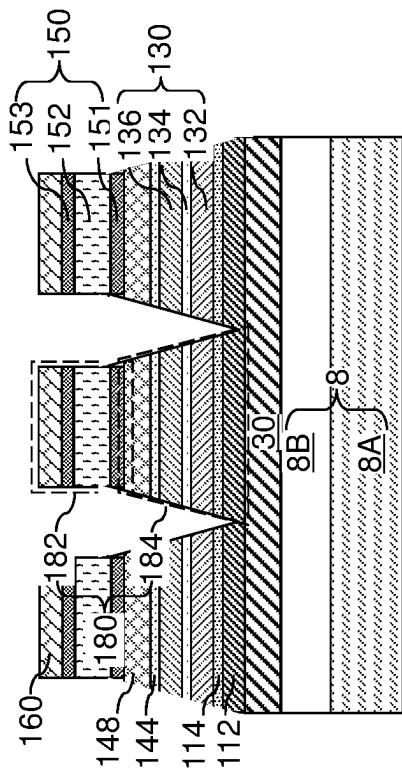


FIG. 47E

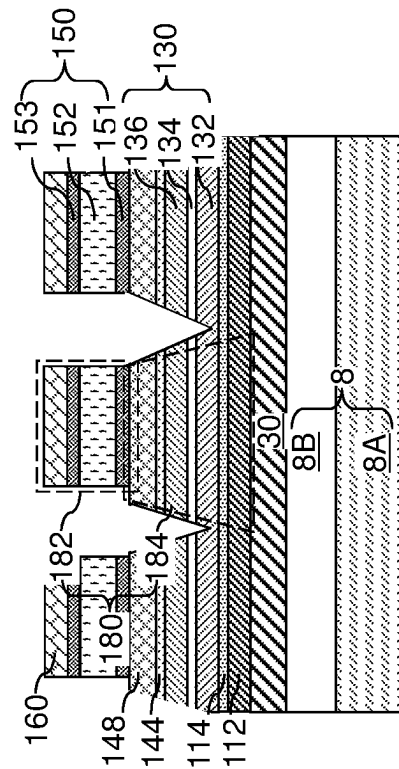


FIG. 47G

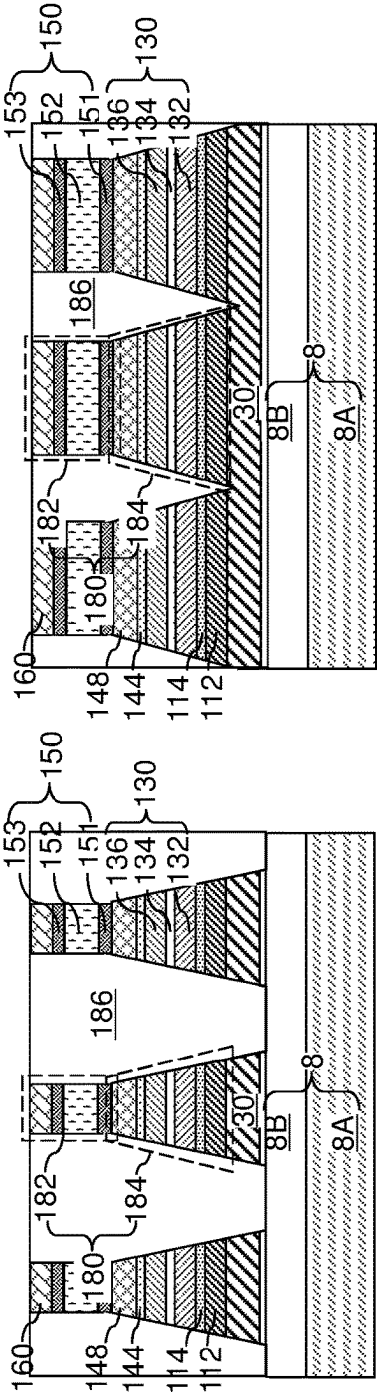
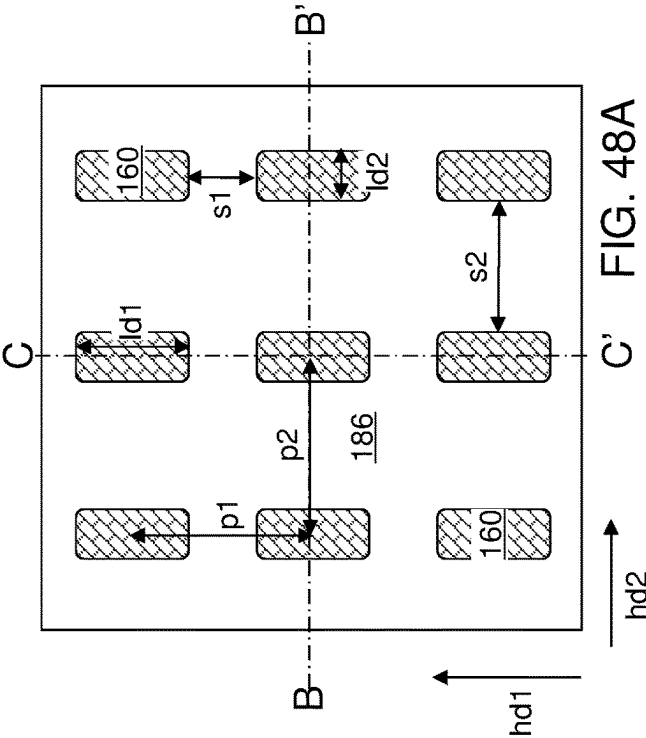
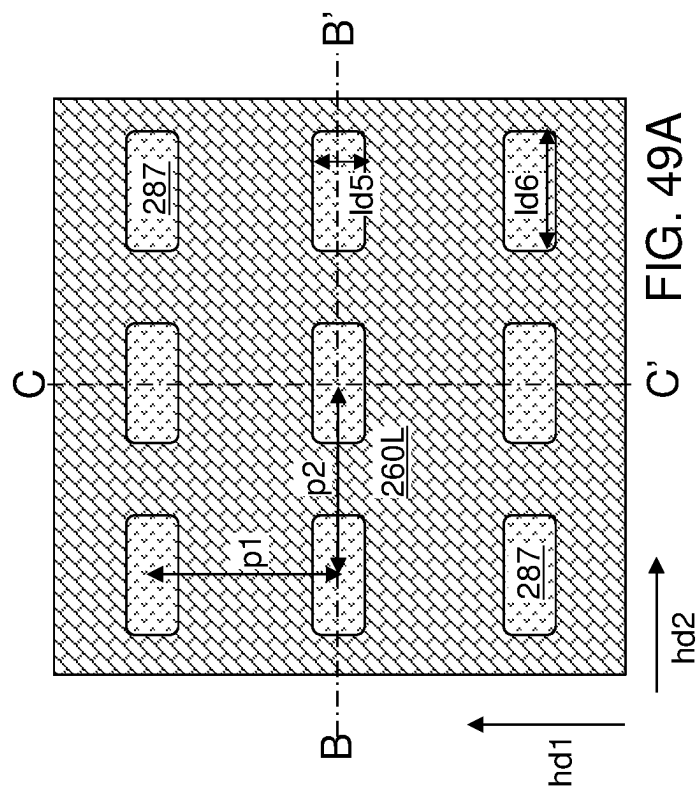


FIG. 48C

FIG. 48B



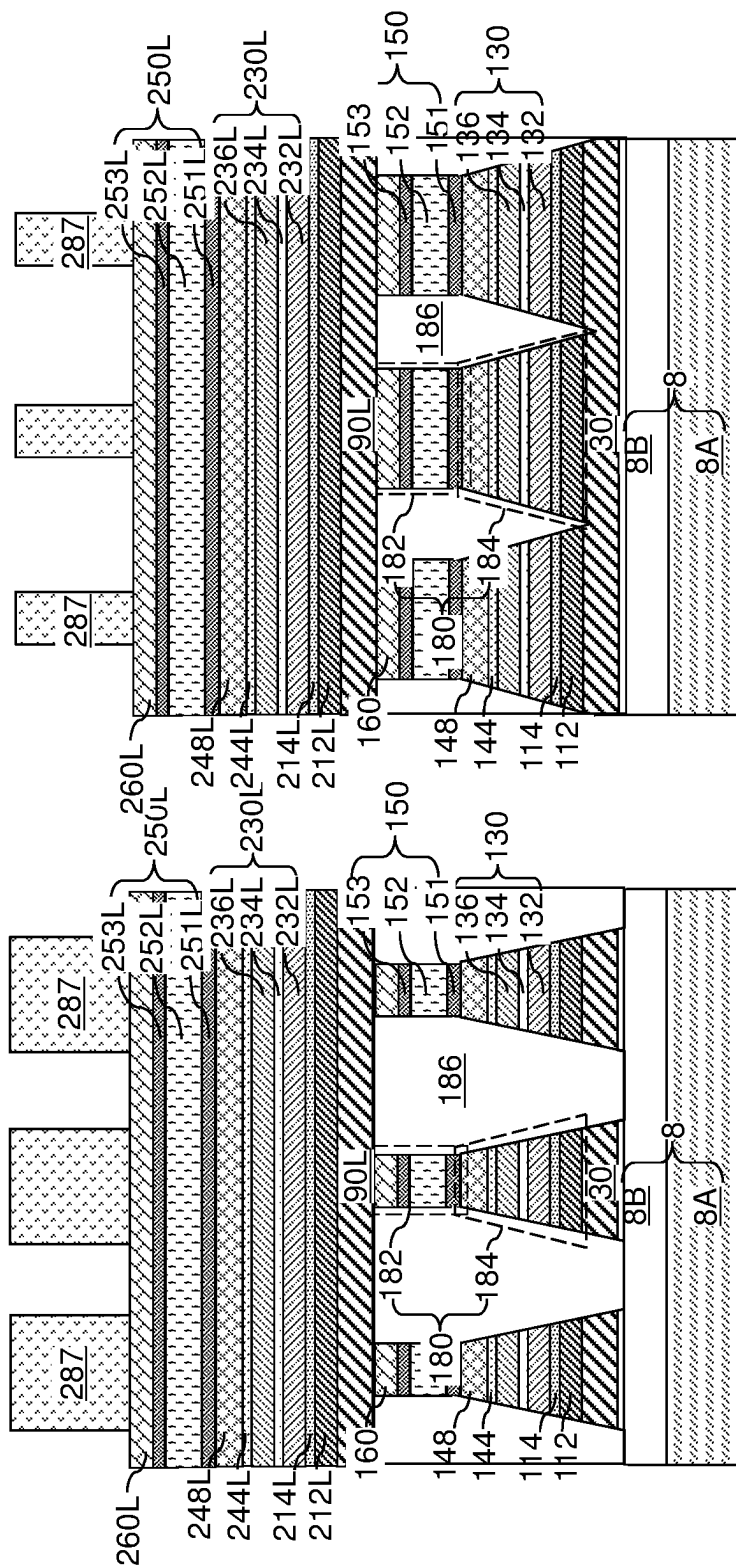
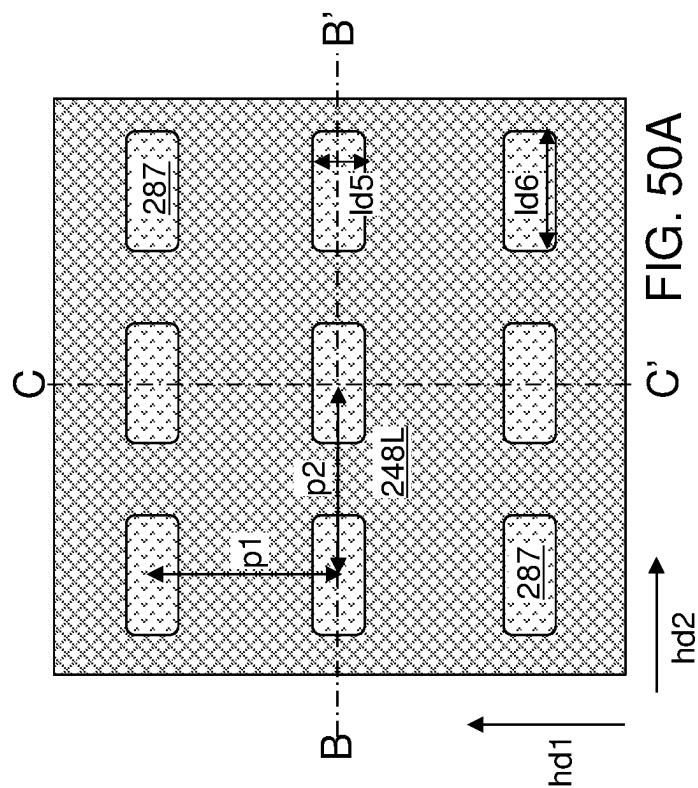


FIG. 49C

FIG. 49B



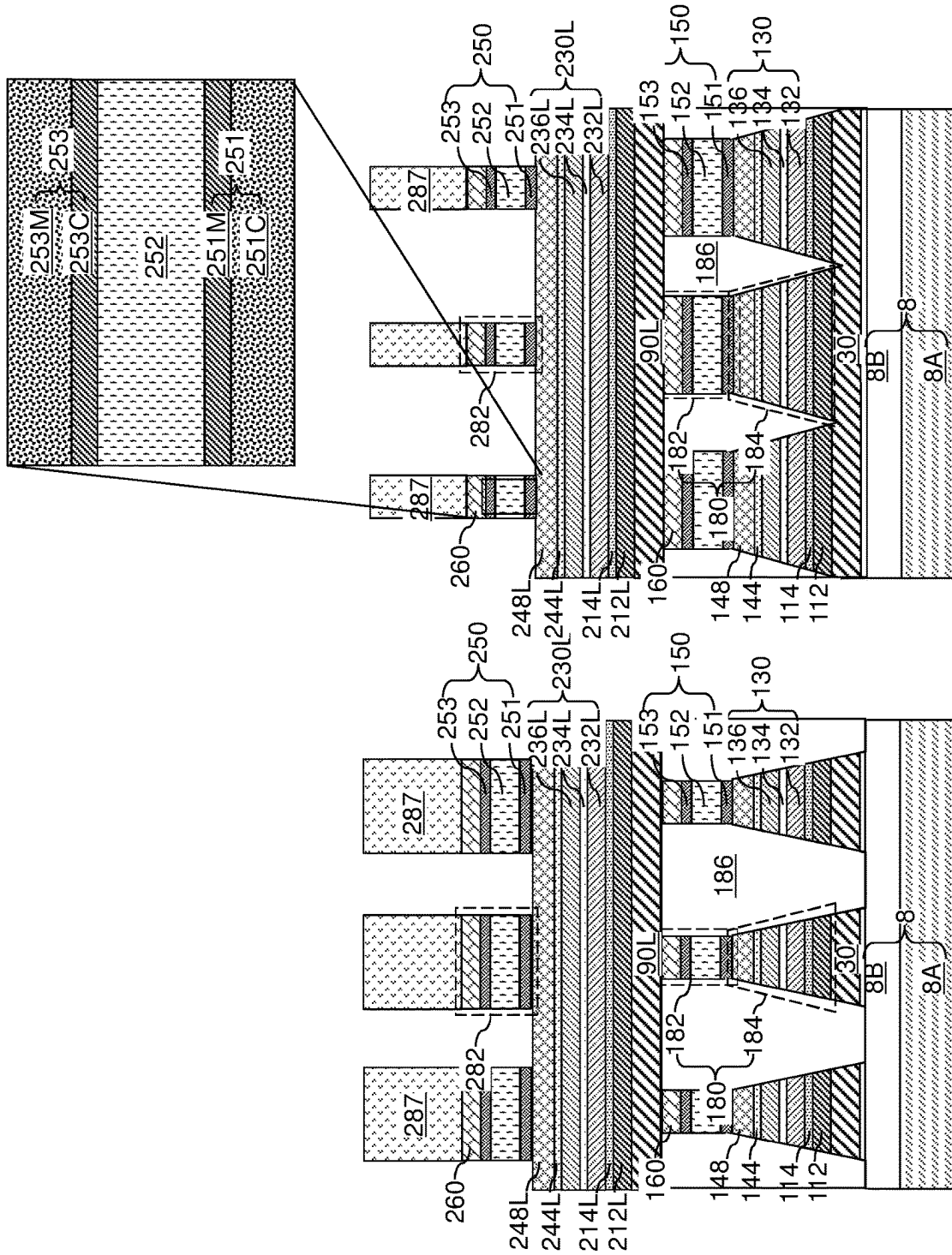
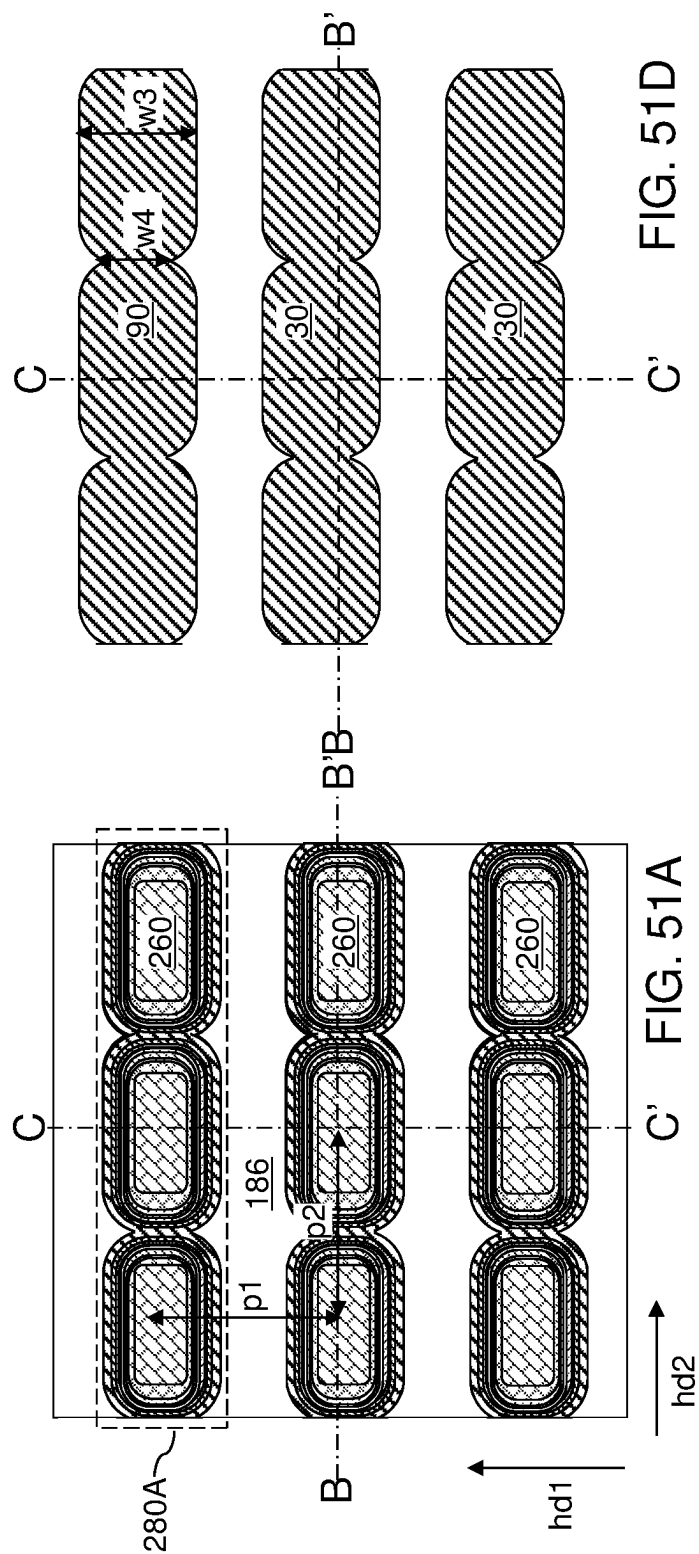


FIG. 50C

FIG. 50B



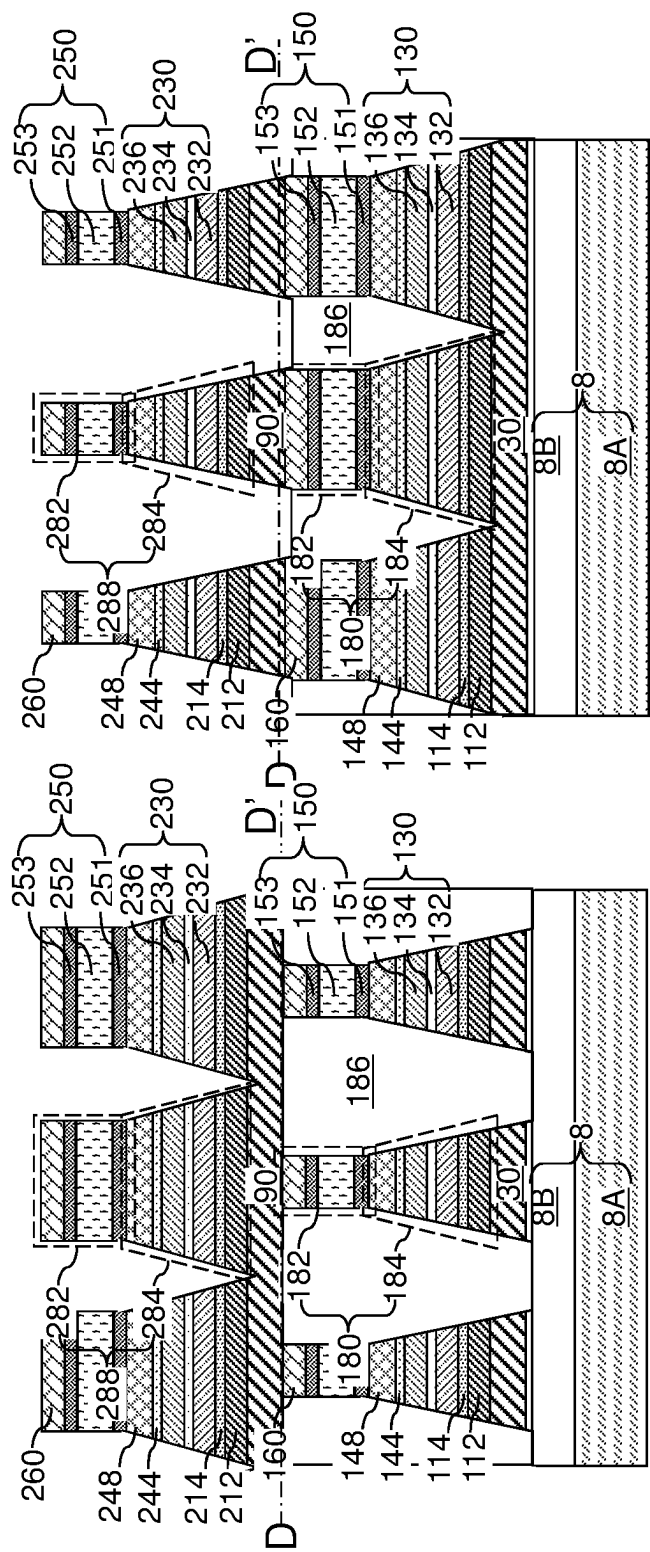
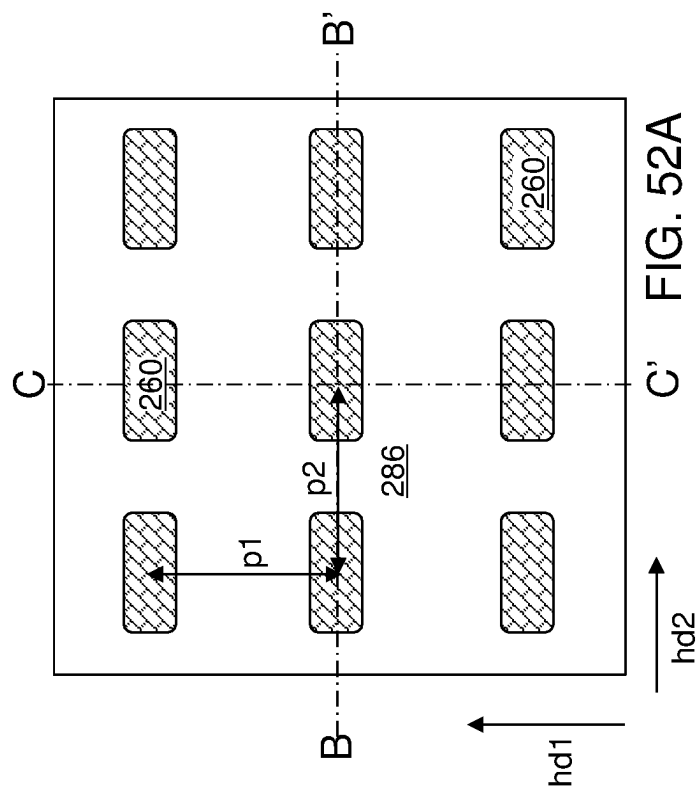


FIG. 51C

FIG. 51B





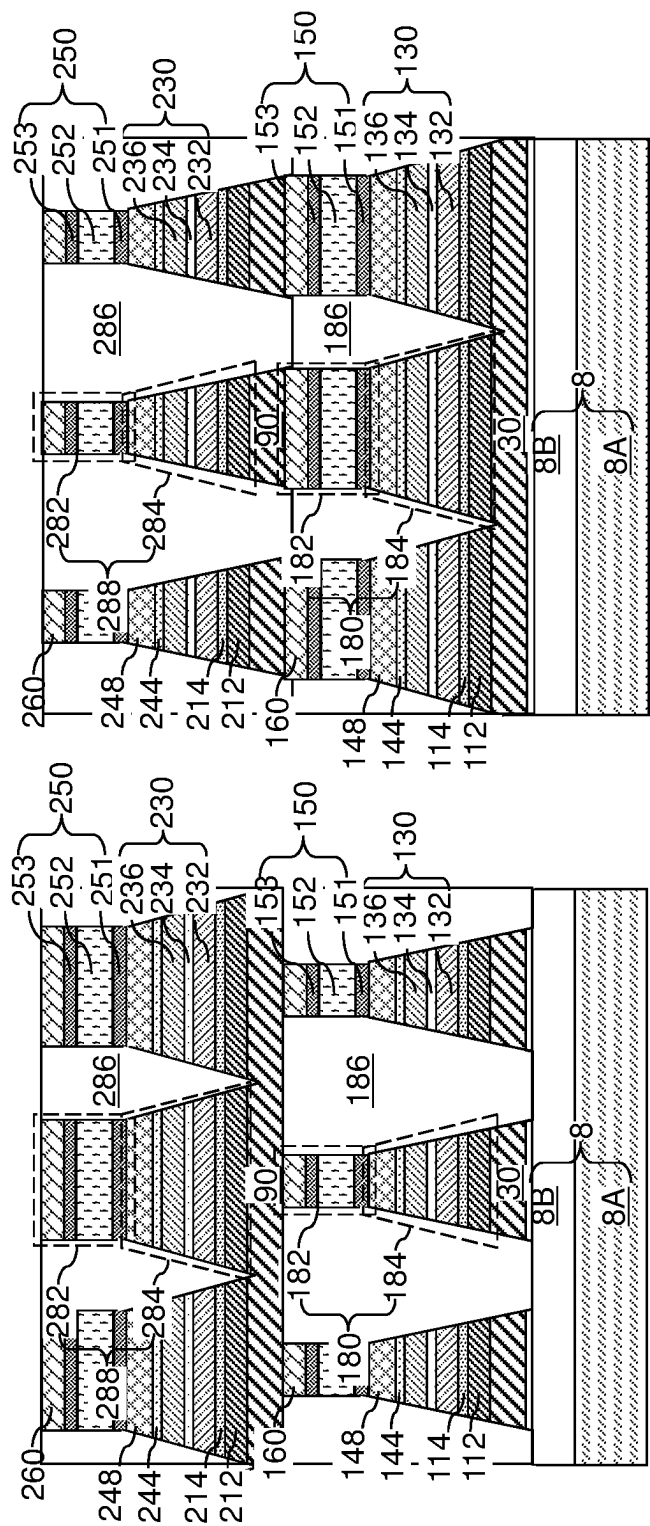
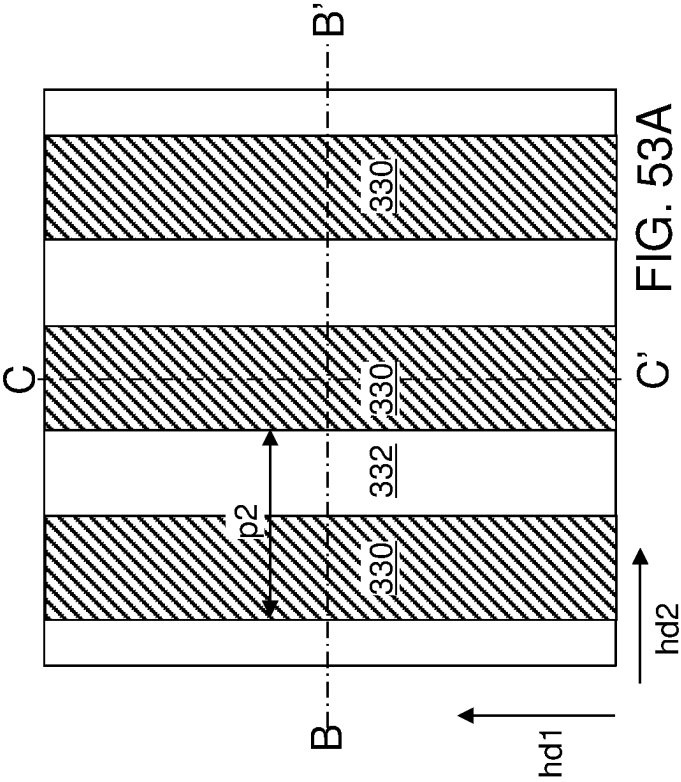


FIG. 52C

FIG. 52B



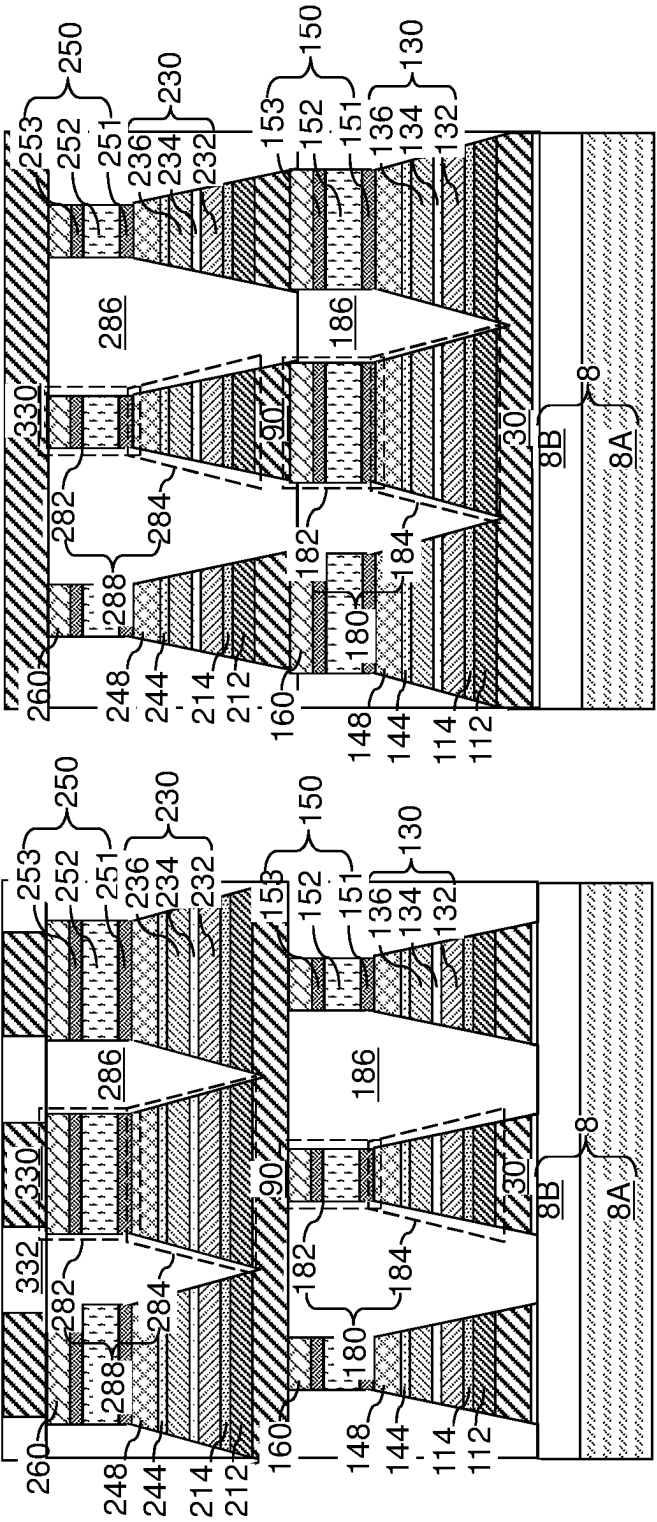
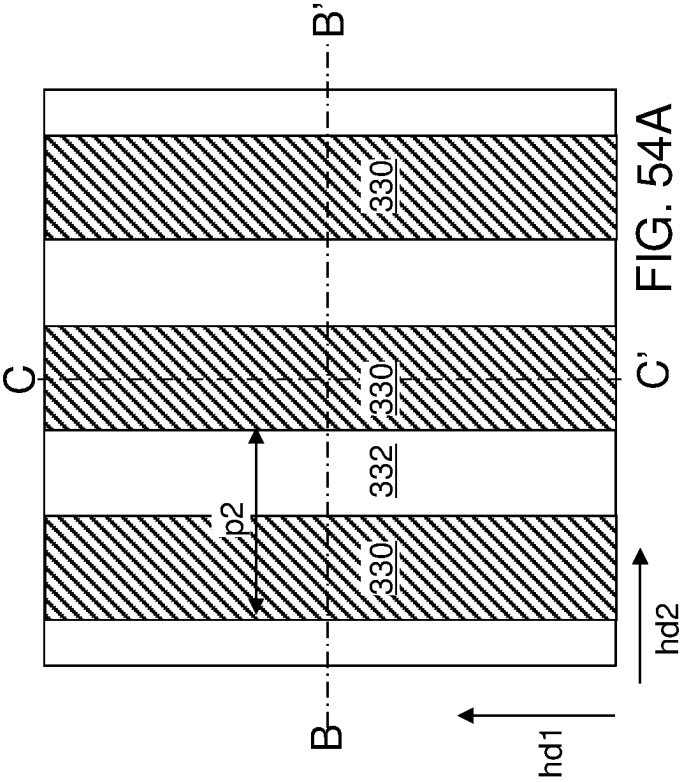


FIG. 53C

FIG. 53B



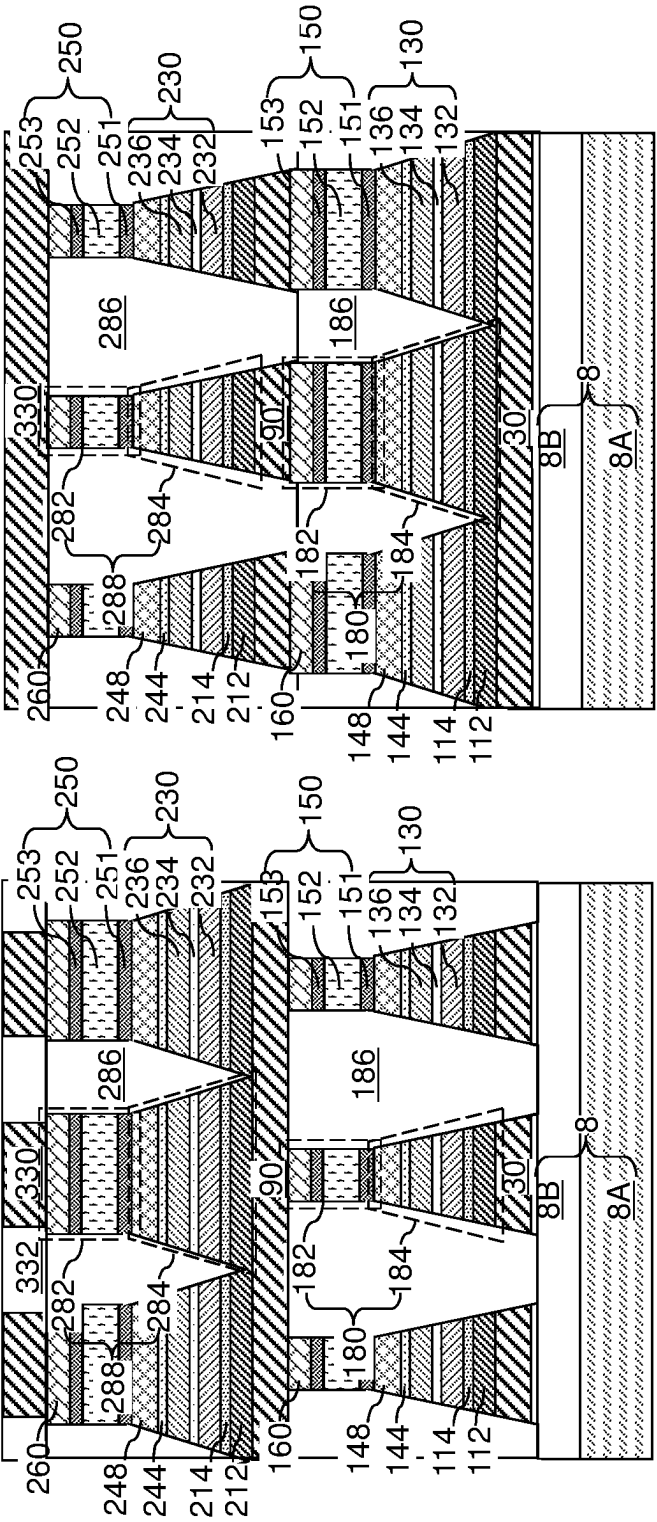
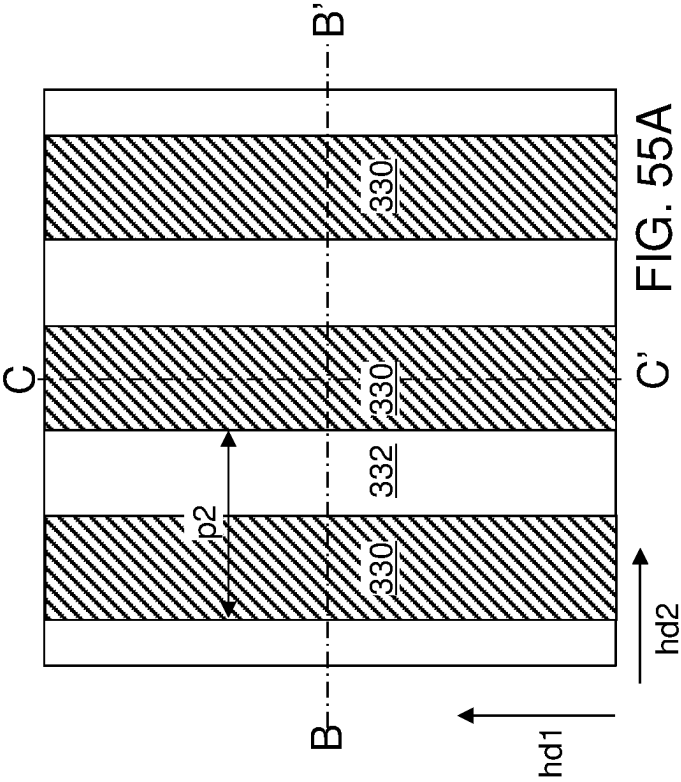


FIG. 54C

FIG. 54B



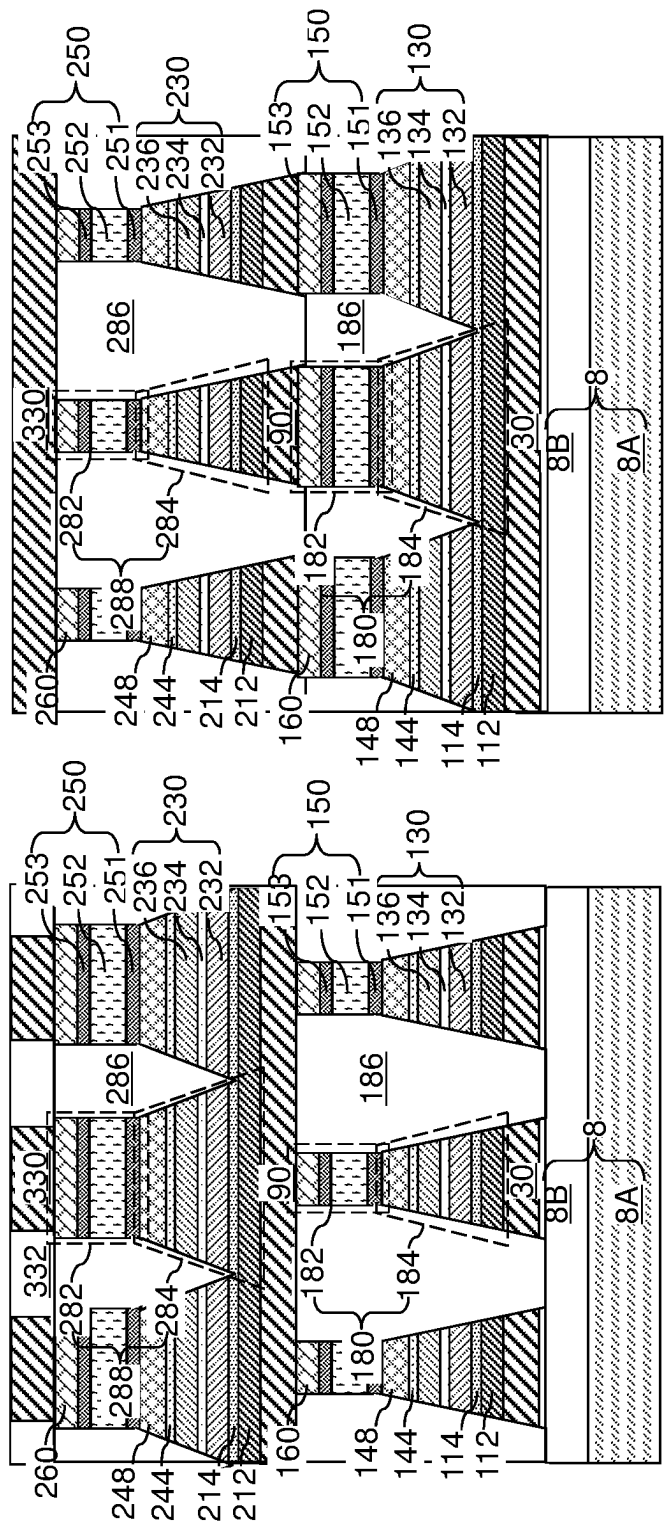
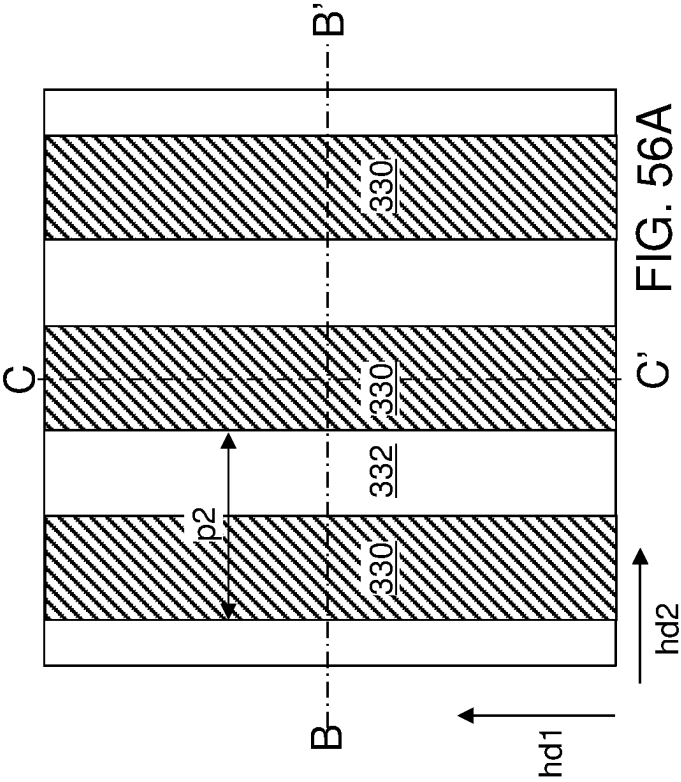


FIG. 55C

FIG. 55B





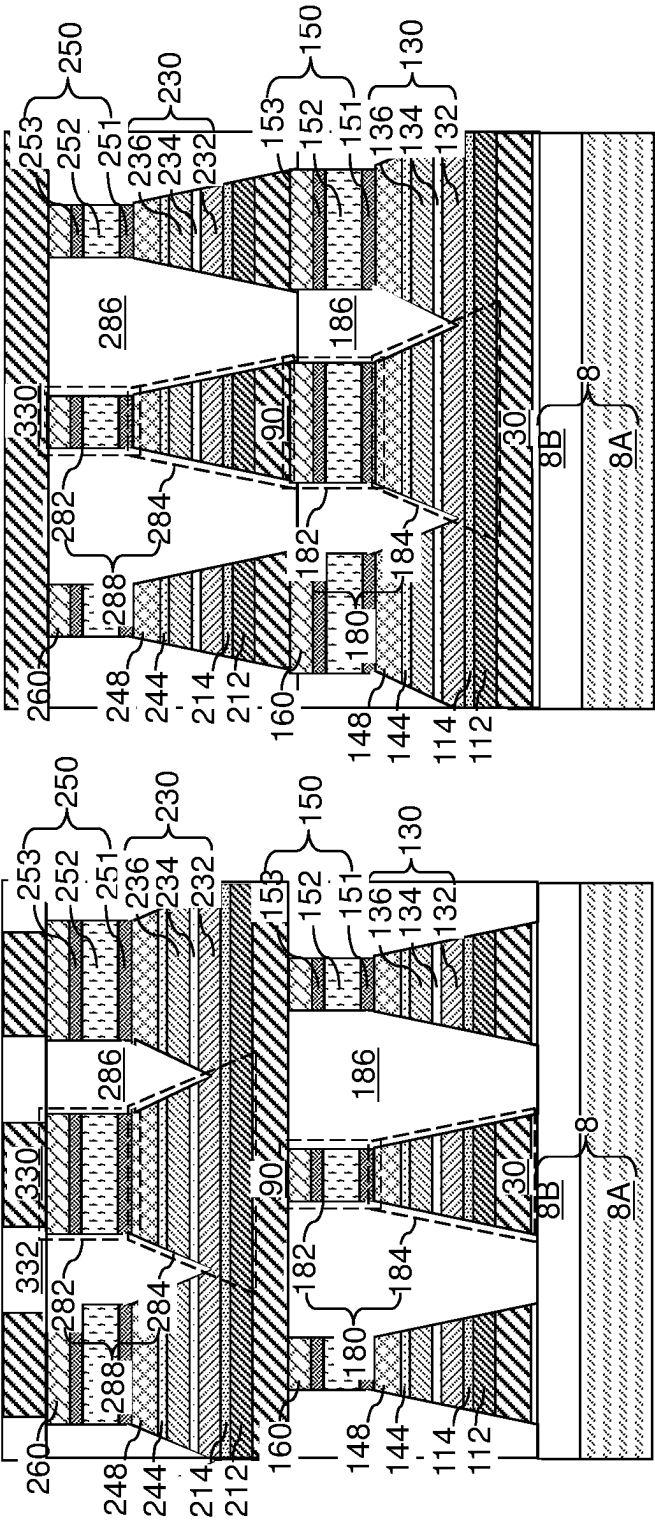
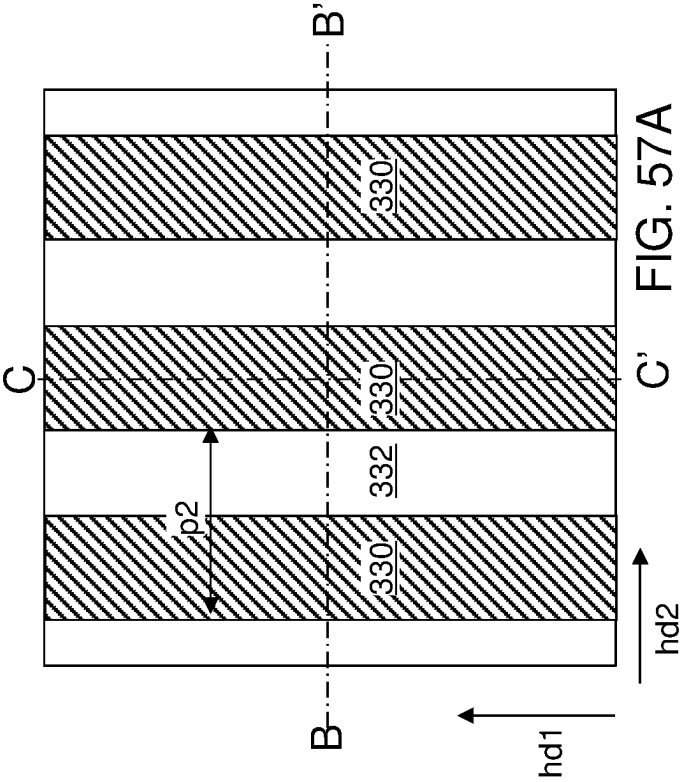


FIG. 56C

FIG. 56B



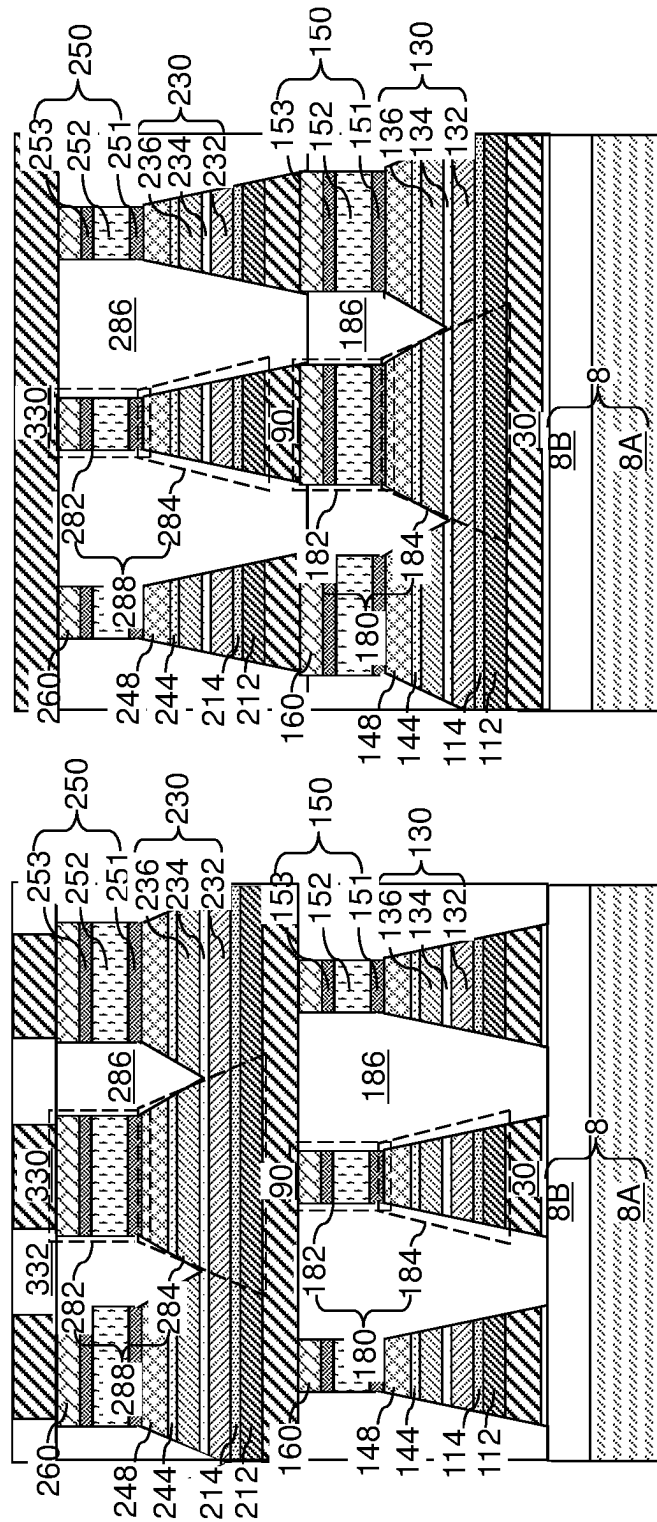


FIG. 57C

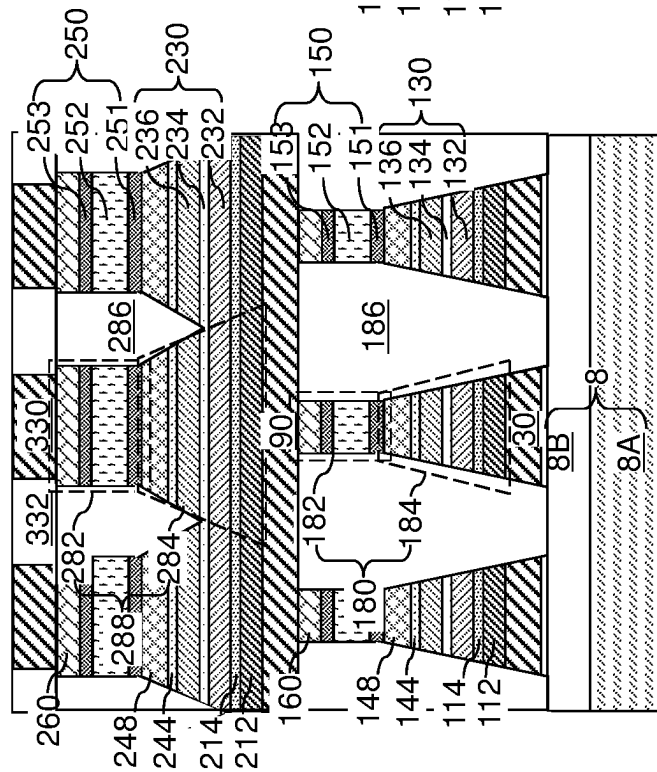


FIG. 57B

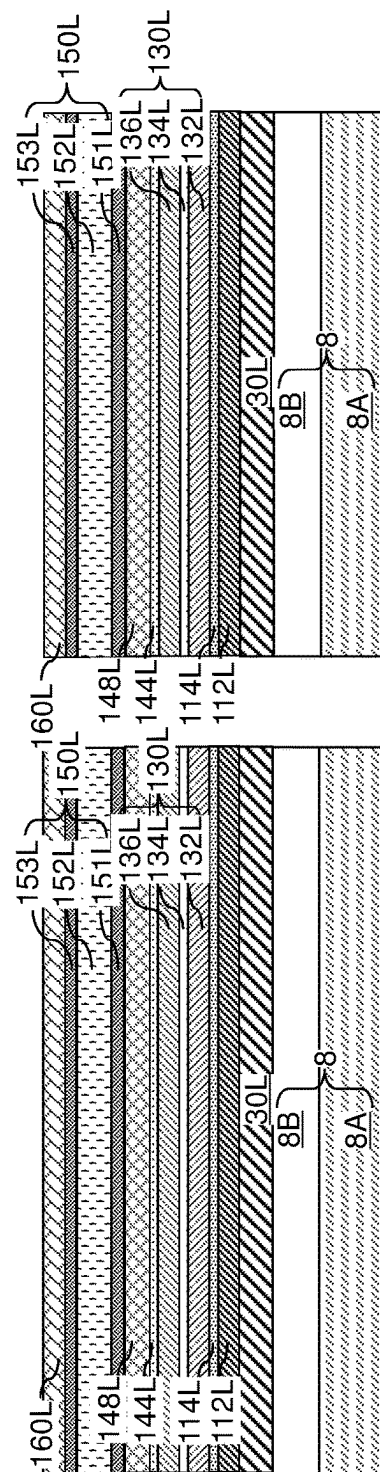
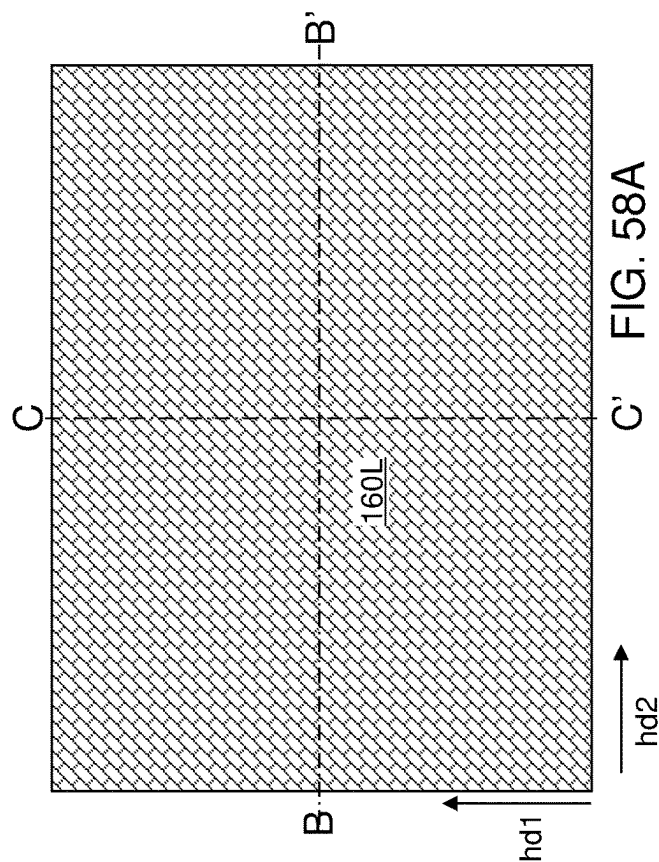


FIG. 58C

FIG. 58B

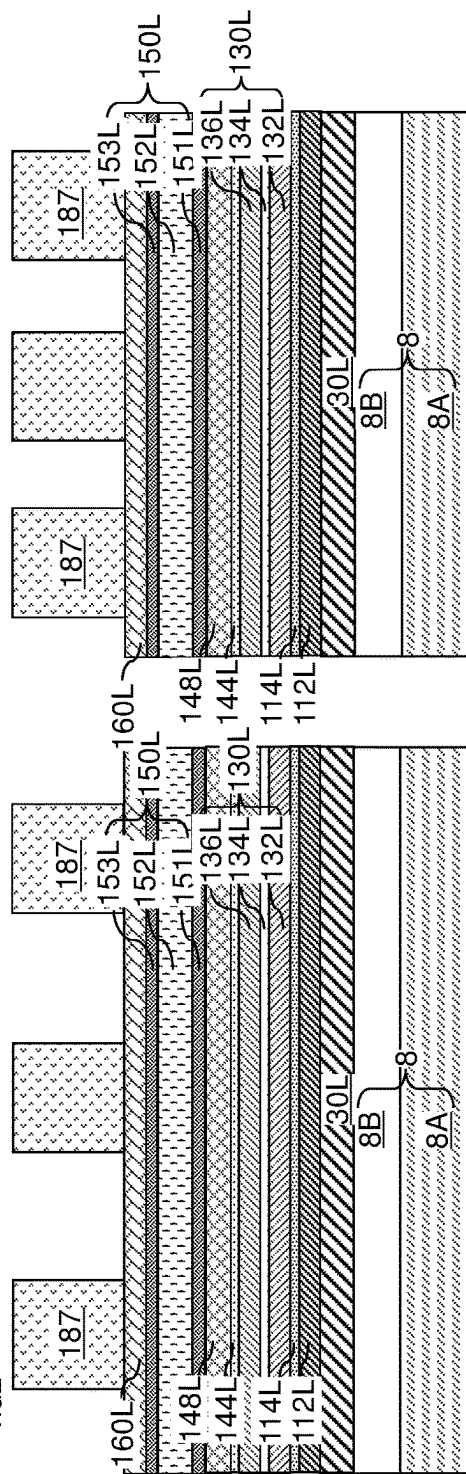
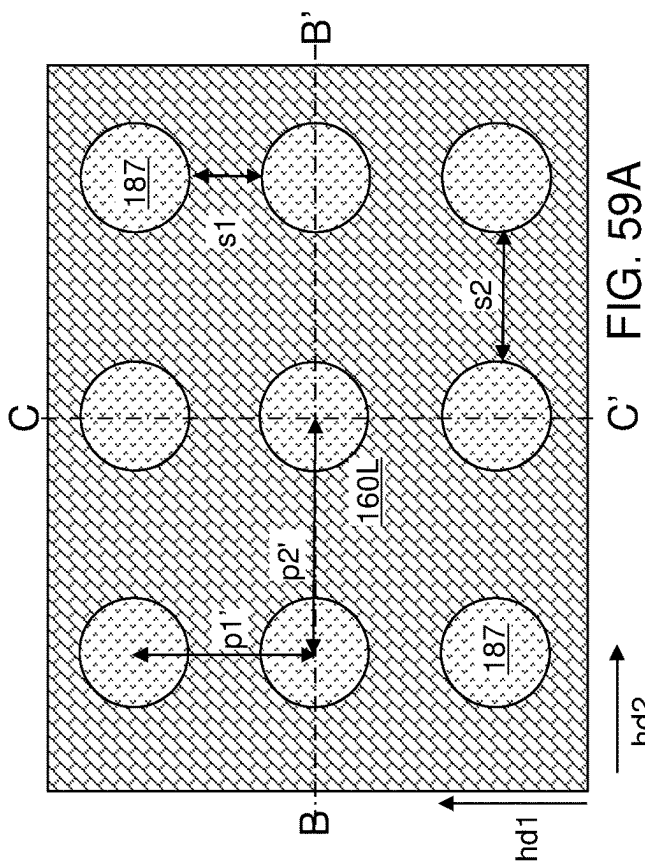
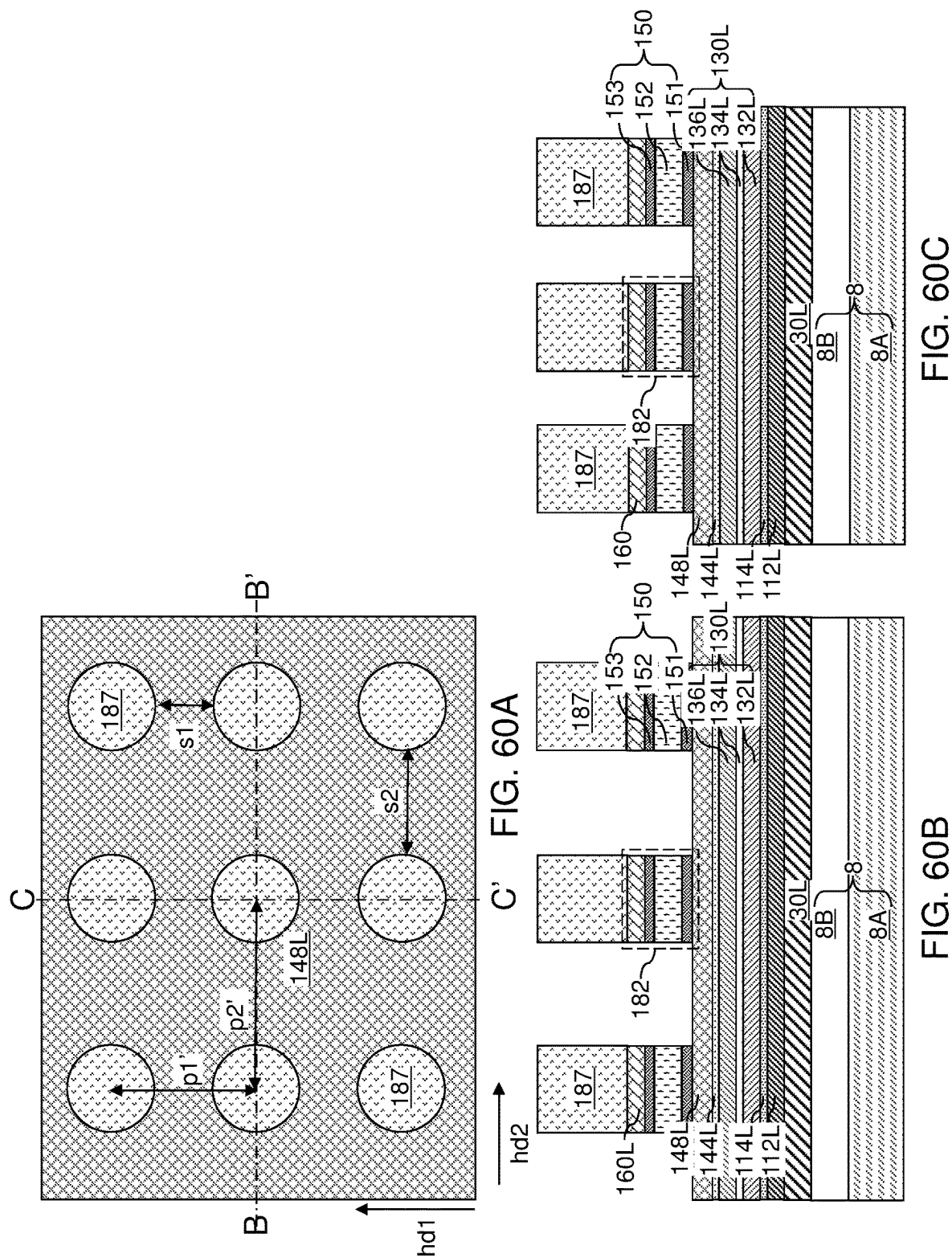
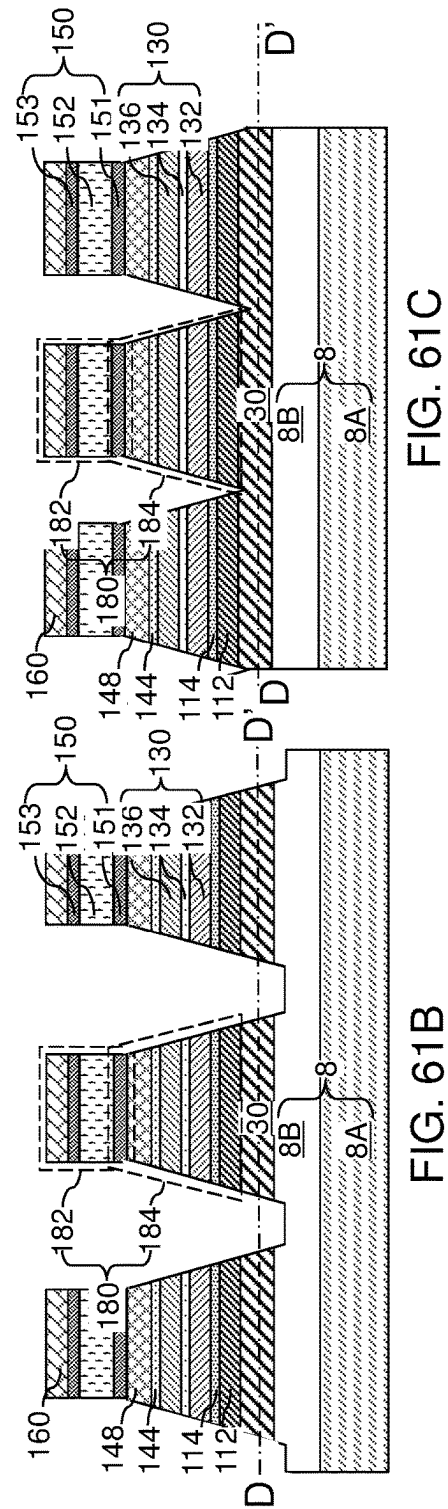
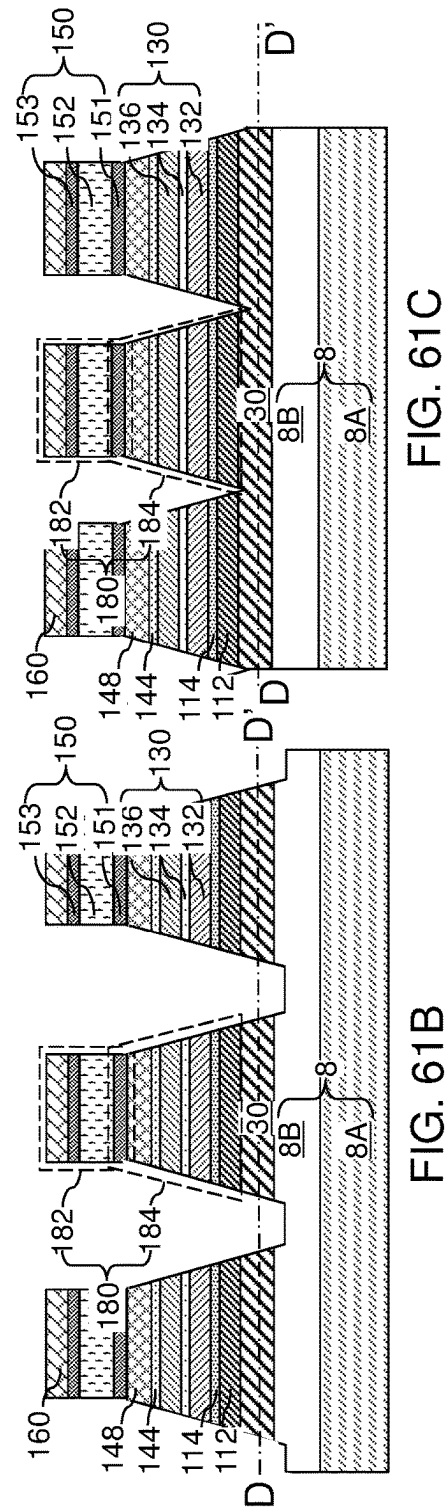
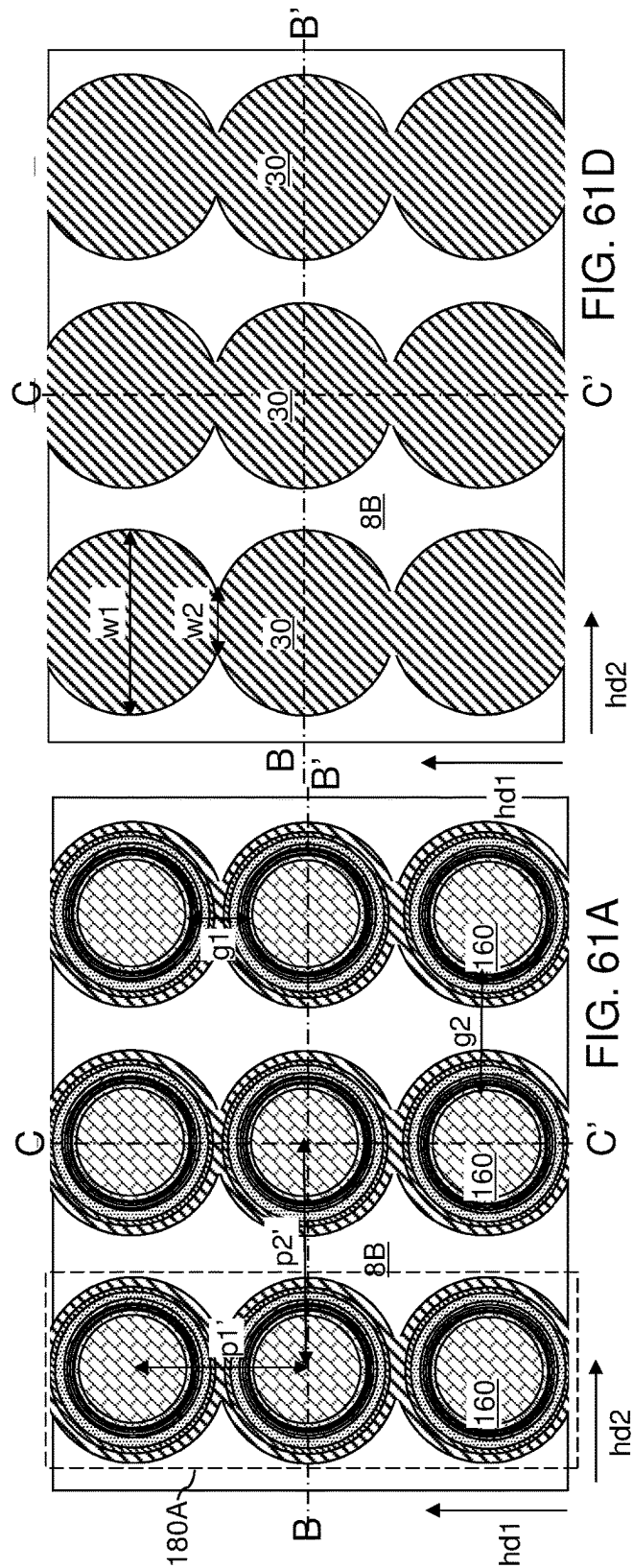


FIG. 59C

FIG. 59B







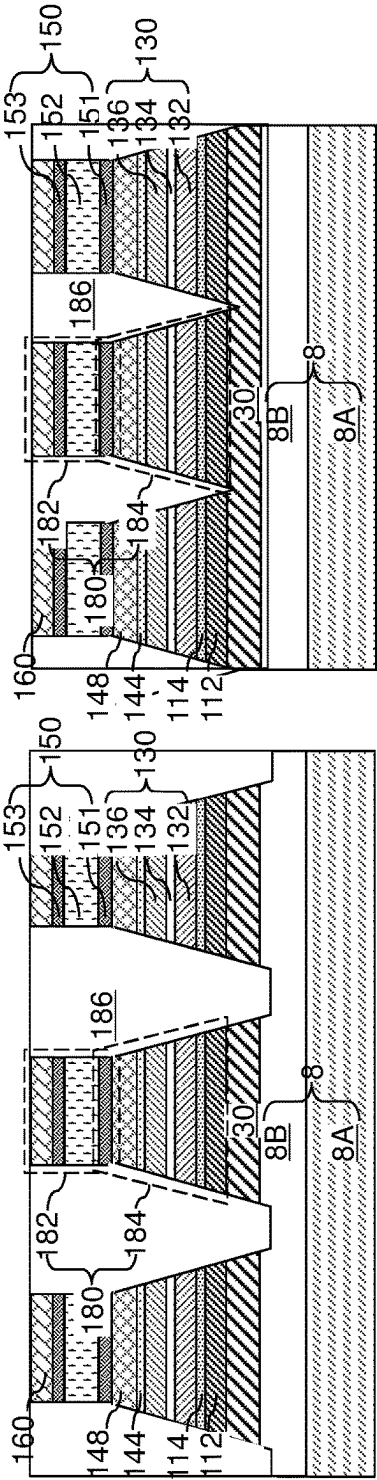
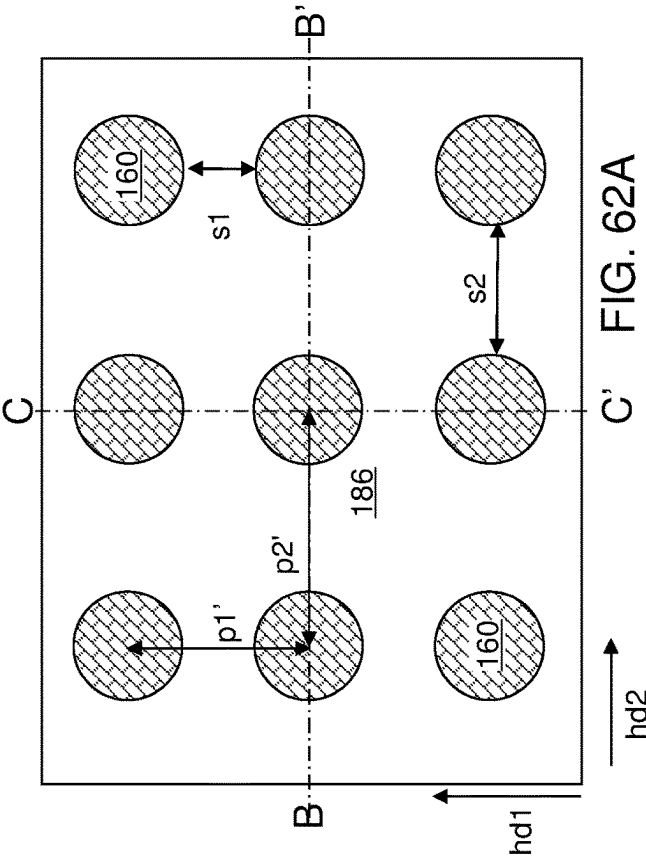
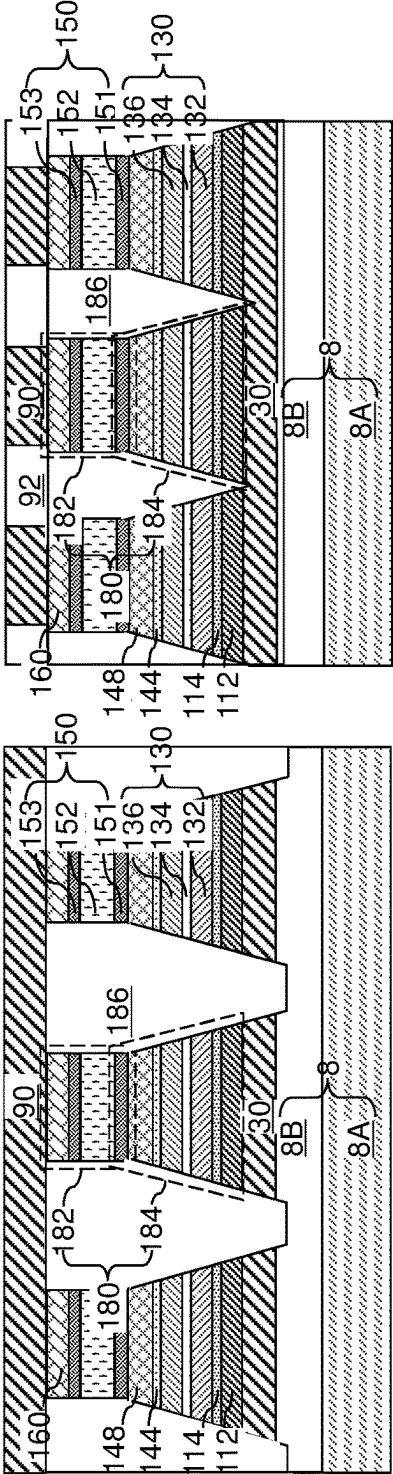
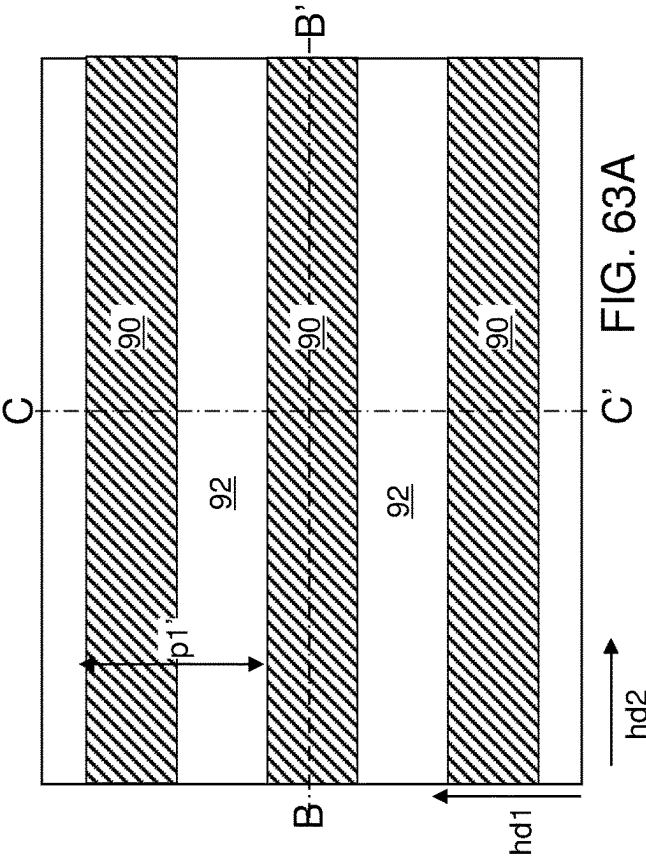
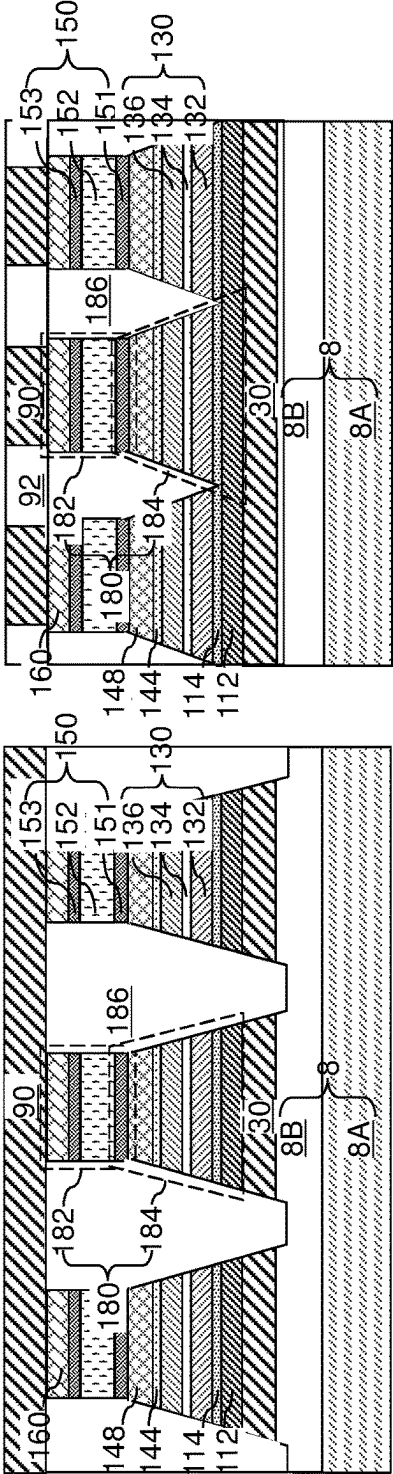
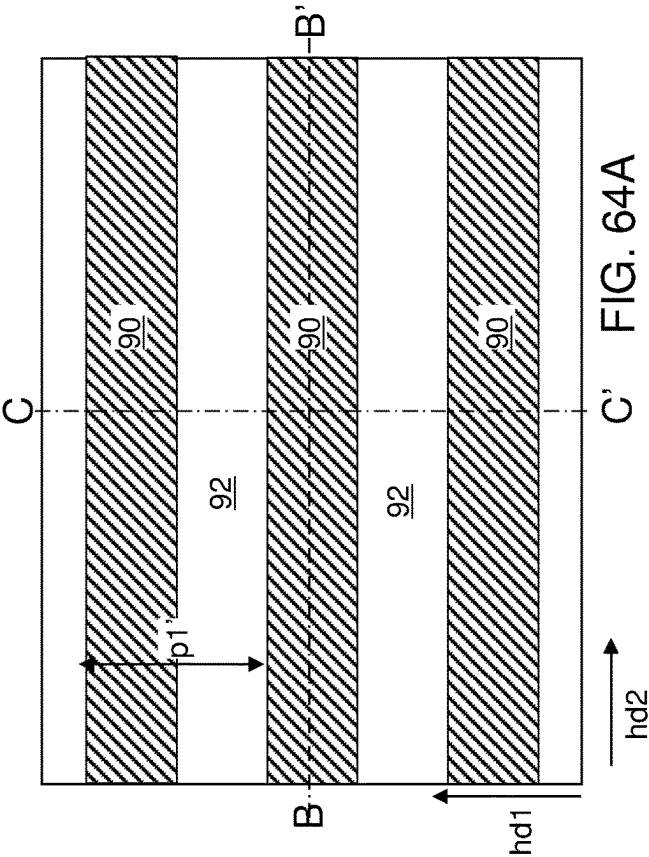
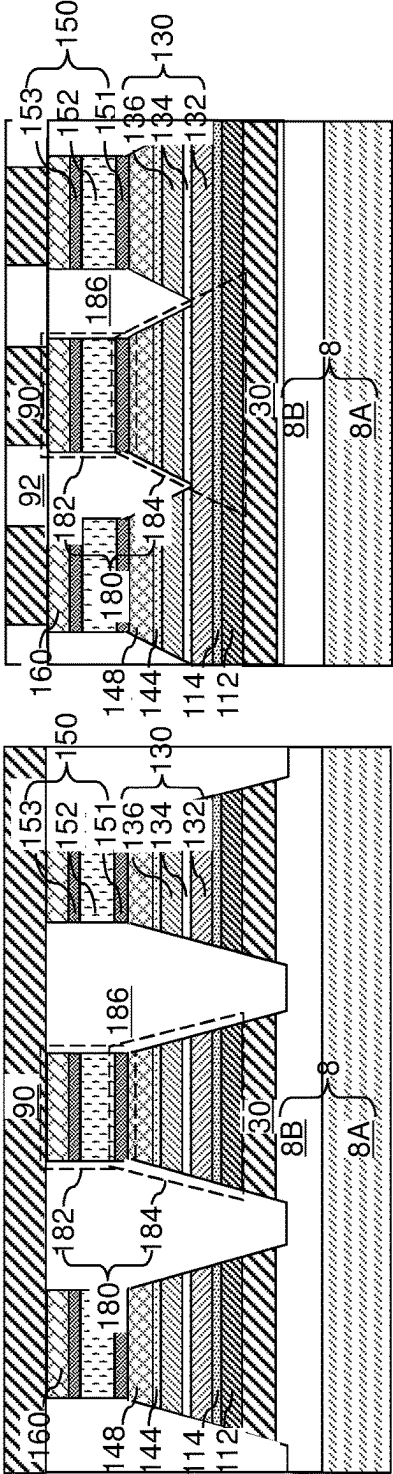
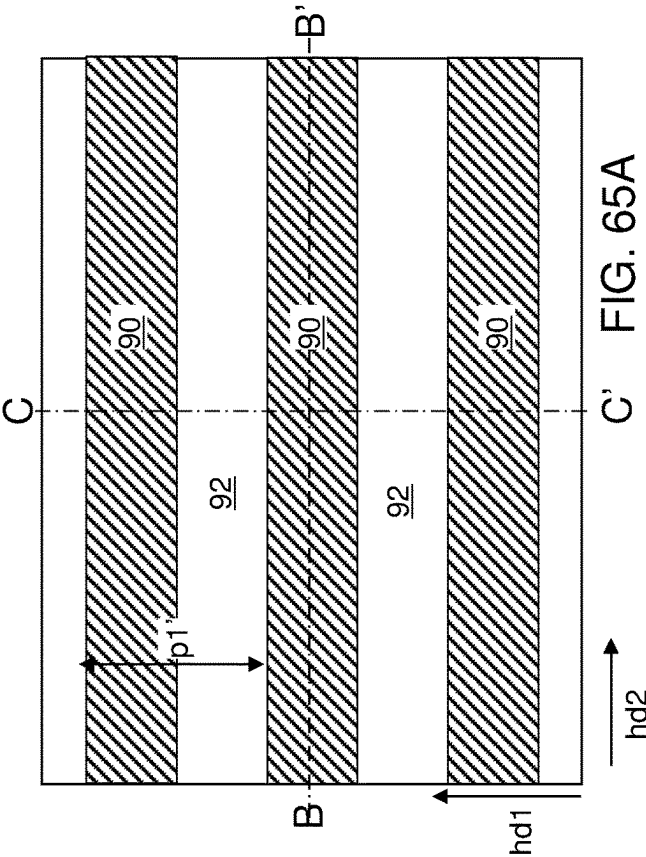


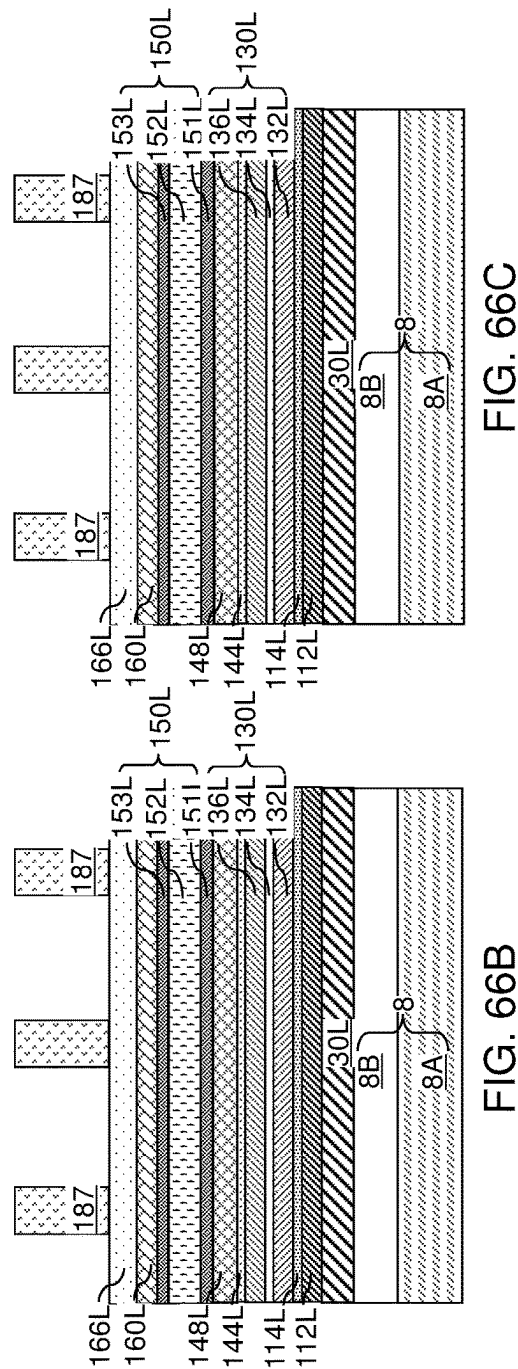
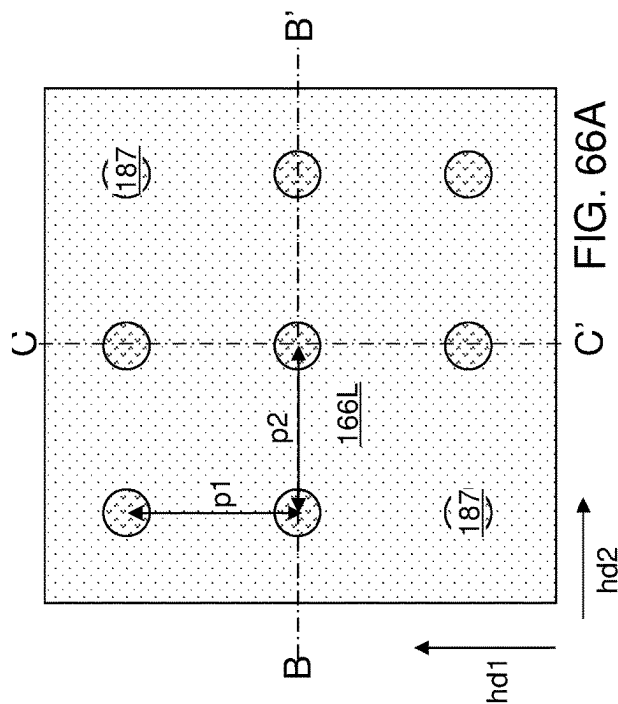
FIG. 62C

FIG. 62B









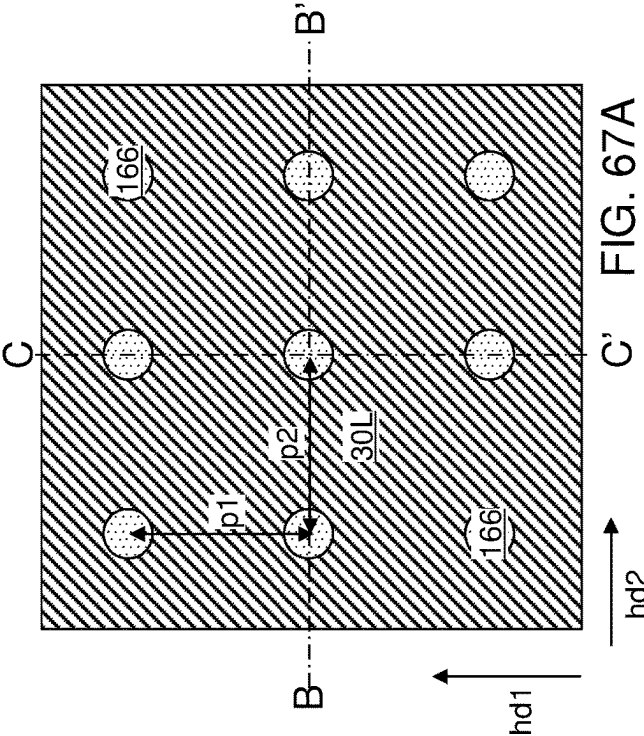


FIG. 67A

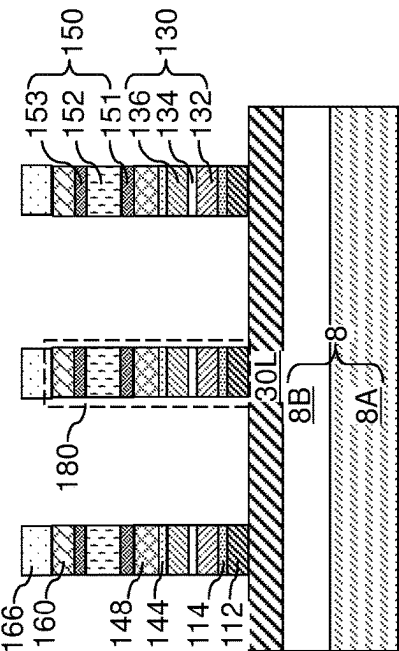


FIG. 67B

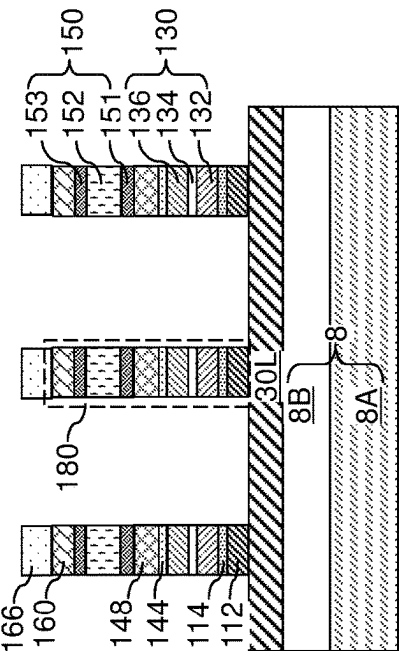


FIG. 67C

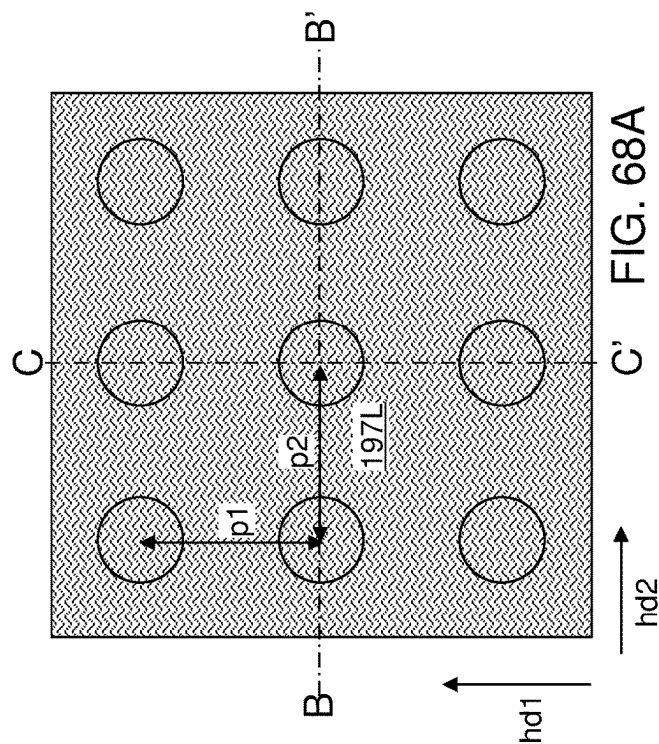


FIG. 68A

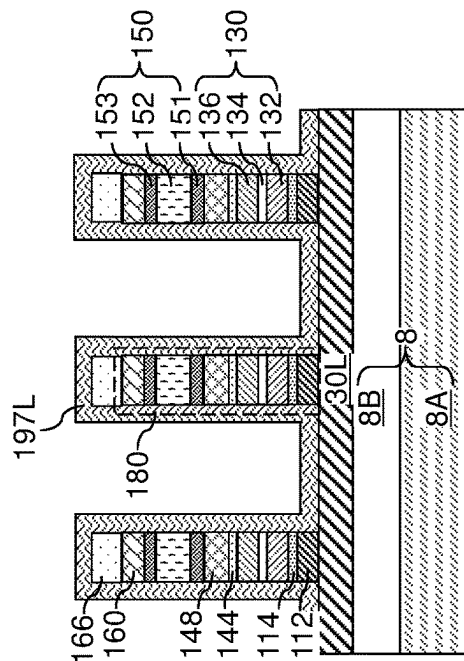


FIG. 68B

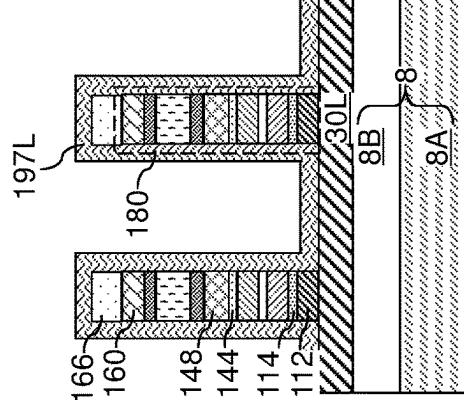


FIG. 68C

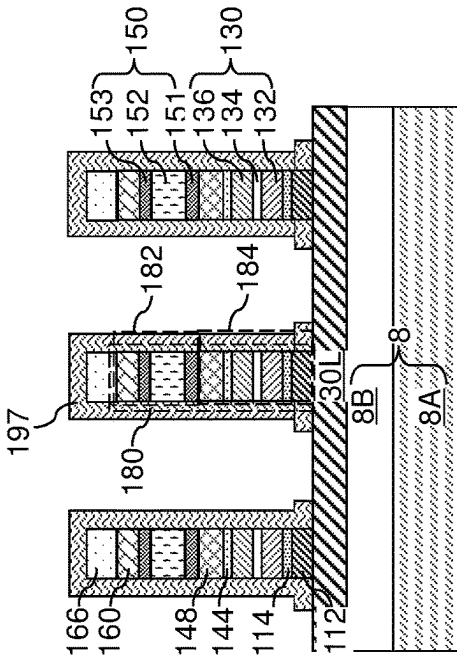
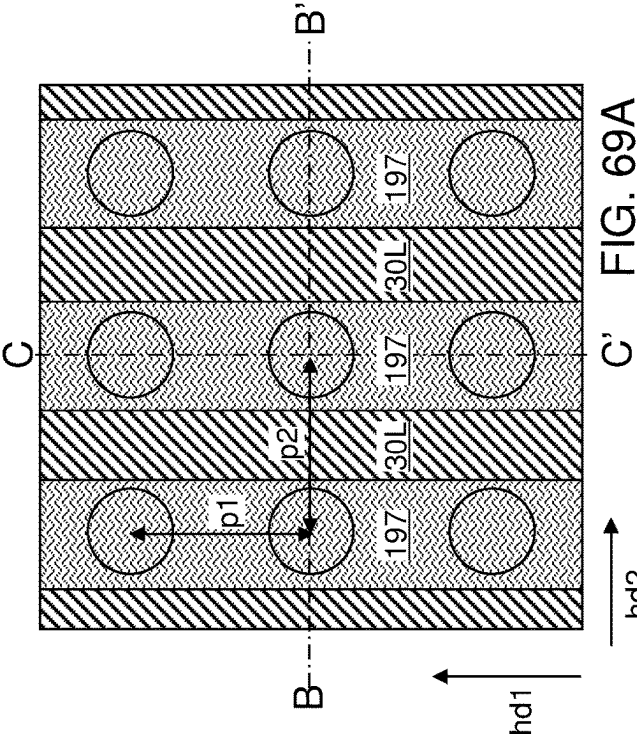


FIG. 69B

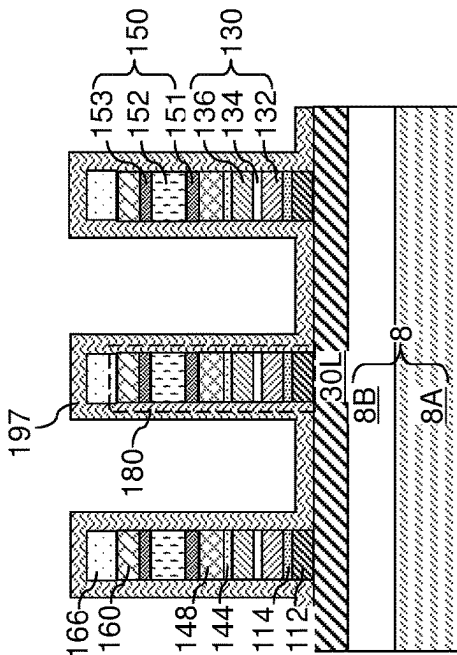
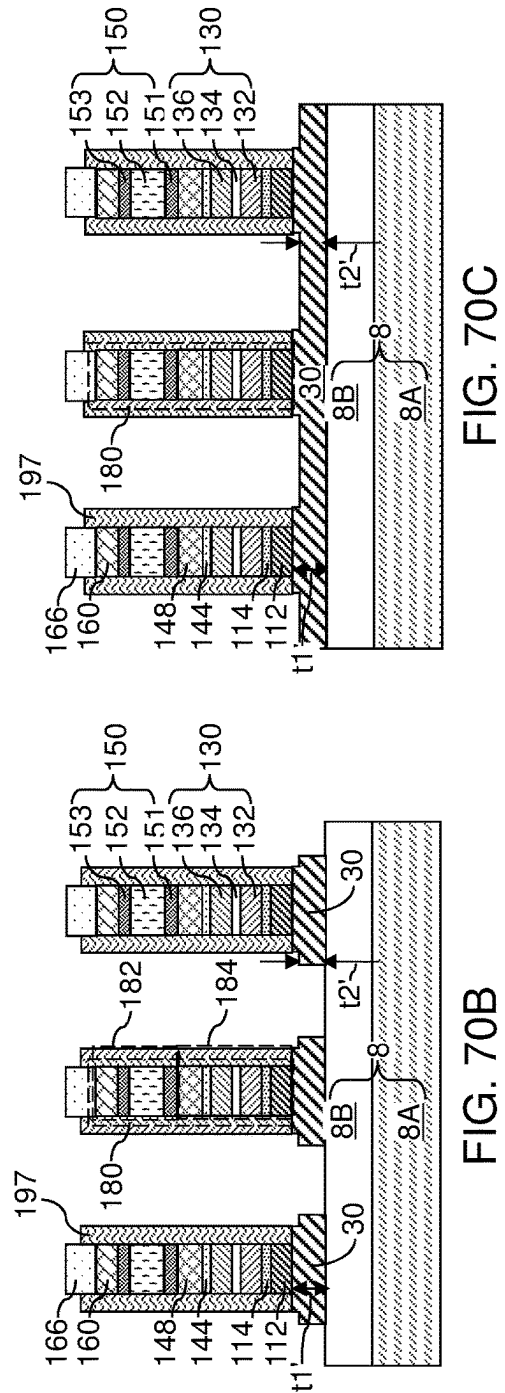
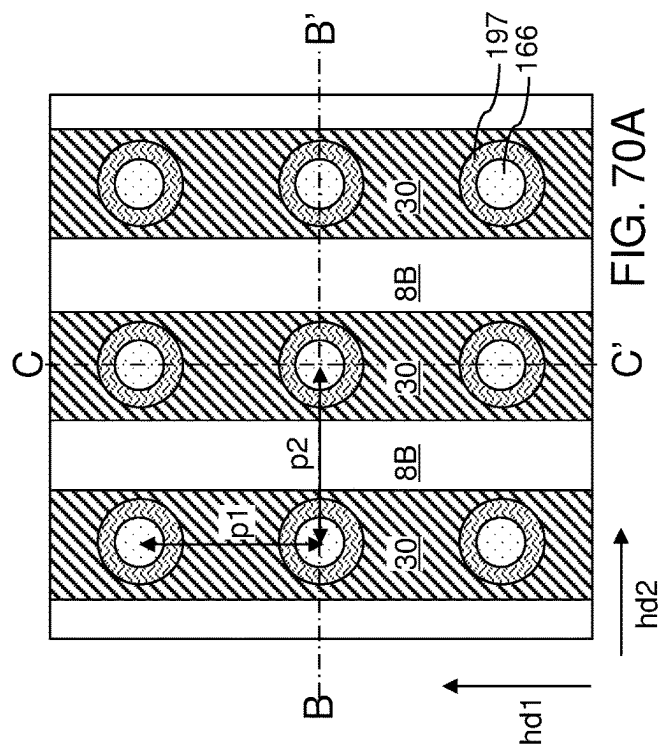


FIG. 69C





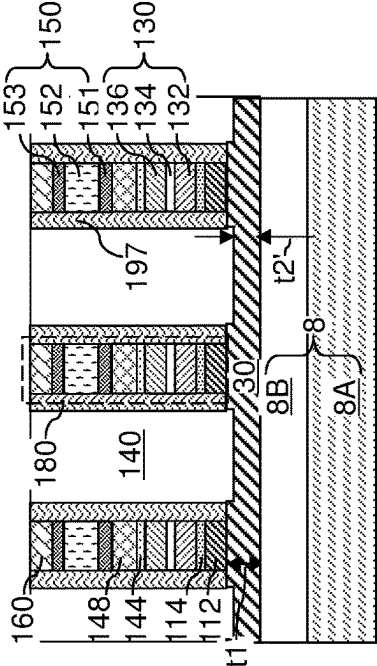
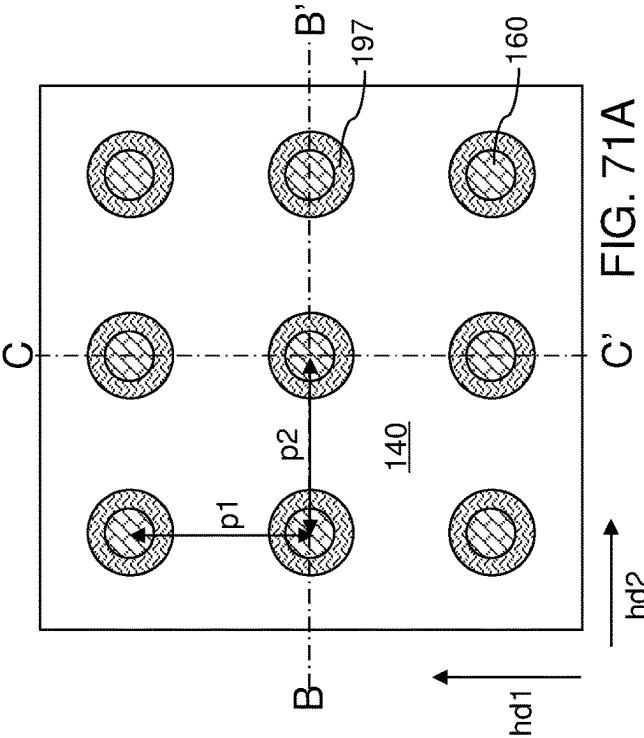


FIG. 71C

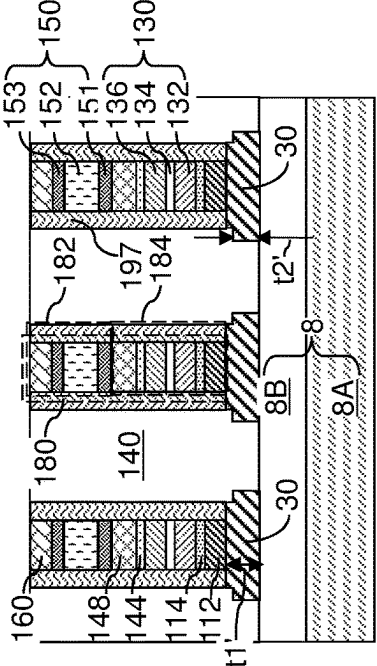


FIG. 71B

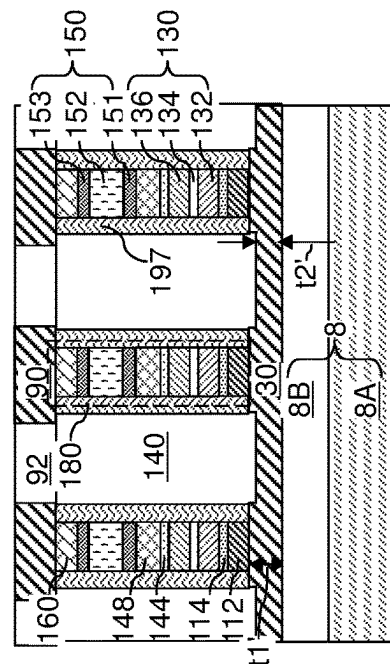
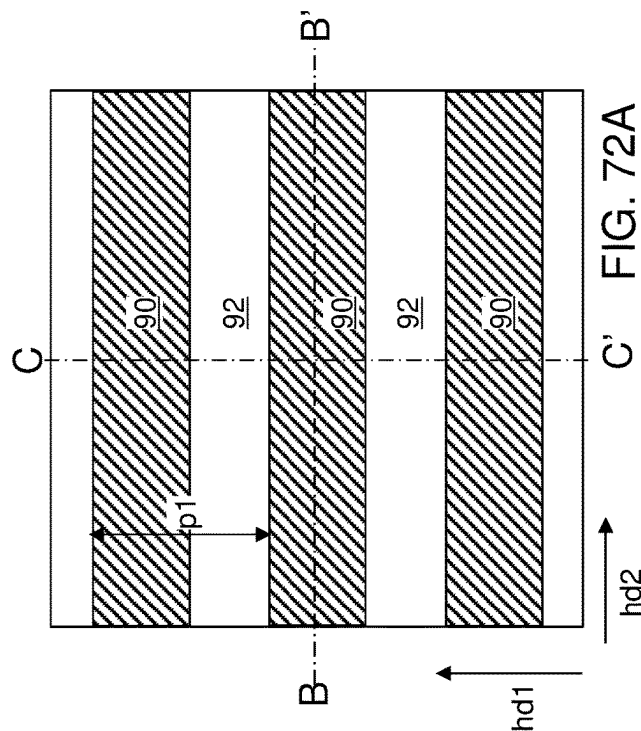


FIG. 72C

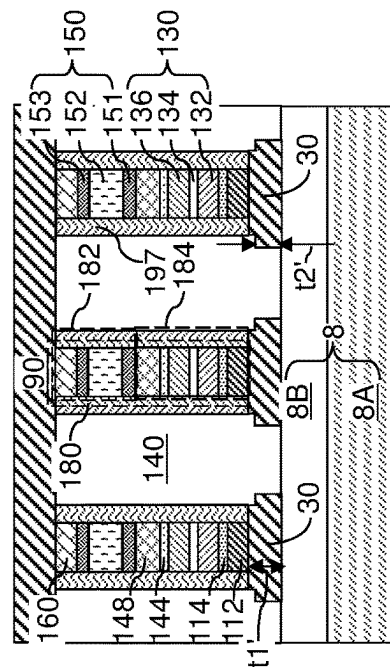


FIG. 72B

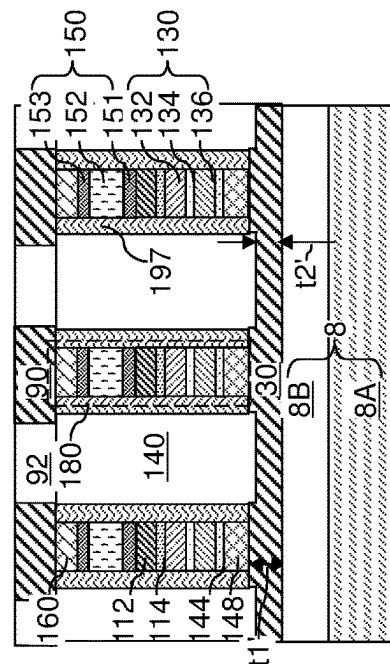
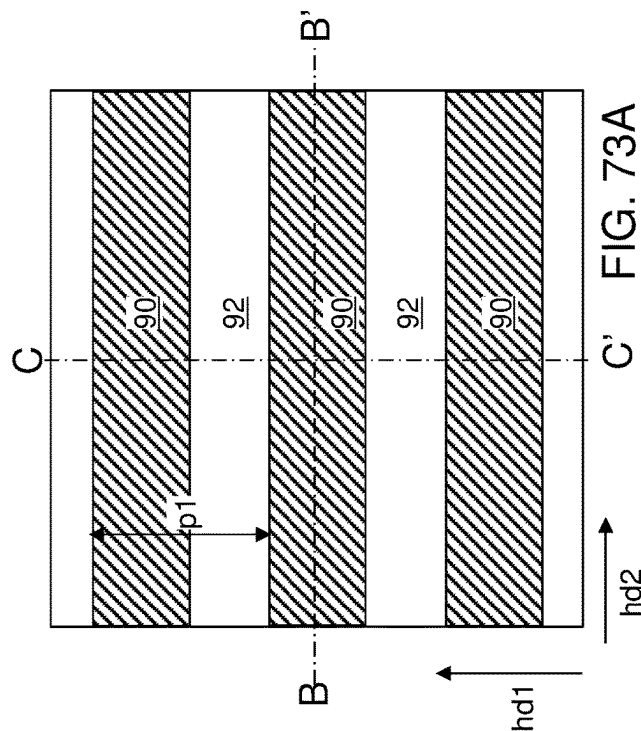
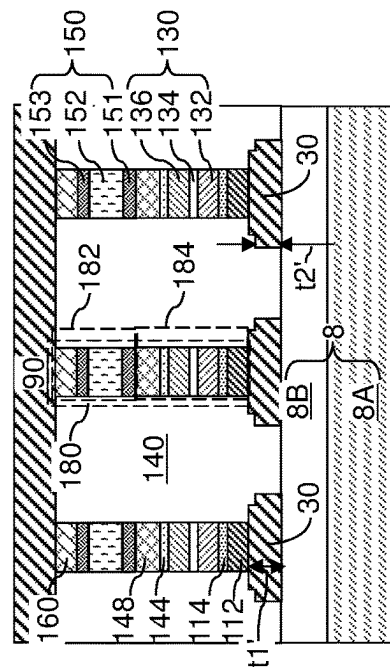
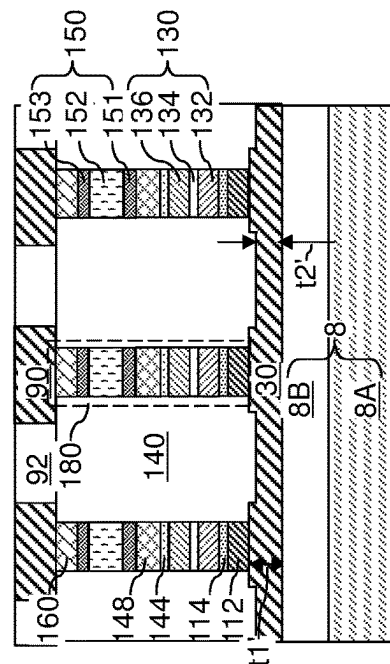
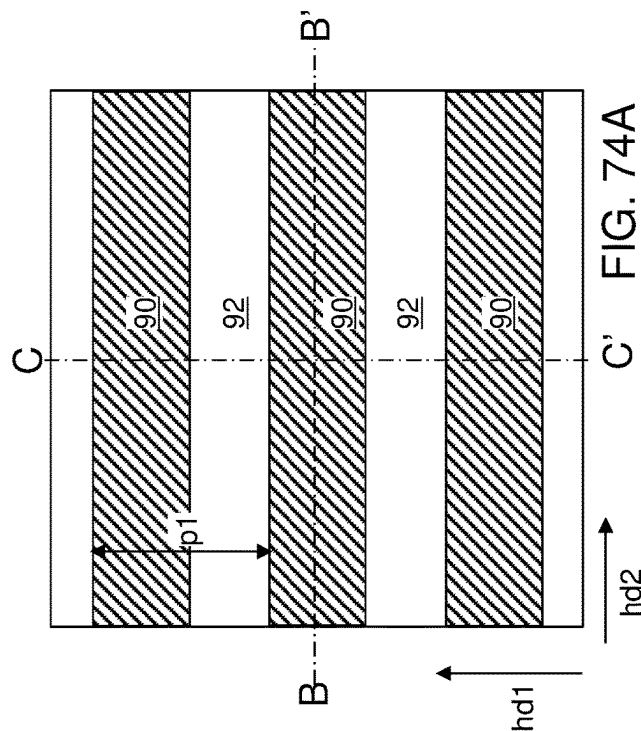
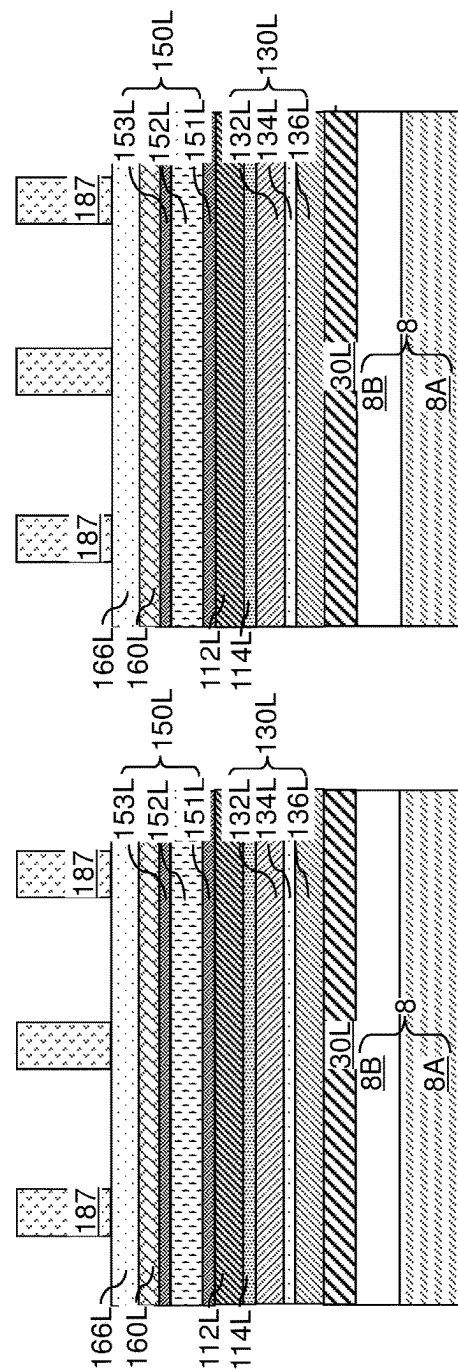
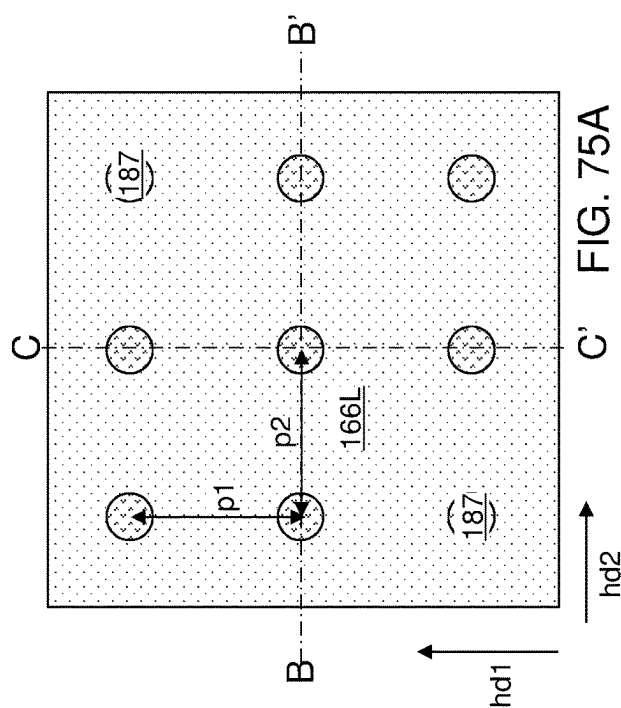


FIG. 73C





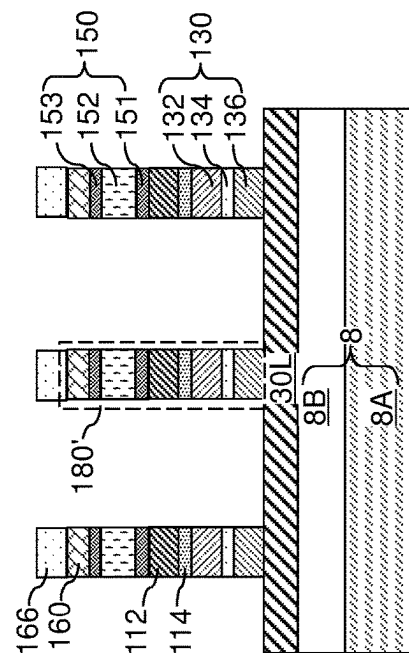
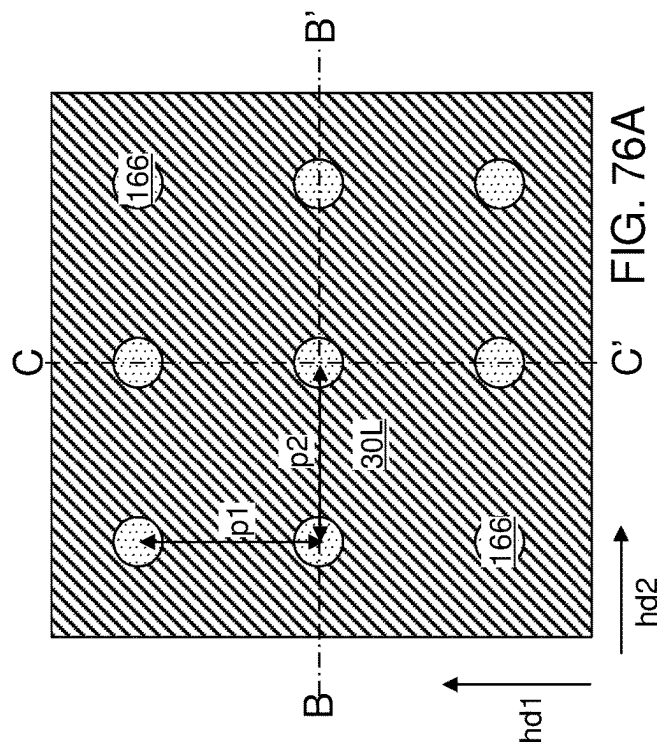


FIG. 76C

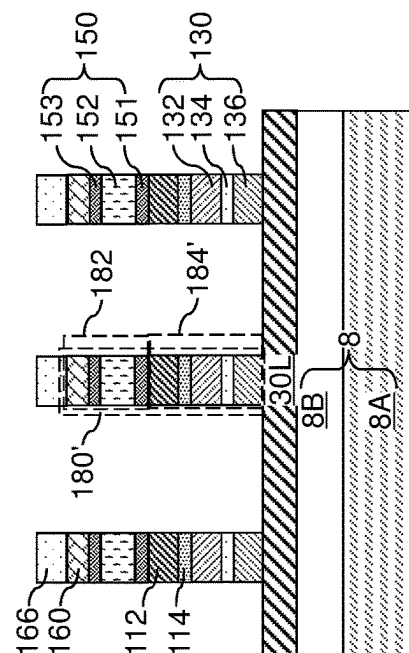


FIG. 76B

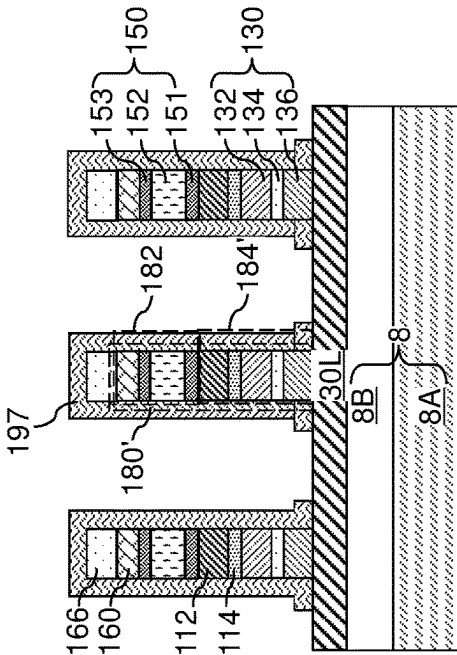
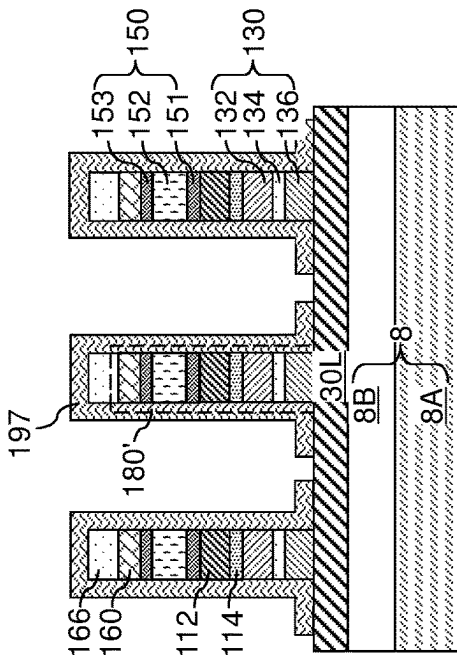
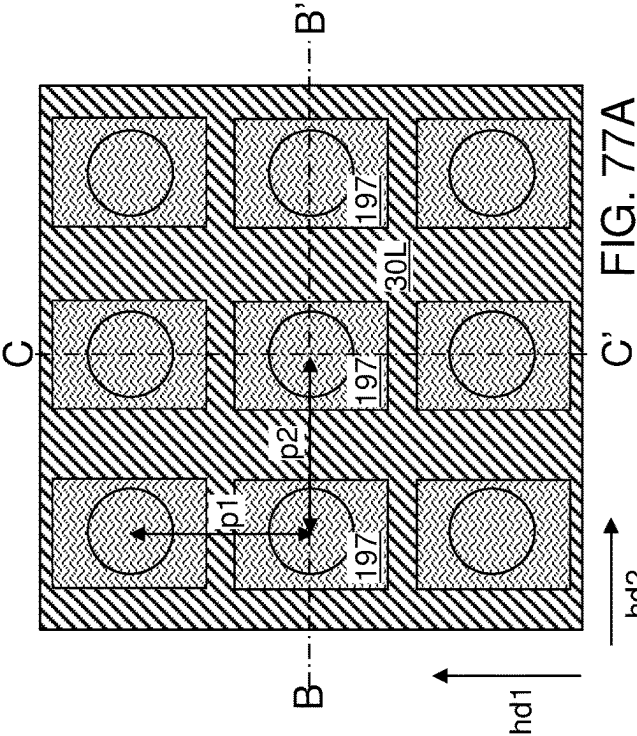


FIG. 77C

FIG. 77B



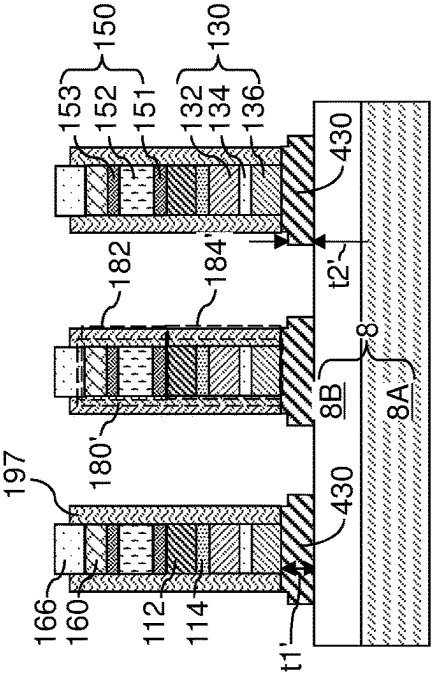
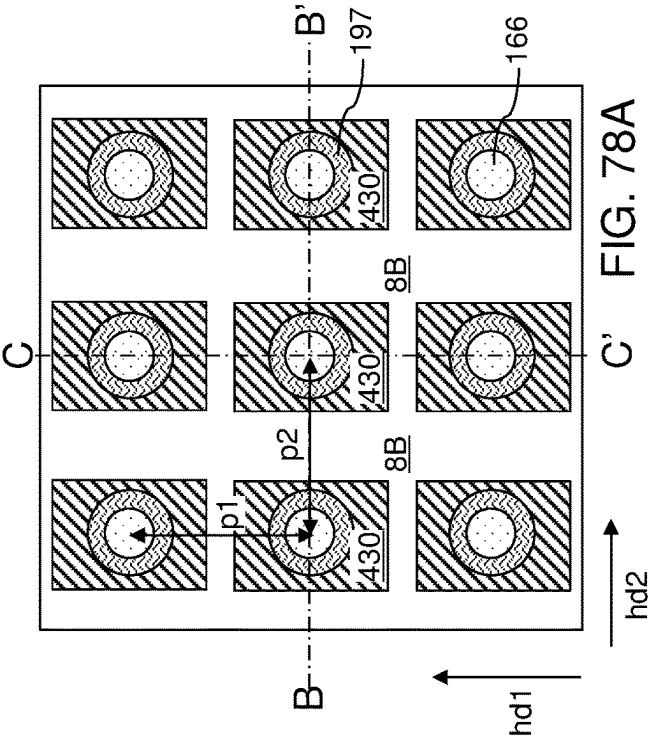


FIG. 78B

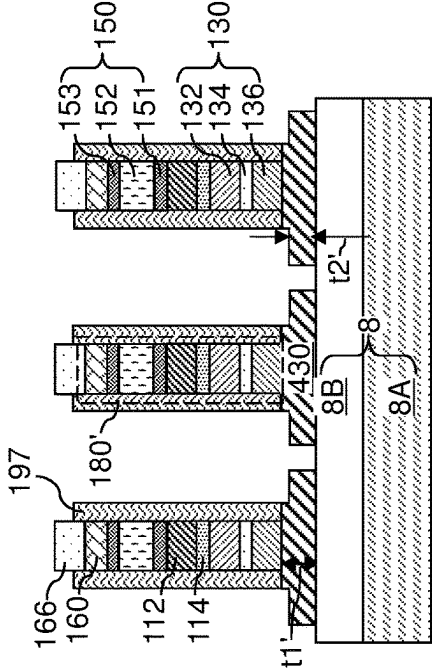


FIG. 78C

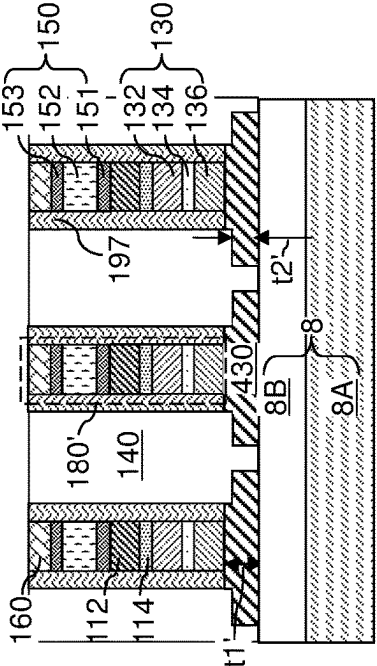
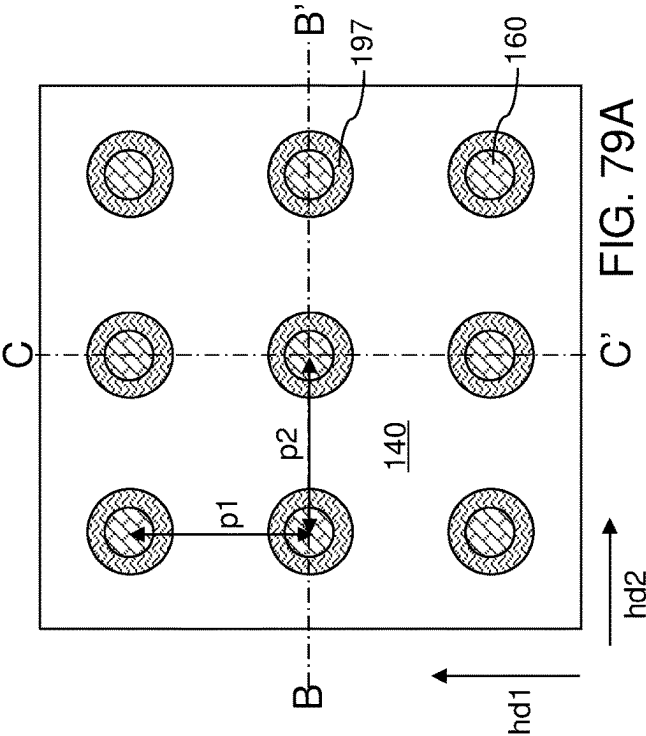


FIG. 79C

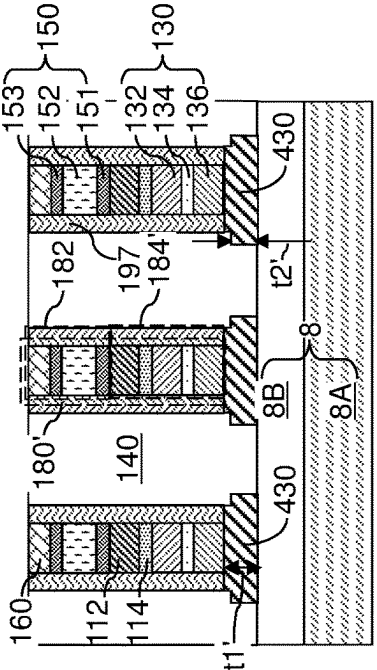


FIG. 79B

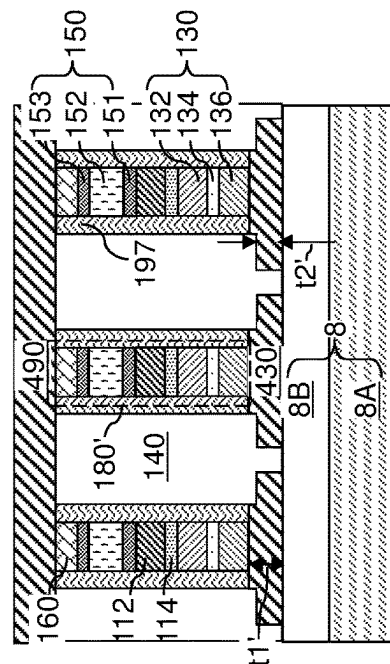
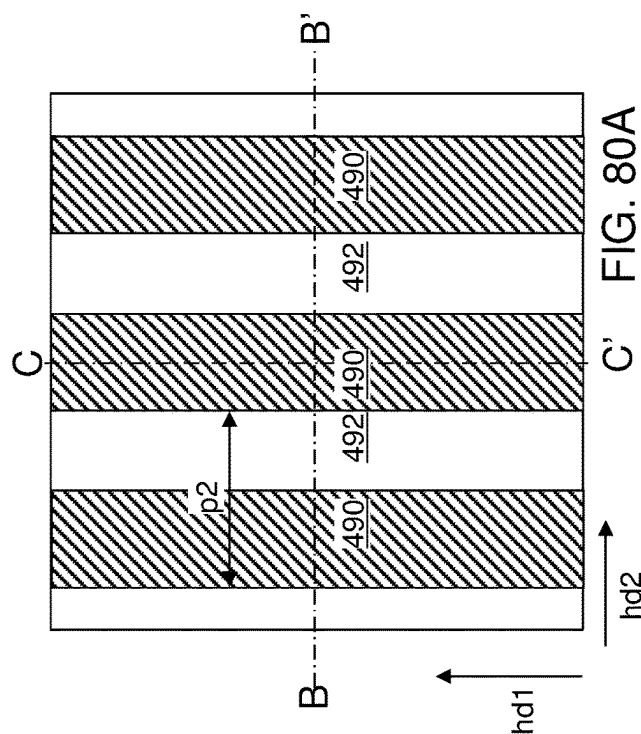


FIG. 80C

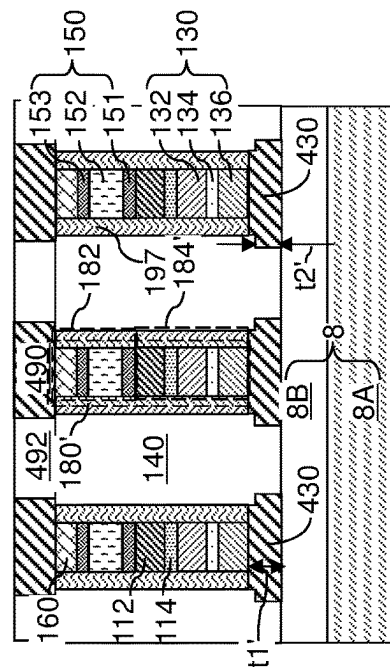


FIG. 80B

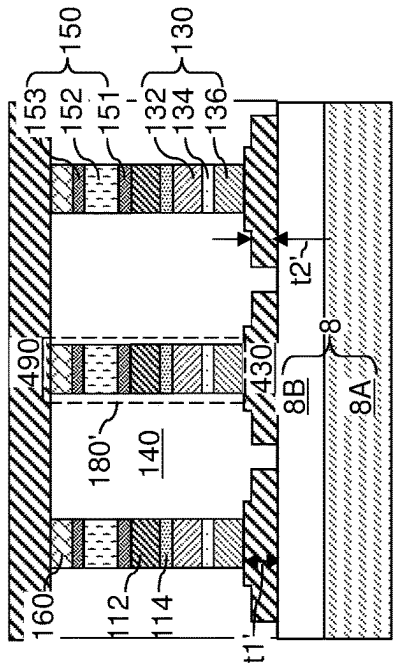
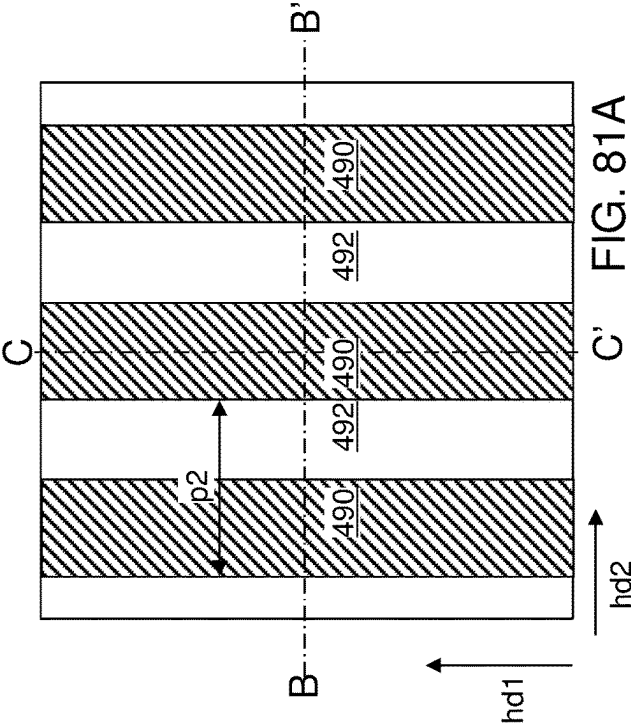


FIG. 81C

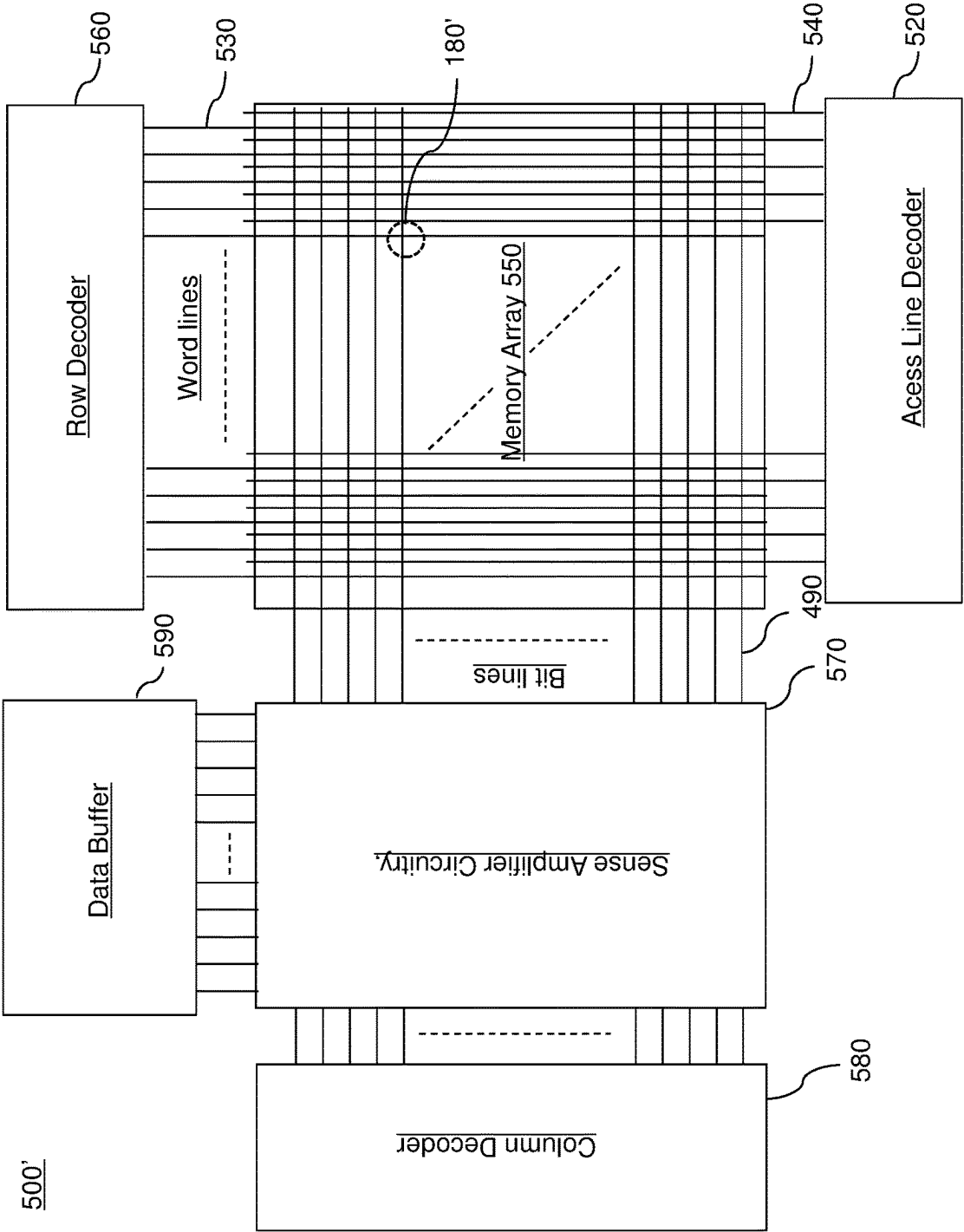
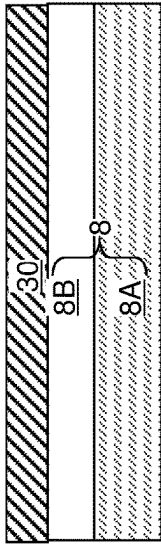
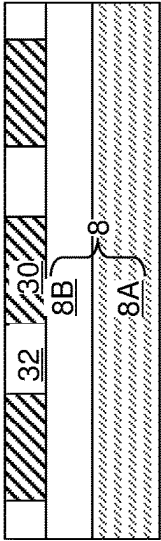
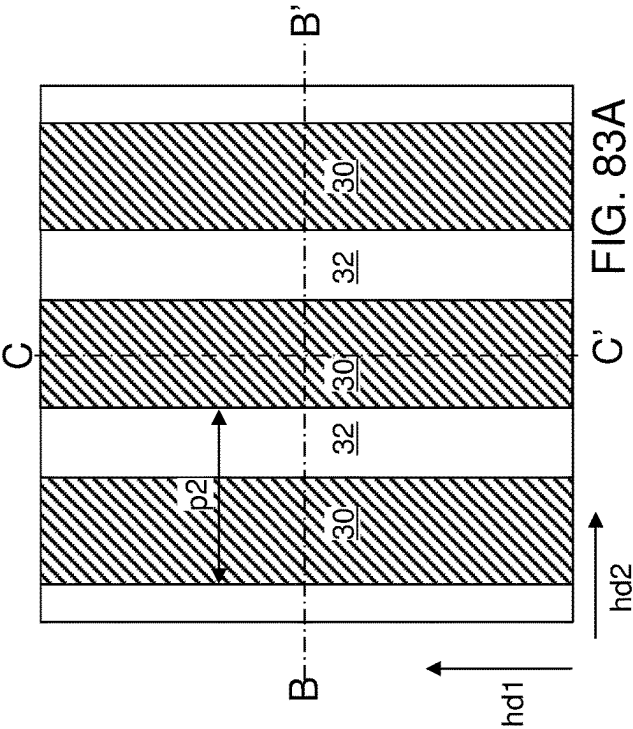
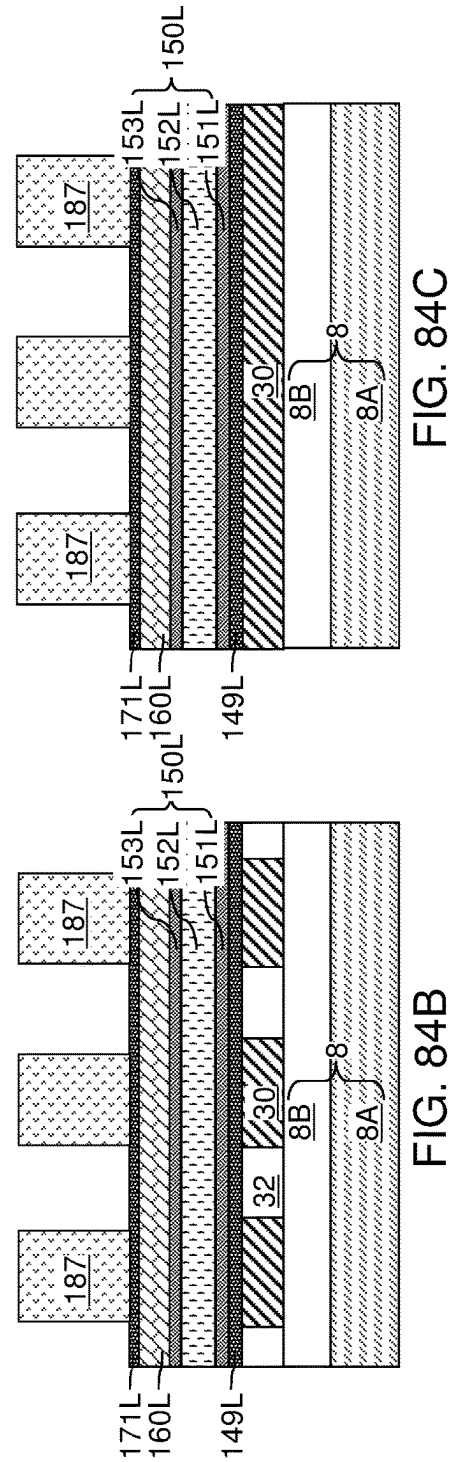
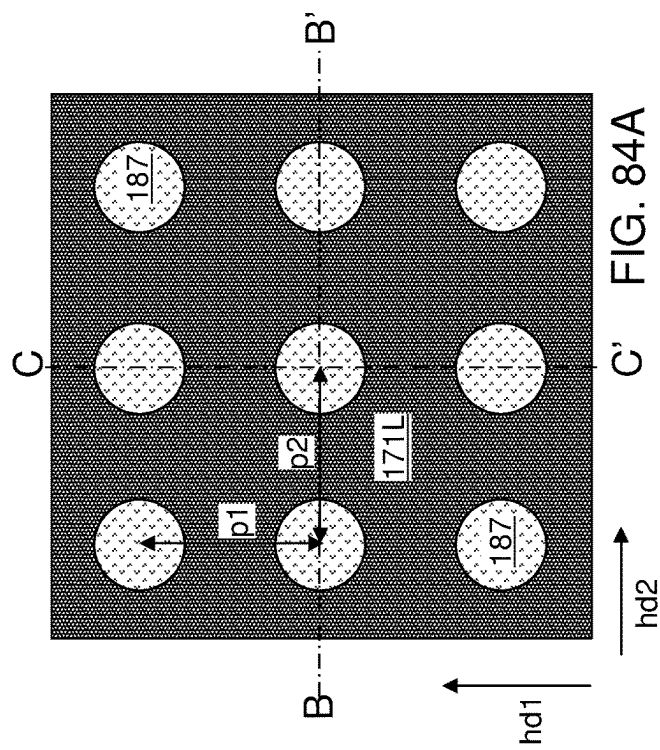
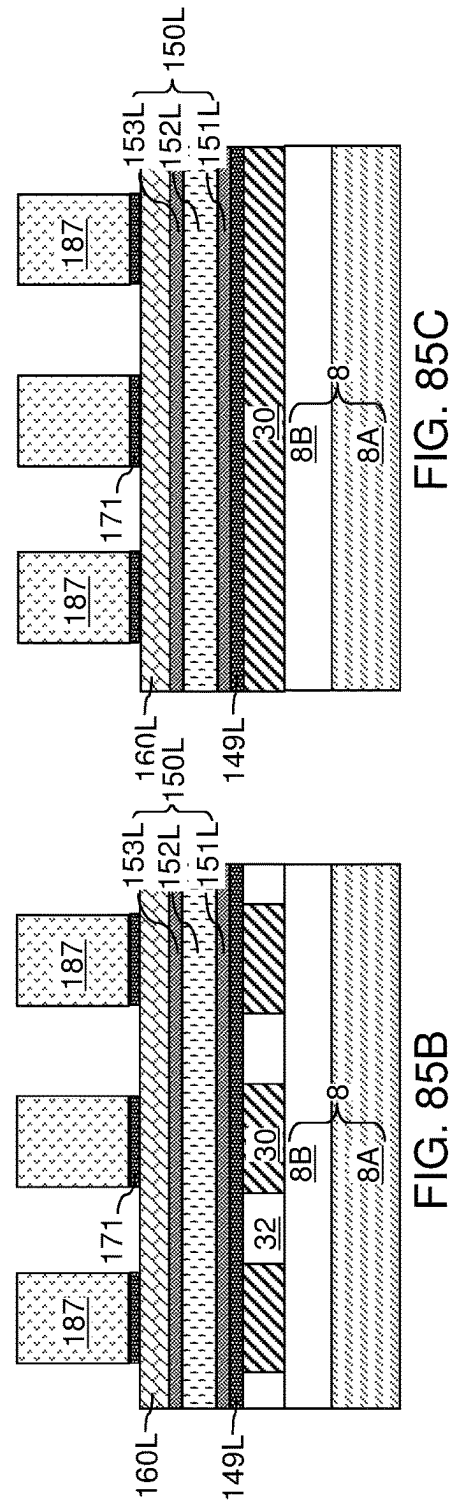
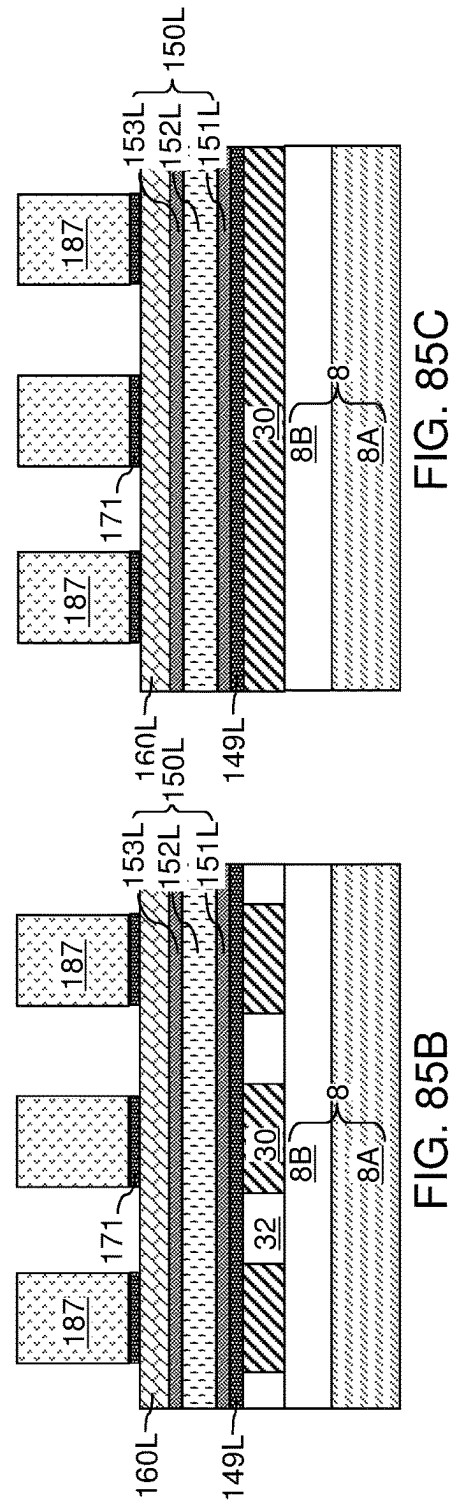
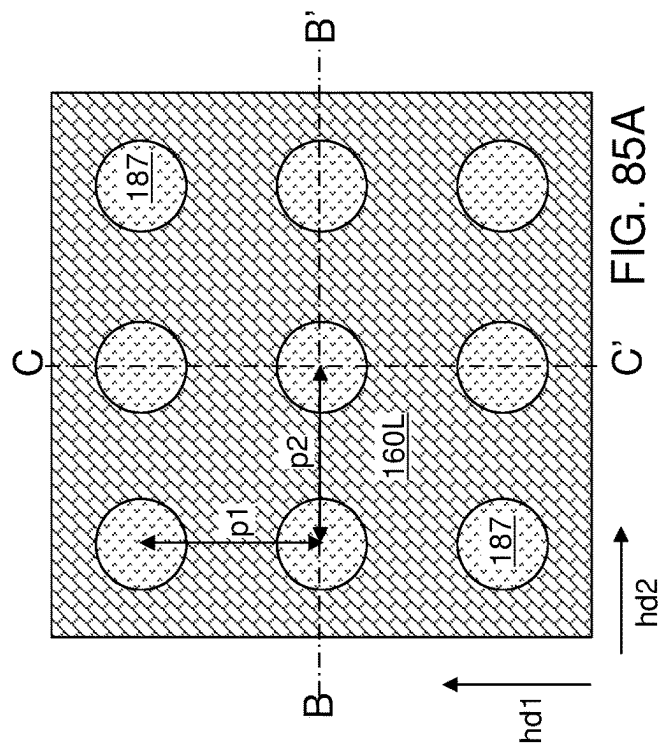


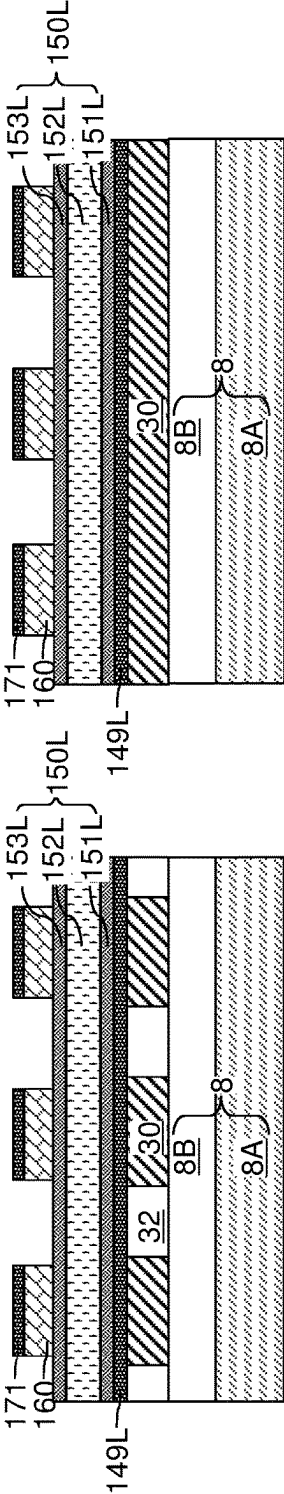
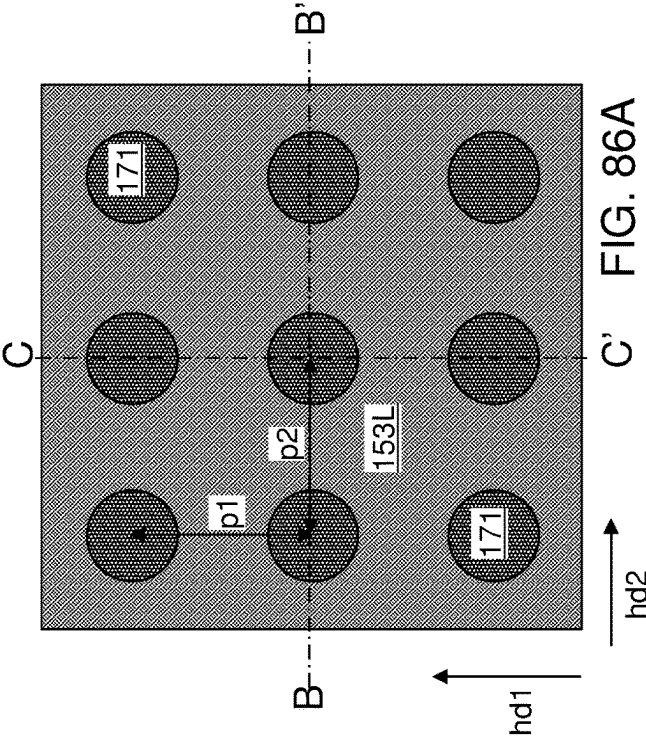
FIG. 82

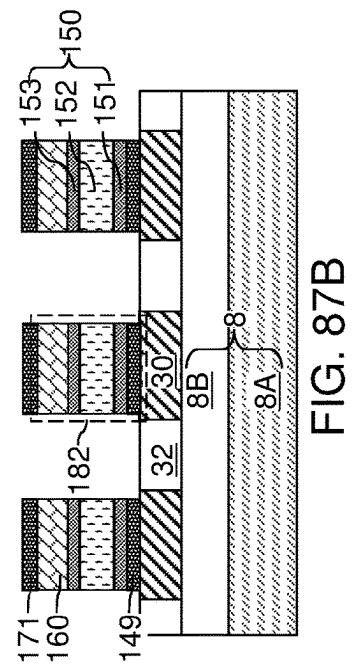
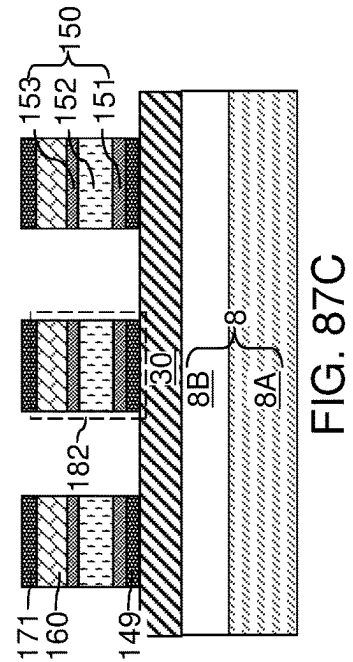
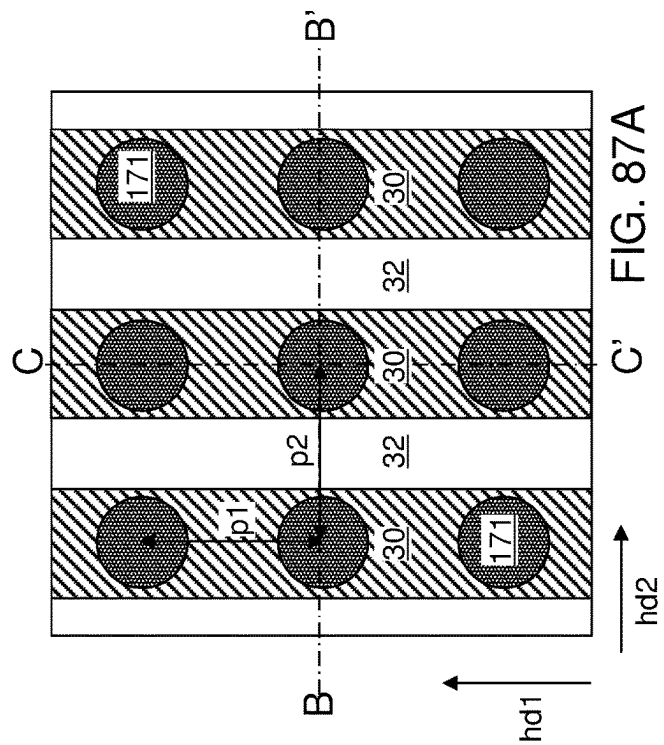


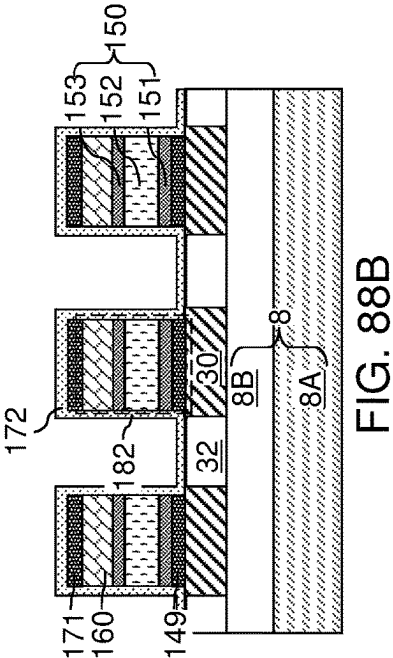
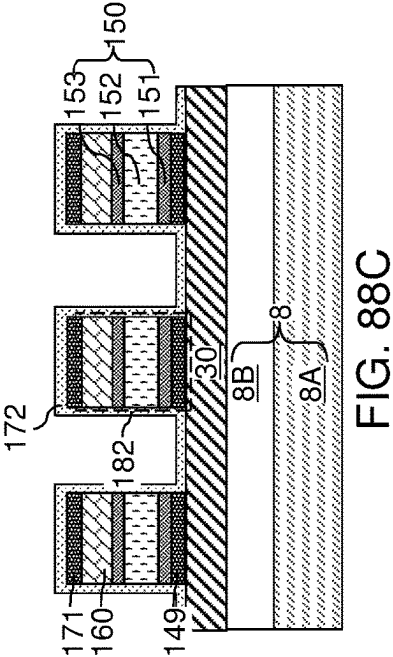
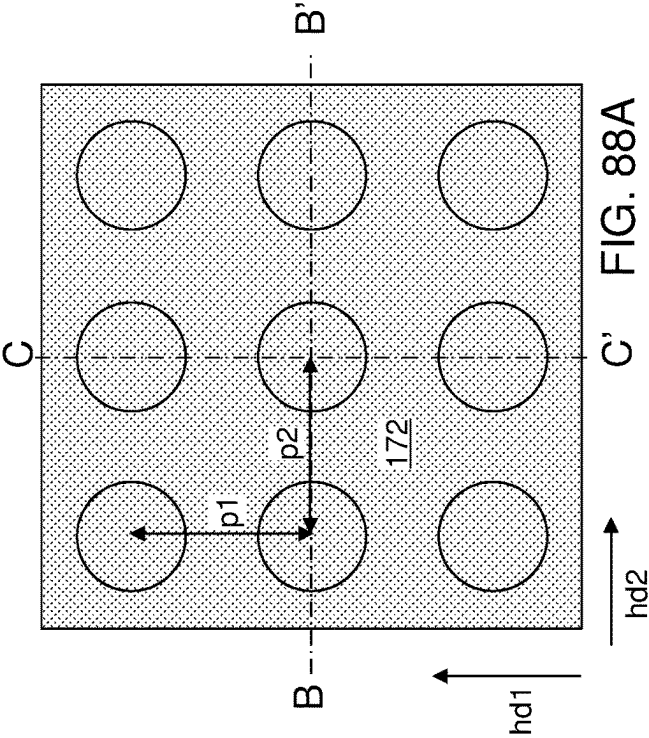












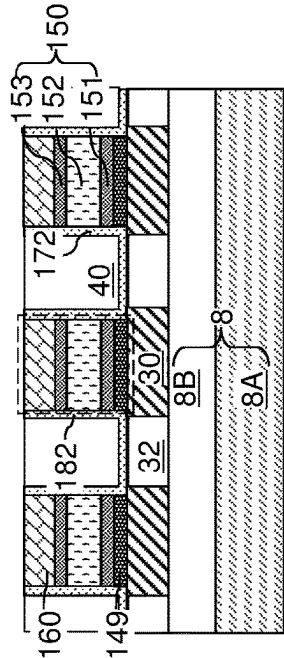
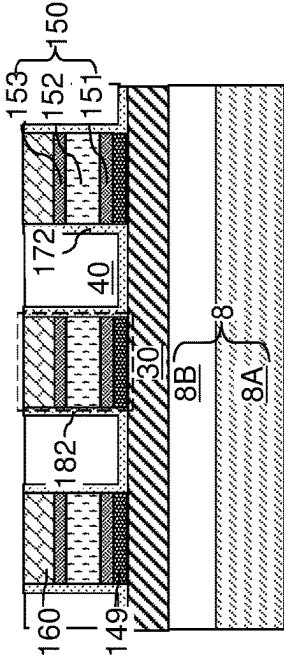
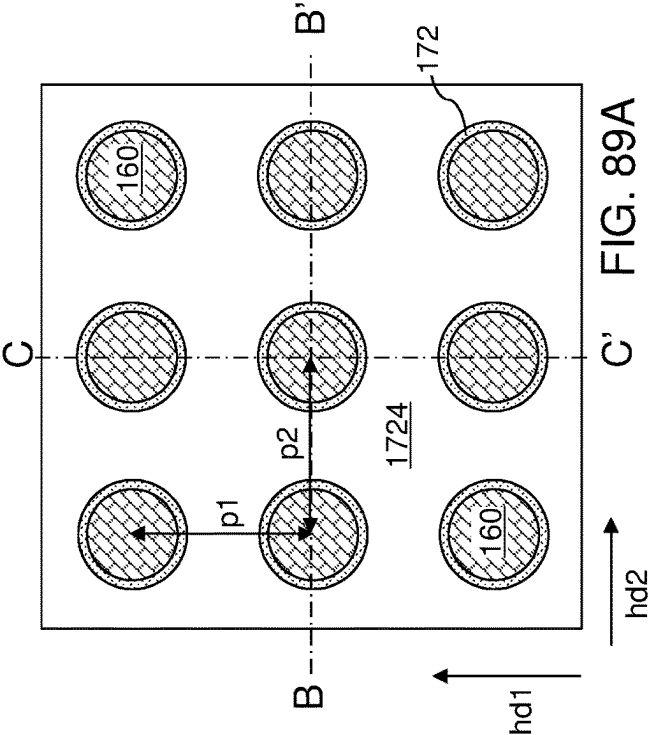
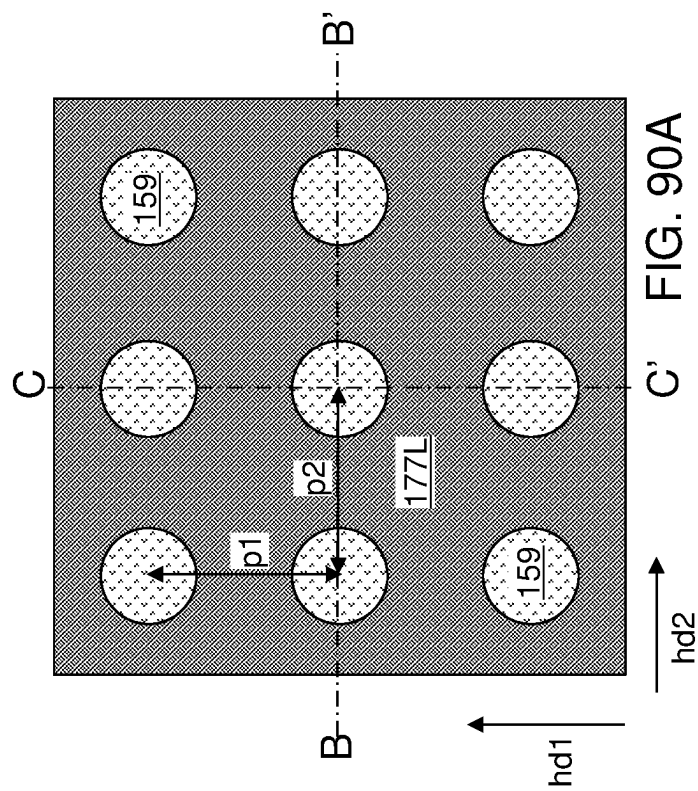


FIG. 89C

FIG. 89B



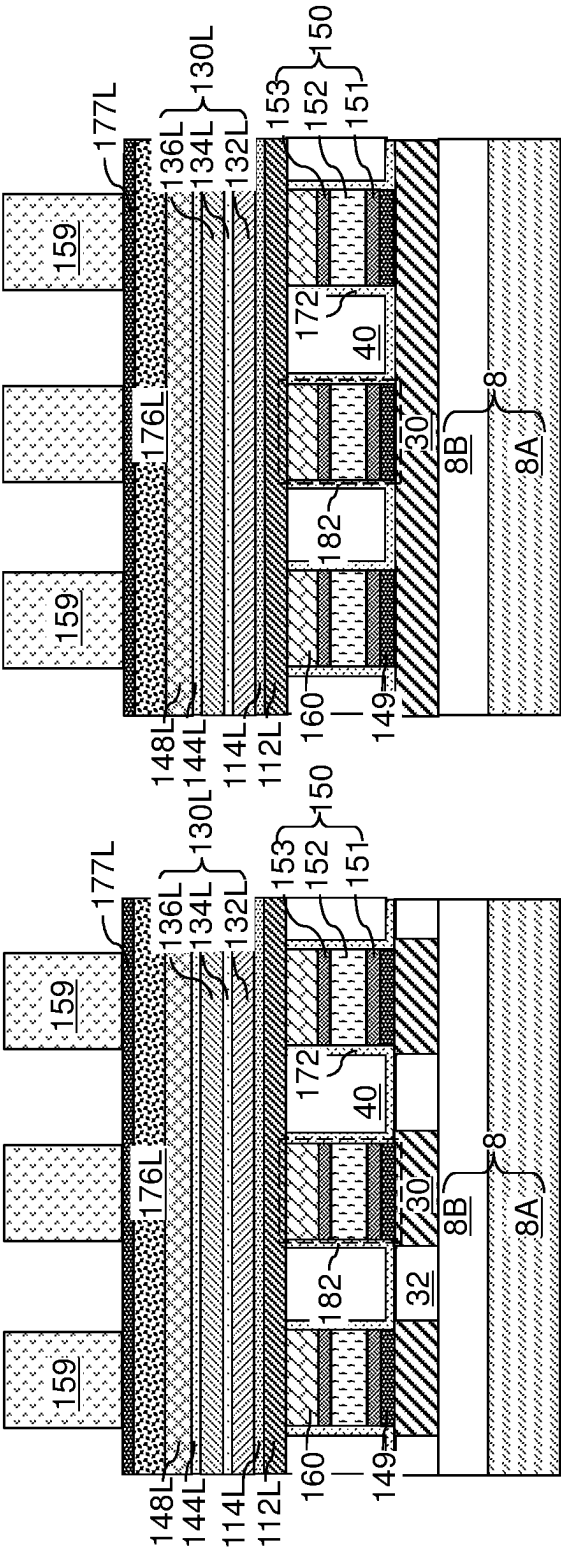
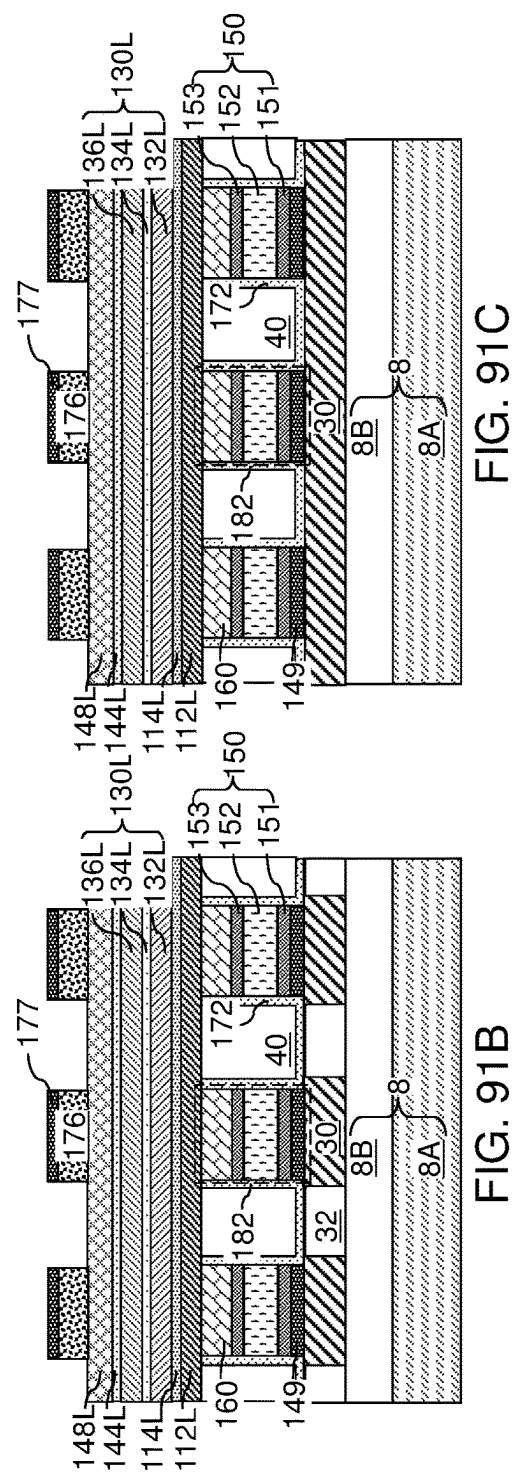
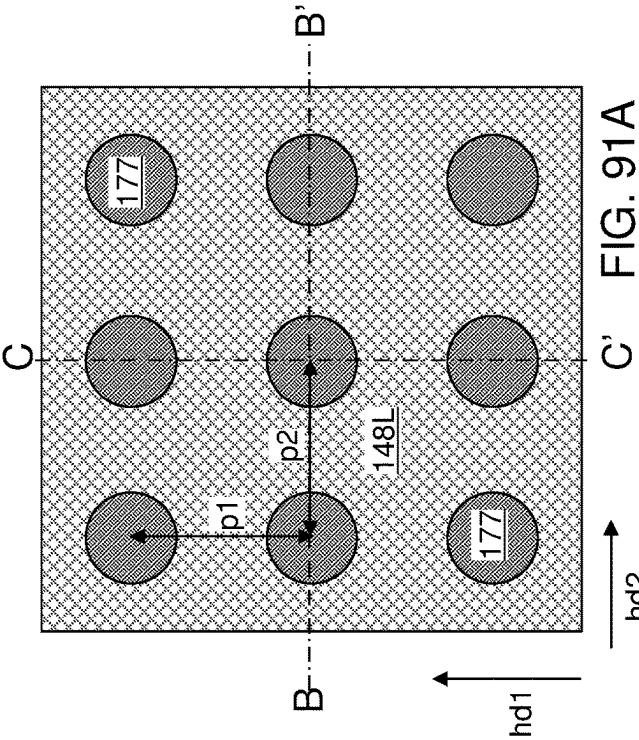
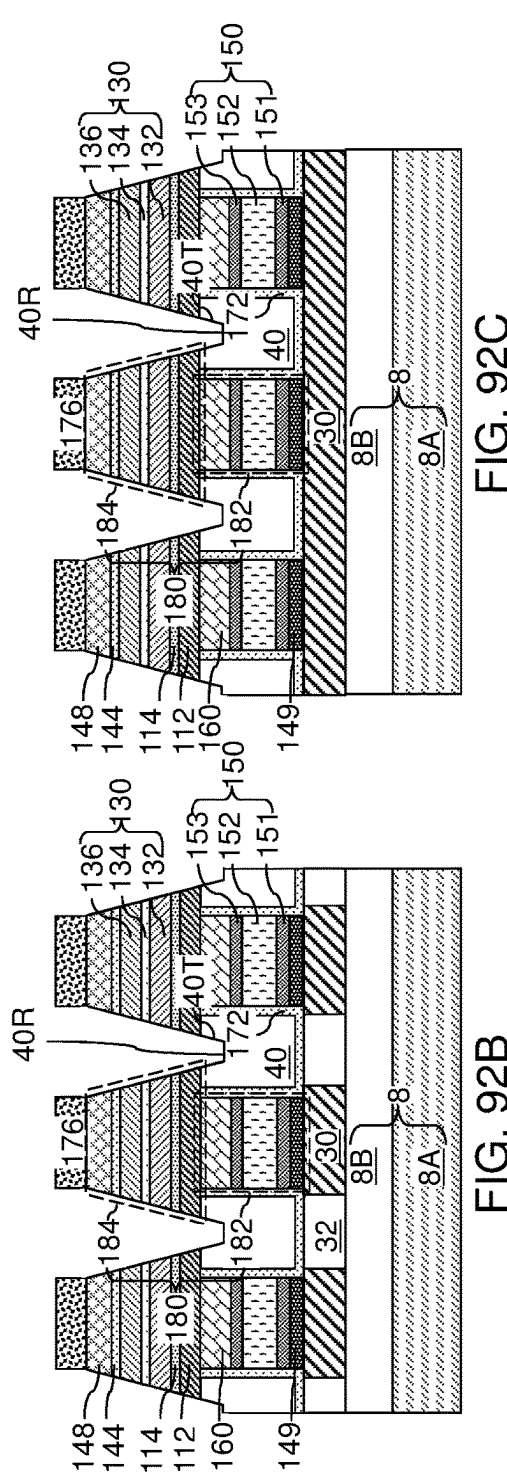
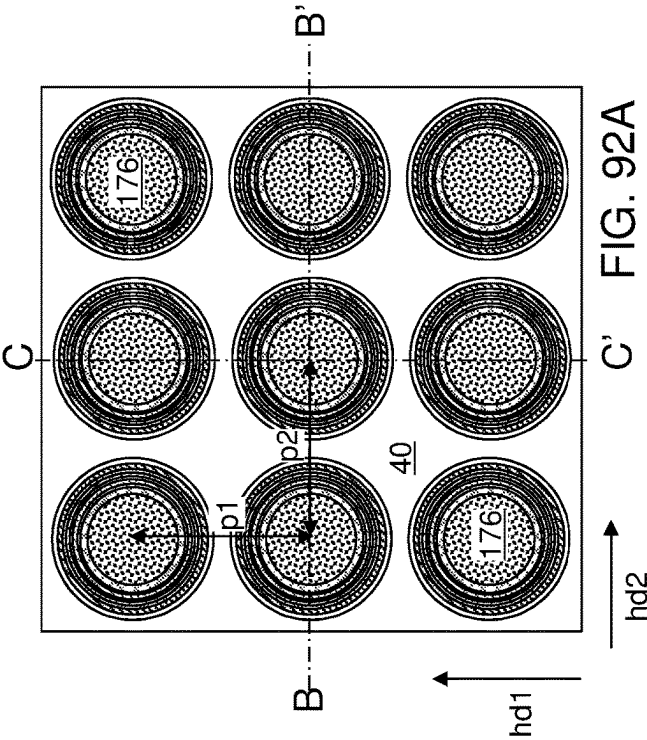


FIG. 90C

FIG. 90B







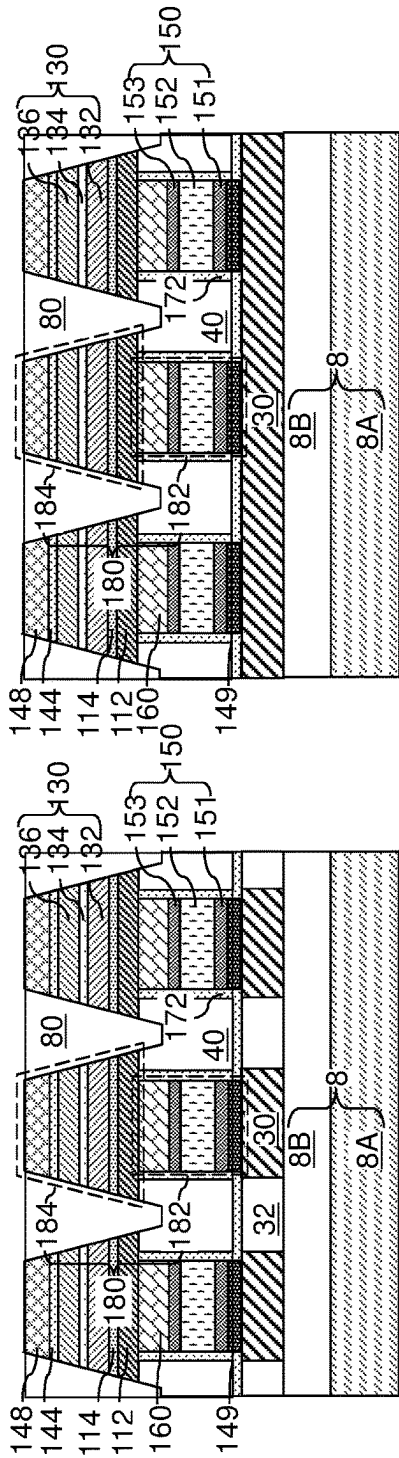
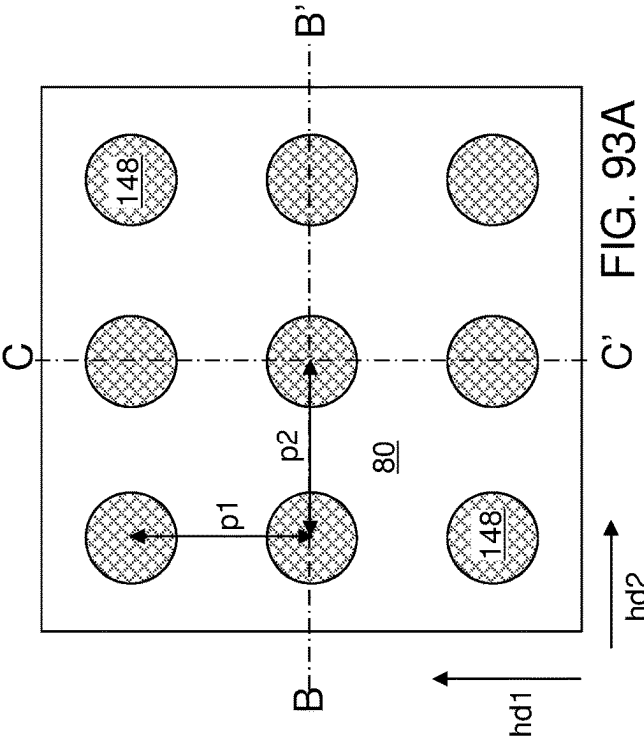


FIG. 93C

FIG. 93B

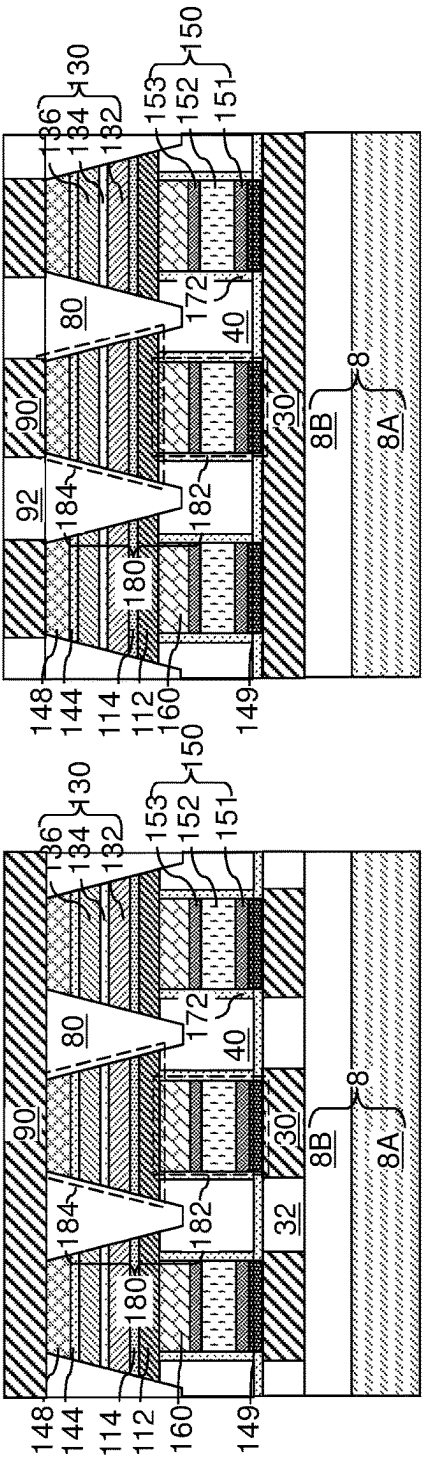
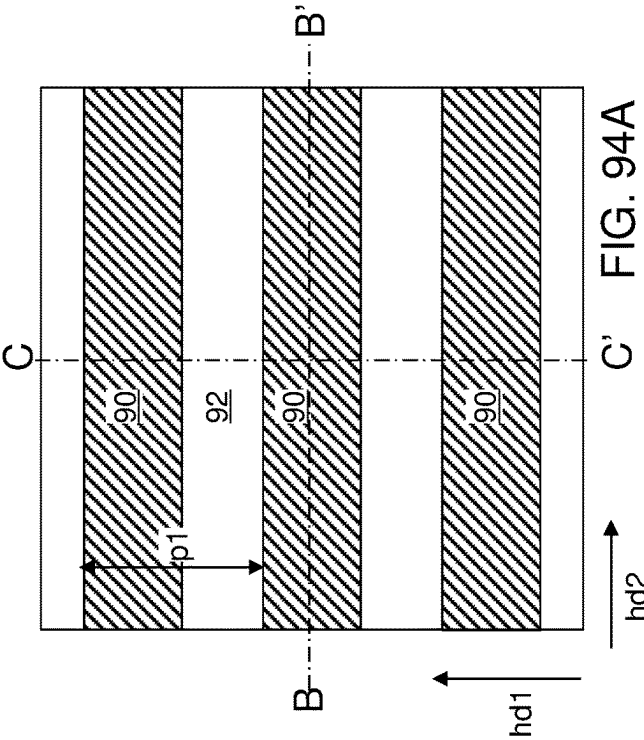


FIG. 94C

FIG. 94B

1

**CROSS-POINT MAGNETORESISTIVE  
RANDOM MEMORY ARRAY AND METHOD  
OF MAKING THEREOF USING  
SELF-ALIGNED PATTERNING**

**RELATED APPLICATIONS**

This application is a continuation-in-part (CIP) application of U.S. application Ser. No. 17/477,958 filed on Sep. 17, 2021, which is a divisional application of U.S. application Ser. No. 16/666,967 filed on Oct. 29, 2019. Further, this application is a CIP application of U.S. application Ser. No. 17/590,561 filed on Feb. 1, 2022, which is a continuation application of U.S. application Ser. No. 16/401,172 filed on May 2, 2019. In addition, this application is a CIP application of U.S. application Ser. No. 17/354,541 filed on Jun. 22, 2021, which is a divisional application of U.S. application Ser. No. 16/460,820 filed on Jul. 2, 2019, which claims priority from U.S. Provisional Application Ser. No. 62/867,590 filed on Jun. 27, 2019. Each of the above-referenced applications are incorporated herein by reference in their entirety.

**FIELD**

The present disclosure relates generally to the field of magnetic memory devices, and particularly to cross-point MRAM arrays and methods of manufacturing the same.

**BACKGROUND**

Spin-transfer torque (STT) refers to an effect in which the orientation of a magnetic layer in a magnetic tunnel junction or spin valve is modified by a spin-polarized current. Generally, electric current is unpolarized with electrons having random spin orientations. A spin polarized current is one in which electrons have a net non-zero spin due to a preferential spin orientation distribution. A spin-polarized current can be generated by passing electrical current through a magnetic polarizer layer. When the spin-polarized current flows through a free layer of a magnetic tunnel junction or a spin valve, the electrons in the spin-polarized current can transfer at least some of their angular momentum to the free layer, thereby producing a torque on the magnetization of the free layer. When a sufficient amount of spin-polarized current passes through the free layer, spin-transfer torque can be employed to flip the orientation of the spin (e.g., change the magnetization) in the free layer. A resistance differential of a magnetic tunnel junction between different magnetization states of the free layer can be employed to store data within the magnetoresistive random access memory (MRAM) cell depending if the magnetization of the free layer is parallel or antiparallel to the magnetization of the polarizer layer, also known as a reference layer.

Spin-orbit-torque (SOT) MRAM devices use switching of magnetization direction of a free magnetic layer by injection of an in-plane current in an adjacent conductive layer, which is referred to as a spin-orbit-torque (SOT) layer. Unlike the STT MRAM devices in which the electrical current is injected along a direction perpendicular to magnetic tunnel junction, the write operation is performed by flowing an electrical current through the SOT layer parallel to the magnetic tunnel junction. The read operation of a SOT memory cell is performed by passing electrical current through the magnetic tunnel junction of the SOT memory cell.

2

**SUMMARY**

According to an aspect of the present disclosure, a memory array is provided, which comprises: first electrically conductive lines laterally extending along a first horizontal direction and having a respective variable width along a second horizontal direction that varies along the first horizontal direction; a two-dimensional array of selector-containing pillar structures located over the first electrically conductive lines and including a respective selector element; a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures located over the two-dimensional array of selector-containing pillar structures and including a respective magnetic tunnel junction (MTJ); and second electrically conductive lines laterally extending along the second horizontal direction and overlying the two-dimensional array of MTJ pillar structures.

According to another aspect of the present disclosure, a method of forming a memory device is provided, which comprises: forming a first electrically conductive layer over a substrate; forming a two-dimensional array of selector-containing pillar structures including a respective selector element over the first electrically conductive layer; forming dielectric spacers around the two-dimensional array of selector-containing pillar structures, wherein each of the dielectric spacers laterally surrounds a respective row of selector-containing pillar structures that are arranged along a first horizontal direction, and the dielectric spacers are laterally spaced from each other along a second horizontal direction; patterning the first electrically conductive layer into first electrically conductive lines by transferring a pattern of lengthwise sidewalls of the dielectric spacers through the first electrically conductive layer; forming a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures over the two-dimensional array of selector-containing pillar structures; and forming second electrically conductive lines laterally extending along the second horizontal direction over the two-dimensional array of MTJ pillar structures.

According to yet another aspect of the present disclosure, a method of forming a memory device is provided, which comprises: forming a first electrically conductive layer over a substrate; forming selector-level material layers over the first electrically conductive layer; forming a two-dimensional array of selector-containing pillar structures including a respective selector element by patterning the selector-level material layers employing one or more pattern transfer processes; patterning the first electrically conductive layer into first electrically conductive lines laterally extending along a first horizontal direction and laterally spaced apart along a second horizontal direction after performing at least one pattern transfer process among the one or more pattern transfer processes; forming dielectric fill material portions between rows of selector-containing pillar structures arranged along the first horizontal direction or between columns of selector-containing pillar structures arranged along the second horizontal direction, wherein top surfaces of the dielectric fill material portions are formed within a horizontal plane including top surfaces of the two-dimensional array of selector-containing pillar structures; forming a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures over the two-dimensional array of selector-containing pillar structures; and forming second electrically conductive lines laterally extending along the second horizontal direction over the two-dimensional array of MTJ pillar structures.

According to still another aspect of the present disclosure, a memory array is provided, which comprises: first electrically

cally conductive lines laterally extending along a first horizontal direction and laterally spaced apart along a second horizontal direction; rows of selector-magnetic tunnel junction (selector-MTJ) assemblies located on a respective one of the first electrically conductive lines, wherein each of the selector-MTJ assemblies comprises a respective row of magnetic tunnel junctions (MTJs) and a respective row of selector-containing pillar structures that are arranged along the first horizontal direction, and a lateral spacing between neighboring pairs of selector-containing pillar structures that are laterally spaced apart along the first horizontal direction is less than a lateral spacing between neighboring pairs of selector-containing pillar structures that are laterally spaced apart along the second horizontal direction; and second electrically conductive lines laterally extending along the second horizontal direction and overlying a respective column of the selector-MTJ assemblies.

According to even another aspect of the present disclosure, a method of forming a memory device is provided, which comprises: forming a first electrically conductive layer, magnetic-tunnel-junction-level (MTJ-level) material layers that include magnetic tunnel junction material layers, and selector-level material layers over a substrate; forming a two-dimensional array of discrete patterned resist material portions over the selector-level material layers, wherein a first nearest-neighbor spacing along a first horizontal direction of the two-dimensional array of discrete patterned resist material portions is less than a second nearest-neighbor spacing along a second horizontal direction of the two-dimensional array of discrete patterned resist material portions; and transferring a pattern in the two-dimensional array of discrete patterned resist material portions through the selector material layers, the magnetic tunnel junction material layers, and the first electrically conductive layer such that physically exposed surfaces of remaining portions of the MTJ-level material layers are formed with taper angles, wherein patterned portions of the selector-level material layers comprise a two-dimensional array of selector-containing pillar structures including a respective selector element, patterned portions of the MTJ-level material layers comprise a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures, and patterned portions of the first electrically conductive layer comprise first electrically conductive lines that laterally extend along the first horizontal direction and laterally spaced apart from each other along the second horizontal direction.

According to aspect of the present disclosure, a method of forming a memory device is provided, which comprises: forming a first electrically conductive layer over a substrate; forming a two-dimensional array of memory cells over the first electrically conductive layer, wherein each of the memory cells comprises a vertical stack including a magnetic tunnel junction pillar structure and a selector-containing pillar structure; coating a continuous resist layer over the two-dimensional array of memory cells such that the continuous resist layer comprises a horizontally-extending planar resist layer overlying the first electrically conductive layer, a two-dimensional array of tubular resist portions laterally surrounding the two-dimensional array of memory cells, and a two-dimensional array of capping resist portions overlying the two-dimensional array of memory cells; patterning the continuous resist layer into discrete resist material portions by lithographic exposure and development, wherein the horizontally-extending planar resist layer is divided into a plurality of horizontally-extending planar resist portions having a respective pair of lengthwise edges laterally extending along a first horizontal direction and

adjoined to a respective set of at least one tubular resist portion; and patterning the first electrically conductive layer into a plurality of first electrically conductive lines by etching portions of the first electrically conductive layer that are not covered by the discrete resist material portions.

According to another aspect of the present disclosure, a memory device is provided, which comprises: first electrically conductive lines laterally extending along a first horizontal direction, laterally spaced apart from each other along a second horizontal direction, and located over a substrate; a two-dimensional array of memory cells located over the first electrically conductive lines, wherein each of the memory cells comprises a vertical stack including a magnetic tunnel junction pillar structure and a selector-containing pillar structure, and each of the first electrically conductive lines contacts a respective row of memory cells arranged along the first horizontal direction; discrete resist material portions having a tubular configuration and laterally surrounds a respective one of the memory cells; second electrically conductive lines contacting top surfaces of a respective subset of the memory cells; and a dielectric matrix layer laterally surrounding the two-dimensional array of discrete resist material portions.

According to yet another aspect of the present disclosure, a memory device is provided, which comprises: first electrically conductive lines laterally extending along a first horizontal direction and laterally spaced apart from each other along a second horizontal direction; a two-dimensional array of selector-containing pillar structures located over the first electrically conductive lines, wherein each of the first electrically conductive lines contacts a respective row of selector-containing pillar structures of the two-dimensional array of selector-containing pillar structures; a protective dielectric liner comprising a two-dimensional array of tubular dielectric liner portions laterally surrounding the two-dimensional array of selector-containing pillar structures; a two-dimensional array of magnetic tunnel junction pillar structures located above the two-dimensional array of selector-containing pillar structures; and second electrically conductive lines laterally extending along the second horizontal direction, laterally spaced apart from each other along the first horizontal direction, and located over the two-dimensional array of magnetic tunnel junction pillar structures.

According to still another aspect of the present disclosure, a method of forming a memory device includes forming a two-dimensional array of selector-containing pillar structures over first electrically conductive lines which extend in a first horizontal direction; depositing a layer stack including a continuous reference layer, a continuous nonmagnetic tunnel barrier layer, and a continuous free layer over the two-dimensional array of selector-containing pillar structures; patterning the layer stack into a two-dimensional array of magnetic tunnel junction pillar structures; and forming second electrically conductive lines over the two-dimensional array of magnetic tunnel junction pillar structures.

#### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic diagram of a memory device including resistive memory cells of the present disclosure in an array configuration.

FIG. 2 illustrates an exemplary STT MRAM cell according to an embodiment of the present disclosure.

FIGS. 3A-3C are various views of a first exemplary structure after formation of a layer stack comprising a first electrically conductive layer, first selector-level material layers, and a first conductive material layer over a substrate

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according to a first embodiment of the present disclosure. FIG. 3A is a top-down view, FIG. 3B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 3A, and FIG. 3C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 3A.

FIGS. 4A-4C are various views of the first exemplary structure after formation of a two-dimensional array of first discrete patterned photoresist material portions over the first conductive material layer according to the first embodiment of the present disclosure. FIG. 4A is a top-down view, FIG. 4B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 4A, and FIG. 4C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 4A.

FIGS. 5A-5C are various views of the first exemplary structure after formation of a two-dimensional array of first selector-containing pillar structures according to the first embodiment of the present disclosure. FIG. 5A is a top-down view, FIG. 5B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 5A, and FIG. 5C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 5A.

FIGS. 6A-6C are various views of the first exemplary structure after formation of a first dielectric spacer material layer according to the first embodiment of the present disclosure. FIG. 6A is a top-down view, FIG. 6B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 6A, and FIG. 6C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 6A.

FIGS. 7A-7C are various views of the first exemplary structure after formation of first dielectric spacers according to the first embodiment of the present disclosure. FIG. 7A is a top-down view, FIG. 7B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 7A, and FIG. 7C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 7A.

FIGS. 8A-8D are various views of the first exemplary structure after patterning the first electrically conductive layer into first electrically conductive lines according to the first embodiment of the present disclosure. FIG. 8A is a top-down view, FIG. 8B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 8A, FIG. 8C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 8A, and FIG. 8D is a horizontal cross-sectional view of the first exemplary structure along the horizontal plane D-D' of FIGS. 8B and 8C.

FIGS. 9A-9C are various views of the first exemplary structure after formation of a first selector-level dielectric matrix layer according to the first embodiment of the present disclosure. FIG. 9A is a top-down view, FIG. 9B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 9A, and FIG. 9C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 9A.

FIGS. 9D-9F are various views of an alternative configuration of the first exemplary structure after formation of a first selector-level dielectric matrix layer according to the first embodiment of the present disclosure. FIG. 9D is a top-down view, FIG. 9E is a vertical cross-sectional view along the vertical plane E-E' of FIG. 9D, and FIG. 9F is a vertical cross-sectional view along the vertical plane E-E' of FIG. 9D.

FIGS. 10A-10C are various views of the first exemplary structure after formation of a first continuous superlattice layer, a first continuous antiferromagnetic coupling layer, a first continuous reference layer, a first continuous nonmagnetic tunnel barrier layer, a first continuous free layer, a first continuous dielectric capping layer, and a first continuous metallic capping layer according to the first embodiment of

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the present disclosure. FIG. 10A is a top-down view, FIG. 10B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 10A, and FIG. 10C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 10A.

FIGS. 11A-11C are various views of the first exemplary structure after formation of a two-dimensional array of second discrete patterned photoresist material portions over the continuous metallic capping layer according to the first embodiment of the present disclosure. FIG. 11A is a top-down view, FIG. 11B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 11A, and FIG. 11C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 11A.

FIGS. 12A-12C are various views of the first exemplary structure after formation of a two-dimensional array of first magnetic tunnel junction pillar structures according to the first embodiment of the present disclosure. FIG. 12A is a top-down view, FIG. 12B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 12A, and FIG. 12C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 12A.

FIGS. 13A-13C are various views of the first exemplary structure after formation of a first magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer according to the first embodiment of the present disclosure. FIG. 13A is a top-down view, FIG. 13B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 13A, and FIG. 13C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 13A.

FIGS. 14A-14C are various views of the second exemplary structure after formation of a layer stack comprising a second electrically conductive layer, second selector-level material layers, and a second conductive material layer according to the second embodiment of the present disclosure. FIG. 14A is a top-down view, FIG. 14B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 14A, and FIG. 14C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 14A.

FIGS. 15A-15C are various views of the first exemplary structure after formation of a two-dimensional array of second selector-containing pillar structures according to the first embodiment of the present disclosure. FIG. 15A is a top-down view, FIG. 15B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 15A, and FIG. 15C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 15A.

FIGS. 16A-16C are various views of the first exemplary structure after formation of a second dielectric spacer material layer according to the first embodiment of the present disclosure. FIG. 16A is a top-down view, FIG. 16B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 16A, and FIG. 16C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 16A.

FIGS. 17A-17C are various views of the first exemplary structure after formation of second dielectric spacers according to the first embodiment of the present disclosure. FIG. 17A is a top-down view, FIG. 17B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 17A, and FIG. 17C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 17A.

FIGS. 18A-18C are various views of the first exemplary structure after patterning the second electrically conductive layer into first electrically conductive lines according to the first embodiment of the present disclosure. FIG. 18A is a top-down view, FIG. 18B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 18A, and FIG. 18C is a vertical cross-sectional view along the vertical plane C-C' of

FIG. 18A, and FIG. 18D is a horizontal cross-sectional view of the first exemplary structure along the horizontal plane D-D' of FIGS. 18B and 18C.

FIGS. 19A-19C are various views of the first exemplary structure after formation of a second selector-level dielectric matrix layer according to the first embodiment of the present disclosure. FIG. 19A is a top-down view, FIG. 19B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 19A, and FIG. 19C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 19A.

FIGS. 20A-20C are various views of the first exemplary structure after formation of a second continuous superlattice layer, a second continuous antiferromagnetic coupling layer, a second continuous reference layer, a second continuous nonmagnetic tunnel barrier layer, a second continuous free layer, a second continuous dielectric capping layer, and a second continuous metallic capping layer according to the first embodiment of the present disclosure. FIG. 20A is a top-down view, FIG. 20B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 20A, and FIG. 20C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 20A.

FIGS. 21A-21C are various views of the first exemplary structure after formation of a two-dimensional array of second magnetic tunnel junction pillar structures according to the first embodiment of the present disclosure. FIG. 21A is a top-down view, FIG. 21B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 21A, and FIG. 21C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 21A.

FIGS. 22A-22C are various views of the first exemplary structure after formation of a second magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer and third electrically conductive lines according to the first embodiment of the present disclosure. FIG. 22A is a top-down view, FIG. 22B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 22A, and FIG. 22C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 22A.

FIGS. 23A-23C are various views of an alternative configuration of the first exemplary structure according to the first embodiment of the present disclosure. FIG. 23A is a top-down view, FIG. 23B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 23A, and FIG. 23C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 23A.

FIGS. 24A-24C are various views of a second exemplary structure after formation of a layer stack comprising a first electrically conductive layer, first selector-level material layers, and a first conductive material layer and formation of a first patterned photoresist layer according to a second embodiment of the present disclosure. FIG. 24A is a top-down view, FIG. 24B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 24A, and FIG. 24C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 24A.

FIGS. 25A-25C are various views of the second exemplary structure after formation of selector rail structures according to the second embodiment of the present disclosure. FIG. 25A is a top-down view, FIG. 25B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 25A, and FIG. 25C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 25A.

FIGS. 26A-26C are various views of the second exemplary structure after formation of first selector-level isolation rails according to the second embodiment of the present disclosure. FIG. 26A is a top-down view, FIG. 26B is a vertical cross-sectional view along the vertical plane B-B' of

FIG. 26A, and FIG. 26C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 26A.

FIGS. 27A-27C are various views of the second exemplary structure after formation of a second patterned photoresist layer according to the second embodiment of the present disclosure. FIG. 27A is a top-down view, FIG. 27B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 27A, and FIG. 27C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 27A.

FIGS. 28A-28C are various views of the second exemplary structure after formation of a two-dimensional array of selector-containing pillar structures and first electrically conductive lines according to the second embodiment of the present disclosure. FIG. 28A is a top-down view, FIG. 28B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 28A, and FIG. 28C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 28A.

FIGS. 29A-29C are various views of the second exemplary structure after formation of second selector-level isolation rails according to the second embodiment of the present disclosure. FIG. 29A is a top-down view, FIG. 29B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 29A, and FIG. 29C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 29A.

FIGS. 30A-30C are various views of the second exemplary structure after formation of a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures according to the second embodiment of the present disclosure. FIG. 30A is a top-down view, FIG. 30B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 30A, and FIG. 30C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 30A.

FIGS. 31A-31C are various views of the second exemplary structure after formation of a magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer according to the second embodiment of the present disclosure. FIG. 31A is a top-down view, FIG. 31B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 31A, and FIG. 31C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 31A.

FIGS. 32A-32C are various views of the second exemplary structure after formation of second electrically conductive lines according to the second embodiment of the present disclosure. FIG. 32A is a top-down view, FIG. 32B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 32A, and FIG. 32C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 32A.

FIGS. 33A-33C are various views of a first alternative configuration of the second exemplary structure according to the second embodiment of the present disclosure. FIG. 33A is a top-down view, FIG. 33B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 33A, and FIG. 33C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 33A.

FIGS. 34A-34C are various views of a second alternative configuration of the second exemplary structure after formation of a layer stack comprising a first electrically conductive layer, first selector-level material layers, and a first conductive material layer and formation of a first patterned photoresist layer according to a second embodiment of the present disclosure. FIG. 34A is a top-down view, FIG. 34B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 34A, and FIG. 34C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 34A.

FIGS. 35A-35C are various views of the second alternative configuration of the second exemplary structure after formation of selector rail structures and first electrically

conductive lines according to the second embodiment of the present disclosure. FIG. 35A is a top-down view, FIG. 35B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 35A, and FIG. 35C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 35A.

FIGS. 36A-36C are various views of the second alternative configuration of the second exemplary structure after formation of first selector-level isolation rails according to the second embodiment of the present disclosure. FIG. 36A is a top-down view, FIG. 36B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 36A, and FIG. 36C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 36A.

FIGS. 37A-37C are various views of the second alternative configuration of the second exemplary structure after formation of a second patterned photoresist layer according to the second embodiment of the present disclosure. FIG. 37A is a top-down view, FIG. 37B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 37A, and FIG. 37C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 37A.

FIGS. 38A-38C are various views of the second alternative configuration of the second exemplary structure after formation of a two-dimensional array of selector-containing pillar structures according to the second embodiment of the present disclosure. FIG. 38A is a top-down view, FIG. 38B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 38A, and FIG. 38C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 38A.

FIGS. 39A-39C are various views of the second alternative configuration of the second exemplary structure after formation of second selector-level isolation rails according to the second embodiment of the present disclosure. FIG. 39A is a top-down view, FIG. 39B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 39A, and FIG. 39C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 39A.

FIGS. 40A-40C are various views of the second alternative configuration of the second exemplary structure after formation of a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures according to the second embodiment of the present disclosure. FIG. 40A is a top-down view, FIG. 40B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 40A, and FIG. 40C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 40A.

FIGS. 41A-41C are various views of the second alternative configuration of the second exemplary structure after formation of a magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer according to the second embodiment of the present disclosure. FIG. 41A is a top-down view, FIG. 41B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 41A, and FIG. 41C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 41A.

FIGS. 42A-42C are various views of the second alternative configuration of the second exemplary structure after formation of second electrically conductive lines according to the second embodiment of the present disclosure. FIG. 42A is a top-down view, FIG. 42B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 42A, and FIG. 42C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 42A.

FIGS. 43A-43C are various views of a third alternative configuration of the second exemplary structure according to the second embodiment of the present disclosure. FIG. 43A is a top-down view, FIG. 43B is a vertical cross-sectional

view along the vertical plane B-B' of FIG. 43A, and FIG. 43C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 43A.

FIGS. 44A-44C are various views of a third exemplary structure after formation of a first electrically conductive layer and first magnetic tunnel junction material layers according to a third embodiment of the present disclosure. FIG. 44A is a top-down view, FIG. 44B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 44A, and FIG. 44C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 44A.

FIGS. 45A-45C are various views of the third exemplary structure after formation of first selector level material layers and a two-dimensional array of discrete photoresist material portions according to the third embodiment of the present disclosure. FIG. 45A is a top-down view, FIG. 45B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 45A, and FIG. 45C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 45A.

FIGS. 46A-46C are various views of the third exemplary structure after formation of a two-dimensional array of first selector-containing pillar structures according to the third embodiment of the present disclosure. FIG. 46A is a top-down view, FIG. 46B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 46A, and FIG. 46C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 46A.

FIGS. 47A-47D are various views of the third exemplary structure after formation of a two-dimensional array of first magnetic tunnel junction pillar structures and first electrically conductive lines according to the third embodiment of the present disclosure. FIG. 47A is a top-down view, FIG. 47B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 47A, FIG. 47C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 47A, and FIG. 47D is a horizontal cross-sectional view along the horizontal plane D-D' of FIGS. 47B and 47C.

FIGS. 47E, 47F, 47G and 47H are vertical cross-sectional view along the vertical plane C-C' of FIG. 47A according to alternative configurations of the third exemplary structure.

FIGS. 48A-48C are various views of the third exemplary structure after formation of a first dielectric matrix layer according to the third embodiment of the present disclosure. FIG. 48A is a top-down view, FIG. 48B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 48A, and FIG. 48C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 48A.

FIGS. 49A-49C are various views of the third exemplary structure after formation of a second electrically conductive layer, second magnetic tunnel junction material layers, second selector level material layers, and a two-dimensional array of discrete photoresist material portions according to the third embodiment of the present disclosure. FIG. 49A is a top-down view, FIG. 49B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 49A, and FIG. 49C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 49A.

FIGS. 50A-50C are various views of the third exemplary structure after formation of a two-dimensional array of second selector-containing pillar structures according to the third embodiment of the present disclosure. FIG. 50A is a top-down view, FIG. 50B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 50A, and FIG. 50C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 50A.

FIGS. 51A-51D are various views of the third exemplary structure after formation of a two-dimensional array of

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second magnetic tunnel junction pillar structures and second electrically conductive lines according to the third embodiment of the present disclosure. FIG. 51A is a top-down view, FIG. 51B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 51A, FIG. 51C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 51A, and FIG. 51D is a horizontal cross-sectional view along the horizontal plane D-D' of FIGS. 51B and 51C.

FIGS. 52A-52C are various views of the third exemplary structure after formation of a second dielectric matrix layer according to the third embodiment of the present disclosure. FIG. 52A is a top-down view, FIG. 52B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 52A, and FIG. 52C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 52A.

FIGS. 53A-53C are various views of the third exemplary structure after formation of third electrically conductive lines according to the third embodiment of the present disclosure. FIG. 53A is a top-down view, FIG. 53B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 53A, and FIG. 53C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 53A.

FIGS. 54A-54C are various views of a first alternative configuration of the third exemplary structure after formation of third electrically conductive lines according to the third embodiment of the present disclosure. FIG. 54A is a top-down view, FIG. 54B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 54A, and FIG. 54C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 54A.

FIGS. 55A-55C are various views of a second alternative configuration of the third exemplary structure after formation of third electrically conductive lines according to the third embodiment of the present disclosure. FIG. 55A is a top-down view, FIG. 55B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 55A, and FIG. 55C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 55A.

FIGS. 56A-56C are various views of a third alternative configuration of the third exemplary structure after formation of third electrically conductive lines according to the third embodiment of the present disclosure. FIG. 56A is a top-down view, FIG. 56B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 56A, and FIG. 56C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 56A.

FIGS. 57A-57C are various views of a fourth alternative configuration of the third exemplary structure after formation of third electrically conductive lines according to the third embodiment of the present disclosure. FIG. 57A is a top-down view, FIG. 57B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 57A, and FIG. 57C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 57A.

FIGS. 58A-58C are various views of a fourth exemplary structure after formation of a first electrically conductive layer, magnetic tunnel junction material layers, and selector-level material layers according to a fourth embodiment of the present disclosure. FIG. 58A is a top-down view, FIG. 58B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 58A, and FIG. 58C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 58A.

FIGS. 59A-59C are various views of a fourth exemplary structure after formation of a two-dimensional array of discrete resist material portions according to a fourth embodiment of the present disclosure. FIG. 59A is a top-down view, FIG. 59B is a vertical cross-sectional view along

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the vertical plane B-B' of FIG. 59A, and FIG. 59C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 59A.

FIGS. 60A-60C are various views of the fourth exemplary structure after formation of a two-dimensional array of first selector-containing pillar structures according to the fourth embodiment of the present disclosure. FIG. 60A is a top-down view, FIG. 60B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 60A, and FIG. 60C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 60A.

FIGS. 61A-61D are various views of the fourth exemplary structure after formation of a two-dimensional array of first magnetic tunnel junction pillar structures and first electrically conductive lines according to the fourth embodiment of the present disclosure. FIG. 61A is a top-down view, FIG. 61B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 61A, FIG. 61C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 61A, and FIG. 61D is a horizontal cross-sectional view along the horizontal plane D-D' of FIGS. 61B and 61C.

FIGS. 62A-62C are various views of the fourth exemplary structure after formation of a first dielectric matrix layer according to the fourth embodiment of the present disclosure. FIG. 62A is a top-down view, FIG. 62B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 62A, and FIG. 62C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 62A.

FIGS. 63A-63C are various views of the fourth exemplary structure after formation of second electrically conductive lines according to the fourth embodiment of the present disclosure. FIG. 63A is a top-down view, FIG. 63B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 63A, and FIG. 63C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 63A.

FIGS. 64A-64C are various views of a first alternative configuration of the fourth exemplary structure after formation of fourth electrically conductive lines according to the fourth embodiment of the present disclosure. FIG. 64A is a top-down view, FIG. 64B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 64A, and FIG. 64C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 64A.

FIGS. 65A-65C are various views of a second alternative configuration of the fourth exemplary structure after formation of fourth electrically conductive lines according to the fourth embodiment of the present disclosure. FIG. 65A is a top-down view, FIG. 65B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 65A, and FIG. 65C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 65A.

FIGS. 66A-66C are various views of a fifth exemplary structure after formation of a first electrically conductive layer, magnetic tunnel junction material layers, selector-level material layers, and a two-dimensional array of discrete resist material portions according to a fifth embodiment of the present disclosure. FIG. 66A is a top-down view, FIG. 66B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 66A, and FIG. 66C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 66A.

FIGS. 67A-67C are various views of the fifth exemplary structure after formation of a two-dimensional array of memory pillar structures according to a fifth embodiment of the present disclosure. FIG. 67A is a top-down view, FIG. 67B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 67A, and FIG. 67C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 67A.



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FIGS. 68A-68C are various views of the fifth exemplary structure after formation of a continuous resist layer according to the fifth embodiment of the present disclosure. FIG. 68A is a top-down view, FIG. 68B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 68A, and FIG. 68C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 68A.

FIGS. 69A-69C are various views of the fifth exemplary structure after patterning the continuous resist layer into discrete resist material portions according to the fifth embodiment of the present disclosure. FIG. 69A is a top-down view, FIG. 69B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 69A, and FIG. 69C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 69A.

FIGS. 70A-70C are various views of the fifth exemplary structure after patterning the first electrically conductive layer into a plurality of first electrically conductive lines according to the fifth embodiment of the present disclosure. FIG. 70A is a top-down view, FIG. 70B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 70A, and FIG. 70C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 70A.

FIGS. 71A-71C are various views of the fifth exemplary structure after formation of a dielectric matrix layer according to the fifth embodiment of the present disclosure. FIG. 71A is a top-down view, FIG. 71B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 71A, and FIG. 71C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 71A.

FIGS. 72A-72C are various views of the fifth exemplary structure after formation of second electrically conductive lines according to the fifth embodiment of the present disclosure. FIG. 72A is a top-down view, FIG. 72B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 72A, and FIG. 72C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 72A.

FIGS. 73A-73C are various views of a first alternative configuration of the fifth exemplary structure after formation of second electrically conductive lines according to the fifth embodiment of the present disclosure. FIG. 73A is a top-down view, FIG. 73B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 73A, and FIG. 73C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 73A.

FIGS. 74A-74C are various views of a second alternative configuration of the fifth exemplary structure after formation of second electrically conductive lines according to the fifth embodiment of the present disclosure. FIG. 74A is a top-down view, FIG. 74B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 74A, and FIG. 74C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 74A.

FIGS. 75A-75C are various views of a sixth exemplary structure after formation of a two-dimensional array of discrete patterned resist material portions according to a sixth embodiment of the present disclosure. FIG. 75A is a top-down view, FIG. 75B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 75A, and FIG. 75C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 75A.

FIGS. 76A-76C are various views of the sixth exemplary structure after formation of a two-dimensional array of memory cells according to the sixth embodiment of the present disclosure. FIG. 76A is a top-down view, FIG. 76B is a vertical cross-sectional view along the vertical plane

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B-B' of FIG. 76A, and FIG. 76C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 76A.

FIGS. 77A-77C are various views of the sixth exemplary structure after formation of a two-dimensional array of discrete resist material portions according to the sixth embodiment of the present disclosure. FIG. 77A is a top-down view, FIG. 77B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 77A, and FIG. 77C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 77A.

FIGS. 78A-78C are various views of the sixth exemplary structure after formation of a two-dimensional array of first electrically conductive lines according to the sixth embodiment of the present disclosure. FIG. 78A is a top-down view, FIG. 78B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 78A, and FIG. 78C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 78A.

FIGS. 79A-79C are various views of the sixth exemplary structure after formation of a dielectric matrix layer according to the sixth embodiment of the present disclosure. FIG. 79A is a top-down view, FIG. 79B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 79A, and FIG. 79C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 79A.

FIGS. 80A-80C are various views of the sixth exemplary structure after formation of second electrically conductive lines according to the sixth embodiment of the present disclosure. FIG. 80A is a top-down view, FIG. 80B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 80A, and FIG. 80C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 80A.

FIGS. 81A-81C are various views of an alternative configuration of the sixth exemplary structure after formation of second electrically conductive lines according to the sixth embodiment of the present disclosure. FIG. 81A is a top-down view, FIG. 81B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 81A, and FIG. 81C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 81A.

FIG. 82 is a schematic side-cross sectional view of a spin-orbit-torque (SOT) magnetoresistive random access memory (MRAM) cell that incorporates an array of memory cells of the sixth exemplary structure illustrated in FIGS. 78A-78C.

FIGS. 83A-83C are various views of a seventh exemplary structure after formation of first electrically conductive lines according to a seventh embodiment of the present disclosure. FIG. 83A is a top-down view, FIG. 83B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 83A, and FIG. 83C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 83A.

FIGS. 84A-84C are various views of the seventh exemplary structure after formation of selector-level material layers, an optional first image transfer assist layer, and a two-dimensional array of first patterned resist material portions according to the seventh embodiment of the present disclosure. FIG. 84A is a top-down view, FIG. 84B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 84A, and FIG. 84C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 84A.

FIGS. 85A-85C are various views of the seventh exemplary structure after formation of optional first etch mask plates according to the seventh embodiment of the present disclosure. FIG. 85A is a top-down view, FIG. 85B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 85A, and FIG. 85C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 85A.

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FIGS. 86A-86C are various views of the seventh exemplary structure after formation of conductive material plates according to the seventh embodiment of the present disclosure. FIG. 86A is a top-down view, FIG. 86B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 86A, and FIG. 86C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 86A.

FIGS. 87A-87C are various views of the seventh exemplary structure after formation of a two-dimensional array of selector-containing pillar structures according to the seventh embodiment of the present disclosure. FIG. 87A is a top-down view, FIG. 87B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 87A, and FIG. 87C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 87A.

FIGS. 88A-88C are various views of the seventh exemplary structure after formation of a protective dielectric liner according to the seventh embodiment of the present disclosure. FIG. 88A is a top-down view, FIG. 88B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 88A, and FIG. 88C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 88A.

FIGS. 89A-89C are various views of the seventh exemplary structure after formation of a selector-level dielectric matrix layer according to the seventh embodiment of the present disclosure. FIG. 89A is a top-down view, FIG. 89B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 89A, and FIG. 89C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 89A.

FIGS. 90A-90C are various views of the seventh exemplary structure after formation of magnetic tunnel junction material layers, an optional an optional patterning film, and an optional second image transfer assist layer according to the seventh embodiment of the present disclosure. FIG. 90A is a top-down view, FIG. 90B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 90A, and FIG. 90C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 90A.

FIGS. 91A-91C are various views of the seventh exemplary structure after formation of optional second etch mask plates and patterning plates according to the seventh embodiment of the present disclosure. FIG. 91A is a top-down view, FIG. 91B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 91A, and FIG. 91C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 91A.

FIGS. 92A-92C are various views of the seventh exemplary structure after formation of a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures according to the seventh embodiment of the present disclosure. FIG. 92A is a top-down view, FIG. 92B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 92A, and FIG. 92C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 92A.

FIGS. 93A-93C are various views of the seventh exemplary structure after formation of a magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer according to the seventh embodiment of the present disclosure. FIG. 93A is a top-down view, FIG. 93B is a vertical cross-sectional view along the vertical plane B-B' of FIG. 93A, and FIG. 93C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 93A.

FIGS. 94A-94C are various views of the seventh exemplary structure after formation of second electrically conductive lines according to the seventh embodiment of the present disclosure. FIG. 94A is a top-down view, FIG. 94B is a vertical cross-sectional view along the vertical plane

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B-B' of FIG. 94A, and FIG. 94C is a vertical cross-sectional view along the vertical plane C-C' of FIG. 94A.

#### DETAILED DESCRIPTION

As discussed above, the present disclosure is directed to a cross-point MRAM array and methods of manufacturing the same, the various aspects of which are discussed herein in detail.

The drawings are not drawn to scale. Multiple instances of an element may be duplicated where a single instance of the element is illustrated, unless absence of duplication of elements is expressly described or clearly indicated otherwise. Same reference numerals refer to the same element or to a similar element. Elements having the same reference numerals are presumed to have the same material composition unless expressly stated otherwise. Ordinals such as "first," "second," and "third" are employed merely to identify similar elements, and different ordinals may be employed across the specification and the claims of the instant disclosure. As used herein, a first element located "on" a second element can be located on the exterior side of a surface of the second element or on the interior side of the second element. As used herein, a first element is located "directly on" a second element if there exist a physical contact between a surface of the first element and a surface of the second element. As used herein, an "in-process" structure or a "transient" structure refers to a structure that is subsequently modified.

As used herein, a "layer" refers to a material portion including a region having a thickness. A layer may extend over the entirety of an underlying or overlying structure, or may have an extent less than the extent of an underlying or overlying structure. Further, a layer may be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer may be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer may extend horizontally, vertically, and/or along a tapered surface. A substrate may be a layer, may include one or more layers therein, and/or may have one or more layer thereupon, thereabove, and/or therebelow.

As used herein, a "layer stack" refers to a stack of layers. As used herein, a "line" or a "line structure" refers to a layer that has a predominant direction of extension, i.e., having a direction along which the layer extends the most.

As used herein, a "conductive material" refers to a material having electrical conductivity greater than  $1.0 \times 10^5$  S/cm. As used herein, an "insulating material" or a "dielectric material" refers to a material having electrical conductivity less than  $1.0 \times 10^{-6}$  S/cm. As used herein, a "metallic material" refers to a conductive material including at least one metallic element therein. All measurements for electrical conductivities are made at the standard condition.

Referring to FIG. 1, a schematic diagram is shown for a magnetic memory device including memory cells 180 of an embodiment of the present disclosure in an array configuration. The magnetic memory device can be configured as a MRAM device 500 containing MRAM cells 180. As used herein, a "RAM device" refers to a memory device containing memory cells that allow random access, e.g., access to any selected memory cell upon a command for reading the contents of the selected memory cell. As used herein, an "MRAM device" refers to a RAM device in which the memory cells are magnetoresistive memory cells.

The MRAM device **500** of an embodiment of the present disclosure includes a memory array region **550** containing an array of the respective MRAM cells **180** located at the intersection of the respective word lines (which may comprise first electrically conductive lines **30** as illustrated or as second electrically conductive lines **90** in an alternate configuration) and bit lines (which may comprise second electrically conductive lines **90** as illustrated or as first electrically conductive lines **30** in an alternate configuration). The MRAM device **500** may also contain a row decoder **560** connected to the word lines, a sense circuitry **570** (e.g., a sense amplifier and other bit line control circuitry) connected to the bit lines, a column decoder **580** connected to the bit lines, and a data buffer **590** connected to the sense circuitry. Multiple instances of the MRAM cells **180** are provided in an array configuration that forms the MRAM device **500**. As such, each of the MRAM cells **180** can be a two-terminal device including a respective first electrode and a respective second electrode. It should be noted that the location and interconnection of elements are schematic and the elements may be arranged in a different configuration. Further, a MRAM cell **180** may be manufactured as a discrete device, i.e., a single isolated device.

Each MRAM cell **180** includes a magnetic tunnel junction or a spin valve having at least two different resistive states depending on the alignment of magnetizations of different magnetic material layers. The magnetic tunnel junction or the spin valve is provided between a first electrode and a second electrode within each MRAM cell **180**. Configurations of the MRAM cells **180** are described in detail in subsequent sections.

Referring to FIG. 2, an exemplary spin-transfer torque (STT) MRAM device is illustrated, which may comprise one MRAM cell **180** within the magnetic memory device illustrated in FIG. 1. The MRAM cell **180** of FIG. 2 can include a first terminal that may be electrically connected to, or comprises, a portion of a first electrically conductive line **30** and a second terminal that may be electrically connected to, or comprises, a portion of a second electrically conductive line **90**. The first terminal can function as a first electrode, and the second terminal can function as a second electrode.

Generally, the MRAM cell **180** includes a magnetic tunnel junction (MTJ) **140**. The magnetic tunnel junction **140** includes a reference layer **132** (which may also be referred to as a “pinned” layer) having a fixed vertical magnetization, a nonmagnetic tunnel barrier layer **134**, and the free layer **136** (which may also be referred to as a “storage” layer) having a magnetization direction that can be programmed. The reference layer **132** and the free layer **136** can be separated by the nonmagnetic tunnel barrier layer **134** (which may be a dielectric layer such as a MgO layer), and have a magnetization direction perpendicular to the interface between the free layer **136** and the nonmagnetic tunnel barrier layer **134**.

In one embodiment, the reference layer **132** is located below the nonmagnetic tunnel barrier layer **134**, while the free layer **136** is located above the nonmagnetic tunnel barrier layer **134**. An electrically conductive capping layer **148** may be formed on top of the free layer **136** in order to provide additional perpendicular anisotropy. A dielectric capping layer **144** may be provided between the free layer **136** and the electrically conductive capping layer **148**. In one embodiment, the reference layer **132** and the free layer **136** have respective positive uniaxial magnetic anisotropy. Positive uniaxial magnetic anisotropy is also referred to as perpendicular magnetic anisotropy (PMA) in which a mini-

mum energy preference for quiescent magnetization is along the axis perpendicular to the plane of the magnetic film.

The configuration in which the reference layer **132** and the free layer **136** have respective perpendicular magnetic anisotropy provides bistable magnetization states for the free layer **136**. The bistable magnetization states include a parallel state in which the free layer **136** has a magnetization (e.g., magnetization direction) that is parallel to the fixed vertical magnetization (e.g., magnetization direction) of the reference layer **132**, and an antiparallel state in which the free layer **136** has a magnetization (e.g., magnetization direction) that is antiparallel to the fixed vertical magnetization (e.g., magnetization direction) of the reference layer **132**.

A data bit can be written in the STT MRAM cell by passing high enough electrical current through the reference layer **132** and the free layer **136** in a programming operation so that spin-transfer torque can set or reset the magnetization state of the free layer **136**. The direction of the magnetization of the free layer **136** after the programming operation depends on the current polarity with respect to magnetization direction of the reference layer **132**. The data bit can be read by passing smaller electrical current through the STT MRAM cell and measuring the resistance of the STT MRAM cell. The data bit “0” and the data bit “1” correspond to low and high resistance states of the STT MRAM cell (or vice versa), which are provided by parallel or antiparallel alignment of the magnetization directions of the free layer **136** and the reference layer **132**, respectively. The relative resistance change between parallel and antiparallel alignment (i.e., orientation) of the magnetization direction is called tunnel magnetoresistance (TMR).

In one embodiment, the reference layer **132** and the free layer **136** may include one or more ferromagnetic layers, such as CoFe or CoFeB. In plural ferromagnetic layers are included in the reference layer **132**, then a thin non-magnetic layer comprised of tantalum or tungsten having a thickness of 0.2 nm–0.5 nm may be located between the ferromagnetic layers. The nonmagnetic tunnel barrier layer **134** can include any tunneling barrier material such as an electrically insulating material, for example magnesium oxide. The thickness of the nonmagnetic tunnel barrier layer **134** can be 0.7 nm to 1.3 nm, such as about 1 nm.

The reference layer **132** may be provided as a component within a synthetic antiferromagnetic structure (SAF structure) **120** which is formed over an optional nonmagnetic metallic seed layer **110**, such as a Ta and/or Pt seed layer. In one embodiment, the SAF structure **120** can include a vertical stack including at least one superlattice **112** and an antiferromagnetic coupling layer **114** located between the reference layer **132** and the at least one superlattice **112**. In one embodiment, the at least one superlattice **112** may comprise a first superlattice and a second superlattice. The antiferromagnetic layer **114** may comprise an Ir or an IrMn alloy layer located between the first and the second superlattices. In one embodiment, the first superlattice comprises N1 repetitions of a first unit layer stack of the first cobalt layer and the first platinum layer, and a first capping cobalt layer, such that N1 of the first platinum layers are interlaced with (N1+1) of the first cobalt layers, where N1 is an integer in a range from 2 to 10. The second superlattice comprises N2 repetitions of a second unit layer stack of the second cobalt layer and the second platinum layer, and a second capping cobalt layer, such that N2 first platinum layers are interlaced with (N2+1) second cobalt layers, where N2 is an integer in a range from 2 to 10. Other SAF structures **120** may be used. For example, a hard-magnetization layer may

be used instead of the at least one superlattice **112**. The hard-magnetization layer **112** includes a ferromagnetic material having perpendicular magnetic anisotropy. The magnetization of the reference layer **132** can be antiferromagnetically coupled to the magnetization of the hard-magnetization layer **112**.

The electrically conductive capping layer **148**, if present, can include a nonmagnetic metal layer or multilayers, such as ruthenium, tungsten and/or tantalum. The electrically conductive capping layer **148** may be a portion of a second electrically conductive line **90**, or may be an electrically conductive structure that underlies the second electrically conductive line **90**.

In one embodiment, the insulating cap layer **144** may comprise a thin magnesium oxide layer that is thin enough to enable tunneling of electrical current, such as a thickness in a range from 4 Angstroms to 10 Angstroms. In one embodiment, the MRAM cell **180** can be a single tunnel junction device that includes only one magnetic tunnel junction **140**.

A selector element **150** can be formed in a series connection with the magnetic tunnel junction **140**. The selector element **150** includes a selector material that provides a bidirectional current flow when the current or voltage exceeds a threshold value. Thus, the selector element **150** is a bidirectional selector device which permits bidirectional current flow when the current or voltage exceeds a threshold value and blocks current flow when the current or voltage is below the threshold value. The selector element **150** may include an ovonic threshold switch (OTS) material that allows flow of electrical current only when a voltage differential thereacross exceeds a threshold voltage value. As used herein, an "ovonic threshold switch material" refers to a material that displays a non-linear resistivity curve under an applied external bias voltage such that the resistivity of the material decreases with the magnitude of the applied external bias voltage. In other words, an ovonic threshold switch material is non-Ohmic, and becomes more conductive under a higher external bias voltage than under a lower external bias voltage. An ovonic threshold switch material can be non-crystalline (for example, by being amorphous) at a non-conductive state, and can remain non-crystalline (for example, by remaining amorphous) at a conductive state, and can revert back to a high resistance state when a high voltage bias thereacross is removed, i.e., when not subjected to a large voltage bias across a layer of the ovonic threshold voltage material. Throughout the resistive state changes, the ovonic threshold switch material can remain amorphous. In one embodiment, the ovonic threshold switch material can comprise a chalcogenide material. The chalcogenide material may be a GeSeAs alloy, a GeSeAsTe alloy, a GeTeAs alloy, a GeSeTe alloy, a GeSe alloy, a SeAs alloy, a AsTe alloy, a GeTe alloy, a SiTe alloy, a SiAsTe alloy, or SiAsSe alloy. The chalcogenide material may be undoped or doped with at least one of N, O, C, P, Ge, As, Te, Se, In, or Si.

The selector element **150** may also include one or more electrically conductive and/or barrier layers, such as tungsten, tungsten nitride, tantalum, tantalum nitride, a carbon-nitrogen layer, etc.). The electrically conductive and/or barrier layers may be located above and/or below the ovonic threshold switch material.

The layer stack including the selector element **150**, the SAF structure **120**, the magnetic tunnel junction **140**, the insulating cap layer **144** and the electrically conductive capping layer **148** can be annealed to induce crystallographic alignment between the crystalline structure of the nonmagnetic tunnel barrier layer **134** (which may include

crystalline MgO having a rock salt crystal structure) and the crystalline structure within the free layer **136**.

In one embodiment, the reference layer **132** has a fixed vertical magnetization that is perpendicular to an interface between the reference layer **132** and the nonmagnetic tunnel barrier layer **134**. The free layer **136** has perpendicular magnetic anisotropy to provide bistable magnetization states that include a parallel state having a magnetization that is parallel to the fixed vertical magnetization and an antiparallel state having a magnetization that is antiparallel to the fixed vertical magnetization. The magnetization direction of the free layer **136** can be flipped (i.e., from upward to downward or vice versa) by flowing electrical current through the discrete patterned layer stack (**120**, **140**, **144**, **148**, **150**, **170**).

Referring to FIGS. 3A-3C, a first exemplary structure for forming a two-dimensional array of STT MRAM cells **180** is illustrated. The first exemplary structure can be provided by forming a layer stack of blanket (unpatterned) layers over a substrate **8**. The substrate **8** may comprise, for example, a semiconductor substrate **8A** and at least one dielectric material layer **8B** formed over the semiconductor substrate **8A**. Alternatively, an insulating substrate **8** (e.g., a ceramic or a glass substrate) or a conductive substrate **8** (e.g., a metal or metal alloy substrate) may be used instead. In one embodiment, various semiconductor devices (not shown) including switching devices and peripheral (i.e., driver) circuits may be formed over the semiconductor substrate **8A**, and metal interconnect structures (not shown) may be formed in the at least one dielectric material layer **8B**. The various semiconductor devices, if present, may comprise the various driver circuits of the MRAM device **500** illustrated in FIG. 1 other than the memory array region **550**, which is subsequently formed in subsequent processing steps.

A layer stack (**30L**, **150L**, **160L**) can be deposited over the substrate **8**. The layer stack (**30L**, **150L**, **160L**) can include, from bottom to top, a first electrically conductive layer **30L**, first selector material layers **150L**, and a first hardmask layer **160L**.

The first electrically conductive layer **30L** includes a first nonmagnetic electrically conductive material, such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or a combination thereof. The thickness of the first electrically conductive layer **30L** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed.

The first selector material layers **150L** can comprise, from bottom to top, a first lower selector electrode material layer **151L**, a first non-Ohmic material layer **152L**, and a first upper selector electrode material layer **153L**. The first lower selector electrode material layer **151L** includes at least one conductive material that may be employed for lower selector electrodes to be subsequently formed. The first non-Ohmic material layer **152L** includes a selector material that exhibits a non-Ohmic switching behavior. The first upper selector electrode material layer **153L** includes at least one conductive material that may be employed for upper selector electrodes to be subsequently formed.

In one embodiment, the first lower selector electrode material layer **151L** may comprise a layer stack including a first lower carbon-based electrode material layer **151C** and a first lower metallic material layer **151M** formed on the first lower carbon-based electrode material layer **151C**. In one embodiment, the first upper selector electrode material layer **153L** may comprise a layer stack including a first upper metallic material layer **153M** and a first upper carbon-based

electrode material layer **153C** formed on the first upper metallic material layer **153M**.

The first lower carbon-based electrode material layer **151C** and the first upper carbon-based electrode material layer **153C** within the selector-level material layers can include a respective carbon-based conductive material including carbon atoms at an atomic concentration greater than 50%. In one embodiment, the first lower carbon-based electrode material layer **151C** and the first upper carbon-based electrode material layer **153C** may include carbon atoms at an atomic concentration in a range from 50% to 100%, such as from 70% to 100% and/or from 80% to 100%. In one embodiment, each of first lower carbon-based electrode material layer **151C** and the first upper carbon-based electrode material layer **153C** comprises a respective material selected from diamond-like carbon (DLC), a carbon nitride material, and a carbon-rich conductive compound of carbon atoms and non-carbon atoms. Each of the first lower carbon-based electrode material layer **151C** and the first upper carbon-based electrode material layer **153C** may have a respective thickness in a range from 3 nm to 300 nm, although lesser and greater thicknesses may also be employed.

The first lower metallic material layer **151M** and the first upper metallic material layer **153M** within the first selector material layers **150L** can include a respective metallic material having electrical conductivity that is greater than the electrical conductivity of the carbon-based conductive materials of the first lower carbon-based electrode material layer **151C** and the first upper carbon-based electrode material layer **153C**. In one embodiment, the first lower metallic material layer **151M** comprises a first metallic material having electrical conductivity that is at least 10 times (which may be at least 30 times and/or at least 100 times and/or at least 1,000 times) the electrical conductivity of the carbon-based conductive material of first lower carbon-based electrode material layer **151C**, and the first upper metallic material layer **153M** comprises a second metallic material having electrical conductivity that is at least 10 times (which may be at least 30 times and/or at least 100 times and/or at least 1,000 times) the electrical conductivity of the carbon-based conductive material of the first upper carbon-based electrode material layer **153C**.

Generally, each of the first lower metallic material layer **151M** and the first upper metallic material layer **153M** may comprise, and/or may consist essentially of, a high-conductivity metallic material that has a high electrical conductivity, and thus, is capable of functioning as a current-spreading material that prevents concentration of electrical current in the non-Ohmic material of the first non-Ohmic material layer **152L**. In one embodiment, the first lower metallic material layer **151M** and/or the first upper metallic material layer **153M** may comprise, and/or may consist essentially of, an elemental metal, a conductive metallic carbide, or a conductive metallic nitride. In one embodiment, the first lower metallic material layer **151M** and/or the first upper metallic material layer **153M** may comprise, and/or may consist essentially of, a respective elemental metal selected from ruthenium, niobium, molybdenum, tantalum, tungsten, or rhenium. In one embodiment, the first lower metallic material layer **151M** and/or the first upper metallic material layer **153M** may comprise, and/or may consist essentially of, a

conductive metallic carbide such as tungsten carbide. In one embodiment, the first lower metallic material layer **151M** and/or the first upper metallic material layer **153M** may comprise, and/or may consist essentially of, a conductive metallic nitride such as tungsten nitride, titanium nitride, or tantalum nitride.

Generally, the first lower metallic material layer **151M** and the first upper metallic material layer **153M** may have a lower thickness than the first lower carbon-based electrode material layer **151C** and the first upper carbon-based electrode material layer **153C**. Each of the first lower metallic material layer **151M** and the first upper metallic material layer **153M** may have a respective thickness in a range from 0.2 nm to 10 nm, such as from 1 nm to 5 nm, although lesser and greater thicknesses may also be employed. In one embodiment, the ratio of the thickness of the first lower carbon-based electrode material layer **151C** to the thickness of the first lower metallic material layer **151M** may be in a range from 3.0 to 500, such as from 10 to 100, although lesser and greater ratios may also be employed. In one embodiment, the ratio of the thickness of the first upper carbon-based electrode material layer **153C** to the thickness of the first upper metallic material layer **153M** may be in a range from 3.0 to 500, such as from 10 to 100, although lesser and greater ratios may also be employed.

In one embodiment, the first non-Ohmic material layer **152L** within the selector material layers **150L** can include any suitable non-Ohmic selector material which exhibits non-linear electrical behavior. For example, the non-Ohmic selector material may comprise the above described OTS material or a volatile conductive bridge material. In another embodiment, the non-Ohmic selector material may comprise at least one non-threshold switch material, such as a tunneling selector material or diode materials (e.g., materials for p-n semiconductor diode, p-i-n semiconductor diode, Schottky diode or metal-insulator-metal diode). Thus, the material layer **152L** may comprise a diode layer stack, such as a layer stack of p-doped semiconductor material layer and an n-doped semiconductor material layer, or a layer stack of a p-doped semiconductor material layer, an intrinsic semiconductor material layer, and an n-doped semiconductor material layer.

The OTS material can be non-crystalline (for example, amorphous) in a high resistivity state, and can remain non-crystalline (for example, remain amorphous) in a low resistivity state during application of a voltage above its threshold voltage across the OTS material. The ovonic threshold switch material can revert back to the high resistivity state when the high voltage above its threshold voltage is lowered below a critical holding voltage. Throughout the resistivity state changes, the ovonic threshold switch material can remain non-crystalline (e.g., amorphous). In one embodiment, the ovonic threshold switch material can comprise a chalcogenide material. The chalcogenide material may be a GeSeAs alloy, a GeSeAsTe alloy, a GeTeAs alloy, a GeSeTe alloy, a GeSe alloy, a SeAs alloy, a AsTe alloy, a GeTe alloy, a SiTe alloy, a SiAsTe alloy, or SiAsSe alloy. The chalcogenide material may be undoped or doped with at least one of N, O, C, P, Ge, As, Te, Se, In, or Si. The thickness of the ovonic threshold selector-level material layers can be, for example, in a range from 1 nm to 50 nm, such as from 5 nm to 25 nm, although lesser and greater thicknesses can also be employed.

The first hardmask layer **160L** includes any suitable hardmask material, such as an insulating, semiconductor or conductive hardmask material which may be used as a patterning mask for the underlying layers during a subse-

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quent patterning step. Insulating hardmask materials include silicon nitride, silicon oxide, silicon oxynitride or insulating metal oxide materials, such as aluminum oxide. Electrically conductive hardmask materials include metals or metal alloys, such as Al, Cu, W, Ru, Mo, Nb, Ta, Ti, TiN, TaN, WN, MoN, or a combination thereof.

Referring to FIGS. 4A-4C, a photoresist layer can be applied over the first hardmask layer 160L, and can be lithographically patterned into a two-dimensional array of first discrete patterned photoresist material portions 157. The two-dimensional array of first discrete patterned photoresist material portions 157 may be a periodic two-dimensional array of first discrete patterned photoresist material portions 157 having a first periodicity along the first horizontal direction hd1 and having a second periodicity along the second horizontal direction hd2 that is perpendicular to the first horizontal direction hd1. In one embodiment, the first periodicity may be a first pitch p1, and the second periodicity may be a second pitch p2. The second pitch p2 may be the same as, or may be different from, the first pitch p1. The first pitch p1 may be in a range from 5 nm to 300 nm, such as from 10 nm to 100 nm, although lesser and greater dimensions may also be employed. The second pitch p2 may be in a range from 5 nm to 300 nm, such as from 10 nm to 100 nm, although lesser and greater dimensions may also be employed.

According to an aspect of the present disclosure, a first nearest-neighbor spacing s1 between neighboring pairs of the first discrete patterned photoresist material portions 157 that are laterally spaced apart along the first horizontal direction hd1 is less than a second nearest-neighbor spacing s2 between neighboring pairs of the first discrete patterned photoresist material portions 157 that are laterally spaced apart along the second horizontal direction hd2. In an illustrative case, each of the first discrete patterned photoresist material portions 157 may have a first lateral dimension ld1 along the first horizontal direction hd1, and may have a second lateral dimension ld2 along the second horizontal direction hd2. Each of the first discrete patterned photoresist material portions 157 may have a respective horizontal cross-sectional shape of a rectangle, a rounded rectangle, an oval, or a circle. The first nearest-neighbor spacing s1 between neighboring pairs of the first discrete patterned photoresist material portions 157 that are laterally spaced apart along the first horizontal direction hd1 can be the difference between the first pitch p1 and the first lateral dimension ld1. The second nearest-neighbor spacing s2 between neighboring pairs of the first discrete patterned photoresist material portions 157 that are laterally spaced apart along the second horizontal direction hd2 can be the second pitch p2 less the second lateral dimension ld2. In this case, p1-ld1 is less than p2-ld2, and thus s1 is less than s2. In one embodiment, the second pitch p2 may be the same as the first pitch p1, and the first lateral dimension ld1 may be greater than the second lateral dimension ld2.

Referring to FIGS. 5A-5C, one or more pattern transfer processes may be performed to pattern the first hardmask layer 160L and the first selector material layers 150L. Specifically, an array-pattern-transfer process can be performed to transfer the pattern of the two-dimensional array of first discrete patterned photoresist material portions 157 through the first hardmask layer 160L and the first selector material layers 150L. For example, an anisotropic etch process can be performed to transfer the pattern in the two-dimensional array of first discrete patterned photoresist material portions 157 through the first conductive material layer 160L and the first selector material layers 150L. The

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patterned remaining portions of the first conductive material layer 160L and the first selector material layers 150L can include two-dimensional array of first selector-containing pillar structures 182.

Each of the first selector-containing pillar structures 182 may comprise a first selector element 150 and a first hardmask plate 160. Each first selector element 150 is a patterned portion of the first selector material layers 150L, and each first hardmask plate 160 is a patterned portion of the first hardmask layer 160L. Each first selector element 150 may include a vertical stack of a first lower selector electrode 151, a first non-Ohmic material plate 152, and a first upper selector electrode 153. Each first lower selector electrode 151 is a patterned portion of the first lower selector electrode material layer 151L. Each first non-Ohmic material plate 152 is a patterned portion of the first non-Ohmic material layer 152L. Each first upper selector electrode 153 is a patterned portion of the first upper selector electrode material layer 153L.

In one embodiment, the two-dimensional array of first selector-containing pillar structures 182 comprises a two-dimensional periodic array of first selector-containing pillar structures 182 having the first pitch p1 along the first horizontal direction hd1 and having the second pitch p2 along the second horizontal direction hd2. The nearest-neighbor spacing s1 between neighboring pairs of the first selector-containing pillar structures 182 that are laterally spaced apart along the first horizontal direction hd1 is less than the nearest-neighbor spacing s2 between neighboring pairs of the first selector-containing pillar structures 182 that are laterally spaced apart along the second horizontal direction hd2.

In one embodiment, each first selector-containing pillar structure 182 within the two-dimensional array of first selector-containing pillar structures 182 has a respective elongated horizontal cross-sectional shape having a first lateral dimension ld1 along the first horizontal direction hd1 and having a second lateral dimension ld2 along the second horizontal direction hd2 that is less than the first lateral dimension ld1. In one embodiment, the ratio of the first lateral dimension ld1 to the second lateral dimension ld2 may be in a range from 1.2 to 4, such as from 1.5 to 3.0. The two-dimensional array of first selector-containing pillar structures 182 can be formed over the first electrically conductive layer 30L. The two-dimensional array of first discrete patterned photoresist material portions 157 can be subsequently removed, for example, by ashing.

Referring to FIGS. 6A-6C, a first dielectric spacer material layer 156L can be formed over the two-dimensional array of first selector-containing pillar structures 182 and the first electrically conductive layer 30L. In one embodiment, the first dielectric spacer material layer 156L can be conformally deposited around the two-dimensional array of selector-containing pillar structures 182 such that the thickness of the first dielectric spacer material layer 156L is greater than one half of the nearest-neighbor spacing s1 between neighboring pairs of the first selector-containing pillar structures 182 that are laterally spaced apart along the first horizontal direction hd1, and is less than one half of the nearest-neighbor spacing s2 between neighboring pairs of the first selector-containing pillar structures 182 that are laterally spaced apart along the second horizontal direction hd2. Vertically-extending portions of the first dielectric spacer material layer 156L merge between neighboring pairs of the first selector-containing pillar structures 182 that are laterally spaced apart along the first horizontal direction hd1 to form vertically extending seams 156S. A two-dimensional

periodic array of vertically-extending seams **156S** can be formed, which can have the first pitch **p1** along the first horizontal direction **hd1** and can have the second pitch **p2** along the second horizontal direction **hd2**. The vertically-extending seams **156S** can be parallel to the second horizontal direction **hd2**, and can be located midway between a respective neighboring pair of first selector-containing pillar structures **182** of the two-dimensional array of first selector-containing pillar structures **182** that are laterally spaced apart along the first horizontal direction **hd1**.

Each vertically-extending portion of the first dielectric spacer material layer **156L** located on a sidewall of a first selector-containing pillar structure **182** that extend along the first horizontal direction **hd1** can have a first thickness **t1**, and can be physically exposed. Each vertically-extending portion of the first dielectric spacer material layer **156L** located on a sidewall of a first selector-containing pillar structure **182** that extend along the second horizontal direction **hd2** can have a second thickness **t2**, which is one half of the lateral spacing between neighboring pairs of first selector-containing pillar structures **182** that are laterally spaced apart along the first horizontal direction **hd1**. The second thickness **t2** is less than the first thickness **t1**.

Referring to FIGS. 7A-7C, a first anisotropic etch process (e.g., a sidewall spacer etch process) can be performed to etch horizontally-extending portions of the first dielectric spacer material layer **156L**. The first anisotropic etch process may be selective to the materials of the first hardmask plates **160**. Each remaining continuous portion of the first dielectric spacer material layer **156L** constitutes a first dielectric spacer **156**. The first dielectric spacers **156** can be formed around the two-dimensional array of first selector-containing pillar structures **182** such that each of the first dielectric spacers **156** laterally surrounds a respective row of first selector-containing pillar structures **182** that are arranged along the first horizontal direction **hd1**. The first selector-containing pillar structures **156** within the respective row of first selector-containing pillar structures **156** are arranged along the first horizontal direction **hd1**. The first dielectric spacers **156** are laterally spaced apart from each other along the second horizontal direction **hd2**.

Generally, a first dielectric spacer formation process can be performed, in which the first dielectric spacers **156** are formed around the two-dimensional array of first selector-containing pillar structures **182**. Each of the first dielectric spacers **156** comprises a respective plurality of vertically-extending seams **156S** that are parallel to the second horizontal direction **hd2** and located midway between a respective neighboring pair of first selector-containing pillar structures **182** of the two-dimensional array of first selector-containing pillar structures **182** that are laterally spaced apart along the first horizontal direction **hd1**.

In one embodiment, each of the first dielectric spacers **156** comprises a pair of contoured lengthwise sidewalls that generally extend along the first horizontal direction **hd1** with a lateral undulation along the second horizontal direction **hd2**. Each lengthwise segment of each of the first dielectric spacers **156** that laterally extends along the first horizontal direction **hd1** and located between a respective first selector-containing pillar structure **182** and a respective outer contoured lengthwise sidewall has a first thickness **t1** along the second horizontal direction **hd2**. Each widthwise segment of each of the first dielectric spacers **156** located between a respective first selector-containing pillar structure **182** and a respective vertically-extending seam **156S** has a second thickness **t2** along the first horizontal direction **hd1** that is less than the first thickness **t1**.

Referring to FIGS. 8A-8D, a second anisotropic etch process can be performed to transfer the pattern of the combination of the two-dimensional array of first selector-containing pillar structures **182** and the first dielectric spacers **156** through the first electrically conductive layer **30L**. In other words, the combination of the two-dimensional array of first selector-containing pillar structures **182** and the first dielectric spacers **156** can be employed as an etch mask for anisotropically etching the first electrically conductive layer **30L**. The first electrically conductive layer **30L** can be patterned into a plurality of first electrically conductive lines **30**, which may be a periodic one-dimensional array of first electrically conductive lines **30** having a periodicity of the second pitch **p2** along the second horizontal direction **hd2**. The first electrically conductive lines **30** may comprise word lines in one embodiment. Optionally, the first dielectric spacers **156** may be recessed during the second anisotropic etch relative to the first hardmask plates **160** depending on their respective materials and the etch gas composition.

The pattern of the combination of the two-dimensional array of first selector-containing pillar structures **182** and the first dielectric spacers **156** includes the pattern of the lengthwise sidewalls of the first dielectric spacers **156**. Thus, the first electrically conductive layer **30L** can be patterned into the first electrically conductive lines **30** by transferring a pattern of lengthwise sidewalls of the first dielectric spacers **156** through the first electrically conductive layer **30L**. The first electrically conductive lines **30** laterally extend along the first horizontal direction **hd1**, and have a respective variable width along the second horizontal direction **hd2** that varies along the first horizontal direction **hd1**. In other words, the first electrically conductive lines **30** have a wiggled profile having alternating wider and narrower sections along the second horizontal direction **hd2**, as shown in FIG. 8D. The narrower sections have a first width **w1** which is smaller than the second width **w2** of the wider sections.

Each of the first electrically conductive lines **30** contacts a bottom surface of a respective one of the first dielectric spacers **156**, and comprises a respective pair of contoured sidewalls that are vertically coincident with sidewalls of the respective one of the first dielectric spacers **156**. As used herein, two surfaces are vertically coincident if one of the two surfaces overlies or underlies the other of the two surfaces and if there exists a vertical plane that contains the two surfaces. The vertical plane may have a straight horizontal cross-sectional profile or a contoured horizontal cross-sectional profile.

Referring to FIGS. 9A-9C, a dielectric fill material can be deposited in cavities between neighboring pairs of first dielectric spacers **156** to form a dielectric matrix layer, which is herein referred to as a first selector-level dielectric matrix layer **40**. The first selector-level dielectric matrix layer **40** may comprise a silicon oxide layer and may comprise the same or different material from the material of the first dielectric spacers **156**. Excess portions of the dielectric fill material of the first selector-level dielectric matrix layer **40** can be removed from above a horizontal plane including top surfaces of the first selector-containing pillar structures **182** by a planarization process, such as a chemical mechanical polishing (CMP) process. A top surface of a remaining portion of the first selector-level dielectric matrix layer **40** is located within a horizontal plane including top surfaces of the two-dimensional array of first selector-containing pillar structures **182**. In this configuration, the first hardmask plates **160** are formed of an electrically conductive material and comprise first conductive material plates **160**, which form the top surfaces of the first



selector-containing pillar structures **182**. The first conductive material plates **160** protect the underlying first upper selector electrodes **153** from CMP damage. Thus, in this configuration, the first upper selector electrodes **153** may comprise a carbon based material described above which may be damaged by CMP.

Referring to FIGS. **9D-9F** in an alternative configuration of the first exemplary according to an alternative aspect of the first embodiment of the present disclosure, the first selector-level dielectric matrix layer **40** is also formed in cavities between neighboring pairs of first dielectric spacers **156**, as described above. However, in this alternative configuration, the CMP process is continued to also remove the first hardmask plates **160** and to expose the upper surface of the first upper selector electrodes **153**. In this alternative configuration, the first hardmask plates **160** may comprise an insulating material, such as silicon nitride or metal oxide, and at least an upper portion of the first upper selector electrodes **153** may comprise a metal or metal alloy rather than a carbon based material. Thus, the CMP does not damage the carbon material of the first upper selector electrodes **153**.

Referring to FIGS. **10A-10C**, first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be formed over the two-dimensional array of first selector-containing pillar structures **182** and the first selector-level dielectric matrix layer **40**. The first MTJ-level material layers contact the top exposed surfaces of the first selector-containing pillar structures **182**. The top exposed surfaces of the first selector-containing pillar structures **182** comprise an electrically conductive material, such as either the first conductive material plates **160** which are shown in FIGS. **9B** and **9C**, or the first upper selector electrodes **153** which are shown in FIGS. **9E** and **9F**. Thus, the first conductive material plates **160** may either be present or omitted at this stage of the process.

The first MTJ-level material layers may comprise, for example, a first continuous superlattice layer **112L**, an optional first continuous antiferromagnetic coupling layer **114L**, first continuous magnetic tunnel junction (MTJ) material layers **130L**, a first continuous dielectric capping layer **144L**, and a first continuous metallic capping layer **148L**. The first MTJ material layers **130L** may comprises a layer stack including a first continuous reference layer **132L**, a first continuous nonmagnetic tunnel barrier layer **134L**, and a first continuous free layer **136L**. The first MTJ-level material layers may also optionally comprise the above described seed layer (i.e., the continuous non-magnetic metal layer (e.g., Pt, Ta, W, etc.) that is subsequently patterned to form the seed layer **110** shown in FIG. **2**) located below the first continuous superlattice layer **112L**.

The first continuous superlattice layer **112L** can have the same material composition as the superlattice layer **112** described with reference to FIG. **2**. The first continuous antiferromagnetic coupling layer **114L**, if present, can have the same material composition as the antiferromagnetic coupling layer **114** described with reference to FIG. **2**. In one embodiment, the first continuous antiferromagnetic coupling layer **114** may comprise ruthenium, iridium or IrMn alloy.

The first continuous reference layer **132L** can have the same material composition as the reference layer **132** described with reference to FIG. **2**. In one embodiment, the first continuous reference layer **132L** can include a CoFe alloy or a CoFeB alloy. Optionally, the first continuous reference layer **132L** may additionally include a thin non-magnetic layer comprised of tantalum or tungsten having a

thickness of 0.2 nm–0.5 nm and a thin CoFeB layer (having a thickness in a range from 0.5 nm to 3 nm). The first continuous nonmagnetic tunnel barrier layer **134L** includes any insulating tunnel barrier material such as magnesium oxide. The thickness of the first continuous nonmagnetic tunnel barrier layer **134L** can be 0.7 nm to 1.3 nm, such as about 1 nm. The first continuous free layer **136L** can have the same material composition as the free layer **136** described with reference to FIG. **2**. In one embodiment, the first continuous free layer **136L** can include a CoFe alloy or a CoFeB alloy. Optionally, the first continuous free layer **136L** may additionally include a thin non-magnetic layer comprised of tantalum or tungsten having a thickness of 0.2 nm–0.5 nm and a thin CoFeB layer (having a thickness in a range from 0.5 nm to 3 nm).

The first continuous dielectric capping layer **144L** can have the same material composition as the dielectric capping layer **144** described with reference to FIG. **2**. The first continuous dielectric capping layer **144L** may comprise a thin magnesium oxide layer that is thin enough to enable tunneling of electrical current, such as a thickness in a range from 0.4 nm to 1.0 nm. The first continuous metallic capping layer **148L** can have the same material composition as the metallic capping layer **144** described with reference to FIG. **2**. The first continuous metallic capping layer **148L** may comprise a non-magnetic, electrically conductive material, such as W, Ti, Ta, WN, TiN, TaN, Ru, and Cu. The thickness of the first continuous metallic capping layer **148L** can be in a range from 10 nm to 100 nm, although lesser and greater thicknesses can also be employed. An optional second hardmask layer may be located over the first continuous metallic capping layer **148L**, and may comprise an insulating, semiconductor or conductive material. Alternatively, the first continuous metallic capping layer **148L** may act as a hardmask during the subsequent patterning step.

Referring to FIGS. **11A-11C**, a two-dimensional array of second discrete patterned photoresist material portions **159** can be formed over the first continuous metallic capping layer **148L**. Each of the second discrete patterned photoresist material portions **159** has an areal overlap with a respective underlying one of the first selector-containing pillar structures **182**. The two-dimensional array of second discrete patterned photoresist material portions **159** can be formed as a periodic array having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. The horizontal cross-sectional shapes of the second discrete patterned photoresist material portions **159** can be different from the horizontal cross-sectional shapes of the first selector-containing pillar structures **182**. In one embodiment, the lateral dimension of each of the second discrete patterned photoresist material portions **159** along the first horizontal direction **hd1** may be the same as the lateral dimension of each of the second discrete patterned photoresist material portions **159** along the second horizontal direction **hd2**. In one embodiment, each of the second discrete patterned photoresist material portions **159** may have a respective horizontal cross-sectional shape of a circle.

Referring to FIGS. **12A-12C**, an anisotropic patterning process can be performed to pattern unmasked portions of the layer stack (**112L**, **114L**, **130L**, **144L**, **148L**) of the first continuous superlattice layer **112L**, the optional first continuous antiferromagnetic coupling layer **114L**, the first continuous magnetic tunnel junction (MTJ) material layers **130L**, the first continuous dielectric capping layer **144L**, and the first continuous metallic capping layer **148L** employing



the two-dimensional array of second discrete patterned photoresist material portions **159** as a mask.

In one embodiment, the anisotropic patterning process may comprise at least one of a reactive ion etch process or an ion beam etch (IBE) process (e.g., an ion milling process). In one embodiment, the second hardmask layer (if present) and/or the first continuous metallic capping layer **148L** and/or the first continuous dielectric capping layer **144L** may be patterned into a two-dimensional periodic array of second hardmask plates and/or first metallic capping layers **148** and/or first dielectric capping layers **144** by performing a reactive ion etch process. The two-dimensional array of second discrete patterned photoresist material portions **159** can be subsequently removed, for example, by ashing. The first continuous MTJ material layers **130L**, the optional first continuous antiferromagnetic coupling layer **114L**, and the first continuous superlattice layer **112L** can be patterned by performing an ion beam etch process that employs the two-dimensional periodic array of the second hardmask plates and/or the first metallic capping layers **148** and/or the first dielectric capping layers **144** as a mask.

Each patterned portion of the layer stack (**112L**, **114L**, **130L**, **144L**, **148L**) comprises a first magnetic tunnel junction (MTJ) pillar structure **184**. A two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures **184** can be formed over the two-dimensional array of first selector-containing pillar structures **182**. Each contiguous combination of a first selector-containing pillar structure **182** and a first MTJ pillar structure **184** constitutes a first memory cell **180**, which can function as a memory cell **180** described with reference to FIG. 2.

Each first MTJ pillar structure **184** comprises a stack of an optional seed layer **110** (shown in FIG. 10), a first superlattice layer **112**, a first antiferromagnetic coupling layer **114**, a first magnetic tunnel junction **130**, a first dielectric capping layer **144**, and a first metallic capping layer **148**. The first magnetic tunnel junction **130** includes a first reference layer **132**, a first tunnel barrier layer **134**, and a first free layer **136**. Each first superlattice layer **112** is a patterned portion of the first continuous superlattice layer **112L**. Each first antiferromagnetic coupling layer **114** is a patterned portion of the first continuous antiferromagnetic coupling layer **114L**. Each first magnetic tunnel junction **130** is a patterned portion of the first magnetic tunnel junction material layers **130L**. Each first dielectric capping layer **144** is a patterned portion of the first continuous dielectric capping layer **144L**. Each first metallic capping layer **148** is a patterned portion of the first continuous metallic capping layer **148L**. Each first reference layer **132** is a patterned portion of the first continuous reference layer **132L**. Each first tunnel barrier layer **134** is a patterned portion of the first continuous tunnel barrier layer **134L**. Each first free layer **136** is a patterned portion of the first continuous free layer **136L**. Sidewalls of each component within an MTJ pillar structure **184** can be vertically coincident.

The two-dimensional array of first MTJ pillar structures **184** can be formed above the top surfaces of the remaining portions of the first selector-level dielectric matrix layer **40** and over the two-dimensional array of the first selector-containing pillar structures **182**.

In one embodiment, each of the two-dimensional array of first selector-containing pillar structures **182** and the two-dimensional array of first MTJ pillar structures **184** has the first pitch **p1** along the first horizontal direction **hd1**, and has the second pitch **p2** along the second horizontal direction **hd2**. In one embodiment, each MTJ pillar structure **184** may have a horizontal cross-sectional shape of a circle, a square,

or a rounded square, i.e., a shape that is derived from a square by rounding the four corners.

In one embodiment shown in FIG. 12A, the maximum lateral dimension **ld1** of each of the first selector-containing pillar structures **182** along the first horizontal direction **hd1** can be greater than the maximum lateral dimension **ld3** of each of the first MTJ pillar structures **184** along the first horizontal direction **hd1**. The maximum lateral dimension **ld2** of each of the first selector-containing pillar structures **182** along the second horizontal direction **hd2** is less than the maximum lateral dimension **ld4** of each of the first MTJ pillar structures **184** along the second horizontal direction **hd2**. In this embodiment, each MTJ pillar structure **184** may have a horizontal cross-sectional shape of a circle, while each of each of the first selector-containing pillar structures **182** may have a shape of rectangle or a rounded rectangle. Thus, each MTJ pillar structure **184** may have a horizontal cross-sectional shape that is different from that of the underlying first selector-containing pillar structure **182**. Thus, at least one sidewall of the MTJ pillar structure **184** may not be vertically coincident with (i.e., may be laterally offset from) the underlying respective sidewall of the first selector-containing pillar structures **182**.

In an alternative embodiment, each MTJ pillar structure **184** within the two-dimensional array of MTJ pillar structures **184** has a respective horizontal cross-sectional shape having a same lateral extent along the first horizontal direction **hd1** and along the second horizontal direction **hd2**. Thus, each MTJ pillar structure **184** may have a horizontal cross-sectional shape that is the same as that of the underlying first selector-containing pillar structure **182**.

Referring to FIGS. 13A-13C, a dielectric fill material can be deposited in the gaps between neighboring pairs of the first MTJ pillar structures **184**, and can be subsequently planarized to remove portions of the dielectric fill material from above the horizontal plane including the top surfaces of the first MTJ pillar structures **184**. The remaining portions of the dielectric fill material comprises a dielectric matrix layer, which is herein referred to as a first magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer **80**.

A second electrically conductive layer **90L** may be deposited over the two-dimensional array of first MTJ pillar structures **184** and be patterned into second electrically conductive lines **90** (e.g., bit lines) which extend perpendicular to the first electrically conductive lines **30** to form a one level MRAM device.

Alternatively, to form a multi-level (e.g., three-dimensional) MRAM device, a layer stack (**90L**, **250L**, **260L**) can be deposited over the two-dimensional array of first MTJ pillar structures **184**, as shown in FIGS. 14A-14C. The layer stack (**90L**, **250L**, **260L**) can include, from bottom to top, a second electrically conductive layer **90L**, second selector material layers **250L**, and a second hardmask layer **260L**.

The second electrically conductive layer **90L** includes a second nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or a combination thereof. The thickness of the second electrically conductive layer **90L** can be in a range from 20 nm to 200 nm, although lesser and greater thicknesses can also be employed.

The second selector material layers **250L** can comprise, from bottom to top, a second lower selector electrode material layer **251L**, a second non-Ohmic material layer **252L**, and a second upper selector electrode material layer **253L**. The second lower selector electrode material layer **251L** includes at least one material that may be employed for lower selector electrodes to be subsequently formed. The

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second non-Ohmic material layer **252L** includes a selector material that exhibits a non-Ohmic switching behavior. The second upper selector electrode material layer **253L** includes at least one material that may be employed upper selector electrodes to be subsequently formed.

In one embodiment, the second lower selector electrode material layer **251L** may comprise the same one or more materials used for the first lower selector electrode material layer **151L**. The second lower selector electrode material layer **251L** may comprise a layer stack including a second lower carbon-based electrode material layer **251C** and a second metallic material layer **251M** formed on the second lower carbon-based electrode material layer **251C**. In one embodiment, the second upper selector electrode material layer **253L** may comprise a layer stack including a second metallic material layer **253M** and a second carbon-based electrode material layer **253C** formed on the second metallic material layer **253M**.

In one embodiment, the second non-Ohmic material layer **252L** within the selector material layers **250L** can include any suitable non-Ohmic selector material which exhibits non-linear electrical behavior. In one embodiment, the second non-Ohmic material layer **252L** may have any material composition that may be employed for the first non-Ohmic material layer **152L**, and have the same thickness range as the first non-Ohmic material layer **152L**. The second hardmask layer **260L** may include any material that may be employed for the first hardmask layer **160L**.

Referring to FIGS. **15A-15C**, a photoresist layer can be applied over the second hardmask layer **260L**, and can be lithographically patterned into a two-dimensional array of third discrete patterned photoresist material portions **257**. The two-dimensional array of third discrete patterned photoresist material portions **257** may be a periodic two-dimensional array of third discrete patterned photoresist material portions **257** having a first periodicity along the first horizontal direction **hd1** and having a second periodicity along the second horizontal direction **hd2** that is perpendicular to the first horizontal direction **hd1**. In one embodiment, the first periodicity may be the first pitch **p1**, and the second periodicity may be the second pitch **p2**. As discussed above, the second pitch **p2** may be the same as, or may be different from, the first pitch **p1**. The first pitch **p1** may be in a range from 5 nm to 300 nm, such as from 20 nm to 200 nm, although lesser and greater dimensions may also be employed. The second pitch **p2** may be in a range from 5 nm to 300 nm, such as from 20 nm to 200 nm, although lesser and greater dimensions may also be employed.

According to an aspect of the present disclosure, a nearest-neighbor spacing **s3** between neighboring pairs of the third discrete patterned photoresist material portions **257** that are laterally spaced apart along the first horizontal direction **hd1** is greater than a nearest-neighbor spacing **s4** between neighboring pairs of the third discrete patterned photoresist material portions **257** that are laterally spaced apart along the second horizontal direction **hd2**. In an illustrative case, each of the third discrete patterned photoresist material portions **257** may have a third lateral dimension **ld5** along the first horizontal direction **hd1**, and may have a fourth lateral dimension **ld6** along the second horizontal direction **hd2**. Each of the third discrete patterned photoresist material portions **257** may have a respective horizontal cross-sectional shape of a rectangle, a rounded rectangle, an oval, or a circle. The nearest-neighbor spacing **s3** between neighboring pairs of the third discrete patterned photoresist material portions **257** that are laterally spaced apart along the first horizontal direction **hd1** can be the

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difference between the first pitch **p1** and the third lateral dimension **ld5**. The nearest-neighbor spacing **s4** between neighboring pairs of the third discrete patterned photoresist material portions **257** that are laterally spaced apart along the second horizontal direction **hd2** can be the second pitch **p2** less the fourth lateral dimension **ld6**. In this case, **p1-ld5** is greater than **p2-ld6**. In one embodiment, the second pitch **p2** may be the same as the first pitch **p1**, and the third lateral dimension **ld5** may be less than the fourth lateral dimension **ld6**. The third discrete patterned photoresist material portions **257** may be elongated along the second horizontal direction **hd2**, while the first photoresist material portions are elongated along the first horizontal direction **hd1**.

One or more pattern transfer process may be performed to pattern the second hardmask layer **260L** and the second selector material layers **250L**. Specifically, an array-pattern-transfer process can be performed to transfer the pattern of the two-dimensional array of third discrete patterned photoresist material portions **257** through the second hardmask layer **260L** and the second selector material layers **250L**. For example, an anisotropic etch process can be performed to transfer the pattern in the two-dimensional array of third discrete patterned photoresist material portions **257** through the second hardmask layer **260L** and the second selector material layers **250L**. The patterned remaining portions of the second hardmask layer **260L** and the second selector material layers **250L** can include two-dimensional array of second selector-containing pillar structures **282**.

Each of the second selector-containing pillar structures **282** may comprise a second selector element **250** and a second hardmask plate **260**. Each second selector element **250** is a patterned portion of the second selector material layers **250L**, and each second hardmask plate **260** is a patterned portion of the second hardmask layer **260L**. Each second selector element **250** may include a vertical stack of a second lower selector electrode **251**, a second non-Ohmic material plate **252**, and a second upper selector electrode **253**. Each second lower selector electrode **251** is a patterned portion of the second lower selector electrode material layer **251L**. Each second non-Ohmic material plate **252** is a patterned portion of the second non-Ohmic material layer **252L**. Each second upper selector electrode **253** is a patterned portion of the second upper selector electrode material layer **253L**.

In one embodiment, the two-dimensional array of second selector-containing pillar structures **282** comprises a two-dimensional periodic array of second selector-containing pillar structures **282** having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. The nearest-neighbor spacing **s3** between neighboring pairs of the second selector-containing pillar structures **282** that are laterally spaced apart along the first horizontal direction **hd1** is greater than the nearest-neighbor spacing **s4** between neighboring pairs of the second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2**.

In one embodiment, each second selector-containing pillar structure **282** within the two-dimensional array of second selector-containing pillar structures **282** has a respective elongated horizontal cross-sectional shape having a third lateral dimension **ld5** along the first horizontal direction **hd1** and having a fourth lateral dimension **ld6** along the second horizontal direction **hd2** that is greater than the third lateral dimension **ld5**. In one embodiment, the ratio of the fourth lateral dimension **ld6** to the third lateral dimension **ld5** may be in a range from 2.2 to 4, such as from 2.5 to 3. The

two-dimensional array of second selector-containing pillar structures **282** can be formed over the second electrically conductive layer **90L**. The two-dimensional array of third discrete patterned photoresist material portions **257** can be subsequently removed, for example, by ashing.

Referring to FIGS. **16A-16C**, a second dielectric spacer material layer **256L** can be formed over the two-dimensional array of second selector-containing pillar structures **282** and the second electrically conductive layer **90L**. In one embodiment, the second dielectric spacer material layer **256L** can be conformally deposited around the two-dimensional array of selector-containing pillar structures **282** such that the thickness of the second dielectric spacer material layer **256L** is greater than one half of the nearest-neighbor spacing between **s4** neighboring pairs of the second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2**, and is less than one half of the nearest-neighbor spacing **s3** between neighboring pairs of the second selector-containing pillar structures **282** that are laterally spaced apart along the first horizontal direction **hd1**. Vertically-extending portions of the second dielectric spacer material layer **256L** merge between neighboring pairs of the second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2** to form vertically extending seams **256S**. A two-dimensional periodic array of vertically-extending seams **256S** can be formed, which can have the first pitch **p1** along the first horizontal direction **hd1** and can have the second pitch **p2** along the second horizontal direction **hd2**. The vertically-extending seams **256S** can be parallel to the first horizontal direction **hd1**, and can be located midway between a respective neighboring pair of second selector-containing pillar structures **282** of the two-dimensional array of second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2**.

Each vertically-extending portion of the second dielectric spacer material layer **256L** located on a sidewall of a second selector-containing pillar structure **282** that extends along the second horizontal direction **hd2** can have a third thickness **t3**, and can be physically exposed. Each vertically-extending portion of the second dielectric spacer material layer **256L** located on a sidewall of a second selector-containing pillar structure **282** that extends along the first horizontal direction **hd1** can have a fourth thickness **t4**, which is one half of the lateral spacing between neighboring pairs of second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2**. The fourth thickness **t4** is less than the third thickness **t3**.

Referring to FIGS. **17A-17C**, a third anisotropic etch process (e.g., a sidewall spacer etch process) can be performed to etch horizontally-extending portions of the second dielectric spacer material layer **256L**. The third anisotropic etch process may be selective to the materials of the second hardmask plates **260**. Each remaining continuous portion of the second dielectric spacer material layer **256L** constitutes a second dielectric spacer **256**. The second dielectric spacers **256** can be formed around the two-dimensional array of second selector-containing pillar structures **282** such that each of the second dielectric spacers **256** laterally surrounds a respective column of second selector-containing pillar structures **282** that are arranged along the second horizontal direction **hd2**. The second dielectric spacers **256** are laterally spaced from each other along the first horizontal direction **hd1**.

Generally, a second dielectric spacer formation process can be performed, in which the second dielectric spacers **256** are formed around the two-dimensional array of second selector-containing pillar structures **282**. Each of the second dielectric spacers **256** comprises a respective plurality of vertically-extending seams **256S** that are parallel to the first horizontal direction **hd1** and located midway between a respective neighboring pair of second selector-containing pillar structures **282** of the two-dimensional array of second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2**.

In one embodiment, each of the second dielectric spacers **256** comprises a pair of contoured lengthwise sidewalls that generally extend along the second horizontal direction **hd2** with a lateral undulation along the first horizontal direction **hd1**. Each lengthwise segment of each of the second dielectric spacers **256** that laterally extend along the second horizontal direction **hd2** and located between a respective second selector-containing pillar structure **282** and a respective outer contoured lengthwise sidewall has a third thickness **t3** along the first horizontal direction **hd1**. Each widthwise segment of each of the second dielectric spacers **256** located between a respective second selector-containing pillar structure **282** and a respective vertically-extending seam **256S** has a fourth thickness **t4** along the second horizontal direction **hd2** that is less than the third thickness **t3**.

Referring to FIGS. **18A-18D**, a fourth anisotropic etch process can be performed to transfer the pattern of the combination of the two-dimensional array of second selector-containing pillar structures **282** and the second dielectric spacers **256** through the second electrically conductive layer **90L**. In other words, the combination of the two-dimensional array of second selector-containing pillar structures **282** and the second dielectric spacers **256** can be employed as an etch mask for anisotropically etching the second electrically conductive layer **90L**. The second electrically conductive layer **90L** can be patterned into a plurality of second electrically conductive lines (e.g., bit lines) **90**, which may be a periodic one-dimensional array of second electrically conductive lines **90** having a periodicity of the first pitch **p1** along the first horizontal direction **hd1**.

The pattern of the combination of the two-dimensional array of second selector-containing pillar structures **282** and the second dielectric spacers **256** includes the pattern of the lengthwise sidewalls of the second dielectric spacers **256**. Thus, the second electrically conductive layer **90L** can be patterned into the second electrically conductive lines **90** by transferring a pattern of lengthwise sidewalls of the second dielectric spacers **256** through the second electrically conductive layer **90L**. The second electrically conductive lines **90** laterally extend along the second horizontal direction **hd2**, and have a respective variable width along the first horizontal direction **hd1** that varies along the second horizontal direction **hd2**. In other words, the second electrically conductive lines **90** have a wiggled profile having alternating wider and narrower sections along the first horizontal direction **hd2**, as shown in FIG. **18D**. The narrower sections have a third width **w3** which is smaller than the fourth width **w4** of the wider sections. Each of the second electrically conductive lines **90** contacts a bottom surface of a respective one of the second dielectric spacers **256**, and comprises a respective pair of contoured sidewalls that are vertically coincident with sidewalls of the respective one of the second dielectric spacers **256**.

Referring to FIGS. **19A-19C**, a dielectric fill material can be deposited in cavities between neighboring pairs of first

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dielectric spacers **256** to form a dielectric matrix layer, which is herein referred to as a second selector-level dielectric matrix layer **240**. Excess portions of the dielectric fill material of the second selector-level dielectric matrix layer **240** can be removed from above a horizontal plane including top surfaces of the first selector-containing pillar structures **282** by a planarization process such as a chemical mechanical polishing process. A top surfaces of a remaining portion of the second selector-level dielectric matrix layer **240** is formed within a horizontal plane including top surfaces of the two-dimensional array of second selector-containing pillar structures **282**. In this embodiment, the second hardmask plates **260** may comprise second electrically conductive plates **260**.

In the alternative configuration of the first exemplary according to the alternative aspect of the first embodiment of the present disclosure, the second selector-level dielectric matrix layer **240** is also formed in cavities between neighboring pairs of second dielectric spacers **256**, as described above. However, similar to the step shown in FIGS. **9D-9F**, in this alternative configuration, the CMP process is continued to also remove the second hardmask plates **260** and to expose the upper surface of the second upper selector electrodes **253**. In this alternative configuration, the second hardmask plates **260** may comprise an insulating material, such as silicon nitride or metal oxide, and at least an upper portion of the second upper selector electrodes **253** may comprise a metal or metal alloy rather than a carbon based material. Thus, the CMP does not damage the carbon material of the second upper selector electrodes **153**.

Referring to FIGS. **20A-20C**, second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) can be formed over the two-dimensional array of second selector-containing pillar structures **282** and the second selector-level dielectric matrix layer **240**. The second MTJ-level material layers may comprise, for example, a second continuous superlattice layer **212L**, an optional second continuous antiferromagnetic coupling layer **214L**, second continuous magnetic tunnel junction (MTJ) material layers **230L**, a second continuous dielectric capping layer **244L**, and a second continuous metallic capping layer **248L**. The second MTJ material layers **230L** may comprises a layer stack including a second continuous reference layer **232L**, a second continuous nonmagnetic tunnel barrier layer **234L**, a second continuous free layer **236L**. The second MTJ-level material layers may also optionally comprise the above described seed layer (i.e., the continuous non-magnetic metal layer (e.g., Pt, Ta, W, etc.)) that is subsequently patterned to form the seed layer located below the second continuous superlattice layer **212L**.

The second continuous superlattice layer **212L** can have the same material composition and the same thickness as the first superlattice layer **112** described above. The second continuous antiferromagnetic coupling layer **214L**, if present, can have the same material composition and the same thickness as the first antiferromagnetic coupling layer **114** described above. The second continuous reference layer **232L** can have the same material composition and the same thickness as the first reference layer **132** described above. The second continuous nonmagnetic tunnel barrier layer **234L** can have the same material composition and the same thickness as the first continuous nonmagnetic tunnel barrier layer **134L** described above. The second continuous free layer **236L** can have the same material composition and the same thickness as the first free layer **136** described above. The second continuous dielectric capping layer **244L** can have the same material composition and the same thickness

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as the first dielectric capping layer **144** described above. The second continuous metallic capping layer **248L** can have the same material composition and the same thickness as the first metallic capping layer **148** described above.

Referring to FIGS. **21A-21C**, a two-dimensional array of fourth discrete patterned photoresist material portions (not shown) can be formed over the second continuous metallic capping layer **248L**. Each of the fourth discrete patterned photoresist material portions has an areal overlap with a respective underlying one of the second selector-containing pillar structures **282**. The two-dimensional array of second discrete patterned photoresist material portions can be formed as a periodic array having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. The horizontal cross-sectional shapes of the fourth discrete patterned photoresist material portions can be different from the horizontal cross-sectional shapes of the second selector-containing pillar structures **282**. In one embodiment, the lateral dimension of each of the fourth discrete patterned photoresist material portions along the first horizontal direction **hd1** may be the same as the lateral dimension of each of the fourth discrete patterned photoresist material portions along the second horizontal direction **hd2**. In one embodiment, each of the fourth discrete patterned photoresist material portions may have a respective horizontal cross-sectional shape of a circle.

An anisotropic etch process can be performed to etch unmasked portions of the layer stack (**212L**, **214L**, **230L**, **244L**, **248L**) including the second continuous superlattice layer **212L**, the optional second continuous antiferromagnetic coupling layer **214L**, the second continuous magnetic tunnel junction (MTJ) material layers **230L**, the second continuous dielectric capping layer **244L**, and the second continuous metallic capping layer **248L** employing the two-dimensional array of fourth discrete patterned photoresist material portions as an etch mask.

In one embodiment, the anisotropic etch process may comprise a combination of a reactive ion etch process and an ion beam etch (IBE) process. In one embodiment, an optional hardmask layer (if present) and/or the second continuous metallic capping layer **248L** and/or the second continuous dielectric capping layer **244L** may be patterned into a two-dimensional periodic array of optional hardmask plates and/or second metallic capping layers **248** and/or second dielectric capping layers **244** by performing a reactive ion etch process. The two-dimensional array of fourth discrete patterned photoresist material portions can be subsequently removed, for example, by ashing. The second continuous MTJ material layers **230L**, the optional second continuous antiferromagnetic coupling layer **214L**, and the second continuous superlattice layer **212L** can be patterned by performing an ion beam etch process that employs the two-dimensional periodic array of the optional hardmask plates and/or the second metallic capping layers **248** and/or the second dielectric capping layers **244** as a mask.

Each patterned portion of the layer stack (**212L**, **214L**, **230L**, **244L**, **248L**) comprises a second magnetic tunnel junction (MTJ) pillar structure **284**. A two-dimensional array of second magnetic tunnel junction (MTJ) pillar structures **284** can be formed over the two-dimensional array of second selector-containing pillar structures **282**. Each contiguous combination of a second selector-containing pillar structure **282** and a second MTJ pillar structure **284** constitutes a second memory cell **280**, which can function as a memory cell **180** described with reference to FIG. **2**.

Each second MTJ pillar structure **284** comprises a stack of a second superlattice layer **212**, a second antiferromagnetic coupling layer **214**, a second magnetic tunnel junction **230**, a second dielectric capping layer **244**, and a second metallic capping layer **248**. The second magnetic tunnel junction **230** includes a second reference layer **232**, a second tunnel barrier layer **234**, and a second free layer **236**. Each second superlattice layer **212** is a patterned portion of the second continuous superlattice layer **212L**. Each third antiferromagnetic coupling layer **214** is a patterned portion of the second continuous antiferromagnetic coupling layer **214L**. Each second magnetic tunnel junction **230** is a patterned portion of the second magnetic tunnel junction material layers **230L**. Each second dielectric capping layer **244** is a patterned portion of the second continuous dielectric capping layer **244L**. Each second metallic capping layer **248** is a patterned portion of the second continuous metallic capping layer **248L**. Each second reference layer **232** is a patterned portion of the second continuous reference layer **232L**. Each second tunnel barrier layer **234** is a patterned portion of the second continuous tunnel barrier layer **234L**. Each second free layer **236** is a patterned portion of the second continuous free layer **236L**. Sidewalls of each layer within an MTJ pillar structure **284** can be vertically coincident.

The two-dimensional array of second MTJ pillar structures **284** can be formed above the top surfaces of the remaining portions of the second selector-level dielectric matrix layer **240** and over the two-dimensional array of the second selector-containing pillar structures **282**. Each of the second MTJ pillar structures **284** comprises a respective magnetic tunnel junction **230**.

In one embodiment, each MTJ pillar structure **284** within the two-dimensional array of MTJ pillar structures **284** has a respective horizontal cross-sectional shape having a same lateral extent along the first horizontal direction **hd1** and along the second horizontal direction **hd2**. In one embodiment, each MTJ pillar structure **284** may have a horizontal cross-sectional shape of a circle, a square, or a rounded square, i.e., a shape that is derived from a square by rounding the four corners.

In one embodiment, each of the two-dimensional array of second selector-containing pillar structures **282** and the two-dimensional array of second MTJ pillar structures **284** has the first pitch **p1** along the first horizontal direction **hd1**, and has the second pitch **p2** along the second horizontal direction **hd2**. In one embodiment, the maximum lateral dimension of each of the second selector-containing pillar structures **282** along the first horizontal direction **hd1** can be less than the maximum lateral dimension of each of the second MTJ pillar structures **284** along the first horizontal direction **hd1**, and the maximum lateral dimension of each of the second selector-containing pillar structures **282** along the second horizontal direction **hd2** is greater than the maximum lateral dimension of each of the second MTJ pillar structures **284** along the second horizontal direction **hd2**.

Referring to FIGS. **22A-22C**, a dielectric fill material can be deposited in the gaps between neighboring pairs of the second MTJ pillar structures **284**, and can be subsequently planarized. The portion of the dielectric fill material located underneath the horizontal plane including the top surfaces of the second MTJ pillar structures **284** comprises a dielectric matrix layer, which is herein referred to as a second magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer **288**. The portion of the dielectric fill material located

above the horizontal plane including the top surfaces of the second MTJ pillar structures **284** comprises a line-level dielectric layer **332**.

Line trenches laterally extending along the first horizontal direction **hd1** can be formed above each row of second MTJ pillar structures **284**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the line-level dielectric layer **332**. Remaining portions of the conductive material filling the line trenches constitute third electrically conductive lines **330** (e.g., additional word lines). The third electrically conductive lines **330** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the third electrically conductive lines **330** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed.

Alternatively, instead of using the above described damascene process to form the third electrically conductive lines **330**, these lines may be formed by a pattern and etch process. In the pattern and etch process, a continuous electrically conductive layer is patterned into the third electrically conductive lines **330** by photolithography and etching. The line-level dielectric layer **332** is then deposited between the third electrically conductive lines **330** and optionally planarized with the top surfaces of the third electrically conductive lines **330**.

Referring to FIGS. **23A-23C**, an alternative configuration of the first exemplary structure may be derived from the first exemplary structure illustrated in FIGS. **22A-22C** by reversing the order of the vertical stack of material layers within each of the first MTJ pillar structures **184** and/or within each of the second MTJ pillar structures **284**.

The method of the first embodiment forms the first electrically conductive lines **30** without using a separate photolithographic mask. Instead, the first selector-containing pillar structures **182** and the surrounding dielectric spacers **156** are used as a mask to pattern the first electrically conductive lines **30**. This reduces the number of photolithography steps, and thus reduces the cost and complexity of the process. Furthermore, the word lines **30** are self-aligned with the selector bits (i.e., the first selector-containing pillar structures **182**), thus avoiding misalignment. Still further, due to the presence of the dielectric spacers **156**, damage to the sidewalls of the first selector-containing pillar structures **182** is reduced or avoided during the etching of the word lines **30**.

Furthermore, since the first selector-containing pillar structures **182** are patterned prior to depositing the layers of the MTJ pillar structures **184**, redeposition of the first selector-containing pillar structure **182** materials on the sidewalls of the MTJ pillar structures **184** during the reactive ion etching of the first selector-containing pillar structures **182** does not occur. This permits the first selector-containing pillar structure **182** to be placed closer together, thus increasing the device density and reducing the device cost. Therefore, very small pitch MRAM cross point arrays may be formed with fewer lithography steps.

The MRAM layer stack is deposited onto a polished dielectric surface (e.g., the surface of layer **40**) with embedded selector-containing pillar structures **182**. Since the MRAM layer stack is deposited on a smooth and flat surface, it may result in improved MRAM device performance. Furthermore, since the MRAM pillars (i.e., MTJ pillar structures **184**) are patterned with mostly dielectric material

(i.e., layer 40) exposed below, it is believed that there will be little undesirable sidewall material redeposition on the MTJ 130 of the MTJ pillar structures 184 during the patterning.

IBE may lead to shadowing effects when etching line shaped features. Therefore, in the first embodiment, IBE is preferably only used to pattern the discrete pillar-shaped MTJ pillar structures 184. Thus, the line shaped word lines 30 do not have to be patterned by IBE to avoid the shadowing effects of IBE. The MRAM pillars (i.e., MTJ pillar structures 184) are patterned into dot (i.e., cylindrical pillar) arrays, which are favorable for dense MRAM array fabrication.

Referring collectively to FIGS. 1-23C, a memory array is provided, which comprises: first electrically conductive lines 30 laterally extending along a first horizontal direction hd1 and having a respective variable width along a second horizontal direction that varies along the first horizontal direction hd1; a two-dimensional array of selector-containing pillar structures 182 located over the first electrically conductive lines 182 and including a respective selector element 150; a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures 184 located over the two-dimensional array of selector-containing pillar structures 182 and including a respective magnetic tunnel junction (MTJ) 130; and second electrically conductive lines 90 laterally extending along the second horizontal direction hd2 and overlying the two-dimensional array of (MTJ) pillar structures 184.

In one embodiment, the memory device comprises dielectric spacers 156 laterally surrounding a respective row of selector-containing pillar structures 182 of the two-dimensional array of selector-containing pillar structures 182, wherein the selector-containing pillar structures 182 within the respective row of selector-containing pillar structures 182 are arranged along the first horizontal direction hd1.

In one embodiment, each of the first electrically conductive lines 30 contacts a bottom surface of a respective one of the dielectric spacers 156, and comprises a respective pair of contoured sidewalls that are vertically coincident with sidewalls of the respective one of the dielectric spacers 156.

In one embodiment, each of the dielectric spacers 156 comprises a respective plurality of vertically-extending seams that are parallel to the second horizontal direction hd2 and located midway between a respective neighboring pair of selector-containing pillar structures 182 of the two-dimensional array of selector-containing pillar structures 182 that are laterally spaced apart along the first horizontal direction hd1. In one embodiment, each of the dielectric spacers 156 comprises a pair of contoured lengthwise sidewalls that generally extend along the first horizontal direction hd1 with a lateral undulation along the second horizontal direction hd2; each lengthwise segment of each of the dielectric spacers 156 that laterally extend along the first horizontal direction hd1 and located between a respective selector-containing pillar structure and a respective contoured lengthwise sidewall has a first thickness t1 along the second horizontal direction hd2; and each widthwise segment of each of the dielectric spacers 156 located between a respective selector-containing pillar structure 182 and a respective vertically-extending seam 156S has a second thickness t2 along the first horizontal direction hd1 that is less than the first thickness t1.

In one embodiment, each selector-containing pillar structure 182 within the two-dimensional array of selector-containing pillar structures 182 has a first lateral dimension ld1 along the first horizontal direction hd1 and has a second

lateral dimension ld2 along the second horizontal direction hd2 that is less than the first lateral dimension ld1. In one embodiment, a ratio of the first lateral dimension ld1 to the second lateral dimension ld2 is in a range from 1.2 to 4.

In one embodiment, the second electrically conductive lines 90 have a respective variable width along the first horizontal direction hd1 that varies along the second horizontal direction hd2.

In one embodiment, each of the two-dimensional array of selector-containing pillar structures 182 and the two-dimensional array of (MTJ) pillar structures 184 has a first pitch p1 along the first horizontal direction hd1 and has a second pitch p2 along the second horizontal direction hd2; the first electrically conductive lines 30 are periodic along the second horizontal direction hd2 and have the second pitch p2 along the second horizontal direction hd2; and the second electrically conductive lines 90 are periodic along the first horizontal direction hd1 and have the first pitch p1 along the first horizontal direction hd1.

In one embodiment, a maximum lateral dimension of each of the selector-containing pillar structures 182 along the first horizontal direction hd1 is greater than a maximum lateral dimension of each of the (MTJ) pillar structures 184 along the first horizontal direction hd1; and a maximum lateral dimension of each of the selector-containing pillar structures 182 along the second horizontal direction hd2 is less than a maximum lateral dimension of each of the MTJ pillar structures along the second horizontal direction hd2.

Referring to FIGS. 24A-24C, a second exemplary structure according to a second embodiment of the present disclosure is illustrated, which can be derived from the first exemplary structure illustrated in FIGS. 3A and 3B by applying a photoresist layer over the first hardmask layer 160L and by patterning the photoresist layer into a first patterned photoresist layer 167. The first patterned photoresist layer 167 may have a line-and-space pattern in which each of the first patterned photoresist layer 167 has a uniform width along a first horizontal direction hd1 and laterally extends along a second horizontal direction hd2. Each neighboring pair of first patterned photoresist layer 167 may be laterally spaced apart by a uniform spacing. The sum of the uniform width and the uniform spacing equals the periodicity of the first patterned photoresist layer 167 along the first horizontal direction hd1, which is herein referred to as a first pitch p1.

Referring to FIGS. 25A-25C, a first line-pattern-transfer process can be performed, which transfers the line-and-space pattern in the first patterned photoresist layer 167 through the first hardmask layer 160L and the first selector material layers 150L. The first line-pattern-transfer process can comprise a first anisotropic etch process that etches the materials of the first hardmask layer 160L and the first selector material layers 150L employing the first patterned photoresist layer 167 as an etch mask. The first electrically conductive layer 30L may be employed as an etch stop layer for the first anisotropic etch process. Patterned portions of the first hardmask layer 160L and the first selector material layers 150L comprise first selector rail structures (150R, 160R) that laterally extend along the second horizontal direction hd2 and are laterally spaced apart along the first horizontal direction hd1. Each first selector rail structure (150R, 160R) comprises a first selector-level rail 150R that is a patterned portion of the first selector material layers 150L, and a first hardmask rail 160R that is a patterned portion of the first hardmask layer 160L. The first patterned photoresist layer 167 can be subsequently removed, for example, by ashing.

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Referring to FIGS. 26A-26C, a dielectric fill material can be deposited in the gaps between neighboring pairs of first selector rail structures (150R, 160R). Excess portions of the dielectric fill material can be removed from above the horizontal plane including the top surfaces of the first selector rail structures (150R, 160R) by performing a planarization process such as a chemical mechanical polishing process. Remaining portions of the dielectric fill material are herein referred to as first selector-level isolation rails 41R.

Referring to FIGS. 27A-27C, a photoresist layer can be applied over the first selector rail structures (150R, 160R) and the first selector-level isolation rails 41R, and can be lithographically patterned into a second patterned photoresist layer 169. The second patterned photoresist layer 169 may have a line-and-space pattern in which each of the second patterned photoresist layer 169 has a uniform width along the second horizontal direction hd2 and laterally extends along the first horizontal direction hd1. Each neighboring pair of second patterned photoresist layer 169 may be laterally spaced apart by a uniform spacing. The sum of the uniform width and the uniform spacing equals the periodicity of the second patterned photoresist layer 169 along the second horizontal direction hd2, which is herein referred to as a second pitch p2. The second pitch p2 may be the same as, or may be different from, the first pitch p1. In one embodiment, the second pitch p2 is the same as the first pitch p1.

Referring to FIGS. 28A-28C, a second line-pattern-transfer process can be performed, which transfers the line-and-space pattern in the second patterned photoresist layer 169 through the first selector rail structures (150R, 160R), the first selector-level isolation rails 41R, and the first electrically conductive layer 30L. The second line-pattern-transfer process can comprise a second anisotropic etch process that etches the materials of the first selector rail structures (150R, 160R), the first selector-level isolation rails 41R, and the first electrically conductive layer 30L employing the second patterned photoresist layer 169 as an etch mask. Patterned portions of the first selector rail structures (150R, 160R) comprise a two-dimensional array of first selector-containing pillar structures 182.

Each first selector-containing pillar structure 182 can include a first selector element 150 and a first hardmask plate 160. Each first selector element 150 is a patterned portion of the first selector material layers 150L, and each first hardmask plate 160 is a patterned portion of the first hardmask layer 160L. Each first selector element 150 may include a vertical stack of a first lower selector electrode 151, a first non-Ohmic material plate 152, and a first upper selector electrode 153. Each first lower selector electrode 151 is a patterned portion of the first lower selector electrode material layer 151L. Each first non-Ohmic material plate 152 is a patterned portion of the first non-Ohmic material layer 152L. Each first upper selector electrode 153 is a patterned portion of the first upper selector electrode material layer 153L.

The patterned portions of the first selector-level isolation rails 41R comprise a two-dimensional periodic array of first selector-level isolation pillars 41, which may have a first periodicity of the first pitch p1 along the first horizontal direction hd1 and may have a second periodicity of the second pitch p2 along the second horizontal direction hd2. Each of the first selector-level isolation pillars 41 may have a respective rectangular horizontal cross-sectional shape. Laterally alternating sequences of first selector-containing pillar structures 182 and first selector-level isolation pillars 41 can be formed. Each laterally alternating sequence of first

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selector-containing pillar structures 182 and first selector-level isolation pillars 41 includes a respective plurality of first selector-containing pillar structures 182 and a respective plurality of first selector-level isolation pillars 41 that are interlaced along the first horizontal direction hd1. Line trenches are present between each neighboring pair of laterally alternating sequences of first selector-containing pillar structures 182 and first selector-level isolation pillars 41. The first selector-containing pillar structures 182 may have a rectangular or square horizontal cross-sectional shape.

The first electrically conductive layer 30L is patterned into a plurality of first electrically conductive lines (e.g., word lines) 30 that laterally extend along the first horizontal direction hd1 and are laterally spaced apart along the second horizontal direction hd2. The first electrically conductive lines 30 may have straight sidewalls that laterally extend along the first horizontal direction hd1, and may be laterally spaced apart along the second horizontal direction hd2. The first electrically conductive lines 30 can be formed as a periodic one-dimensional array of first electrically conductive lines 30 having the periodicity of the second pitch p2 along the second horizontal direction. The second patterned photoresist layer 169 can be subsequently removed, for example, by ashing.

Generally, one or more pattern transfer processes can be employed to form a two-dimensional array of first selector-containing pillar structures 182. In one embodiment, the one or more pattern transfer processes may comprise a first line-pattern-transfer process and a second line-pattern-transfer process. In one embodiment, a first line-and-space pattern is transferred during the first line-pattern-transfer process through the first selector material layers 150L to pattern the first selector-level material layers 150L into first selector rail structures 150R that laterally extend along the second horizontal direction hd2 and are laterally spaced apart along the first horizontal direction hd1. During the second line-pattern-transfer process, a second line-and-space pattern is transferred through the first selector rail structures 150R and the first electrically conductive layer 30L to pattern the first selector rail structures 150R into a two-dimensional array of first selector elements 150 and to pattern the first electrically conductive layer 30L into the first electrically conductive lines 30.

Referring to FIGS. 29A-29C, a dielectric fill material can be deposited in the line trenches between each neighboring pairs of laterally alternating sequences of first selector-containing pillar structures 182 and first selector-level isolation pillars 41. Excess portions of the dielectric fill material can be removed from above the horizontal plane including the top surfaces of the first selector-containing pillar structures 182 by performing a planarization process, such as a chemical mechanical polishing process. Remaining portions of the dielectric fill material are herein referred to as second selector-level isolation rails 42R. Each second selector-level isolation rail 42R may laterally extend along the first horizontal direction hd1, and may be laterally spaced apart along the second horizontal direction hd2. The second selector-level isolation rails 42R may be arranged as a one-dimensional periodic array having the periodicity of the second pitch p2 along the second horizontal direction hd2.

Generally, dielectric fill material portions (41R, 42R) can be formed during processing steps for manufacturing the second exemplary structure between rows of selector-containing pillar structures 182 arranged along the first horizontal direction hd1, and/or between columns of selector-



containing pillar structures **182** arranged along the second horizontal direction **hd2**. Top surfaces of the dielectric fill material portions (**41R**, **42R**) are formed within a horizontal plane including top surfaces of the two-dimensional array of selector-containing pillar structures **182**.

In an alternative configuration, the CMP process is continued to also remove the first hardmask plates **160** and to expose the upper surface of the first upper selector electrodes **153**, similar to the step shown in FIGS. **9D-9F**. In this alternative configuration, the first hardmask plates **160** may comprise an insulating material, such as silicon nitride or metal oxide, and at least an upper portion of the first upper selector electrodes **153** may comprise a metal or metal alloy rather than a carbon based material. Thus, the CMP does not damage the carbon material of the first upper selector electrodes **153**.

Referring to FIGS. **30A-30C**, the processing steps of FIGS. **10A-10C**, **11A-11C**, and **12A-12C** can be performed to form a two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures **184** over the two-dimensional array of first selector-containing pillar structures **182**. Each contiguous combination of a first selector-containing pillar structure **182** and a first MTJ pillar structure **184** constitutes a first memory cell **180**, which can function as a memory cell **180** described with reference to FIG. **2**. Generally, first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be formed over the two-dimensional array of first selector-containing pillar structures **182** and the second selector-level isolation rails **42R**. The first MTJ-level material layers may comprise, for example, a first continuous superlattice layer **112L**, an optional first continuous antiferromagnetic coupling layer **114L**, first continuous magnetic tunnel junction (MTJ) material layers **130L**, a first continuous dielectric capping layer **144L**, and a first continuous metallic capping layer **148L**. The first MTJ material layers **130L** may comprise a layer stack including a first continuous reference layer **132L**, a first continuous nonmagnetic tunnel barrier layer **134L**, a first continuous free layer **136L**. A two-dimensional array of discrete patterned photoresist material portions **159** can be employed as an etch mask, and can be subsequently removed after an anisotropic etch process that forms the two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures **184**.

Referring to FIGS. **31A-31C**, a dielectric fill material can be deposited in the gaps between neighboring pairs of the first MTJ pillar structures **184**. Portions of the dielectric fill material underlying the horizontal plane including the top surfaces of the first MTJ pillar structures **184** can be removed by a planarization process such as a chemical mechanical polishing process. The remaining contiguous portion of the dielectric fill material located underneath the horizontal plane including the top surfaces of the first MTJ pillar structures **184** comprises a dielectric matrix layer, which is herein referred to as a first magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer **80**.

Referring to FIGS. **32A-32C**, a dielectric material can be deposited over the two-dimensional array of first MTJ pillar structures **184** to form a line-level dielectric layer **92**. Line trenches laterally extending along the second horizontal direction **hd2** can be formed through the line-level dielectric layer **92** above each column of MTJ pillar structures **184** arranged along the second horizontal direction **hd2**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the line-level dielectric layer **92**. Remaining portions of the

conductive material filling the line trenches constitute second electrically conductive lines **90**. The second electrically conductive lines **90** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the second electrically conductive lines **90** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed.

Alternatively, instead of using the above described damascene process to form the second electrically conductive lines **90**, these lines may be formed by a pattern and etch process. In the pattern and etch process, a continuous electrically conductive layer is patterned into the second electrically conductive lines **90** by photolithography and etching.

Alternatively, the processing steps of FIGS. **24A-24C**, **25A-25C**, **26A-26C**, **27A-27C**, **28A-28C**, **29A-29C**, **30A-30C**, and **31A-31C** can be performed with a 90 degree rotation in all patterns to form a two-dimensional array of second memory cells (not illustrated) over the two-dimensional array of first memory cells **180**. In this case, the processing steps of FIGS. **22A-22C** can be subsequently performed to form a second magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer (not shown), a line-level dielectric layer (not shown), and third electrically conductive lines (not shown).

Referring to FIGS. **33A-33C**, a first alternative configuration of the second exemplary structure may be derived from the second exemplary structure illustrated in FIGS. **32A-32C** by reversing the order of the vertical stack of material layers within each of the first MTJ pillar structures **184**.

Referring to FIGS. **34A-34C**, a second alternative configuration of the second exemplary structure is shown, which may be derived from the second exemplary structure illustrated in FIGS. **24A-24C** by modifying the pattern of the first patterned photoresist layer **167**. Specifically, the pattern of the first patterned photoresist layer **167** employed in the second alternative configuration of the second exemplary structure at the processing steps of FIGS. **34A-34C** can be the same as the pattern of the second patterned photoresist layer **169** that is employed at the processing steps of FIGS. **27A-27C**.

Referring to FIGS. **35A-35C**, the processing steps of FIGS. **25A-25C** can be performed with a modification to the first anisotropic etch process. Specifically, the first anisotropic etch process includes an additional etch step that patterns the first electrically conductive layer **30L** into first electrically conductive lines **30**. Patterned portions of the first hardmask layer **160L** and the first selector material layers **150L** comprise first selector rail structures (**150R**, **160R**) that laterally extend along the first horizontal direction **hd1** and are laterally spaced apart along the second horizontal direction **hd2**. Each first selector rail structure (**150R**, **160R**) comprises a first selector-level rail **150R** that is a patterned portion of the first selector material layers **150L**, and a first hardmask rail **160R** that is a patterned portion of the first hardmask layer **160L**.

The first electrically conductive lines **30** laterally extend along the first horizontal direction **hd1** and are laterally spaced apart along the second horizontal direction **hd2**. The first electrically conductive lines **30** may have straight sidewalls that laterally extend along the first horizontal direction **hd1**, and may be laterally spaced apart along the second horizontal direction **hd2**. The first electrically conductive lines **30** can be formed as a periodic one-dimen-



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sional array of first electrically conductive lines **30** having the periodicity of the second pitch **p2** along the second horizontal direction. The first patterned photoresist layer **167** can be subsequently removed, for example, by ashing.

Referring to FIGS. **36A-36C**, a dielectric fill material can be deposited in the gaps between neighboring pairs of first selector rail structures (**150R**, **160R**). Excess portions of the dielectric fill material can be removed from above the horizontal plane including the top surfaces of the first selector rail structures (**150R**, **160R**) by performing a planarization process such as a chemical mechanical polishing process. Remaining portions of the dielectric fill material are herein referred to as first selector-level isolation rails **41R**.

Referring to FIGS. **37A-37C**, the processing steps of FIGS. **27A-27C** can be performed to form second patterned photoresist layer **169**, which can have the same pattern as the first patterned photoresist layer **167** illustrated in FIGS. **24A-24C**.

Referring to FIGS. **38A-38C**, the processing steps of FIGS. **28A-28C** can be performed with a modification such that the second anisotropic etch process etches the materials of the first selector rail structures (**150R**, **160R**) and the first selector-level isolation rails **41R** selective to the material of the first electrically conductive lines **30** employing the second patterned photoresist layer **169** as an etch mask.

Patterned portions of the first selector rail structures (**150R**, **160R**) comprise a two-dimensional array of first selector-containing pillar structures **182**. Each first selector-containing pillar structure **182** can include a first selector element **150** and a first hardmask plate **160**. Each first selector element **150** is a patterned portion of the first selector material layers **150L**, and each first hardmask plate **160** is a patterned portion of the first hardmask layer **160L**. Each first selector element **150** may include a vertical stack of a first lower selector electrode **151**, a first non-Ohmic material plate **152**, and a first upper selector electrode **153**. Each first lower selector electrode **151** is a patterned portion of the first lower selector electrode material layer **151L**. Each first non-Ohmic material plate **152** is a patterned portion of the first non-Ohmic material layer **152L**. Each first upper selector electrode **153** is a patterned portion of the first upper selector electrode material layer **153L**.

The first selector-level isolation rails **41R** can be patterned such that each of the first selector-level isolation rails **41R** includes a repetition of indentations having the periodicity of the first pitch **p1** along the first horizontal direction **hd1**. The indentations can be formed in each area of the first selector-level isolation rails **41R** that are not masked by the second patterned photoresist layer **169**. Line trenches are present between each neighboring columns of first selector-containing pillar structures **182** that are arranged along the second horizontal direction **hd2**. The second patterned photoresist layer **169** can be subsequently removed, for example, by ashing.

Generally, one or more pattern transfer processes can be employed to form a two-dimensional array of first selector-containing pillar structures **182**. In one embodiment, the one or more pattern transfer processes may comprise a first line-pattern-transfer process and a second line-pattern-transfer process. In one embodiment, a first line-and-space pattern is transferred during the first line-pattern-transfer process through the first selector material layers **150L** and the first electrically conductive layer **30L** into first selector rail structures **150R** that laterally extend along the first horizontal direction **hd1** and are laterally spaced apart along the second horizontal direction **hd2** and to pattern the first

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electrically conductive layer **30L** into the first electrically conductive lines **30**. During the second line-pattern-transfer process, a second line-and-space pattern is transferred through the first selector rail structures **150R** to pattern the first selector rail structures **150R** into a two-dimensional array of selector elements **150**.

Referring to FIGS. **39A-39C**, a dielectric fill material can be deposited in the line trenches between each neighboring columns of first selector-containing pillar structures **182** that are laterally spaced apart along the first horizontal direction **hd1**. Excess portions of the dielectric fill material can be removed from above the horizontal plane including the top surfaces of the first selector-containing pillar structures **182** by performing a planarization process such as a chemical mechanical polishing process. Remaining portions of the dielectric fill material are herein referred to as second selector-level isolation rails **42R**. Each second selector-level isolation rail **42R** may laterally extend along the second horizontal direction **hd2**, and may be laterally spaced apart along the first horizontal direction **hd1**. The second selector-level isolation rails **42R** may be arranged as a one-dimensional periodic array having the periodicity of the first pitch **p1** along the first horizontal direction **hd1**.

In an alternative configuration, the CMP process is continued to also remove the first hardmask plates **160** and to expose the upper surface of the first upper selector electrodes **153**, similar to the step shown in FIGS. **9D-9F**. In this alternative configuration, the first hardmask plates **160** may comprise an insulating material, such as silicon nitride or metal oxide, and at least an upper portion of the first upper selector electrodes **153** may comprise a metal or metal alloy rather than a carbon based material. Thus, the CMP does not damage the carbon material of the first upper selector electrodes **153**.

Generally, dielectric fill material portions (**41R**, **42R**) can be formed during processing steps for manufacturing the second exemplary structure between rows of selector-containing pillar structures **182** arranged along the first horizontal direction **hd1**, and/or between columns of selector-containing pillar structures **182** arranged along the second horizontal direction **hd2**. Top surfaces of the dielectric fill material portions (**41R**, **42R**) are formed within a horizontal plane including top surfaces of the two-dimensional array of selector-containing pillar structures **182**.

Referring to FIGS. **40A-40C**, the processing steps of FIGS. **10A-10C**, **11A-11C**, and **12A-12C** can be performed to form a two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures **184** over the two-dimensional array of first selector-containing pillar structures **182**. Each contiguous combination of a first selector-containing pillar structure **182** and a first MTJ pillar structure **184** constitutes a first memory cell **180**, which can function as a memory cell **180** described with reference to FIG. **2**.

Referring to FIGS. **41A-41C**, a dielectric fill material can be deposited in the gaps between neighboring pairs of the first MTJ pillar structures **184**. Portions of the dielectric fill material underlying the horizontal plane including the top surfaces of the first MTJ pillar structures **184** can be removed by a planarization process such as a chemical mechanical polishing process. The remaining contiguous portion of the dielectric fill material located underneath the horizontal plane including the top surfaces of the first MTJ pillar structures **184** comprises a dielectric matrix layer, which is herein referred to as a first magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer **80**.

Referring to FIGS. **42A-42C**, a dielectric material can be deposited over the two-dimensional array of first MTJ pillar

structures **184** to form a line-level dielectric layer **92**. Line trenches laterally extending along the second horizontal direction **hd2** can be formed through the line-level dielectric layer **92** above each column of MTJ pillar structures **184** arranged along the second horizontal direction **hd2**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the line-level dielectric layer **92**. Remaining portions of the conductive material filling the line trenches constitute second electrically conductive lines **90**. The second electrically conductive lines **90** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the second electrically conductive lines **90** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed.

Alternatively, instead of using the above described damascene process to form the second electrically conductive lines **90**, these lines may be formed by a pattern and etch process. In the pattern and etch process, a continuous electrically conductive layer is patterned into the second electrically conductive lines **90** by photolithography and etching.

Alternatively, the processing steps of FIGS. **24A-24C**, **25A-25C**, **26A-26C**, **27A-27C**, **28A-28C**, **29A-29C**, **30A-30C**, and **31A-31C** can be performed with a 90 degree rotation in all patterns to form a two-dimensional array of second memory cells (not illustrated) over the two-dimensional array of first memory cells **180**. In this case, the processing steps of FIGS. **22A-22C** can be subsequently performed to form a second magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer (not shown), a line-level dielectric layer (not shown), and third electrically conductive lines (not shown).

Referring to FIGS. **43A-43C**, a third alternative configuration of the second exemplary structure may be derived from the second exemplary structure illustrated in FIGS. **42A-42C** by reversing the order of the vertical stack of material layers within each of the first MTJ pillar structures **184**.

Referring to FIGS. **44A-44C**, a third exemplary structure according to a third embodiment of the present disclosure can be derived from the first exemplary structure illustrated in FIGS. **3A-3C** by forming first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) described with reference to FIGS. **10A-10C** in lieu of selector-level material layers (**150L**, **160L**) described with reference to FIGS. **3A-3C**. As described above, the first MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) comprise first magnetic tunnel junction (MTJ) material layers **130L**. The first MTJ material layers **130L** may comprise a layer stack including a first continuous reference layer **132L**, a first continuous nonmagnetic tunnel barrier layer **134L**, a first continuous free layer **136L**. An optional seed layer **110** described above with respect to FIG. **2** may be formed below the superlattice layer **112L**.

Referring to FIGS. **45A-45C**, first selector-level material layers (**150L**, **160L**) including first selector material layers **150L** and a first hardmask layer **160L** described with reference to FIGS. **3A-3C** can be formed over the first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**). In this embodiment, the first hardmask layer **160L** may comprise a first conductive material layer which is retained in the final device. A photoresist layer can be deposited over the first conductive material

layer **160L**, and can be lithographically patterned to form a two-dimensional array of first discrete patterned photoresist material portions **187**.

According to an aspect of the present disclosure, the two-dimensional array of first discrete patterned photoresist material portions **187** can be a periodic two-dimensional array having a first pitch **p1** along a first horizontal direction **hd1** and having a second pitch **p2** along a second horizontal direction **hd2**. The first nearest-neighbor spacing **s1** along the first horizontal direction **hd1** of the two-dimensional array of first discrete patterned photoresist material portions **187** is less than the second nearest-neighbor spacing **s2** along the second horizontal direction **hd2** of the two-dimensional array of first discrete patterned photoresist material portions **187**.

In an illustrative case, each of the first discrete patterned photoresist material portions **187** may have a first lateral dimension **ld1** along the first horizontal direction **hd1**, and may have a second lateral dimension **ld2** along the second horizontal direction **hd2**. Each of the first discrete patterned photoresist material portions **187** may have a respective horizontal cross-sectional shape of a rectangle, a rounded rectangle, an oval, or a circle. The nearest-neighbor spacing **s1** between neighboring pairs of the first discrete patterned photoresist material portions **187** that are laterally spaced apart along the first horizontal direction **hd1** can be the difference between the first pitch **p1** and the first lateral dimension **ld1**. The nearest-neighbor spacing **s2** between neighboring pairs of the first discrete patterned photoresist material portions **187** that are laterally spaced apart along the second horizontal direction **hd2** can be the second pitch **p2** less the second lateral dimension **ld2**. In this case, **p1-ld1** is less than **p2-ld2**. In one embodiment, the second pitch **p2** may be the same as the first pitch **p1**, and the first lateral dimension **ld1** may be greater than the second lateral dimension **ld2**. In one embodiment, the pattern of the first discrete patterned photoresist material portions **187** may be the same as the pattern of the first discrete patterned photoresist material portions **157** described with reference to FIGS. **4A-4C**.

Referring to FIGS. **46A-46C**, an array-pattern-transfer process can be performed to transfer the pattern of the two-dimensional array of first discrete patterned photoresist material portions **187** through the first conductive material layer **160L** and the first selector material layers **150L**. For example, an anisotropic etch process can be performed to transfer the pattern in the two-dimensional array of first discrete patterned photoresist material portions **187** through the first conductive material layer **160L** and the first selector material layers **150L**. The patterned remaining portions of the first conductive material layer **160L** and the first selector material layers **150L** can include two-dimensional array of first selector-containing pillar structures **182**.

Each of the first selector-containing pillar structures **182** may comprise a first selector element **150** and a first conductive material plate **160**. Each first selector element **150** is a patterned portion of the first selector material layers **150L**, and each first conductive material plate **160** is a patterned portion of the first conductive material layer **160L**. Each first selector element **150** may include a vertical stack of a first lower selector electrode **151**, a first non-Ohmic material plate **152**, and a first upper selector electrode **153**. Each first lower selector electrode **151** is a patterned portion of the first lower selector electrode material layer **151L**. Each first non-Ohmic material plate **152** is a patterned portion of the first non-Ohmic material layer **152L**. Each first upper selec-

tor electrode **153** is a patterned portion of the first upper selector electrode material layer **153L**.

In one embodiment, the two-dimensional array of first selector-containing pillar structures **182** comprises a two-dimensional periodic array of first selector-containing pillar structures **182** having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. The nearest-neighbor spacing **s1** between neighboring pairs of the first selector-containing pillar structures **182** that are laterally spaced apart along the first horizontal direction **hd1** is less than the nearest-neighbor spacing **s2** between neighboring pairs of the first selector-containing pillar structures **182** that are laterally spaced apart along the second horizontal direction **hd2**.

In one embodiment, each first selector-containing pillar structure **182** within the two-dimensional array of first selector-containing pillar structures **182** has a respective elongated horizontal cross-sectional shape having a first lateral dimension **ld1** along the first horizontal direction **hd1** and having a second lateral dimension **ld2** along the second horizontal direction **hd2** that is less than the first lateral dimension **ld1**. In one embodiment, the ratio of the first lateral dimension **ld1** to the second lateral dimension **ld2** may be in a range from 1.2 to 4, such as from 1.5 to 3. The two-dimensional array of first selector-containing pillar structures **182** can be formed over the first electrically conductive layer **30L**.

Referring to FIGS. 47A-47D, an ion beam etch process can be performed to etch unmasked portions of the layer stack (**112L**, **114L**, **130L**, **144L**, **148L**) employing the two-dimensional array of second discrete patterned photoresist material portions **187** and/or the array of first conductive material plates (i.e., hardmask plates) **160** as an etch mask. In other words, the second discrete patterned photoresist material portions **187** may be removed during or after formation of the first conductive material plates **160**. The ion beam etch of the layer stack then proceeds using the first conductive material plates **160** as a mask. Alternatively, the second discrete patterned photoresist material portions **187** are retained as a mask during the ion beam etch of the layer stack.

In one embodiment shown in FIG. 47C, the ion beam etch process etches the first continuous metallic capping layer **148L**, the first continuous dielectric capping layer **144L**, the first continuous MTJ material layers **130L**, the optional first continuous antiferromagnetic coupling layer **114L**, and the first continuous superlattice layer **112L**. Each patterned portion of the layer stack (**112L**, **114L**, **130L**, **144L**, **148L**) comprises a first magnetic tunnel junction (MTJ) pillar structure **184**. A two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures **184** can be formed underneath the two-dimensional array of first selector-containing pillar structures **182**. Each contiguous combination of a first selector-containing pillar structure **182** and a first MTJ pillar structure **184** constitutes a first memory cell **180**, which can function as a memory cell **180** described with reference to FIG. 2.

Each first MTJ pillar structure **184** comprises a stack of a first superlattice layer **112**, a first antiferromagnetic coupling layer **114**, a first magnetic tunnel junction **130**, a first dielectric capping layer **144**, and a first metallic capping layer **148**. The first magnetic tunnel junction **130** includes a first reference layer **132**, a first tunnel barrier layer **134**, and a first free layer **136**. Each first superlattice layer **112** is a patterned portion of the first continuous superlattice layer **112L**. Each first antiferromagnetic coupling layer **114** is a

patterned portion of the first continuous antiferromagnetic coupling layer **114L**. Each first magnetic tunnel junction **130** is a patterned portion of the first magnetic tunnel junction material layers **130L**. Each first dielectric capping layer **144** is a patterned portion of the first continuous dielectric capping layer **144L**. Each first metallic capping layer **148** is a patterned portion of the first continuous metallic capping layer **148L**. Each first reference layer **132** is a patterned portion of the first continuous reference layer **132L**. Each first tunnel barrier layer **134** is a patterned portion of the first continuous tunnel barrier layer **134L**. Each first free layer **136** is a patterned portion of the first continuous free layer **136L**.

According to an aspect of the present disclosure, the transfer of the pattern in the two-dimensional array of first discrete patterned photoresist material portions **187** via the pattern in the two-dimensional array of first conductive material plates **160** through the first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be performed employing an aspect-ratio-dependent ion beam etch process that etches materials of the first MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) with dependency on the aspect ratio of the local geometry. An aspect ratio is the ratio of the depth of an etched region to the width of the etched region. The ions used in ion beam etching are generally not perfectly collimated, but have an angular distribution around the primary direction of the ion beam (which may be a downward vertical direction). Due to the finite angular distribution of the ions, the percentage of the ions that impinge on sidewalls of an etched region increases with the increase in the aspect ratio. In other words, a lower fraction of the ions impinge on the bottom surface of the etched region if the aspect ratio is high, and a higher fraction of the ions impinge on the bottom surface of the etched region if the aspect ratio is low. This effect is referred to as a "shadowing effect". In this case, areas having a greater lateral distance between neighboring pairs of first electrically conductive plates **160** are etched at a higher etch rate than areas having a smaller lateral distance between neighboring pairs of first electrically conductive plates **160**. Generally, the aspect-ratio-dependent etch process can etch the materials of the first MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) at a variable etch rate that decreases with a local aspect ratio.

In one embodiment, the two-dimensional array of first electrically conductive plates **160** may comprise rows of first electrically conductive plates **160** that are arranged along the first horizontal direction **hd1**. Each row of first electrically conductive plates **160** may comprise a respective subset of first electrically conductive plates **160** that are arranged along the first horizontal direction **hd1**. The gap (i.e., spacing **s1**) between neighboring pairs of first electrically conductive plates **160** within each row of first electrically conductive plates **160** along direction **hd1** may be less than the gap (i.e., spacing **s2**) between neighboring rows of first electrically conductive plates **160** along direction **hd2**. In this case, the etch rate of the materials of the first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) between neighboring rows of first electrically conductive plates **160** can be higher than the etch rate of the materials of the first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) between neighboring pairs of first electrically conductive plates **160** within each row of first electrically conductive plates **160**.

According to an aspect of the present disclosure, the ion beam etch process can anisotropically etch the first magnetic

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tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) and the first electrically conductive layer 30L such that physically exposed surfaces of remaining portions of the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) are formed with taper angles. The taper angles can be measured with respect to the vertical direction that is perpendicular to the top surface of the substrate 8. In one embodiment, the taper angle may be in a range from 3 degrees to 30 degrees, such as from 6 degrees to 20 degrees, although lesser and greater taper angles may also be employed.

Generally, patterned portions of the first selector-level material layers (150L, 160L) comprise a two-dimensional array of selector-containing pillar structures 182 including a respective selector element 150 and a respective first conductive material plate 160, and patterned portions of the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) comprise a two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures 184. Patterned portions of the first magnetic tunnel junction material layers 130L comprise a two-dimensional array of first magnetic tunnel junctions (MTJs) 130.

According to an aspect of the present disclosure, the duration of the ion beam etch process can be selected such that portions of the first electrically conductive layer 30L located in areas between neighboring rows of first conductive material plates 160 are etched through, while portions of the first electrically conductive material layer 30L located in areas between neighboring pairs of first conductive material plates 160 within each row of first conductive material plates 160 are not etched through. The difference in the etch depth between the two types of areas is caused by the tapered profile of the sidewalls of the etched portions of the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) and the shadowing effect. Thus, the first electrically conductive layer 30L is divided into multiple disjointed patterned electrically conductive strips that laterally extend along the first horizontal direction hd1, which constitute first electrically conductive lines 30. In other words, the patterned portions of the first electrically conductive layer 30L comprise first electrically conductive lines (e.g., word lines) 30 that laterally extend along the first horizontal direction hd1 and laterally spaced apart from each other along the second horizontal direction hd2 and have a respective variable width along the second horizontal direction hd2 that varies along the first horizontal direction hd1. In other words, the first electrically conductive lines 30 have a wiggled profile having alternating wider and narrower sections along the second horizontal direction hd2, as shown in FIG. 47D. The narrower sections have a first width w1 which is smaller than the second width w2 of the wider sections.

According to an aspect of the present disclosure, each of the first electrically conductive lines 30 may be patterned with a respective pair of contoured and tapered lengthwise sidewalls that laterally extend along the first horizontal direction hd1. Each of the first electrically conductive lines 30 may be formed with a respective bottom surface having a respective variable width along the second horizontal direction hd2 that varies along the first horizontal direction hd1. In one embodiment, each of the first electrically conductive lines 30 may comprise a periodic repetition of uniform-width regions and neck regions with a periodicity of the first pitch p1.

The two-dimensional array of first MTJ pillar structures 184 comprises rows of first MTJ pillar structures 184 that are

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arranged along the first horizontal direction hd1. In some embodiments, the tapered sidewalls of each neighboring pair of first MTJ pillar structures 184 within each row of first MTJ pillar structures 184 may be adjoined to each other at a respective edge. According to an aspect of the present disclosure, the edges are formed above a horizontal plane including top surfaces of the first electrically conductive lines 30 during patterning of the first MTJ-level material layers (112L, 114L, 130L, 144L, 148L). The merged edges laterally extend along the second horizontal direction hd2. The edges at which a respective pair of tapered sidewalls of the first MTJ pillar structures 184 are merged (i.e., joined to each other) are formed below the horizontal plane including the bottom surfaces of the first free layers 136.

In one embodiment illustrated in FIG. 47E, the edges at which a respective pair of tapered sidewalls of the MTJ pillar structures 184 merge are formed within the superlattice layers 112. In alternative embodiments, the edges at which a respective pair of tapered sidewalls of the first MTJ pillar structures 184 merge may be formed below the first reference layer 132 (e.g., in the antiferromagnetic layer) 114, within the first continuous reference layer 132L, or within the first continuous nonmagnetic tunnel barrier layer 134L, as shown in FIGS. 47F, 47G and 47H respectively. In the embodiments of FIGS. 47G and 47H, the first continuous reference layer 132L is not split into separate reference layers 132 along the first horizontal direction hd1. In the embodiment of FIG. 47H, the first continuous tunnel barrier layer 134L is not split into separate tunnel barrier layers 134 along the first horizontal direction hd1.

Each vertical stack of a first selector-containing pillar structure 182 and a first MTJ pillar structure 184 constitutes a first memory cell 180. According to an aspect of the present disclosure, each row of first memory cells 180 that are arranged along the first horizontal direction hd1 are merged with each other below the horizontal plane including the bottom surfaces of the first free layers 136 within the respective row of first memory cells 180. Each merged row of first memory cells 180 is herein referred to as first selector-magnetic tunnel junction (selector-MTJ) assembly 180A, as shown in FIG. 47A. The third exemplary structure can include rows of first selector-magnetic tunnel junction (selector-MTJ) assemblies 180A located on a respective one of the first electrically conductive lines 30. Each of the selector-MTJ assemblies 180A comprises a respective row of first magnetic tunnel junction (MTJ) pillar structures 184 and a respective row of first selector-containing pillar structures 182 that are arranged along the first horizontal direction hd1. Tapered sidewalls of each neighboring pair of first MTJ pillar structures 184 within the respective row of first MTJ pillar structures 184 are adjoined to each other at a respective edge laterally extending along the second horizontal direction hd2 and located above a horizontal plane including top surfaces of the first electrically conductive lines 30.

In one embodiment, the rows of first selector-containing pillar structures 182 are arranged as a two-dimensional periodic array of first selector-containing pillar structures 182 having a first pitch p1 along the first horizontal direction hd1 and having a second pitch p2 along the second horizontal direction hd2. A lateral spacing s1 between neighboring pairs of first selector-containing pillar structures 182 that are laterally spaced apart along the first horizontal direction hd1 is less than a lateral spacing s2 between neighboring pairs of first selector-containing pillar structures 182 that are laterally spaced apart along the second horizontal direction hd2.

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In one embodiment, each of the selector-containing pillar structures **182** has an elongated horizontal cross-sectional shape having a greater lateral dimension along the first horizontal direction **hd1** than along the second horizontal direction **hd2**, and the rows of selector-containing pillar structures **182** are arranged as a two-dimensional periodic array of selector-containing pillar structures **182** having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. In one embodiment, the second pitch **p2** is the same as the first pitch **p1**.

In one embodiment, each of the first electrically conductive lines **30** comprises a laterally alternating sequence of uniform thickness segments (e.g., regions **w1**) that underlie a respective one of the first MTJ pillar structures **184** and indented segments (e.g., regions **w2**) that includes a V-shaped indentation. In one embodiment, each of the first electrically conductive lines **30** comprises a pair of contoured lengthwise sidewalls that generally extend along the first horizontal direction **hd1** with a lateral undulation along the second horizontal direction **hd2**, and each of the contoured lengthwise sidewalls comprises straight segments located in a respective uniform width region **w1** and laterally extending along the first horizontal direction **hd1**, and pairs of adjoined convex sidewalls adjoined at a respective vertically-extending edge located at a respective neck region **w2**. Regions **w2** have a narrower width along the second horizontal direction **hd2** than regions **w1**.

Each of the first MTJ pillar structures **184** comprises a vertical stack including a first reference layer **132**, a first tunnel barrier layer **134**, and a first free layer **136**. First free layers **136** within the respective row of first MTJ pillar structures are laterally spaced apart from each other along the first horizontal direction **hd1** and do not contact one another.

In one embodiment, each free layer **136** within the two-dimensional array of MTJ pillar structures **184** has a respective horizontal cross-sectional shape having a same lateral extent along the first horizontal direction **hd1** and along the second horizontal direction **hd2**. In one embodiment, each free layer **136** may have a horizontal cross-sectional shape of a circle, a square, or a rounded square, i.e., a shape that is derived from a square by rounding the four corners.

Referring to FIGS. **48A-48C**, a dielectric fill material can be deposited in gaps between neighboring pairs of first memory cells **180** to form a dielectric matrix layer, which is herein referred to as a first dielectric matrix layer **186**. Excess portions of the dielectric fill material of the first dielectric matrix layer **186** can be removed from above a horizontal plane including top surfaces of the first selector-containing pillar structures **182** by a planarization process such as a chemical mechanical polishing process. A top surfaces of a remaining portion of the first dielectric matrix layer **186** is formed within a horizontal plane including top surfaces of the two-dimensional array of first selector-containing pillar structures **182**. In an alternative configuration, the CMP process is continued to also remove the first hardmask plates **160** and to expose the upper surface of the first upper selector electrodes **153**, similar to the step shown in FIGS. **9D-9F**. The above described second electrically conductive lines (e.g., bit lines) **90** are subsequently formed over the first selector-containing pillar structures **182**.

Optionally, additional device levels may be formed above the second electrically conductive lines **90**. Referring to FIGS. **49A-49C**, a second electrically conductive layer **90L** and second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) can be

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deposited over the two-dimensional array of first memory cells **180** and the first dielectric matrix layer **186**. The second MTJ-level material layers (**212L**, **214L**, **230L**, **244L**, **248L**) may be the same as the first MTJ-level material layers (**112**, **114L**, **130L**, **144L**, **148L**). As described above, the second MTJ-level material layers (**212L**, **214L**, **230L**, **244L**, **248L**) comprise second magnetic tunnel junction (MTJ) material layers **230L**. The second MTJ material layers **230L** may comprise a layer stack including a second continuous reference layer **232L**, a second continuous nonmagnetic tunnel barrier layer **234L**, a second continuous free layer **236L**.

A layer stack (**250L**, **260L**) including second selector material layers **250L** and a second conductive material layer (e.g., second hardmask layer) **260L** described with reference to FIGS. **14A-14C** can be formed over the second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**). A photoresist layer can be deposited over the second conductive material layer **260L**, and can be lithographically patterned to form a two-dimensional array of second discrete patterned photoresist material portions **287**.

According to an aspect of the present disclosure, the two-dimensional array of second discrete patterned photoresist material portions **287** can be a periodic two-dimensional array having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. A first nearest-neighbor spacing along the first horizontal direction **hd1** of the two-dimensional array of second discrete patterned photoresist material portions **287** is greater than a second nearest-neighbor spacing along the second horizontal direction **hd2** of the two-dimensional array of second discrete patterned photoresist material portions **287**. In one embodiment, the pattern of the second discrete patterned photoresist material portions **287** may be the same as the pattern of the second discrete patterned photoresist material portions **257** described with reference to FIGS. **15A-15C**.

Referring to FIGS. **50A-50C**, an array-pattern-transfer process can be performed to transfer the pattern of the two-dimensional array of second discrete patterned photoresist material portions **287** through the second conductive material layer **260L** and the second selector material layers **250L**. For example, an anisotropic etch process can be performed to transfer the pattern in the two-dimensional array of second discrete patterned photoresist material portions **287** through the second conductive material layer **260L** and the second selector material layers **250L**. The patterned remaining portions of the second conductive material layer **260L** and the second selector material layers **250L** can include two-dimensional array of second selector-containing pillar structures **282**.

Each of the second selector-containing pillar structures **282** may comprise a second selector element **250** and a second conductive material plate **260**. Each second selector element **250** is a patterned portion of the second selector material layers **250L**, and each second conductive material plate **260** is a patterned portion of the second conductive material layer **260L**. Each second selector element **250** may include a vertical stack of a second lower selector electrode **252**, a second non-Ohmic material plate **252**, and a second upper selector electrode **253**. Each second lower selector electrode **252** is a patterned portion of the second lower selector electrode material layer **252L**. Each second non-Ohmic material plate **252** is a patterned portion of the second non-Ohmic material layer **252L**. Each second upper selector

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electrode **253** is a patterned portion of the second upper selector electrode material layer **253L**.

In one embodiment, the two-dimensional array of second selector-containing pillar structures **282** comprises a two-dimensional periodic array of second selector-containing pillar structures **282** having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. The nearest-neighbor spacing between neighboring pairs of the second selector-containing pillar structures **282** that are laterally spaced apart along the first horizontal direction **hd1** is greater than the nearest-neighbor spacing between neighboring pairs of the second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2**.

In one embodiment, each second selector-containing pillar structure **282** within the two-dimensional array of second selector-containing pillar structures **282** has a respective elongated horizontal cross-sectional shape having a third lateral dimension **ld5** along the first horizontal direction **hd1** and having a fourth lateral dimension **ld6** along the second horizontal direction **hd2** that is greater than the third lateral dimension **ld5**. In one embodiment, the ratio of the fourth lateral dimension **ld6** to the third lateral dimension **ld5** may be in a range from 1.2 to 4, such as from 1.5 to 3. The two-dimensional array of second selector-containing pillar structures **282** can be formed over the second electrically conductive layer **90L**.

Referring to FIGS. **51A-51D**, an anisotropic etch process can be performed to etch unmasked portions of the layer stack (**212L**, **214L**, **230L**, **244L**, **248L**) of the second continuous superlattice layer **212L**, the optional second continuous antiferromagnetic coupling layer **214L**, the second continuous magnetic tunnel junction (MTJ) material layers **230L**, the second continuous dielectric capping layer **244L**, and the second continuous metallic capping layer **248L** employing the two-dimensional array of second discrete patterned photoresist material portions **287** as an etch mask. In one embodiment, the anisotropic etch process may comprise an beam etch (IBE) process.

Each patterned portion of the layer stack (**212L**, **214L**, **230L**, **244L**, **248L**) comprises a second magnetic tunnel junction (MTJ) pillar structure **284**. A two-dimensional array of second magnetic tunnel junction (MTJ) pillar structures **284** can be formed underneath the two-dimensional array of second selector-containing pillar structures **282**. Each contiguous combination of a second selector-containing pillar structure **282** and a second MTJ pillar structure **284** constitutes a second memory cell **280**, which can function as a memory cell **180** described with reference to FIG. **2**.

Each second MTJ pillar structure **284** comprises a stack of a second superlattice layer **212**, a second antiferromagnetic coupling layer **214**, a second magnetic tunnel junction **230**, a second dielectric capping layer **244**, and a second metallic capping layer **248**. The second magnetic tunnel junction **230** includes a second reference layer **232**, a second tunnel barrier layer **234**, and a second free layer **236**. Each second superlattice layer **212** is a patterned portion of the second continuous superlattice layer **212L**. Each second antiferromagnetic coupling layer **214** is a patterned portion of the second continuous antiferromagnetic coupling layer **214L**. Each second magnetic tunnel junction **230** is a patterned portion of the second magnetic tunnel junction material layers **230L**. Each second dielectric capping layer **244** is a patterned portion of the second continuous dielectric capping layer **244L**. Each second metallic capping layer **248** is a patterned portion of the second continuous metallic cap-

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ping layer **248L**. Each second reference layer **232** is a patterned portion of the second continuous reference layer **232L**. Each second tunnel barrier layer **234** is a patterned portion of the second continuous tunnel barrier layer **234L**. Each second free layer **236** is a patterned portion of the second continuous free layer **236L**.

According to an aspect of the present disclosure, the transfer of the pattern in the two-dimensional array of second discrete patterned photoresist material portions **287** via the pattern in the two-dimensional array of second conductive material plates **260** through the second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) can be performed employing an aspect-ratio-dependent ion beam etch process that etches materials of the second MTJ-level material layers (**212L**, **214L**, **230L**, **244L**, **248L**) with dependency on the aspect ratio of the local geometry.

In one embodiment, the ion beam etch process etches an area with a smaller aspect ratio at a higher etch rate than an area with a large aspect ratio. In this case, areas having a greater lateral distance between neighboring pairs of second electrically conductive plates **260** are etched at a higher etch rate than areas having a lesser lateral distance between neighboring pairs of second electrically conductive plates **260**.

In one embodiment, the two-dimensional array of second electrically conductive plates **260** may comprise columns of second electrically conductive plates **260** that are arranged along the second horizontal direction **hd2**. Each column of second electrically conductive plates **260** may comprise a respective subset of second electrically conductive plates **260** that are arranged along the second horizontal direction **hd2**. The gap between neighboring pairs of second electrically conductive plates **260** within each column of second electrically conductive plates **260** may be less than the gap between neighboring columns of second electrically conductive plates **260**. In this case, the etch rate of the materials of the second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) between neighboring columns of second electrically conductive plates **260** can be higher than the etch rate of the materials of the second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) between neighboring pairs of second electrically conductive plates **260** within each column of second electrically conductive plates **260**.

According to an aspect of the present disclosure, the ion beam etch process can anisotropically etch the second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) and the second electrically conductive layer **90L** such that physically exposed surfaces of remaining portions of the second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) are formed with taper angles. The taper angles can be measured with respect to the vertical direction that is perpendicular to the top surface of the substrate **8**. In one embodiment, the taper angle may be in a range from 3 degrees to 30 degrees, such as from 6 degrees to 20 degrees, although lesser and greater taper angles may also be employed.

Generally, patterned portions of the second selector-level material layers (**250L**, **260L**) comprise a two-dimensional array of selector-containing pillar structures **282** including a respective selector element **250** and a respective second conductive material plate **260**, and patterned portions of the second magnetic tunnel junction-level (MTJ-level) material layers (**212L**, **214L**, **230L**, **244L**, **248L**) comprise a two-

dimensional array of second magnetic tunnel junction (MTJ) pillar structures **284**. Patterned portions of the second magnetic tunnel junction material layers **230L** comprise a two-dimensional array of second magnetic tunnel junctions (MTJs) **230**.

According to an aspect of the present disclosure, the duration of the etch process can be selected such that portions of the second electrically conductive layer **90L** located in areas between neighboring columns of second conductive material plates **260** are etched through, while portions of the second electrically conductive material layer **90L** located in areas between neighboring pairs of second conductive material plates **260** within each column of second conductive material plates **260** are not etched through. Thus, the second electrically conductive layer **90L** is divided into multiple disjointed patterned electrically conductive strips that laterally extend along the second horizontal direction **hd2**, which constitute second electrically conductive lines **90**. In other words, the patterned portions of the second electrically conductive layer **90L** comprise second electrically conductive lines **90** that laterally extend along the second horizontal direction **hd2** and laterally spaced apart from each other along the first horizontal direction **hd1**. The second electrically conductive lines **90** include wider portions **w3** and narrower neck portions **w4** which have a narrower width than portions **w3** in the first horizontal direction. In one embodiment, each of the second electrically conductive lines **90** comprises a laterally alternating sequence of uniform thickness segments that underlie a respective one of the second MTJ pillar structures **284** and indented segments that includes a V-shaped indentation in a respective top surface segment.

According to an aspect of the present disclosure, each of the second electrically conductive lines **90** may be patterned with a respective pair of contoured and tapered lengthwise sidewalls that laterally extend along the second horizontal direction **hd2**. Each of the second electrically conductive lines **90** may be formed with a respective bottom surface having a respective variable width along the first horizontal direction **hd1** that varies along the second horizontal direction **hd2**. In one embodiment, each of the second electrically conductive lines **90** may comprise a periodic repetition of uniform-width regions **w3** and neck regions **w4** with a periodicity of the second pitch **p2**.

According to an aspect of the present disclosure, each of the second electrically conductive lines **90** may be patterned with a respective pair of contoured and tapered lengthwise sidewalls that laterally extend along the second horizontal direction **hd2**. Each of the second electrically conductive lines **90** may be formed with a respective bottom surface having a respective variable width along the first horizontal direction **hd1** that varies along the second horizontal direction **hd2**. In one embodiment, each of the second electrically conductive lines **90** may comprise a periodic repetition of uniform-width regions **w3** and neck regions **w4** with a periodicity of the second pitch **p2**.

In one embodiment, each of the second electrically conductive lines **90** comprises a pair of contoured lengthwise sidewalls that generally extend along the second horizontal direction **hd2** with a lateral undulation along the first horizontal direction **hd1**, and each of the contoured lengthwise sidewalls comprises straight segments located in a respective uniform width region and laterally extending along the second horizontal direction **hd2**, and pairs of adjoining convex sidewalls adjoining at a respective vertically-extending edge located at a respective neck region.

The two-dimensional array of second MTJ pillar structures **284** comprises columns of MTJ pillar structures **284** that are arranged along the second horizontal direction **hd2**. In one embodiment, tapered sidewalls of each neighboring pair of MTJ pillar structures **284** within each column of second MTJ pillar structures **284** are adjoined to each other at a respective edge. According to an aspect of the present disclosure, the edges are formed above a horizontal plane including top surfaces of the second electrically conductive lines **90** during patterning of the second MTJ-level material layers (**212L**, **214L**, **230L**, **244L**, **248L**). The edges laterally extending along the first horizontal direction **hd1**.

In one embodiment, the edges at which a respective pair of tapered sidewalls of the second MTJ pillar structures **284** merge (i.e., are joined to each other) are formed below the horizontal plane including the bottom surfaces of the second free layers **236**. In one embodiment, the edges at which a respective pair of tapered sidewalls of the second MTJ pillar structures **284** merge may be formed within the second continuous nonmagnetic tunnel barrier layer **234L**, within the second continuous reference layer **232L**, or below the second continuous reference layer **232L**, similar to the embodiments illustrated in FIGS. **47E-47H**.

Each vertical stack of a second selector-containing pillar structure **282** and a second MTJ pillar structure **284** constitutes a second memory cell **280**. According to an aspect of the present disclosure, each column of second memory cells **280** that are arranged along the second horizontal direction **hd2** are merged with each other below the horizontal plane including the bottom surfaces of the second free layers **236** within the respective column of second memory cells **280**. Each merged column of second memory cells **280** is herein referred to as second selector-magnetic tunnel junction (selector-MTJ) assembly **280A**. The third exemplary structure can include columns of second selector-magnetic tunnel junction (selector-MTJ) assemblies **280A** located on a respective one of the second electrically conductive lines **90**. Each of the selector-MTJ assemblies **280A** comprises a respective column of second magnetic tunnel junction (MTJ) pillar structures **284** and a respective column of second selector-containing pillar structures **282** that are arranged along the second horizontal direction **hd2**. Tapered sidewalls of each neighboring pair of second MTJ pillar structures **284** within the respective column of second MTJ pillar structures **284** are adjoined to each other at a respective edge laterally extending along the second horizontal direction **hd1** and located above a horizontal plane including top surfaces of the second electrically conductive lines **90**.

Each of the second MTJ pillar structures **284** comprises a vertical stack including a second reference layer **232**, a second tunnel barrier layer **234**, and a second free layer **236**. First free layers **236** within the respective row of second MTJ pillar structures are laterally spaced apart among one another along the second horizontal direction **hd2** and do not contact one another.

In one embodiment, the columns of second selector-containing pillar structures **282** are arranged as a two-dimensional periodic array of second selector-containing pillar structures **282** having a first pitch **p1** along the first horizontal direction **hd1** and having a second pitch **p2** along the second horizontal direction **hd2**. A lateral spacing between neighboring pairs of second selector-containing pillar structures **282** that are laterally spaced apart along the first horizontal direction **hd1** is greater than a lateral spacing between neighboring pairs of second selector-containing pillar structures **282** that are laterally spaced apart along the second horizontal direction **hd2**.



In one embodiment, each of the selector-containing pillar structures **282** has an elongated horizontal cross-sectional shape having a greater lateral dimension along the second horizontal direction **hd2** than along the first horizontal direction **hd1**, and the columns of selector-containing pillar structures **282** are arranged as a two-dimensional periodic array of selector-containing pillar structures **282** having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. In one embodiment, the second pitch **p2** is the same as the first pitch **p1**.

In one embodiment, each free layer **236** within the two-dimensional array of MTJ pillar structures **284** has a respective horizontal cross-sectional shape having a same lateral extent along the first horizontal direction **hd1** and along the second horizontal direction **hd2**. In one embodiment, each free layer **236** may have a horizontal cross-sectional shape of a circle, a square, or a rounded square, i.e., a shape that is derived from a square by rounding the four corners.

Referring to FIGS. **52A-52C**, a dielectric fill material can be deposited in gaps between neighboring pairs of second memory cells **280** to form a dielectric matrix layer, which is herein referred to as a second dielectric matrix layer **286**. Excess portions of the dielectric fill material of the second dielectric matrix layer **286** can be removed from above a horizontal plane including top surfaces of the second selector-containing pillar structures **282** by a planarization process such as a chemical mechanical polishing process. A top surface of a remaining portion of the second dielectric matrix layer **286** is formed within a horizontal plane including top surfaces of the two-dimensional array of second selector-containing pillar structures **282**. In an alternative configuration, the CMP process is continued to also remove the second hardmask plates **260** and to expose the upper surface of the second upper selector electrodes **253**, similar to the step shown in FIGS. **9D-9F**.

Referring to FIGS. **53A-53C**, a line-level dielectric layer **332** can be formed by depositing a dielectric material over the two-dimensional array of second memory cells **280**. Line trenches laterally extending along the first horizontal direction **hd1** can be formed above each row of second MTJ pillar structures **284**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the line-level dielectric layer **332**. Remaining portions of the conductive material filling the line trenches constitute third electrically conductive lines **330**. The third electrically conductive lines **330** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the third electrically conductive lines **330** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed. Alternatively, instead of using the above described damascene process to form the second electrically conductive lines **90**, these lines may be formed by a pattern and etch process.

Referring to FIGS. **54A-54C**, a first alternative configuration of the third exemplary structure can be derived from the third exemplary structure illustrated in FIGS. **53A-53C** by modifying the ion beam etching process that patterns the second MTJ pillar structures **284**. In this case, the height of the edges of the V-shaped indentations in the first selector-MTJ assemblies **180A** may be changed. The edges of the V-shaped indentations in the second selector-MTJ assemblies **280A** may be formed anywhere between the horizontal plane including the top surfaces of the second electrically

conductive lines **90** and the horizontal plane including the bottom surfaces of the second free layers **236**.

In one embodiment, a second superlattice layer **212** is located underneath and is magnetically coupled to a second reference layer **232** within each second MTJ pillar structure **284** in a second selector-MTJ assembly **280A**. The second superlattice layers **212** within the respective column of MTJ pillar structures **284** in each second selector-MTJ assembly **280A** may be interconnected as a single continuous structure, and may extend underneath the respective column of selector-containing pillar structures **282**. In one embodiment, the edges at which tapered sidewalls of a respective neighboring pair of MTJ pillar structures **284** are adjoined may be located at V-shaped indentations in a top surface of the single continuous structure.

Referring to FIGS. **55A-55C**, a second alternative configuration of the third exemplary structure can be derived from the third exemplary structure illustrated in FIGS. **53A-53C** by modifying the ion beam etch process that patterns the second MTJ pillar structures **284**. In one embodiment, the second antiferromagnetic coupling layer **214** is located underneath the second reference layer **232** within each second MTJ pillar structure **284** in the second selector-MTJ assembly **280A**. The second antiferromagnetic coupling layers **214** within the respective column of MTJ pillar structures **284** in each second selector-MTJ assembly **280A** may be interconnected as a single continuous antiferromagnetic coupling structure, and may extend underneath the respective column of selector-containing pillar structures **282**. In one embodiment, the edges at which tapered sidewalls of a respective neighboring pair of MTJ pillar structures **284** are adjoined may be located at V-shaped indentations in a top surface of the single continuous antiferromagnetic coupling structure.

Referring to FIGS. **56A-56C**, a third alternative configuration of the third exemplary structure can be derived from the third exemplary structure illustrated in FIGS. **53A-53C** by modifying the ion beam etch process that patterns the second MTJ pillar structures **284**. In one embodiment, the second reference layers **232** within the respective column of second MTJ pillar structures **284** in the second selector-MTJ assembly **280A** are interconnected as a single continuous reference structure underlying the respective row of selector-containing pillar structures **282**. In one embodiment, the edges at which tapered sidewalls of a respective neighboring pair of second MTJ pillar structures **284** are adjoined are located at V-shaped indentations in a top surface of the single continuous reference structure.

Referring to FIGS. **57A-57C**, a fourth alternative configuration of the third exemplary structure can be derived from the third exemplary structure illustrated in FIGS. **53A-53C** by modifying the ion beam etch process that patterns the second MTJ pillar structures **284**. In one embodiment, the second nonmagnetic tunnel barrier layers **234** within the respective column of second MTJ pillar structures **284** in the second selector-MTJ assembly **280A** are merged as a single continuous nonmagnetic tunnel barrier structure underlying the respective column of second selector-containing pillar structures **282**. The edges at which tapered sidewalls of a respective neighboring pair of second MTJ pillar structures **284** are adjoined are located at V-shaped indentations in a top surface of the single continuous nonmagnetic tunnel barrier structure.

Referring to FIGS. **58A-58C**, a fourth exemplary structure according to a fourth embodiment of the present disclosure may be the same as the third exemplary structure illustrated



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in FIGS. 45A-45C prior to formation of the two-dimensional array of first discrete patterned photoresist material portions 187.

Referring to FIGS. 59A-59C, a two-dimensional array of first discrete patterned resist material portions 187 can be formed over the top surface of the first conductive material layer 160L. The resist material portions may comprise electron beam resist or photoresist material portions. According to an aspect of the present disclosure, the two-dimensional array of first discrete patterned resist material portions 187 can be a periodic two-dimensional array having a first pitch p1' along a first horizontal direction hd1 and having a second pitch p2' along a second horizontal direction hd2. A first nearest-neighbor spacing s1 along the first horizontal direction hd1 of the two-dimensional array of first discrete patterned resist material portions 187 is less than a second nearest-neighbor spacing s2 along the second horizontal direction hd2 of the two-dimensional array of first discrete patterned resist material portions 187.

Each of the first discrete patterned resist material portions 187 may have a respective horizontal cross-sectional shape of a circle. In one embodiment, the maximum lateral dimension of each of the first discrete patterned resist material portions 187 along the first horizontal direction hd1 may be the same as, or may be substantially the same as, the maximum lateral dimension of each of the first discrete patterned resist material portions 187 along the second horizontal direction hd2. In this case, the first pitch p1' is different from the second pitch p2'. In one embodiment, the second pitch p2' is greater than the first pitch p1'. The ratio of the second pitch p2' to the first pitch p1' may be in a range from 1.2 to 4, such as from 1.5 to 3, although lesser and greater ratios may also be employed.

Referring to FIGS. 60A-60C, the processing steps of FIGS. 46A-46C can be performed to form the two-dimensional array of first selector-containing pillar structures 182. Each first selector-containing pillar structure 182 may comprise a first selector element 150 and a first conductive material plate 160. Each first selector element 150 is a patterned portion of the first selector material layers 150L, and each first conductive material plate 160 is a patterned portion of the first conductive material layer 160L. Each first selector element 150 may include a vertical stack of a first lower selector electrode 151, a first non-Ohmic material plate 152, and a first upper selector electrode 153. Each first lower selector electrode 151 is a patterned portion of the first lower selector electrode material layer 151L. Each first non-Ohmic material plate 152 is a patterned portion of the first non-Ohmic material layer 152L. Each first upper selector electrode 153 is a patterned portion of the first upper selector electrode material layer 153L. In one embodiment, the two-dimensional array of first selector-containing pillar structures 182 comprises a two-dimensional periodic array of first selector-containing pillar structures 182 having the first pitch p1' along the first horizontal direction hd1 and having the second pitch p2' along the second horizontal direction hd2. In one embodiment, each of the first selector-containing pillar structures 182 may have a same lateral extent along the second horizontal direction hd2 as along the first horizontal direction hd1.

Referring to FIGS. 61A-61D, the ion beam etch process can be performed to etch unmasked portions of the layer stack (112L, 114L, 130L, 144L, 148L) employing the two-dimensional array of second discrete patterned resist material portions 187 and/or the first conductive material plates

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(i.e., hard mask plates) 160 as an etch mask. The processing steps described with reference to FIGS. 47A-47D may be performed.

Each patterned portion of the layer stack (112L, 114L, 130L, 144L, 148L) comprises a first magnetic tunnel junction (MTJ) pillar structure 184. A two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures 184 can be formed underneath the two-dimensional array of first selector-containing pillar structures 182. Each contiguous combination of a first selector-containing pillar structure 182 and a first MTJ pillar structure 184 constitutes a first memory cell 180, which can function as a memory cell 180 described with reference to FIG. 2.

The gap g1 (which may be about the same as spacing s1) between neighboring pairs of first electrically conductive plates 160 within each row of first electrically conductive plates 160 may be less than the gap g2 (which may be about the same as spacing s2) between neighboring rows of first electrically conductive plates 160. In this case, the etch rate of the materials of the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) between neighboring rows of first electrically conductive plates 160 can be higher than the etch rate of the materials of the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) between neighboring pairs of first electrically conductive plates 160 within each row of first electrically conductive plates 160 due to the IBE shadowing effect. The first MTJ pillar structures 184 can be formed with tapered sidewalls employing an aspect-ratio-dependent ion beam etch process. The lateral dimensions of the first conductive material plates 160, the first pitch p1', and the second pitch p2' are selected such that portions of the first electrically conductive layer 30L located between neighboring rows of first conductive material plates 160 are removed while portions of the first electrically conductive layer within each row of the first conductive material plates 160 are not removed. In one embodiment, the tapered surfaces of neighboring pairs of first MTJ pillar structures 184 within each row of first MTJ pillar structures 184 arranged along the first horizontal direction hd1 merge above the horizontal plane including the top surface of the first electrically conductive layer 30L, as described above.

The ion beam etch process can anisotropically etch the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) and the first electrically conductive layer 30L such that physically exposed surfaces of remaining portions of the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) are formed with taper angles. The taper angles can be measured with respect to the vertical direction that is perpendicular to the top surface of the substrate 8. In one embodiment, the taper angle may be in a range from 3 degrees to 30 degrees, such as from 6 degrees to 20 degrees, although lesser and greater taper angles may also be employed.

Generally, patterned portions of the first selector-level material layers (150L, 160L) comprise a two-dimensional array of selector-containing pillar structures 182 including a respective selector element 150 and a respective first conductive material plate 160, and patterned portions of the first magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L) comprise a two-dimensional array of first magnetic tunnel junction (MTJ) pillar structures 184. Patterned portions of the first magnetic tunnel junction material layers 130L comprise a two-dimensional array of first magnetic tunnel junctions (MTJs) 130.

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In one embodiment, the edges at which a respective pair of tapered sidewalls of the first MTJ pillar structures **184** merge (i.e., are joined to each other) are formed below the horizontal plane including the bottom surfaces of the first free layers **136**. In one embodiment, the edges at which a respective pair of tapered sidewalls of the first MTJ pillar structures **184** merge may be formed within the first continuous nonmagnetic tunnel barrier layer **134L**, within the first continuous reference layer **132L**, or below the first continuous reference layer **132L**, as described above with respect to FIGS. **47E** to **47G** above. Alternatively, in the illustrated example of FIGS. **61A-61D**, the edges at which a respective pair of tapered sidewalls of the MTJ pillar structures **184** are not merged above the top surface of the first electrically conductive lines **30**.

According to an aspect of the present disclosure, each of the first electrically conductive lines **30** may be patterned with a respective pair of contoured and tapered lengthwise sidewalls that laterally extend along the first horizontal direction **hd1**. Each of the first electrically conductive lines **30** may be formed with a respective bottom surface having a respective variable width along the second horizontal direction **hd2** that varies along the first horizontal direction **hd1**. In one embodiment shown in FIG. **61D**, each of the first electrically conductive lines **30** may comprise a periodic repetition of wider bulging regions **w1** and narrower neck regions **w2** with a periodicity of the first pitch **p1**. In one embodiment, each bulging region may have a bottom surface having a uniform radius of curvature (i.e., having a shape of an arc of a circle).

In one embodiment, each of the first electrically conductive lines **30** comprises a pair of contoured lengthwise sidewalls that generally extend along the first horizontal direction **hd1** with a lateral undulation along the second horizontal direction **hd2**, and each of the contoured lengthwise sidewalls comprises curved segments **w1** having a uniform radius of curvature at any given height, and pairs of adjoined convex sidewalls adjoined at a respective vertically-extending edge located at a respective neck region **w2**.

Each vertical stack of a first selector-containing pillar structure **182** and a first MTJ pillar structure **184** constitutes a first memory cell **180**. According to an aspect of the present disclosure, each row of first memory cells **180** that are arranged along the first horizontal direction **hd1** are merged with each other below the horizontal plane including the bottom surfaces of the first free layers **136** within the respective row of first memory cells **180**. Each merged row of first memory cells **180** is herein referred to as first selector-magnetic tunnel junction (selector-MTJ) assembly **180A**. The fourth exemplary structure can include rows of first selector-magnetic tunnel junction (selector-MTJ) assemblies **180A** located on a respective one of the first electrically conductive lines **30**. Each of the selector-MTJ assemblies **180A** comprises a respective row of first magnetic tunnel junction (MTJ) pillar structures **184** and a respective row of first selector-containing pillar structures **182** that are arranged along the first horizontal direction **hd1**. Tapered sidewalls of each neighboring pair of first MTJ pillar structures **184** within the respective row of first MTJ pillar structures **184** are adjoined to each other at a respective edge laterally extending along the second horizontal direction **hd2** and located above a horizontal plane including bottom surfaces of the first electrically conductive lines **30**.

In one embodiment, the rows of first selector-containing pillar structures **182** are arranged as a two-dimensional periodic array of first selector-containing pillar structures **182** having a first pitch **p1'** along the first horizontal direc-

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tion **hd1** and having a second pitch **p2'** long the second horizontal direction **hd2**. A lateral spacing **g1** between neighboring pairs of first selector-containing pillar structures **182** that are laterally spaced apart along the first horizontal direction **hd1** is less than a lateral spacing **g2** between neighboring pairs of first selector-containing pillar structures **182** that are laterally spaced apart along the second horizontal direction **hd2**.

In one embodiment, each of the selector-containing pillar structures **182** has an elongated horizontal cross-sectional shape having a greater lateral dimension along the first horizontal direction **hd1** than along the second horizontal direction **hd2**, and the rows of selector-containing pillar structures **182** are arranged as a two-dimensional periodic array of selector-containing pillar structures **182** having the first pitch **p1'** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2'**. In one embodiment, the second pitch **p2** is greater than the first pitch **p1**.

Each of the first MTJ pillar structures **184** comprises a vertical stack including a first reference layer **132**, a first tunnel barrier layer **134**, and a first free layer **136**. First free layers **136** within the respective row of first MTJ pillar structures are laterally spaced apart from each other along the first horizontal direction **hd1** and do not contact one another.

In one embodiment, each free layer **136** within the two-dimensional array of MTJ pillar structures **184** has a respective horizontal cross-sectional shape having a same lateral extent along the first horizontal direction **hd1** and along the second horizontal direction **hd2**. In one embodiment, each free layer **136** may have a horizontal cross-sectional shape of a circle.

Referring to FIGS. **62A-62C**, a dielectric fill material can be deposited in gaps between neighboring pairs of first memory cells **180** to form a dielectric matrix layer, which is herein referred to as a first dielectric matrix layer **186**. Excess portions of the dielectric fill material of the first dielectric matrix layer **186** can be removed from above a horizontal plane including top surfaces of the first selector-containing pillar structures **182** by a planarization process such as a chemical mechanical polishing process. A top surface of a remaining portion of the first dielectric matrix layer **186** is formed within a horizontal plane including top surfaces of the two-dimensional array of first selector-containing pillar structures **182**. In an alternative configuration, the CMP process is continued to also remove the first hardmask plates **160** and to expose the upper surface of the first upper selector electrodes **153**, similar to the step shown in FIGS. **9D-9F**.

Referring to FIGS. **63A-63C**, a line-level dielectric layer **92** can be formed by depositing a dielectric material over the two-dimensional array of first memory cells **180**. Line trenches laterally extending along the second horizontal direction **hd2** can be formed above each column of first MTJ pillar structures **184**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the line-level dielectric layer **92**. Remaining portions of the conductive material filling the line trenches constitute second electrically conductive lines **90**. The second electrically conductive lines **90** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the second electrically conductive lines **90** can be in a range from 20 nm to 100 nm, although lesser and greater

thicknesses can also be employed. Alternatively, instead of using the above described damascene process to form the second electrically conductive lines **90**, these lines may be formed by a pattern and etch process.

Generally, the height of the edges of the V-shaped indentations in the first selector-MTJ assemblies **180A** may be changed as in the alternative configurations of the third exemplary structure. Specifically, the edges of the V-shaped indentations in the first selector-MTJ assemblies **180A** may be formed anywhere between the horizontal plane including the top surfaces of the first electrically conductive lines **30** and the horizontal plane including the bottom surfaces of the first free layers **136**.

Referring to FIGS. **64A-64C**, a first alternative configuration of the fourth exemplary structure can be derived from the fourth exemplary structure illustrated in FIGS. **63A-63C** by modifying at least one of the ion beam etch processes that patterns the first MTJ pillar structures **184** and/or the second MTJ pillar structures **284**. In one embodiment, a first anti-ferromagnetic coupling layer **114** is located underneath a first reference layer **132** within each first MTJ pillar structure **184** in a first selector-MTJ assembly **180A**. The first anti-ferromagnetic coupling layers **114** within the respective row of MTJ pillar structures **184** in each first selector-MTJ assembly **180A** may be interconnected as a single continuous anti-ferromagnetic coupling structure, and may extend underneath the respective row of selector-containing pillar structures **182**. In one embodiment, the edges at which tapered sidewalls of a respective neighboring pair of MTJ pillar structures **184** are adjoined may be located at V-shaped indentations in a top surface of the single continuous anti-ferromagnetic coupling structure.

Referring to FIGS. **65A-65C**, a second alternative configuration of the fourth exemplary structure can be derived from the fourth exemplary structure illustrated in FIGS. **63A-63C** by modifying at least one of the ion beam etch processes that patterns the first MTJ pillar structures **184** and/or the second MTJ pillar structures **284**. In one embodiment, the first nonmagnetic tunnel barrier layers **134** within the respective row of first MTJ pillar structures **184** in a first selector-MTJ assembly **180A** are merged as a single continuous nonmagnetic tunnel barrier structure underlying the respective row of first selector-containing pillar structures **182**. The edges at which tapered sidewalls of a respective neighboring pair of first MTJ pillar structures **184** are adjoined are located at V-shaped indentations in a top surface of the single continuous nonmagnetic tunnel barrier structure.

In other alternative embodiments, the edges at which tapered sidewalls of a respective neighboring pair of first MTJ pillar structures **184** are adjoined are located at V-shaped indentations in a top surface of the single continuous superlattice structure, as described above with respect to FIG. **47E**, or in a top surface of the single continuous reference structure, as described above with respect to FIG. **47G**.

In the third and fourth embodiments, the bottom electrically conductive lines (e.g., word lines) are formed without an additional lithography step and are self-aligned with the memory bits (e.g., MRAM cells **180**). The MRAM/selector film stack may be deposited onto a polished dielectric substrate without potential roughness/topography caused by deposition onto pre-patterned bottom word lines. By using a rectangular lattice, this method can significantly boost the areal density of MRAM die.

Referring collectively to FIGS. **1, 2, 44A-65C**, and other related drawings of the instant application, a memory array

is provided, which comprises: first electrically conductive lines **30** laterally extending along a first horizontal direction **hd1** and laterally spaced apart along a second horizontal direction **hd2**; rows of selector-magnetic tunnel junction (selector-MTJ) assemblies **180A** located on a respective one of the first electrically conductive lines **30**, wherein each of the selector-MTJ assemblies **180A** comprises a respective row of magnetic tunnel junction (MTJ) pillar structures **184** and a respective row of selector-containing pillar structures **182** that are arranged along the first horizontal direction **hd1**, and a lateral spacing between neighboring pairs of selector-containing pillar structures **182** that are laterally spaced apart along the first horizontal direction **hd1** is less than a lateral spacing between neighboring pairs of selector-containing pillar structures **182** that are laterally spaced apart along the second horizontal direction **hd2**; and second electrically conductive lines **90** laterally extending along the second horizontal direction **hd2** and overlying a respective column of the selector-MTJ assemblies **180A**.

In one embodiment, each of the first electrically conductive lines **30** has a respective variable width along the second horizontal direction **hd2** that varies along the first horizontal direction **hd1**.

In one embodiment, each of the selector-MTJ assemblies **180A** includes a respective selector-containing pillar structure **182** overlying a respective one of the MTJ pillar structures **184**; and the selector-containing pillar structures **182** are arranged as a two-dimensional periodic array of the selector-containing pillar structures having a first pitch (**p1** or **p1'**) along the first horizontal direction **hd1** and having a second pitch (**p2** or **p2'**) along the second horizontal direction **hd2**.

In the third embodiment, each of the selector-containing pillar structures **182** has a first lateral dimension **ld1** along the first horizontal direction **hd1** and has a second lateral dimension **ld2** along the second horizontal direction **hd2** that is less than the first lateral dimension **ld1**; and the second pitch **p2** is the same as the first pitch **p1**.

In the fourth embodiment, each of the selector-containing pillar structures **182** has a same lateral extent along the second horizontal direction **hd2** as along the first horizontal direction **hd1**; and the second pitch **p2'** is greater than the first pitch **p1'**. Each of the selector-containing pillar structures **182** has a circular horizontal cross-sectional shape.

In one embodiment, tapered sidewalls of each neighboring pair of MTJ pillar structures **184** within the respective row of MTJ pillar structures **184** are adjoined to each other at a respective edge located above a horizontal plane including top surfaces of the first electrically conductive lines **30**. Each of the MTJ pillar structures **184** comprises a vertical stack including a reference layer **132**, a tunnel barrier layer **134**, and a free layer **136**; and free layers **136** within the respective row of MTJ pillar structures **184** are laterally spaced apart from each other along the first horizontal direction **hd1** and do not contact one another.

In one embodiment, reference layers **132** within the respective row of MTJ pillar structures **184** are interconnected as a single continuous reference structure underlying the respective row of selector-containing pillar structures **182**. The edges at which tapered sidewalls of a respective neighboring pair of MTJ pillar structures **184** are adjoined are located at V-shaped indentations in a top surface of the single continuous reference structure.

In one embodiment, nonmagnetic tunnel barrier layers **134** within the respective row of MTJ pillar structures **184** are interconnected as a single continuous nonmagnetic tunnel barrier structure underlying the respective row of selec-

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tor-containing pillar structures **182**. The edges at which tapered sidewalls of a respective neighboring pair of MTJ pillar structures **184** are adjoined are located at V-shaped indentations in a top surface of the single continuous non-magnetic tunnel barrier structure.

In one embodiment, a superlattice layer **112** is located underneath the reference layer **132** within each vertical stack, and superlattice layers **112** within the respective row of MTJ pillar structures **184** are interconnected as a single continuous superlattice structure that extends underneath the respective row of selector-containing pillar structures **182**. In one embodiment, the edges at which tapered sidewalls of a respective neighboring pair of MTJ pillar structures **184** are adjoined are located at V-shaped indentations in a top surface of the single continuous superlattice structure.

In one embodiment, an antiferromagnetic coupling layer **114** is located underneath the reference layer **132** within each vertical stack, and antiferromagnetic coupling layers **114** within the respective row of MTJ pillar structures **184** are interconnected as a single continuous antiferromagnetic coupling structure that extends underneath the respective row of selector-containing pillar structures **182**. The edges at which tapered sidewalls of a respective neighboring pair of MTJ pillar structures **184** are adjoined are located at V-shaped indentations in a top surface of the single continuous antiferromagnetic coupling structure.

In one embodiment, each of the first electrically conductive lines **30** comprises a laterally alternating sequence of uniform thickness segments that underlie a respective one of the MTJ pillar structures and indented segments that includes a V-shaped indentation in a respective top surface.

In the third embodiment, each of the first electrically conductive lines **30** comprises a pair of contoured lengthwise sidewalls that generally extend along the first horizontal direction **hd1** with a lateral undulation along the second horizontal direction **hd2**. Each of the contoured lengthwise sidewalls comprises straight segments laterally extending along the first horizontal direction **hd1** and pairs of adjoined convex sidewalls adjoined at a respective vertically-extending edge.

In the fourth embodiment, each of the first electrically conductive lines **30** comprises a periodic repetition of wider bulging regions **w1** and narrower neck regions **w2**; and each bulging region **w1** has a bottom surface having a uniform radius of curvature.

Referring to FIGS. **66A-66C**, a fifth exemplary structure according to a fifth embodiment of the present disclosure is illustrated, which can be derived from the fourth exemplary structure illustrated in FIGS. **58A-58C** by forming an optional sacrificial capping material layer **166L** over the selector-level material layers (**150L**, **160L**). If layer **160L** comprises a conductive material layer which remains in the final device, then the sacrificial capping material layer **166L** comprises a sacrificial material that may be subsequently employed as hardmask material during a subsequent anisotropic etch process that patterns the first electrically conductive layer **30L**. Alternatively, if layer **160L** comprises a sacrificial hardmask material, then the sacrificial capping material layer **166L** may be omitted. In one embodiment, the sacrificial capping material layer **166L** comprises a dielectric material such as silicon oxide, silicon nitride, or a metal oxide (e.g., aluminum oxide). The thickness of the sacrificial capping material layer **166L** may be in a range from 3 nm to 100 nm, such as from 10 nm to 30 nm, although lesser and greater thicknesses may also be employed.

A photoresist layer may be applied over the sacrificial capping material layer **166L** (if present), and can be litho-

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graphically patterned into a two-dimensional array of discrete photoresist material portions **187**. The two-dimensional array of discrete photoresist material portions **187** may be formed as a periodic two-dimensional array having a first pitch **p1** along a first horizontal direction **hd1** and having a second pitch **p2** along a second horizontal direction **hd2**. The second pitch **p2** may be the same as, or may be different from, the first pitch **p1**. Each of the discrete patterned photoresist material portions **187** may have a respective horizontal cross-sectional shape of a rectangle, a rounded rectangle, an oval, or a circle. The dimensions and shapes of the discrete patterned photoresist material portions **187** may be selected to provide a geometry that is conducive to subsequent patterning of the first electrically conductive layer **30L**. In one embodiment, each of the discrete patterned photoresist material portions **187** may have a circular horizontal cross-sectional shape or an elongated horizontal cross-sectional shape. The lateral dimensions of each discrete patterned photoresist material portions **187** may be in a range from 3 nm to 300 nm, such as from 10 nm to 100 nm, although lesser and greater lateral dimensions may also be employed. The first pitch **p1** and the second pitch **p2** may be in a range from 6 nm to 200 nm, such as from 20 nm to 80 nm, although lesser and greater dimensions may also be employed.

Referring to FIGS. **67A-67C**, one or more pattern transfer processes may be performed transfer the pattern in the two-dimensional array of discrete photoresist material portions **187** through the sacrificial capping material layer **166L** (if present), the selector-level material layers (**150L**, **160L**), and the magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**). The first electrically conductive layer **30L** can be employed as an etch stop material layer. The sacrificial capping material layer **166L** may be patterned into a two-dimensional array of sacrificial capping material plates **166**. The selector-level material layers (**150L**, **160L**) is patterned into a two-dimensional array of selector-containing pillar structures **182**. The MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be patterned into a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures **184** containing a two-dimensional array of magnetic tunnel junctions **130**.

In one embodiment, the two-dimensional array of discrete photoresist material portions **187** may be employed as an etch mask throughout the one or more pattern transfer processes. Alternatively, the two-dimensional array of discrete photoresist material portions **187** may be consumed during the one or more pattern transfer processes, and the two-dimensional array of sacrificial capping material plates **166** may be employed as an etch mask at least during a terminal step of the one or more pattern transfer processes. Yet alternatively, the two-dimensional array of discrete photoresist material portions **187** may be removed after etching a subset of layers within the sacrificial capping material layer **166L**, the selector-level material layers (**150L**, **160L**), or the magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**), and the two-dimensional array of sacrificial capping material plates **166** may be employed as an etch mask at least during a terminal step of the one or more pattern transfer processes. In an illustrative example, a reactive ion etch process may be employed to pattern the sacrificial capping material layer **166L** and the selector-level material layers (**150L**, **160L**), the two-dimensional array of discrete photoresist material portions **187** may be removed, for example, by ashing, and an ion beam etch process employing the two-dimensional array of sacrificial capping material plates **166** may be performed

to pattern the magnetic tunnel junction-level (MTJ-level) material layers (112L, 114L, 130L, 144L, 148L).

Generally, the sacrificial capping material layer 166L can be patterned into sacrificial capping material plates 166 by transferring the pattern in the two-dimensional array of discrete photoresist material portions 187 through the sacrificial capping material layer 166L. The pattern in the two-dimensional array of discrete photoresist material portions 187 can be subsequently transferred through the selector-level material layers (150L, 160L) and the magnetic-tunnel-junction-level material layers (112L, 114L, 130L, 144L, 148L). Remaining portions of the selector-level material layers (150L, 160L) comprise the two-dimensional array of selector-containing pillar structures 182, and remaining portions of the magnetic-tunnel-junction-level material layers (112L, 114L, 130L, 144L, 148L) comprise the magnetic tunnel junction pillar structures 184. The magnetic tunnel junction material layers 130L is patterned into a two-dimensional array of magnetic tunnel junctions 130. Each contiguous combination of a magnetic tunnel junction pillar structure 184 and a selector-containing pillar structure 182 constitutes a memory cell 180. Sidewalls of each component within a memory cell 180 may be vertically coincident among one another. A two-dimensional array of memory cells 180 can be formed over the first electrically conductive layer 30L. Each of the memory cells 180 comprises a vertical stack including a magnetic tunnel junction pillar structure 184 and a selector-containing pillar structure 182.

Referring to FIGS. 68A-68C, a continuous resist layer 197L can be deposited over the two-dimensional array of memory cells 180 by a conformal deposition process. The conformal deposition process may comprise an atomic layer deposition process or a CVD process. In one embodiment, the continuous resist layer 197L comprises a dry electron beam (e-beam) resist material or a dry extreme ultraviolet (EUV) lithography (e.g., 13.5 nm wavelength lithography) resist that lacks a solvent. In one embodiment, the continuous resist layer 197L comprises a negative resist material, i.e., a resist material that becomes chemical insoluble upon exposure to an electron beam or UV radiation.

In an alternative embodiment, the continuous resist layer 197L comprises, and/or consists essentially of, a hydrogen silsesquioxane-based polymer material. In one embodiment, the continuous resist layer 197L may be composed primarily of, and/or may consist essentially of, and/or may consist of, hydrogen silsesquioxane including a polymerized chain of  $[\text{HSiO}_{3/2}]_n$ , in which n is an integer in a range from 4 to 10,000.

The continuous resist layer 197L comprises a horizontally-extending planar resist layer overlying the first electrically conductive layer 30L, a two-dimensional array of tubular resist portions laterally surrounding the two-dimensional array of memory cells 180, and a two-dimensional array of capping resist portions overlying the two-dimensional array of memory cells 180. The horizontally-extending planar resist layer may have uniform vertical thickness. The uniform thickness of the horizontally-extending planar resist layer may be in a range from 10 nm to 200 nm, although lesser and greater thicknesses may also be employed. In one embodiment, the two-dimensional array of tubular resist portions have a respective lateral thickness between an inner sidewall and an outer sidewall that is in a range from 50% to 100%, such as from 80% to 100%, of the uniform vertical thickness of the horizontally-extending planar resist layer. The two-dimensional array of tubular resist portions are spaced apart and are not in direct contact with each other.

Referring to FIGS. 69A-69C, a lithographic exposure process and a lithographic development process can be performed to pattern the continuous resist layer 197L. The lithographic exposure process may comprise an e-beam exposure or an EUV exposure process. In one embodiment, the continuous resist layer 197L comprises a negative resist material, such as a negative e-beam resist material. The lithographic exposure comprises lithographically exposing the two-dimensional array of tubular resist portions, the two-dimensional array of capping resist portions, and first regions of the horizontally-extending planar resist layer adjoined to a respective one of the tubular resist portions, without lithographically exposing second regions of the horizontally-extending planar resist layer that are subsequently removed during the subsequent development process. In one embodiment, the second regions of the horizontally-extending planar resist layer that are not lithographically exposed, i.e., are not irradiated by UV radiation or an e-beam, may have a pattern of straight line strips that are located between neighboring rows of memory cells 180 arranged along the first horizontal direction hd1. The irradiated regions of the resist layer are cross-linked. The unirradiated second regions of the horizontally-extending planar resist layer that are not irradiated are subsequently removed using a developer, leaving the cross-linked irradiated resist material portions 197 of the resist layer 197L over the memory cells 180.

Generally, the continuous resist layer 197L can be patterned into discrete resist material portions 197 by lithographic exposure and development. In one embodiment, the horizontally-extending planar resist layer is divided into a plurality of horizontally-extending planar resist portions having a respective pair of lengthwise edges laterally extending along a first horizontal direction hd1 and adjoined to a respective set of at least one tubular resist portion within the same discrete resist material portions 197. In one embodiment, the respective set of at least one tubular resist portion within the same discrete resist material portions 197 may include a plurality of tubular resist portions arranged along the first horizontal direction hd1. In one embodiment, the discrete resist material portions 197 can comprise a periodic one-dimensional array of discrete resist material portions 197 that are repeated along the second horizontal direction hd2 with the second pitch p2. The width of each discrete resist material portion 197 along the second horizontal direction hd2 may be uniform or substantially uniform, and may be in a range from 20% to 80%, such as from 40% to 60%, of the second pitch p2.

Referring to FIGS. 70A-70C, an anisotropic etch process can be performed to etch portions of the first electrically conductive layer 30L that are not masked by the discrete resist material portions 197. The first electrically conductive layer 30L can be patterned into a plurality of first electrically conductive lines 30 by etching portions of the first electrically conductive layer 30L that are not covered by the discrete resist material portions 197. In one embodiment, the anisotropic etch process may employ at least one of a reactive ion etch process or an ion beam etch process.

Each of plurality of first electrically conductive lines 30 extends underneath and contacts a respective row of memory cells 180 that are arranged along the first horizontal direction hd1. Horizontally-extending portions of the discrete resist material portions 197 may be collaterally removed during the anisotropic etch process. Thus, the remaining portions of the discrete resist material portions 197. In one embodiment, the

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sacrificial capping material plates 166 can be physically exposed during the anisotropic etch process, and may be employed as protective cover structures that protect the two-dimensional array of memory cells 180.

In one embodiment, the discrete resist material portions 197 have a tubular configuration that laterally surrounds a respective one of the memory cells 180. In one embodiment, portions of the first electrically conductive lines 30 that are not covered by the discrete resist material portions 197 may be recessed relative to portions of the first electrically conductive lines 30 that are covered by the discrete resist material portions 197. In one embodiment, portions of the first electrically conductive lines 30 that are covered by the two-dimensional array of memory cells 180 and the two-dimensional array of discrete resist material portions 197 may have a first thickness  $t1'$ , and portions of the first electrically conductive lines 30 that are not covered by the two-dimensional array of memory cells 180 and the two-dimensional array of discrete resist material portions 197 may have a second thickness  $t2'$  that is less than the first thickness  $t1'$ . The difference between the first thickness  $t1'$  and the second thickness may be in a range from 0.1 nm to 30 nm, such as from 0.3 nm to 10 nm. In this case, each of the first electrically conductive lines 30 may have a contoured top surface including a plurality of raised horizontal surface segments, a recessed horizontal surface segment, and cylindrical surface segments connecting the plurality of raised horizontal surface segments to the recessed horizontal surface segment.

Referring to FIGS. 71A-71C, a dielectric fill material can be deposited around the two-dimensional array of discrete resist material portions 197. A planarization process, such as a chemical mechanical polishing process can be performed to remove portions of the dielectric fill material from above the horizontal plane including the top surfaces of the memory cells 180. The sacrificial capping material plates 166 can be collaterally removed during the planarization process. Remaining portions of the dielectric fill material constitute a dielectric matrix layer 140. The top surface of the dielectric matrix layer 140 may be formed within the horizontal plane including the top surfaces of the memory cells 180 (such as the top surfaces of the conductive material plates 160).

Generally, in one embodiment, the dielectric matrix layer 140 can be formed around and directly on the discrete resist material portions 197 after patterning the first electrically conductive layer 30L into the plurality of first electrically conductive lines 30. The dielectric matrix layer 140 can be planarized such that a top surface of the dielectric matrix layer 140 is formed within a horizontal plane including top surfaces of the two-dimensional array of memory cells 180. The sacrificial capping material plates 166 can be removed after patterning the first electrically conductive layer 30L into the plurality of first electrically conductive lines 30. After the planarization process that planarizes the dielectric matrix layer 140, the discrete resist material portions 197 may comprise annular top surfaces located within the horizontal plane including the top surface of the two-dimensional array of memory cells 180. As discussed above, the discrete resist material portions 197 may comprise a dry e-beam or EUV resist material.

Referring to FIGS. 72A-72C, a dielectric material can be deposited over the two-dimensional array of memory cells 180 to form a line-level dielectric layer 92. Line trenches laterally extending along the second horizontal direction  $hd2$  can be formed through the line-level dielectric layer 92 above each column of memory cells 180 arranged along the

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second horizontal direction  $hd2$ . A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the line-level dielectric layer 92. Remaining portions of the conductive material filling the line trenches constitute second electrically conductive lines 90. The second electrically conductive lines 90 comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the second electrically conductive lines 90 can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed. Alternatively, instead of using the above described damascene process to form the second electrically conductive lines 90, these lines may be formed by a pattern and etch process.

Generally, the second electrically conductive lines 90 can be formed over the dielectric matrix layer 140 such that each of the second electrically conductive lines 90 contacts top surfaces of a respective subset of the memory cells 180. In one embodiment, each of the second electrically conductive lines 90 contacts top surfaces of a respective column of memory cells 180 that are arranged along a second horizontal direction  $hd2$  that is perpendicular to the first horizontal direction  $hd1$ .

Referring to FIGS. 73A-73C, a first alternative configuration of the fifth exemplary structure can be derived from the fifth exemplary structure illustrated in FIGS. 72A-72C by reversing the order of material portions within each magnetic tunnel junction pillar structure 184.

Referring to FIGS. 74A-74C, a second alternative configuration of the fifth exemplary structure can be derived from the fifth exemplary structure illustrated in FIGS. 72A-72C or from the first alternative configuration thereof illustrated in FIGS. 73A-73C by removing the discrete resist material portions 197 after patterning the first electrically conductive layer 30L into the plurality of first electrically conductive lines 30 and prior to formation of the dielectric matrix layer 140. In this embodiment, the dielectric matrix layer 140 contacts the sidewalls of the memory cells 180.

In the first through fifth embodiments, two terminal MRAM memory cells 180, such as STT-MRAM cells are formed. In the sixth embodiment, three terminal MRAM memory cells 180, such as SOT-MRAM cells are formed instead.

Referring to FIGS. 75A-75C, a sixth exemplary structure according to the sixth embodiment of the present disclosure can be derived from the fifth exemplary structure illustrated in FIGS. 66A-66C by modifying the magnetic-tunnel-junction-level material layers (112L, 114L, 130L, 144L, 148L) such that the modified magnetic-tunnel-junction-level material layers (130L, 114L, 112L) includes, from bottom to top, continuous magnetic tunnel junction material layers 130L, a continuous antiferromagnetic coupling layer 114L, and a continuous superlattice layer 112L. Further, the order of layers within the magnetic tunnel junction material layers 130L can be, from bottom to top, a continuous free layer 136L, a continuous nonmagnetic tunnel barrier layer 134L, and a continuous reference layer 132L. In addition, the material of the first electrically conductive layer 30L may be selected to increase the spin-orbit-torque charge-to-spin conversion ratio and to facilitate programming of a free layer in each spin-orbit-torque memory cell to be subsequently formed. Thus, the first electrically conductive layer 30L comprises a nonmagnetic heavy metal SOT layer with strong spin-orbit coupling with and in contact with the

continuous free layer **136L**. When an electric write current laterally passes through the SOT layer, spin current is generated in a direction perpendicular to the electrical current via the spin Hall effect (SHE). The spin current exerts a torque on the magnetization of the ferromagnetic free layer. Thus, the SOT layer assists in the transition of the magnetization direction in the free layer through the spin Hall effect. The SOT layer is also referred to as metallic assist layer, i.e., a metallic layer that assists the magnetic transition in the free layer. For example, the first electrically conductive layer **30L** may comprise, and/or may consist essentially of, a transition metal element or metal alloy having an atomic number greater than 56, and/or greater than 70. The first electrically conductive layer **30L** may be made of a material having large spin-orbit coupling strength, such as Pt, Ta, W, Hf, Ir, CuBi, CuIr, AuPt, AuW, PtPd, or PtMgO.

In addition, metal interconnect structures (not shown) for contacting first electrically conductive lines can be embedded in the topmost dielectric material layer within the at least one dielectric material layer **8B**. The metal interconnect structures may include contact via structures for contacting two end portions of each electrically conductive layer to be subsequently patterned from the first electrically conductive layer **30L**. In other words, a pair of contact via structures can be formed for each first electrically conductive line (i.e., SOT layer) to be subsequently patterned from the first electrically conductive layer **30L**. In one embodiment, the contact via structures within each pair of contact via structures may be laterally spaced apart along the first horizontal direction **hd1**. The pairs of contact via structures may be repeated with the first periodicity of the first pitch **p1** along the first horizontal direction **hd1**, and may be repeated with the second periodicity of the second pitch **p2** along the second horizontal direction **hd2**. The metal interconnect structures including the contact via structures can be employed to provide electrical connection to each of the first electrically conductive lines (i.e., SOT layers) to be subsequently formed.

Referring to FIGS. **76A-76C**, the processing steps of FIGS. **67A-67C** can be performed, with any needed changes, in view of the change in the sequence of the material layers overlying the first electrically conductive layer **30L**, to form a two-dimensional periodic array of memory cells **180'**. Each memory cell **180'** may include a vertical stack of a magnetic tunnel junction pillar structure **184'** including a respective magnetic tunnel junction **130**, and a selector-containing pillar structure **182**. Each magnetic tunnel junction pillar structure **184'** can include, from bottom to top, a free layer **136**, a nonmagnetic tunnel barrier layer **134**, a reference layer **132**, an optional antiferromagnetic coupling layer **114**, and an optional superlattice layer **112**.

Referring to FIGS. **77A-77C**, the processing steps of FIGS. **68A-68C** can be performed to form a continuous resist material layer **197L**, and the processing steps of FIGS. **69A-69B** can be performed with a change in the pattern of lithographic exposure to pattern the continuous resist layer **197L** into a plurality of discrete resist material portions **197**. According to an aspect of the present disclosure, the plurality of discrete resist material portions **197** can be formed as a two-dimensional periodic array of resist material portions **197**.

Specifically, a lithographic exposure process and a lithographic development process can be performed to pattern the continuous resist layer **197L**. The lithographic exposure process may comprise an e-beam or EUV exposure process.

In one embodiment, the continuous resist layer **197L** comprises a negative resist material, such as a negative e-beam resist material. The lithographic exposure comprises lithographically exposing the two-dimensional array of tubular resist portions, the two-dimensional array of capping resist portions, and first regions of the horizontally-extending planar resist layer adjoined to a respective one of the tubular resist portions without lithographically exposing second regions of the horizontally-extending planar resist layer that are subsequently removed during the subsequent development process. In one embodiment, the second regions of the horizontally-extending planar resist layer that are not lithographically exposed, i.e., are not irradiated by UV radiation or an e-beam, may have a grid shaped pattern (i.e., straight line strips that are located between neighboring rows and columns of memory cells **180'** arranged along the first horizontal direction **hd1** and along the second horizontal direction **hd2**). The irradiated regions of the resist layer **197L** are cross-linked. The unirradiated second regions of the horizontally-extending planar resist layer that are not irradiated are subsequently removed using a developer, leaving the cross-linked irradiated resist material portions **197** of the resist layer **197L** over the memory cells **180'**.

Generally, the continuous resist layer **197L** can be patterned into discrete rectangular resist material portions **197** by lithographic exposure and development. In one embodiment, the horizontally-extending planar resist layer is divided into a plurality of horizontally-extending planar resist portions having a respective pair of lengthwise edges laterally extending along a first horizontal direction **hd1** and adjoined to a respective tubular resist portion within the same discrete resist material portions **197**. In one embodiment, the discrete resist material portions **197** can comprise a periodic two-dimensional array (such as a rectangular array) of discrete resist material portions **197** that are repeated along the first horizontal direction **hd1** with the first pitch **p1** and along the second horizontal direction **hd2** with the second pitch **p2**. The length of each discrete resist material portion **197** along the first horizontal direction **hd1** may be in a range from 20% to 80%, such as from 40% to 60%, of the first pitch **p1**. The width of each discrete resist material portion **197** along the second horizontal direction **hd2** may be uniform or substantially uniform, and may be in a range from 20% to 80%, such as from 40% to 60%, of the second pitch **p2**.

Referring to FIGS. **78A-78C**, an anisotropic etch process can be performed to etch portions of the first electrically conductive layer **30L** that are not masked by the discrete resist material portions **197**. The first electrically conductive layer **30L** can be patterned into a plurality of first electrically conductive lines (i.e., SOT layers) **430** by etching portions of the first electrically conductive layer **30L** that are not covered by the discrete resist material portions **197**. In one embodiment, the anisotropic etch process may employ at least one of a reactive ion etch process and an ion beam etch process.

Each of plurality of first electrically conductive lines **430** is formed underneath and contacts a respective memory cell **180'**. Horizontally-extending portions of the discrete resist material portions **197** can be collaterally removed during the anisotropic etch process. Thus, the remaining portions of the discrete resist material portions may consist of a two-dimensional array of cylindrical discrete resist material portions **197**. In one embodiment, the sacrificial capping material plates **166** can be physically exposed during the



anisotropic etch process, and may be employed as protective cover structures that protect the two-dimensional array of memory cells **180'**.

In one embodiment, the first electrically conductive lines **430** may be formed as a two-dimensional periodic array of first electrically conductive lines **430** having a first periodicity of the first pitch **p1** along the first horizontal direction **hd1** and having a second periodicity of the second pitch **p2** along the second horizontal direction **hd2**. In one embodiment, each of the first electrically conductive lines **430** may have a respective rectangular horizontal cross-sectional shape. The lateral extent of each first electrically conductive line **430** along the first horizontal direction **hd1** is less than the first pitch **p1**, and the lateral extent of each first electrically conductive line **430** along the second horizontal direction **hd2** is less than the second pitch **p2**.

In one embodiment, the discrete resist material portions **197** have a tubular configuration that laterally surrounds a respective one of the memory cells **180'**. In one embodiment, portions of the first electrically conductive lines **430** that are not covered by the discrete resist material portions **197** may be recessed relative to portions of the first electrically conductive lines **430** that are covered by the discrete resist material portions **197**. In one embodiment, portions of the first electrically conductive lines **430** that are covered by the two-dimensional array of memory cells **180'** and the two-dimensional array of discrete resist material portions **197** may have a first thickness **t1'**, and portions of the first electrically conductive lines **430** that are not covered by the two-dimensional array of memory cells **180** and the two-dimensional array of discrete resist material portions **197** may have a second thickness **t2'** that is less than the first thickness **t1'**. The difference between the first thickness **t1'** and the second thickness may be in a range from 0.1 nm to 430 nm, such as from 0.3 nm to 10 nm. In this case, each of the first electrically conductive lines **430** may have a contoured top surface including a raised horizontal surface segment, a recessed horizontal surface segment, and cylindrical surface segments connecting the raised horizontal surface segment to the recessed horizontal surface segment.

Referring to FIGS. **79A-79C**, a dielectric fill material can be deposited around the two-dimensional array of discrete resist material portions **197**. A planarization process such as a chemical mechanical polishing process can be performed to remove portions of the dielectric fill material from above the horizontal plane including the top surfaces of the memory cells **180'**. The sacrificial capping material plates **166** can be collaterally removed during the planarization process. Remaining portions of the dielectric fill material constitute a dielectric matrix layer **140**. The top surface of the dielectric matrix layer **140** may be formed within the horizontal plane including the top surfaces of the memory cells **180'** (such as the top surfaces of the conductive material plates **160**).

Generally, the dielectric matrix layer **140** can be formed around and directly on the discrete resist material portions **197** after patterning the first electrically conductive layer **30L** into the plurality of first electrically conductive lines **430**. The dielectric matrix layer **140** can be planarized such that a top surface of the dielectric matrix layer **140** is formed within a horizontal plane including top surfaces of the two-dimensional array of memory cells **180'**. The sacrificial capping material plates **166** can be removed after patterning the first electrically conductive layer **30L** into the plurality of first electrically conductive lines **430**. After the planarization process that planarizes the dielectric matrix layer **140**, the discrete resist material portions **197** may comprise

annular top surfaces located within the horizontal plane including the top surface of the two-dimensional array of memory cells **180'**. As discussed above, the discrete resist material portions **197** may comprise an e-beam resist material.

Referring to FIGS. **80A-80C**, a dielectric material can be deposited over the two-dimensional array of memory cells **180'** to form a line-level dielectric layer **492**. Line trenches laterally extending along the first horizontal direction **hd1** can be formed through the line-level dielectric layer **492** above each row of memory cells **180** arranged along the first horizontal direction **hd1**. Alternatively, line trenches laterally extending along the second horizontal direction **hd2** can be formed through the line-level dielectric layer **492** above each column of memory cells **180** arranged along the second horizontal direction **hd2**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the line-level dielectric layer **492**. Remaining portions of the conductive material filling the line trenches constitute second electrically conductive lines **490**. The second electrically conductive lines **490** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the second electrically conductive lines **490** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed. Alternatively, instead of using the above described damascene process to form the second electrically conductive lines **490**, these lines may be formed by a pattern and etch process.

Generally, the second electrically conductive lines **490** can be formed over the dielectric matrix layer **140** such that each of the second electrically conductive lines **490** contacts top surfaces of a respective subset of the memory cells **180'**. In one embodiment, each of the second electrically conductive lines **490** contacts top surfaces of a respective row of memory cells **180'** that are arranged along a first horizontal direction **hd1** that is perpendicular to the second horizontal direction **hd2**. Alternatively, each of the second electrically conductive lines **490** contacts top surfaces of a respective column of memory cells **180'** that are arranged along a second horizontal direction **hd2** that is perpendicular to the first horizontal direction **hd1**.

Referring to FIGS. **81A-81C**, an alternative configuration of the sixth exemplary structure can be derived from the sixth exemplary structure illustrated in FIGS. **80A-80C** by removing the discrete resist material portions **197** after patterning the first electrically conductive layer **30L** into the plurality of first electrically conductive lines **430** and prior to formation of the dielectric matrix layer **140**.

The sixth exemplary structure or alternative configurations thereof may comprise a memory device including a two-dimensional array of spin-orbit-torque (SOT) magnetoresistive random access memory cells **180'**.

Referring to FIG. **82**, a schematic diagram is shown for a magnetoresistive random access memory (MRAM) device **500'** including a two-dimensional array of spin-orbit-torque (SOT) magnetoresistive random access memory cells, which may include a two-dimensional array of memory cells **180'** of the sixth exemplary structure. The MRAM device **500'** includes a memory array region **550** containing an array of memory cells **180'** located at intersections of word lines and bit lines.

In one embodiment, the word lines **530** may be electrically connected to first end portions of a respective plurality



of first electrically conductive lines (i.e., SOT layers) **430** that are arranged along the first horizontal direction **hd1**, and the bit lines may comprise second electrically conductive lines **490** laterally extending along the second horizontal direction **hd2** and electrically contacting a respective column of memory cells **180'**. In this case, access lines **540** may be electrically connected to second end portions of a respective plurality of first electrically conductive lines (i.e., SOT layers) **430** that are arranged along the first horizontal direction **hd1**.

Alternatively, the word lines **530** may be electrically connected to first end portions of a respective plurality of first electrically conductive lines **430** that are arranged along the second horizontal direction **hd2**, and the bit lines may comprise the second electrically conductive lines **490** laterally extending along the first horizontal direction **hd1** and contacting a respective row of memory cells **180'**. In this case, the access lines **540** may be electrically connected to second end portions of a respective plurality of first electrically conductive lines **430** that are arranged along the second horizontal direction **hd2**.

The MRAM device **500'** contains a row decoder **560** connected to the word lines **530**, sense circuitry **570** (e.g., a sense amplifier and other bit line control circuitry) and a column decoder **580** connected to the bit lines **490**, and a data buffer **590** connected to the sense circuitry. In one embodiment, the MRAM device **500'** can contain an access line decoder **520** connected to the access lines **540**. Generally, each memory cell **180'** of the sixth exemplary structure may be configured as a three terminal device in which a word line **530** is electrically connected to a first end of a first electrically conductive line **430**, an access line **540** is electrically connected to a second end of the first electrically conductive line **430**, and a bit line comprises, or is electrically connected to, a second electrically conductive line **490**. During a read operation, the access line may be electrically floating, and a read bias voltage can be applied between the bit line and the word line. During a programming operation, the bit line may be grounded or may be electrically floating, and a programming bias voltage can be applied between the word line and the access line.

In the fifth and sixth embodiments, the number of process steps is reduced by utilizing direct lithography patterning on a non-planarized surface (e.g., a resist layer located over the protruding pillar shaped memory cells). A much thinner resist layer serves as the etching mask for the underlying electrically conductive layer **30L**, which significantly decreases the shadowing and loading effects of IBE. The embodiment methods may be carried out using EUV or e-beam lithography, in which flat underlayers are not required. The dry resist layer **197L** can be coated by CVD or ALD, and therefore will coat and protect the side walls of the MRAM pillars. The embodiment methods can boost the areal density of a STT-MRAM cross point array or a SOT-MRAM bit array.

Referring collectively to FIGS. **1**, **2**, and **66A-82** and related drawings, a memory device is provided, which comprises: first electrically conductive lines **30** laterally extending along a first horizontal direction **hd1**, laterally spaced apart from each other along a second horizontal direction **hd2**, and located over a substrate **8**; a two-dimensional array of memory cells (**180** or **180'**) located over the first electrically conductive lines (**30** or **430**), wherein each of the memory cells (**180** or **180'**) comprises a vertical stack including a magnetic tunnel junction pillar structure (**184** or **184'**) and a selector-containing pillar structure **182**, and each of the first electrically conductive lines (**30** or **430**) contacts

a respective row of memory cells (**180** or **180'**) arranged along the first horizontal direction **hd1**; discrete resist material portions **197** having a tubular configuration and laterally surrounding a respective one of the memory cells (**180** or **180'**); second electrically conductive lines (**90** or **490**) contacting top surfaces of a respective subset of the memory cells (**180** or **180'**); and a dielectric matrix layer **140** laterally surrounding the two-dimensional array of discrete resist material portions **197**.

In one embodiment, the discrete resist material portions **197** comprise annular top surfaces located within the horizontal plane including the top surface of the two-dimensional array of memory cells (**180** or **180'**). The dielectric matrix layer **140** has a top surface located within a horizontal plane including top surfaces of the two-dimensional array of memory cells, and contacts the second electrically conductive lines (**90** or **490**). The discrete resist material portions **197** comprise a dry e-beam resist material or a dry EUV resist material.

Referring to FIGS. **83A-83C**, a seventh exemplary structure according to a seventh embodiment of the present disclosure is illustrated, which comprises a substrate **8** that may be the same as, or may be similar to, the substrate **8** described above. A first line-level dielectric layer **32** can be deposited over the substrate **8**, and line trenches laterally extending along the first horizontal direction **hd1** can be formed through the first line-level dielectric layer **32**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the first line-level dielectric layer **32**. Remaining portions of the conductive material filling the line trenches constitute first electrically conductive lines **30**. The first electrically conductive lines **30** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the first electrically conductive lines **30** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed. The first electrically conductive lines **30** laterally extend along the first horizontal direction **hd1**, and are laterally spaced apart among one another along a second horizontal direction **hd2**. The first electrically conductive lines **30** may be formed as a one-dimensional periodic array of first electrically conductive lines **30** having a second pitch **p2** along the second horizontal direction **hd2**. In one embodiment, the remaining portions of the first line-level dielectric layer **32** may comprise first dielectric rails laterally extending along the first horizontal direction **hd1**, and interlaced with the first electrically conductive lines **30** along the second horizontal direction **hd2**. Alternatively, instead of using the above described damascene process to form the first electrically conductive lines **32**, these lines may be formed by a pattern and etch process.

Referring to FIGS. **84A-84C**, an optional metallic adhesion layer **149L**, selector-level material layers (**150L**, **160L**), and an optional first image transfer assist layer **171L** can be formed over the first electrically conductive lines **30**. The optional metallic adhesion layer **149L** comprises a metallic material that promotes adhesion of the selector-level material layers (**150L**, **160L**). For example, the optional metallic adhesion material layer **149L** may comprise an electrically conductive metal or metal alloy, such as Ta, Ti, TaN, TiN, or WN. The selector-level material layers (**150L**, **160L**) may be the same as in the previously described embodiments. In one embodiment, the selector-level material layers (**150L**, **160L**) may comprise a layer stack of selector material layers **150L**

and a conductive material layer **160L**. The conductive material layer **160L** includes a nonmagnetic conductive material, which may comprise, for example, TiN, TaN, WN, MoN, W, Ru, Mo, Nb, Ti, Ta, or a combination thereof.

The optional first image transfer assist layer **171L** includes a material that can provide a high etch resistance for an anisotropic etch process to be subsequently employed with respect to the material of the conductive material layer **160L**, thereby enabling high etch selectivity for the etch process that patterns the conductive material layer **160L**. For example, the optional first image transfer assist layer **171L** may comprise a metal, such as Cr or Ru. The thickness of the first image transfer assist layer **171L** may be in a range from 1 nm to 30 nm, such as from 2 nm to 10 nm, although lesser and greater thicknesses may also be employed.

A resist layer can be deposited over the optional first image transfer assist layer **171L** and/or the conductive material layer **160L**, and can be lithographically patterned to form a two-dimensional array of first discrete patterned resist material portions **187**. The two-dimensional array of first discrete patterned resist material portions **187** can be a periodic two-dimensional array having a first pitch **p1** along a first horizontal direction **hd1** and having a second pitch **p2** along a second horizontal direction **hd2**. Each of the first discrete patterned resist material portions **187** may have a respective horizontal cross-sectional shape of a rectangle, a rounded rectangle, an oval, or a circle.

Referring to FIGS. **85A-85C**, an etch process can be performed to transfer the pattern in the two-dimensional array of first discrete patterned resist material portions **187** through the optional first image transfer assist layer **171L**. The optional first image transfer assist layer **171L** can be patterned into a two-dimensional array of first etch mask plates **171** having the same pattern as the two-dimensional array of first discrete patterned resist material portions **187**. The etch process may comprise an anisotropic etch process such as a reactive ion etch process.

Referring to FIGS. **86A-86C**, an anisotropic etch process can be performed to transfer the pattern in the two-dimensional array of first discrete patterned resist material portions **187** through the conductive material layer **160L**. The conductive material layer **160L** can be patterned into a two-dimensional array of conductive material plates **160** having the same pattern as the two-dimensional array of first discrete patterned resist material portions **187**. Optionally, the two-dimensional array of first discrete patterned resist material portions **187** may be removed, for example, by ashing.

Referring to FIGS. **87A-87C**, another anisotropic etch process can be performed to transfer the pattern in the two-dimensional array of first etch mask plates **171** and the two-dimensional array of conductive material plates **160** through the selector material layers **150L** and the optional metallic adhesion layer **149L**. The selector material layers **150L** are patterned into selector elements **150**. Each selector element **150** may include a vertical stack of a lower selector electrode **151**, a non-Ohmic material plate **152**, and an upper selector electrode **153**. The optional metallic adhesion layer **149L** may be patterned into a two-dimensional array of metallic adhesion plates **149**. Vertical sidewalls of structural elements within each vertical stack of a metallic adhesion plate **149**, a selector element **150**, a conductive material plate **160**, and a first etch mask plate **171** may be vertically coincident. In one embodiment, the first etch mask plates **171** may be collaterally consumed during the anisotropic etch process that patterns the selector elements **150**.

A two-dimensional array of selector-containing pillar structures **182** can be formed over the first electrically conductive lines **30**. Each selector-containing pillar structure **182** can include, from bottom to top, a metallic adhesion plate **149**, a selector element **150**, and a conductive material plate **160**, and may optionally include a first etch mask plate **171**. Each row of the selector-containing pillar structures **182** that is arranged along the first horizontal direction **hd1** may be formed on a top surface of a respective one of the first electrically conductive lines **30**. Each of the first electrically conductive lines **30** contacts a respective row of selector-containing pillar structures **182** of the two-dimensional array of selector-containing pillar structures **182**.

Referring to FIGS. **88A-88C**, a protective dielectric liner **172** can be deposited over the two-dimensional array of selector-containing pillar structures **182**. The protective dielectric liner **172** includes a dielectric material that can prevent or reduce lateral diffusion of the non-Ohmic material plates **152**. In one embodiment, the protective dielectric liner **172** may comprise, and/or may consist essentially of, a dielectric material selected from silicon nitride, silicon oxynitride, silicon carbide nitride (i.e., silicon carbonitride) or a metal oxide, such as aluminum oxide, hafnium oxide or tantalum oxide. The protective dielectric liner **172** may be deposited by a conformal deposition process, such as a chemical vapor deposition process or an atomic layer deposition process. The thickness of the protective dielectric liner **172** may be in a range from 0.5 nm to 20 nm, such as from 2 nm to 10 nm, although lesser and greater thicknesses may also be employed.

The protective dielectric liner **172** comprises a horizontally-extending portion contacting top surfaces of the first electrically conductive lines **30** and the first line-level dielectric layer **32**, a two-dimensional array of tubular dielectric liner portions laterally surrounding the two-dimensional array of selector-containing pillar structures **182**, and a two-dimensional array of horizontal dielectric capping portions overlying the two-dimensional array of selector-containing pillar structures **182**. The horizontally-extending portion of the protective dielectric liner **172** is adjoined to a bottom periphery of each of the tubular dielectric liner portions of the protective dielectric liner **172**. The two-dimensional array of horizontal dielectric capping portions is adjoined to a top periphery of a respective one of the tubular dielectric liner portions of the protective dielectric liner **172**.

The horizontally-extending portion of the protective dielectric liner **172** contacts top surfaces of the first electrically conductive lines **30** within a horizontal plane including interfaces between the first electrically conductive lines **30** and the two-dimensional array of selector-containing pillar structures **182**. The first dielectric rails (which are portions of the first line-level dielectric layer **32** located between neighboring pairs of first electrically conductive lines **30**, laterally extend along the first horizontal direction **hd1**, and are interlaced with the first electrically conductive lines **30** along the second horizontal direction **hd2**) contact a bottom surface of the horizontally-extending portion of the protective dielectric liner **172**.

Referring to FIGS. **89A-89C**, a dielectric fill material, such as silicon oxide can be deposited around the protective dielectric liner **172**. In one embodiment, the dielectric fill material may comprise a different material from the material of the protective dielectric liner **172**. Excess portions of the dielectric fill material can be removed from above the horizontal plane including the top surfaces of the conductive material plates **160** by a planarization process such as a

chemical mechanical polishing process. Remaining portions of the dielectric fill material constitute a dielectric matrix layer, which is herein referred to as a selector-level dielectric matrix layer **40**. Thus, the selector-level dielectric matrix layer **40** is formed over the protective dielectric liner **172**. The sacrificial capping material plates **166** and portions of the protective dielectric liner **172** that overlies the horizontal plane including the top surfaces of the conductive material plates **160** can be collaterally removed during the planarization process that planarizes the selector-level dielectric matrix layer **40**. Generally, the selector-level dielectric matrix layer **40** and the protective dielectric liner **172** can be planarized by removing portions of the selector-level dielectric matrix layer **40** and the protective dielectric liner **172** from above the horizontal plane including top surfaces of the two-dimensional array of selector-containing pillar structures **182**. The selector-level dielectric matrix layer **40** laterally surrounds the two-dimensional array of tubular dielectric liner portions of the protective dielectric liner **172**, and overlies the horizontally-extending portion of the protective dielectric liner **172**.

Referring to FIGS. **90A-90C**, the processing steps of FIGS. **10A-10C** can be performed to form magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**). The magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be the same as the first magnetic tunnel junction-level (MTJ-level) material layers (**112L**, **114L**, **130L**, **144L**, **148L**) described with reference to FIGS. **10A-10C**. The MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) continuous magnetic tunnel junction (MTJ) material layers **130L**, which include a layer stack containing a continuous reference layer **132L**, a continuous nonmagnetic tunnel barrier layer **134L**, and a continuous free layer **136L**.

An optional patterning film **176L** and/or an optional second image transfer assist layer **177L** can be formed over the MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**). The optional patterning film **176L** may comprise a carbon-based material that can enhance pattern fidelity during subsequent anisotropic etch processes. For example, the optional patterning film **176L** may be composed primarily of amorphous carbon or diamond-like carbon. The optional second image transfer assist layer **177L** includes a material that can provide a high etch resistance for an anisotropic etch process to be subsequently employed with respect to the material of the patterning film **176L** and/or with respect to the material of the metallic capping layer **148L** in case the patterning film **176L** is not employed. For example, the optional second image transfer assist layer **177L** may comprise a metal, such as Cr or Ru. The thickness of the second image transfer assist layer **177L** may be in a range from 1 nm to 30 nm, such as from 2 nm to 10 nm, although lesser and greater thicknesses may also be employed.

A two-dimensional array of second discrete patterned resist material portions **159** can be formed over the MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**), the optional patterning film **176L**, and the optional second image transfer assist layer **177L**. Each of the second discrete patterned resist material portions **159** has an areal overlap with a respective underlying one of the selector-containing pillar structures **182**. The two-dimensional array of second discrete patterned resist material portions **159** can be formed as a periodic array having the first pitch **p1** along the first horizontal direction **hd1** and having the second pitch **p2** along the second horizontal direction **hd2**. The horizontal cross-sectional shapes of the second discrete patterned resist material portions **159** can be the same as, or can be different

from, the horizontal cross-sectional shapes of the first selector-containing pillar structures **182**. In one embodiment, the area of each second discrete patterned resist material portions **159** may be located entirely within the area of a respective underlying selector-containing pillar structure **182** in a plan view (such as a top-down view). Alternatively, the area of each second discrete patterned resist material portions **159** may coincide within the area of a respective underlying selector-containing pillar structure **182** in the plan view. Yet alternatively, the area of each second discrete patterned resist material portions **159** may include all of, and may be greater than, the area of a respective underlying selector-containing pillar structure **182** in the plan view. In one embodiment, the lateral dimension of each of the second discrete patterned resist material portions **159** along the first horizontal direction **hd1** may be the same as the lateral dimension of each of the second discrete patterned resist material portions **159** along the second horizontal direction **hd2**. In one embodiment, each of the second discrete patterned resist material portions **159** may have a respective horizontal cross-sectional shape of a circle.

Referring to FIGS. **91A-91C**, the pattern in the two-dimensional array of second discrete patterned resist material portions **159** can be transferred through the second image transfer assist layer **177L** and the patterning film **176L** by performing an anisotropic etch process such as a reactive ion etch process. The second image transfer assist layer **177L** can be divided into a two-dimensional array of second etch mask plates **177**. The patterning film **176L** can be divided into a two-dimensional array of patterning film plates **176**. The two-dimensional array of second discrete patterned resist material portions **159** can be subsequently removed, for example, by ashing. A two-dimensional array of discrete masking structures (i.e., hardmask structures) (**176**, **177**) can be formed. Each discrete masking structure (**176**, **177**) may comprise a patterning film plate **176** and/or a second etch mask plate **177**.

Referring to FIGS. **92A-92C**, an anisotropic etch process can be performed to transfer the pattern in the two-dimensional array of discrete masking structures (**176**, **177**) through the MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**). The anisotropic etch process may comprise an ion beam etch (i.e., ion milling) process. The MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be patterned into a two-dimensional array of magnetic tunnel junction (MTJ) pillar structures **184**. According to an aspect of the present disclosure, physically exposed surfaces of the MTJ pillar structures **184** are formed with taper angles. The taper angles can be measured with respect to the vertical direction that is perpendicular to the top surface of the substrate **8**. In one embodiment, the taper angle may be in a range from 3 degrees to 30 degrees, such as from 6 degrees to 20 degrees, although lesser and greater taper angles may also be employed.

Generally, the layer stack including the MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be patterned into the two-dimensional array of magnetic tunnel junction pillar structures **184** by anisotropically etching the layer stack employing the two-dimensional array of discrete masking structures (**176**, **177**) as an etch mask. The two-dimensional array of magnetic tunnel junction pillar structures **184** can be formed above the two-dimensional array of selector-containing pillar structures **182**. The MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) can be patterned such that each of the magnetic tunnel junction pillar structures **184** is formed with a respective tapered sidewall such that bottoms of the magnetic tunnel junction

pillar structures **184** are wider than the tops of the magnetic tunnel junction pillar structures **184**. The total thickness of the MTJ-level material layers (**112L**, **114L**, **130L**, **144L**, **148L**) and the taper angle can be selected such that each of the magnetic tunnel junction pillar structures **184** has a respective bottom surface having a periphery that is laterally offset outward from a periphery of a top surface of a respective underlying selector-containing pillar structure **182** within the two-dimensional array of selector-containing pillar structures **182**.

In one embodiment, the anisotropic etch process may collaterally etch portions of the selector-level dielectric matrix layer **40** during and/or after formation of the two-dimensional array of magnetic tunnel junction pillar structures **184**. A recessed horizontal surface **40R** of the selector-level dielectric matrix layer **40** may be formed, which can be adjoined to annular tapered sidewall segments **40T** of the selector-level dielectric matrix layer **40**. In one embodiment, each selector-containing pillar structure **182** within the two-dimensional array of selector-containing pillar structures **182** has a respective top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure **184** within the two-dimensional array of magnetic tunnel junction pillar structures **184**. In one embodiment, a periphery of the respective top surface of the selector-containing pillar structures **182** can be laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure **184**. The metal layers at the base of the MTJ pillar structures **184** are electrically separated from neighboring MTJ pillar structures **184** of adjacent MRAM memory cells (i.e., bits).

Due to the tapered profile ion beam etching creates at the base of the MTJ pillar structures **184**, the selector-containing pillar structures **182** are located beneath the MTJ pillar structures **184**, which means they are protected from damage during the ion beam etching. Thus, the selector elements **150** are protected from the ions that are employed during the ion beam etching process employed to pattern the MTJ pillar structures **184** by the protective dielectric layer **172**, by the conductive material plates **160** and by portions of the selector-level dielectric matrix layer **40**. Furthermore, once the MTJ pillar structures **184** are etched through, the etching continues into a dielectric material of the dielectric matrix layer **40** rather than into metal, which reduces the potential for shunting the MTJ pillar structures **184** due to metal redeposition.

In one embodiment, each tubular dielectric liner portion within the two-dimensional array of tubular dielectric liner portions of the protective dielectric liner **172** may have an annular top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure **184** within the two-dimensional array of magnetic tunnel junction pillar structures **184**. In one embodiment, an outer periphery of the annular top surface of each tubular dielectric liner portion of the protective dielectric liner **172** is laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure **184**. Generally, sidewalls of the two-dimensional array of magnetic tunnel junction pillar structures **184** may have a greater taper angle relative to a vertical direction than sidewalls of the two-dimensional array of selector-containing pillar structures **182**.

In one embodiment, the selector-level dielectric matrix layer **40** can have a contoured top surface that includes a two-dimensional array of annular horizontal surface seg-

ments in contact with bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures **184**, a continuous recessed surface **40R** located below a horizontal plane including the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures **184**, and a two-dimensional array of annular tapered surface segments **40T** connecting outer peripheries of the annular horizontal surface segments to the continuous recessed surface **40R**. In one embodiment, the continuous recessed surface **40R** is laterally spaced from the protective dielectric liner **172** by portions of the selector-level dielectric matrix layer **40** that contact the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures **184**.

Referring to FIGS. **93A-93C**, a dielectric fill material can be deposited in the gaps between neighboring pairs of the MTJ pillar structures **184**, and can be subsequently planarized to remove portions of the dielectric fill material from above the horizontal plane including the top surfaces of the MTJ pillar structures **184**. The remaining portions of the dielectric fill material comprises a dielectric matrix layer, which is herein referred to as a magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer **80**. The MTJ-level dielectric matrix layer **80** laterally surrounds the two-dimensional array of magnetic tunnel junction pillar structures **184**, and overlies the selector-level dielectric matrix layer **40**. The MTJ-level dielectric matrix layer **80** comprises downward-protruding portions that extend downward below a horizontal plane including bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures **184**, and have tapered surfaces contacting the selector-level dielectric matrix layer **40**.

Referring to FIGS. **94A-94C**, a dielectric material can be deposited over the two-dimensional array of MTJ pillar structures **184** to form a second line-level dielectric layer **92**. Line trenches laterally extending along the second horizontal direction **hd2** can be formed through the second line-level dielectric layer **92** above each column of MTJ pillar structures **184** arranged along the second horizontal direction **hd2**. A conductive material can be deposited in the line trenches, and excess portions of the conductive material can be removed from above the horizontal plane including the top surface of the second line-level dielectric layer **92**. Remaining portions of the conductive material filling the line trenches constitute second electrically conductive lines **90**. The second electrically conductive lines **90** comprise, and/or consist essentially of, a nonmagnetic electrically conductive material such as Al, Cu, W, Ru, Mo, Nb, Ti, Ta, TiN, TaN, WN, MoN, or combinations thereof. The thickness of the second electrically conductive lines **90** can be in a range from 20 nm to 100 nm, although lesser and greater thicknesses can also be employed. Alternatively, instead of using the above described damascene process to form the second electrically conductive lines **490**, these lines may be formed by a pattern and etch process.

Referring to FIGS. **1**, **2**, and **83A-94C** and related drawings, a memory device is provided, which comprises: first electrically conductive lines **30** laterally extending along a first horizontal direction **hd1** and laterally spaced apart from each other along a second horizontal direction **hd2**; a two-dimensional array of selector-containing pillar structures **182** located over the first electrically conductive lines **30**, wherein each of the first electrically conductive lines **30** contacts a respective row of selector-containing pillar structures **182** of the two-dimensional array of selector-containing pillar structures **182**; a protective dielectric liner **172** comprising a two-dimensional array of tubular dielectric

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liner portions laterally surrounding the two-dimensional array of selector-containing pillar structures **182**; a two-dimensional array of magnetic tunnel junction pillar structures **184** located above the two-dimensional array of selector-containing pillar structures **182**; and second electrically conductive lines **90** laterally extending along the second horizontal direction **hd2**, laterally spaced apart from each other along the first horizontal direction **hd1**, and located over the two-dimensional array of magnetic tunnel junction pillar structures **184**.

In one embodiment, each selector-containing pillar structure **182** within the two-dimensional array of selector-containing pillar structures **182** has a respective top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure **184** within the two-dimensional array of magnetic tunnel junction pillar structures **184**; and a periphery of the respective top surface is laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure **184**.

In one embodiment, the protective dielectric liner **172** further comprises a horizontally-extending portion adjoined to a bottom periphery of each of the tubular dielectric liner portions; and each tubular dielectric liner portion within the two-dimensional array of tubular dielectric liner portions of the protective dielectric liner **172** has an annular top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure **184** within the two-dimensional array of magnetic tunnel junction pillar structures **184**.

In one embodiment, the protective dielectric liner **172** does not surround the magnetic tunnel junction pillar structures **184**; and an outer periphery of the annular top surface of each tubular dielectric liner portion is laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure **184**.

In one embodiment, the memory device comprises a selector-level dielectric matrix layer **40** laterally surrounding the two-dimensional array of tubular dielectric liner portions and overlying the horizontally-extending portion of the protective dielectric liner. In one embodiment, a contoured top surface of the selector-level dielectric matrix layer **40** comprises: a two-dimensional array of annular horizontal surface segments in contact with bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures **184**; a continuous recessed surface **40R** located below a horizontal plane including the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures **184**; and a two-dimensional array of annular tapered surface segments **40T** connecting outer peripheries of the annular horizontal surface segments to the continuous recessed surface **40R**.

In one embodiment, the continuous recessed surface **40R** is laterally spaced from the protective dielectric liner **172** by portions of the selector-level dielectric matrix layer **40** that contact the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures **184**. A magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer **80** may laterally surround the two-dimensional array of magnetic tunnel junction pillar structures **184**, and may overlie the selector-level dielectric matrix layer **40**. In one embodiment, the MTJ-level dielectric matrix layer **80** comprises downward-protruding portions that extend downward below a horizontal plane including bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar struc-

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tures **184** and have tapered surfaces contacting the selector-level dielectric matrix layer **40**.

In one embodiment, sidewalls of the two-dimensional array of magnetic tunnel junction pillar structures **184** have a greater taper angle relative to a vertical direction than sidewalls of the two-dimensional array of selector-containing pillar structures **182**.

In one embodiment, the horizontally-extending portion of the protective dielectric liner **172** contacts top surfaces of the first electrically conductive lines **30** within a horizontal plane including interfaces between the first electrically conductive lines **30** and the two-dimensional array of selector-containing pillar structures **182**.

In one embodiment, the memory device comprises first dielectric rails (which are portions of the first line-level dielectric layer **32**) laterally extending along the first horizontal direction **hd1**, interlaced with the first electrically conductive lines **30** along the second horizontal direction **hd2**, and contacting a bottom surface of the horizontally-extending portion of the protective dielectric liner **172**. In one embodiment, the protective dielectric liner **172** comprises a dielectric material selected from silicon nitride, silicon oxynitride, silicon carbide nitride or metal oxide.

A method of forming a memory device according to the seventh embodiment includes forming a two-dimensional array of selector-containing pillar structures **182** over first electrically conductive lines **30** which extend in a first horizontal direction **hd1**; depositing a layer stack **130L** including a continuous reference layer **132L**, a continuous nonmagnetic tunnel barrier layer **134L**, and a continuous free layer **136L** over the two-dimensional array of selector-containing pillar structures **182**; patterning the layer stack **130L** into a two-dimensional array of magnetic tunnel junction pillar structures **184**; and forming second electrically conductive lines **90** over the two-dimensional array of magnetic tunnel junction pillar structures **184**.

In one embodiment, the method also includes depositing a protective dielectric liner **172** over the two-dimensional array of selector-containing pillar structures **182** prior to depositing the layer stack **130L**. The method may also include forming a selector-level dielectric matrix layer **40** over the protective dielectric liner **172**, wherein the layer stack **130L** is deposited above the selector-level dielectric matrix layer **40**.

In one embodiment, the method also includes planarizing the selector-level dielectric matrix layer and the protective dielectric liner by removing portions of the selector-level dielectric matrix layer and the protective dielectric liner from above a horizontal plane including top surfaces of the two-dimensional array of selector-containing pillar structures prior to depositing the layer stack.

In one embodiment, the method also includes forming a two-dimensional array of discrete masking structures (**176**, **177**) over the layer stack **130L**; and anisotropically etching the layer stack **130L** by ion beam etching using the two-dimensional array of discrete masking structures as an etch mask to pattern the layer stack **130L** into the two-dimensional array of magnetic tunnel junction pillar structures **184**.

The various embodiments of the present disclosure may be employed to provide a magnetoresistive memory array including a two-dimensional array of memory cells (**180** or **180'**) with enhanced performance and/or with cost-effective manufacturing processing sequences.

Although the foregoing refers to particular preferred embodiments, it will be understood that the disclosure is not so limited. It will occur to those of ordinary skill in the art

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that various modifications may be made to the disclosed embodiments and that such modifications are intended to be within the scope of the disclosure. Where an embodiment employing a particular structure and/or configuration is illustrated in the present disclosure, it is understood that the present disclosure may be practiced with any other compatible structures and/or configurations that are functionally equivalent provided that such substitutions are not explicitly forbidden or otherwise known to be impossible to one of ordinary skill in the art. All of the publications, patent applications and patents cited herein are incorporated herein by reference in their entirety.

What is claimed is:

1. A memory device, comprising:

first electrically conductive lines laterally extending along a first horizontal direction, and laterally spaced apart from each other along a second horizontal direction;

a two-dimensional array of selector-containing pillar structures located over the first electrically conductive lines, wherein each of the first electrically conductive lines contacts a respective row of selector-containing pillar structures of the two-dimensional array of selector-containing pillar structures;

a protective dielectric liner comprising a two-dimensional array of tubular dielectric liner portions laterally surrounding the two-dimensional array of selector-containing pillar structures;

a two-dimensional array of magnetic tunnel junction pillar structures located above the two-dimensional array of selector-containing pillar structures; and

second electrically conductive lines laterally extending along the second horizontal direction, laterally spaced apart from each other along the first horizontal direction, and located over the two-dimensional array of magnetic tunnel junction pillar structures,

wherein:

the protective dielectric liner further comprises a horizontally-extending portion adjoined to a bottom periphery of each of the tubular dielectric liner portions;

each tubular dielectric liner portion within the two-dimensional array of tubular dielectric liner portions has an annular top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure within the two-dimensional array of magnetic tunnel junction pillar structures;

the memory device further comprises a selector-level dielectric matrix layer laterally surrounding the two-dimensional array of tubular dielectric liner portions and overlying the horizontally-extending portion of the protective dielectric liner; and

a contoured top surface of the selector-level dielectric matrix layer comprises:

a two-dimensional array of annular horizontal surface segments in contact with bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures,

a continuous recessed surface located below a horizontal plane including the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures, and

a two-dimensional array of annular tapered surface segments connecting outer peripheries of the annular horizontal surface segments to the continuous recessed surface.

2. The memory device of claim 1, wherein:

each selector-containing pillar structure within the two-dimensional array of selector-containing pillar struc-

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tures has a respective top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure within the two-dimensional array of magnetic tunnel junction pillar structures; and

a periphery of the respective top surface is laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure.

3. The memory device of claim 1, wherein:

the protective dielectric liner does not surround the magnetic tunnel junction pillar structures; and

an outer periphery of the annular top surface of each tubular dielectric liner portion is laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure.

4. The memory device of claim 1, wherein the continuous recessed surface is laterally spaced from the protective dielectric liner by portions of the selector-level dielectric matrix layer that contact the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures.

5. The memory device of claim 1, further comprising a magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer laterally surrounding the two-dimensional array of magnetic tunnel junction pillar structures and overlying the selector-level dielectric matrix layer.

6. The memory device of claim 5, wherein the MTJ-level dielectric matrix layer comprises downward-protruding portions that extend downward below a horizontal plane including bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures and have tapered surfaces contacting the selector-level dielectric matrix layer.

7. The memory device of claim 1, wherein the horizontally-extending portion of the protective dielectric liner contacts top surfaces of the first electrically conductive lines within a horizontal plane including interfaces between the first electrically conductive lines and the two-dimensional array of selector-containing pillar structures.

8. The memory device of claim 7, further comprising first dielectric rails laterally extending along the first horizontal direction, interlaced with the first electrically conductive lines along the second horizontal direction, and contacting a bottom surface of the horizontally-extending portion of the protective dielectric liner.

9. The memory device of claim 1, wherein sidewalls of the two-dimensional array of magnetic tunnel junction pillar structures have a greater taper angle relative to a vertical direction than sidewalls of the two-dimensional array of selector-containing pillar structures.

10. The memory device of claim 1, wherein the protective dielectric liner comprises a dielectric material selected from silicon nitride, silicon oxynitride, silicon carbide nitride or metal oxide.

11. A memory device, comprising:

first electrically conductive lines laterally extending along a first horizontal direction, and laterally spaced apart from each other along a second horizontal direction;

a two-dimensional array of selector-containing pillar structures located over the first electrically conductive lines, wherein each of the first electrically conductive lines contacts a respective row of selector-containing pillar structures of the two-dimensional array of selector-containing pillar structures;

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a protective dielectric liner comprising a two-dimensional array of tubular dielectric liner portions laterally surrounding the two-dimensional array of selector-containing pillar structures;

a two-dimensional array of magnetic tunnel junction pillar structures located above the two-dimensional array of selector-containing pillar structures;

second electrically conductive lines laterally extending along the second horizontal direction, laterally spaced apart from each other along the first horizontal direction, and located over the two-dimensional array of magnetic tunnel junction pillar structures;

a selector-level dielectric matrix layer laterally surrounding the two-dimensional array of tubular dielectric liner portions and overlying the horizontally-extending portion of the protective dielectric liner; and

a magnetic-tunnel-junction-level (MTJ-level) dielectric matrix layer laterally surrounding the two-dimensional array of magnetic tunnel junction pillar structures and overlying the selector-level dielectric matrix layer, wherein:

the protective dielectric liner further comprises a horizontally-extending portion adjoined to a bottom periphery of each of the tubular dielectric liner portions;

each tubular dielectric liner portion within the two-dimensional array of tubular dielectric liner portions has an annular top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure within the two-dimensional array of magnetic tunnel junction pillar structures; and

the MTJ-level dielectric matrix layer comprises downward-protruding portions that extend downward below a horizontal plane including bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures and have tapered surfaces contacting the selector-level dielectric matrix layer.

**12.** The memory device of claim **11**, wherein:

each selector-containing pillar structure within the two-dimensional array of selector-containing pillar structures has a respective top surface that contacts a bottom surface of a respective overlying magnetic tunnel junction pillar structure within the two-dimensional array of magnetic tunnel junction pillar structures; and

a periphery of the respective top surface is laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure.

**13.** The memory device of claim **11**, wherein:

the protective dielectric liner does not surround the magnetic tunnel junction pillar structures; and

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an outer periphery of the annular top surface of each tubular dielectric liner portion is laterally offset inward from and does not contact a periphery of the bottom surface of the respective overlying magnetic tunnel junction pillar structure.

**14.** The memory device of claim **11**, wherein a contoured top surface of the selector-level dielectric matrix layer comprises:

a two-dimensional array of annular horizontal surface segments in contact with bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures;

a continuous recessed surface located below a horizontal plane including the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures; and

a two-dimensional array of annular tapered surface segments connecting outer peripheries of the annular horizontal surface segments to the continuous recessed surface.

**15.** The memory device of claim **11**, wherein the continuous recessed surface is laterally spaced from the protective dielectric liner by portions of the selector-level dielectric matrix layer that contact the bottom surfaces of the two-dimensional array of magnetic tunnel junction pillar structures.

**16.** The memory device of claim **11**, wherein the horizontally-extending portion of the protective dielectric liner contacts top surfaces of the first electrically conductive lines within a horizontal plane including interfaces between the first electrically conductive lines and the two-dimensional array of selector-containing pillar structures.

**17.** The memory device of claim **16**, further comprising first dielectric rails laterally extending along the first horizontal direction, interlaced with the first electrically conductive lines along the second horizontal direction, and contacting a bottom surface of the horizontally-extending portion of the protective dielectric liner.

**18.** The memory device of claim **11**, wherein sidewalls of the two-dimensional array of magnetic tunnel junction pillar structures have a greater taper angle relative to a vertical direction than sidewalls of the two-dimensional array of selector-containing pillar structures.

**19.** The memory device of claim **11**, wherein the protective dielectric liner comprises a dielectric material selected from silicon nitride, silicon oxynitride, silicon carbide nitride or metal oxide.

\* \* \* \* \*